

Water Resource Systems Planning and Management (2017)

Exercises and Solutions

Chapter 1.

1.1 How would you define ‘Integrated Water Resources Management’ and what distinguishes it from “Sustainable Water Resources Management”?

- Lots of possibilities here but some notion of multi-resource (land, water), multi-sector (agricultural, industrial, energy, transportation, educational, health, tourist), multi-objective (economic, environmental, social, political) regional comprehensive systems-wide (basin and coastal region) planning and management in a way that maximizes social welfare as well as protects and enhances our vital ecosystems.
- Key words: Balanced, holistic, integration, stakeholder participation, consultation, gender awareness, consensus, sustainability, economic value of water, demand management.
- Sustainability: Meeting the needs or objectives of current and future generations to the maximum extent possible without degrading the supporting hydrologic and ecosystems.

1.2 Can you identify some common water management issues that are found in many parts of the world?

- Resource availability and changes in availability associated with climate change.
- Water supply scarcity
- Water quality deterioration
- Competition for scarce resources and resulting conflicts
- Water extremes (both floods and droughts) increasing in magnitudes and frequencies.
- Urban – rural water supply and quality issues
- Security from natural and man-made hazards
- Infrastructure costs, deferred maintenance.
- Invasive species altering ecosystems and increasing control costs.

1.3 Comment on the common practice of governments giving aid to those in drought or flood areas without any incentives to alter land use management practices in anticipation of the next drought or flood.

- Continuing unrestricted land development leads to more damage or hardship when droughts or floods occur in the future. However benefits of development occur when not in a drought or flood condition.

1.4 What tools and information are available for developing integrated water resources plans and management policies?

- Legislative directives to achieve policies and goals

- Financing and economic incentive structures
- Institutional capacity building and continuing education.
- Demand management – using water more effectively and efficiently
- Conflict resolution methods
- Regulatory instruments
- Predictive hydrologic, economic, environmental, ecosystem modeling
- Communication via all available media including Internet
- Stakeholder participation
- Institutional development (e.g., watershed or river basin authorities).
- Monitoring and adaptive management
- Rationing and conservation measures in times of scarcity
- Flood warning systems and damage mitigation and evacuation policies

1.5 What structural and non-structural measures can be taken to address water resources issues?

- Reservoirs, canals, wells and pumps, weirs, wastewater treatment plants, irrigation systems, hydropower plants, diversion gates, instream aerators, monitoring gages, levees, river training works
- Land use zoning, flood and drought insurance, alternative agricultural practices, urban and golf course fertilizer, herbicide and pesticide management

1.6 Find the following statistics:

- Percent of all freshwater resources worldwide available for drinking: 1
- number of people who die each year from diseases associated with unsafe drinking water: 2.2 million
- Percent of total freshwater resources in polar regions: 97
- Per capita annual withdrawal of cubic meters of freshwater in US: 1,844
- Average world per capita annual withdrawal of cubic meters of freshwater: 644
- tons of pollutants entering lakes and rivers daily in various regions: 500,000
- Average number of gallons of water consumed by humans in a lifetime: 16,000
- Average number of kilometers per day a woman in a developing country must walk to fetch fresh water: 6

<http://www.pbs.org/now/science/unwater.html>

1.7 Identify and briefly describe the 6 greatest rivers in the world.

- o **The Amazon River** is the second longest river in the world, and the largest in terms of the size of its watershed, the number of tributaries, and the volume of water discharged into the sea. No bridge crosses the river along its entire length.

The Amazon and its tributaries flow through the countries of Peru, Bolivia, Venezuela, Colombia, Ecuador, and Brazil before emptying into the Atlantic Ocean 6,437 kilometers (4,000 miles) from the Amazon's headwaters high in the Andes Mountains of

Peru. This huge watershed includes the largest tropical rainforest in the world as well as areas of dry grassland, or savannah.

The rainforest's climate has heavy rainfalls and continuous high temperatures. These forests are so huge that scientists believe they actually help control the world's climate in important ways. Destruction of the forests as settlers clear the land for farming and companies harvest trees for lumber is believed to be contributing to the problem of global warming.

o **The Zambezi River**

The Bundu people of Zambia believe the Zambezi River has a spirit called Nyami. This spirit brings them water to grow crops and fish to eat so they call the river "the river of life". The Zambezi originates in Northwestern Zambia, formerly Rhodesia, and flows through Angola, Botswana, Zimbabwe, and finally to Mozambique. Here it empties into the Indian Ocean 1600 miles from its headwaters. The climate along the river is considered tropical and the rainy season lasts from October through April. The river flow changes at different spots along it, going from a placid flow in sandy plains to annually flooded plains to a tumbling river of falls and rapids. Perhaps the most spectacular spot along the river is in Zimbabwe at Victoria Falls. These falls are two times the size of Niagara Falls at 355 feet high and 5500 feet wide!

The Zambezi River is widely used by wildlife and humans alike. Hippos, crocodiles, baboons, elephants, hyenas and lions are some examples of wildlife you might find along the Zambezi. Humans use the river for transportation, irrigation, tourism and hydropower. The people who live along the river differ from country to country. English is the official language of the area though the different tribes along the banks of the Zambezi speak more than 70 other languages. Currently the countries along the river are at odds as to how to manage the river. In the future these countries must agree to manage "the river of life" so that its people can survive and its spirit can thrive.

- o **The Volga River** is the longest river in Europe. Its basin lies entirely within the Russian federation. It is often called Mother Volga by the Russians. It is thought of as the life blood of Russia. Over 40% of Russia's people live near it and its tributaries. Half of Russia's farmers live and farm near the Volga and over half of Russia's industry is located within its drainage. Because of the building of dams for hydroelectric power, the Volga is navigable for most of its length.

But the river has suffered because it cannot flush itself out and so it suffers from pollution. The Volga rises or starts northwest of Moscow in the Valdei Hills. From there it flows north above Moscow, and then south and southeast to the Caspian Sea. The Volga flows through the Russian autonomous regions of Churashia, Mari El, and Tartarstan. Its basin is broad and often swampy as it flows through low hills that are the leftover piles of rock and dirt (moraines) of glaciers.

Where it starts the weather is similar to that of the midwest U.S., cold winters and hot summers. But near its end in the Caspian Sea the weather is more like the southwest U.S., hot and dry. At the Caspian Sea the Volga is an important source of water for the sea and its famous sturgeon fishery. The Beluga sturgeon is the largest fish found in the Volga.

But the water that flows into the Caspian has been used many times upstream by the factories and the farmers.

- o **The Yangtze River** is the longest river in Asia and third longest in the world. The headwaters of the Yangtze are situated at an elevation of about 16,000 feet in the Kunlun Mountains in the southwestern section of Qinghai. It flows generally south through Sichuan into Yuanan then northeast and east across central China through Sichuan, Hubei, Anhui, and Jiangsu provinces to its mouth, 3,720 miles, in the East China Sea north of Shanghai. The river has over 700 tributaries but the principal tributaries are the Hun, Yalong, Jialing, Min, Tuo Jiang, and Wu Jiang. The climate ranges from 96 degrees in the summer to cold, brisk temperatures in the winter. Precipitation is high due to the height of the mountains.

China covers an area of almost 3.7 million square miles and is the most populous country on earth having more than one billion people. The bordering countries include Mongolia, Kazakhstan, India, Nepal, Burma, Vietnam and Korea. The People's Republic is led by Chinese Communist Party and governed by the State Council. While the main language spoken is the Mandarin dialect of Chinese, there are more than 150 languages and dialects spoken throughout the country. Approximately 350 million people inhabit the region of the Yangtze. There is plenty of wildlife along the river as well which include the Tibetan antelopes, Mongolian gazelles, and snow leopards, just to name a few.

Waters of the Yangtze are often used for rice and wheat irrigation. It also has enormous and inexhaustible hydroelectric resources. In 1995 construction began on the Three Gorges Dam near Yichang and is scheduled for completion in 2009. The dam will measure about 600 feet and about 1.5 miles wide. The dam is expected to help control the flooding of the Yangtze River Valley. The Three Gorges (the Qutang Gorge, the Wuxia Gorge, and Xiling Gorge) will also be the largest electricity generating facility in the world. The hydroelectric generators will provide 1/9 of China's total power output.

There is a lot of controversy surrounding the construction of the dams. There are 140 villages, 1,600 factories and over 80,000 acres of tillable land that will be destroyed. It may also cause animals to become extinct. The area is very prone to earthquakes because it lies directly on a fault line. Actually, the plateau and gorges were created by collision of the Indian-Australian plate, drifting northward, with the Eurasian plate that began over 40 million years ago and continues today.

The Thames is the river at the heart of English and British History. It is fairly short by world standards, only 220 miles long from its beginning at Thames Head to its end at the lighthouse at Nore. The Thames is actually the combination of four rivers, the Isis, the Churn, the Coln and the Leach. But along its banks are many royal residences of past and present English kings, Hampton Court, Windsor Castle and Richmond Park. As well there are many famous towns alongside its waters too, Oxford, Henley, Eton, London and Greenwich, home of the zero meridian of longitude. There are a series of locks up and down the Thames which made it an important highway for transporting goods up and down the river. The docks in south London, though they are falling apart now, once saw ships from all over the world bring exotic goods (silk, spices and others) from around the world in the holds of square-rigged sailing ships of the world's navies.

The country alongside the Thames is mostly rolling hills with farming and grazing being the main uses of the land until you get to London. The moderate climate of this part of England allows farms to be busy year around. The Thames is a tidal river in the city of London and below. With the tides ebbing and flowing four times a day, the danger from a giant storm surge or tide coming up the Thames and flooding the city of London caused the British government to build the Thames Barrier at Woolrich, the world's largest movable flood barrier. But a greater threat to the city of London and the Thames has been pollution. In the mid-1800s the stench was so bad that they closed the House of Parliament, and built the first sewage treatment plant in the city of London. The Thames has benefited greatly from river cleanup programs started in 1963.

- o **The Mississippi** is the centerpiece of the second largest watershed in the world, covering over 1.2 million square miles, and includes tributary rivers from 33 states and two Canadian provinces. It begins as a tiny brook and 2,350 miles later empties into the Gulf of Mexico. Millions of people each year use the Mississippi River for recreation, but the Mississippi is, and always has been a working river. An average of 175 million tons of freight is shipped each year on the Upper Mississippi. The 29 lock and dams on the Upper Mississippi make that shipping possible, allowing for navigation from St. Louis, Missouri, to Saint Paul, Minnesota, a total distance of 854 miles.

There are records of human habitation along the Mississippi river that date back more than five thousand years. Four thousand years ago, American Indians in the Lower Mississippi Valley began establishing communities with large, elaborate earthen architecture. Much later around 1000 A.D., larger and more elaborate complexes of mounds were constructed by a culture referred to as Mississippian. Typically, these towns contained anywhere from 1 to 20 mounds, which often were used as platforms for temples or the residence of leaders.

The Ojibwa Indians of northern Minnesota called it "Messy" or "Big River," and it was also known as the "Me-zee-see-bee" or the "Father of Waters." European explorers who mapped all the river's channels and backwater areas called it a "gathering of waters." The Native Americans of different tribes who originally lived near the Mississippi and used it for canoe transportation, hunting and fishing often viewed the great river as the center of the universe.

The Mississippi river basin was formed by glaciers, moving and melting millions of years ago, which left in their wake sometimes miles-wide floodplains. These floodplains still fill up occasionally, covering towns, roads, farms, and everything else that might stand in the way of its mighty waters.

- o **Ob River**, The combined Ob-Irtysh system, the third-longest river system of Asia (after China's Yangtze and Yellow rivers), is 5,410 km (3,362 mi) long. The largest river port is on the Irtysh in Omsk, with a link to the trans-Siberian Railway. With the Irtysh River, its chief tributary, it is c.3,460 mi (5,600 km) long and is the world's fourth longest river. Formed by the junction of the Biya and Katun rivers (both of which rise in the Altai range) SW of Biysk, the upper Ob flows NW, then NE past Barnaul and Novosibirsk through the W Siberian lowlands to be joined by the Tom River. The middle Ob flows northwest through the swampy forests in the Tomsk and Naryn regions and then is joined by the Chulym, Ket, and Irtysh rivers. The lower Ob consists of the Great Ob and the Small Ob and flows N, then E into Ob Bay, an estuary

and shallow arm (c.500 mi/800 km long and 35–50 mi/56–80 km wide) of the Arctic Ocean between the Yamal and Gyda peninsulas. The width of the Ob increases downstream to c.25 mi (40 km) near its mouth. The valley of the middle Ob is subject to flooding each spring as the thaw occurs in the upper Ob basin before the ice in the lower course of the Ob has melted. The Ob is used mostly for irrigation, drinking water, hydroelectric energy, and fishing; the river has more than 50 species of fish. Although frozen from five to six months of the year, the Ob is an important trade and transport route; Novosibirsk, Barnaul, Kamen-na-Obi, and Mogochin are the chief ports. There is a large hydroelectric power station at Novosibirsk. The largest oil and gas deposits in Russia are found in the basin of the middle and lower Ob. Severe pollution in the lower Ob has damaged the river's formerly famous fisheries. In the late 19th century, a system of canals, using the Ket River, 560 mi long in all, was built to connect the Ob with the Yenisei, but soon abandoned as being uncompetitive with the railway. The river basin of the Ob exists mostly of steppe, taiga, swamps, tundra, and semi-desert topography. The floodplains of the Ob are characterized by many tributaries and lakes. The Ob is ice-bound at southern Barnaul from early in November to near the end of April, and at northern Salekhard, 100 miles above its mouth, from the end of October to the beginning of June. Its middle reaches have been navigated by steamboats since 1845.

1.8. Identify some of the major water resource management issues in the region where you live. What management alternatives might effectively reduce some of the problems or provide additional economic, environmental, or social benefits.

There are many such systems operating where anyone lives. These can include water supply source, treatment and distribution systems, wastewater collection and treatment systems, non-point pollution control measures together with receiving water bodies, multiple-purpose reservoirs for water supplies, hydropower, recreation, and for providing flows for ecosystem enhancement or preservation and quality control, irrigation systems, navigation, etc.

1.9. Describe some water resource systems consisting of various interdependent natural, physical and social components. What are the inputs to the systems and what are their outputs? How did you decide what to include in the system and what not to include?

Components: Natural: Rivers, lakes, estuaries, watersheds, etc. Physical: Diversion infrastructure, reservoirs, hydropower plants, groundwater pumps, water and wastewater treatment facilities, etc. Social: Stakeholder costs, recreational opportunities, flood risk reduction, etc.

Inputs: Flows, pollutant loads, economic benefit and cost functions, etc. **Outputs:** Water quantity, quality, costs, benefits, design parameter values, operating policies, etc.

What to consider: Much depends on the decisions to be made or the issues being addressed, as well as the available data.

1.10. Sustainability is a concept applied to renewable resource management. In your words define what that means and how it can be used in a changing and uncertain environment both with respect to water supplies and demands. Over what space and time scales is it applicable, and how can one

decide whether or not some plan or management policy will be sustainable? How does this concept relate to the adaptive management concept?

Note: See text, especially Sections 4 and 5.5.

1.11. Identify and discuss briefly some of the major issues and challenges facing water managers today.

See text for examples as well as articles in the media regarding specific issues in specific areas. In general: water may be too little, too much, too polluted, or too expensive. In addressing these issues, structural and non-structural alternatives with respect to managing supplies, qualities and demands need to be considered.

A major issue is how to reach a stakeholder or political consensus on what should be done to address any specific problem when the issues are complex. How to best obtain the involvement and cooperation of stakeholders.

Chapter 2

2.1 What is a system?

A collection or set of interdependent components acting together to provide a product or service.

2.2. What is systems analysis?

The search for component designs and/or operating policies of a system that best satisfy one or more performance objectives or goals.

2.3. What is a mathematical model?

A mathematical representation of a system that can be used to evaluate system performance under alternative assumptions.

2.4 Why develop and use models?

Either to better understand the system and how it functions, or for predicting the performance of the system under alternative inputs and other assumptions, including designs and operating policies.

2.5 What is a decision support system?

A computer program or model having a graphics based interface that allows the user to interactively input and edit data, control the operations of the model, and display the outputs in meaningful ways.

In a broader sense, a DSS is any tool that provides information to a decision making process.

2.6 What is shared vision modeling and planning?

Involving stakeholders in model building and operation, giving them the sense of model ownership, and providing them a way to test their assumptions regarding input data and system parameter values, all in an effort to have them gain a better understanding of how their (modeled) system works and perhaps how they would like it to work. This is commonly done with generic decision support systems suited for the particular analyses needed.

2.7 What characteristics of water resources planning or management problems make them suitable for analysis using quantitative systems analysis techniques?

- A decision has to be made with regard to a problem or an issue or opportunity.
- There exists many possible ways to solve or address the problem or opportunity.
- The best alternative is not obvious.
- Many aspects of the problem can be quantified.

Many water resources planning problems involve interdependent structural and nonstructural components whose interactions and performance can be quantified and whose design and operation affect the performance of the entire water resource system. It should also be clear that there are often many possible designs and operating policies and the best one is not always obvious. In such situations systems analysis techniques, if applied properly, can help identify designs and operating policies that best meet performance goals or objectives.

2.8 Identify some specific water resource systems planning problems and for each problem specify in words possible objectives, the unknown decision variables whose values need to be determined, and the constraints or that must be met by any solution of the problem.

The possibilities here are enormous. This question is intended to have one describe in words some planning models for some typical water quantity/or quality management problems that will later be expressed mathematically. This is a first step in model building.

2.9 From a review of the recent issues of various journals pertaining to water resources and the appropriate areas of engineering, economics, planning and operations research, identify those journals that contain articles on water resources systems planning and analysis, and the topics or problems currently being discussed.

English language journals likely to contain the results of water resources systems analysis research or applications include Water Resources Research, the journals of the Hydraulic Engineering, Environmental Engineering, Irrigation and Drainage, and Water Resources Planning and Management Divisions of the American Society of Civil Engineers, Journal of the American Water Resources Association, Advances in Water Resources, the Journal of River Basin Management, Water Resources (translation of Russian), Journal of the American Water Works Association, Journal of Water Resources Management, Water International, Journal of Environmental Systems, Applied Mathematical Modeling, and various management science or operations research journals. A distinction should be made between studies involving mathematical modeling aimed at a better understanding of some physical, biological or chemical process or state as opposed to models for planning and policy (that may include the former but at a different level of detail). The latter are of primary interest in this book.

2.10. Many water resource systems planning problems involve considerations that are very difficult if not impossible to quantify, and hence they cannot easily be incorporated into any mathematical model for defining and evaluating various alternative solutions. Briefly discuss what value these admittedly incomplete quantitative models may have in the planning process when non-quantifiable aspects are also important. Can you identify some planning problems that have such intangible objectives?

Identifying tradeoffs among those quantitative aspects can be helpful even though there are qualitative aspects that may influence the decision. (This is typically the case in most water resources management decisions.)

Building a dam in a beautiful canyon where aesthetic considerations are a factor is one example. Another is the process itself of decision making. The process of coming to a decision can be even more important to some stakeholders than the decision itself.

One example could be the possibility of damming up the Grand Canyon for hydropower production (quantitative) and preserving its scenic value (qualitative). Quantitative models for the analysis and evaluation of alternative plans can provide decision makers with useful information concerning those quantifiable aspects of the system performance and the possible tradeoffs between system performance measures that are quantitative and constraints imposed due to qualitative considerations. Note that quantification is not limited to the use of only monetary units. (This question is not intended to generate discussion on quantitative multi-objective planning).

2.11 Define integrated water management and what that entails as distinct from just water management.

See Section 5. In today's world of increasing demand for fresh water sources, whether it be for drinking, irrigation, hydropower or industrial use, the need for efficient and intelligent planning of the available resources is becoming increasingly important. Do we have adequate and sustainable sources to meet all these demands in future as well as today? Are the available resources being effectively used among all the users? What other alternatives are available? How is the quality of the sources maintained? What are the plans for protecting against natural disasters such as droughts and floods? Are the natural states of water ecology and hydrology being adversely affected due to the human intervention and what are the actual causes? What are the social impacts of implementing a particular water resource project?

It is difficult to find one single answer for each of these questions as all are interconnected. Many other issues such as water rights, bureaucracy, and most of all the economic and political viability of the alternatives for management can make the planning effort more complex. A collective effort involving all the stakeholders is needed for both long and short-term planning.

Conventionally water resource management is thought to be a problem of engineering, hydrology and project construction. The focus in the past has been less multidisciplinary and less inclusive of all stakeholders. The integrated system approach requires taking a long-term basin-wide or regional view that considers multiple objectives and includes all the stakeholders in all the stages of the planning aimed at balancing trade-offs and finding sustainable, robust solutions.

2.12. Water resource systems serve many purposes and can satisfy many objectives. What is the difference between purposes and objectives?

Purposes include water supply, flood control, hydropower, navigation, recreation, wildlife habitat.

Objectives include economic benefits; environmental quality; reliability, resilience and vulnerability measures with respect to water flow, level, volume or quality targets.

2.13. How would you characterize the steps of a planning process aimed at solving a particular problem?

- Determine the objectives to be achieved.
- Determine the resources available to meet the objectives.
- Define alternative ways objectives can be accomplished.
- Determine the best alternative.
- Obtain agreement (a shared vision) among all interested stakeholders.

Note: There are feedbacks among all these steps.

2.14. Suppose you live in an area where the only source of water (at a reasonable cost) is from an aquifer that receives no recharge. Briefly discuss how you might develop a plan for its use over time.

Groundwater mining is common in many semi-arid regions of the world. The challenge is to determine the best rates of abstraction over time realizing that the resource is finite. One strategy is to attempt to equate the present value of marginal net benefits, however measured, over time, and hope that improvements in technology will make it possible to obtain water in other ways once the groundwater resource is depleted, but estimating future benefits is tough, and will always be uncertain, and thus the abstractions over time must clearly be adaptive to new knowledge and conditions.

Chapter 3

3.1 Briefly outline why multiple disciplines are needed to efficiently and effectively manage water resources in major river basins, or even in local watersheds.

No single discipline has all the knowledge needed to manage water within a region. Economists, ecologists, engineers, planners, lawyers, sociologists and anthropologists, and political scientists all have useful inputs to water resources planning and management.

3.2 Describe in a page or two what some of the water management issues are in the region where you live.

The intent here is to focus on the management aspects that are both technical, economic and social or political.

3.3 Define adaptive management, shared vision modeling, and sustainability.

Adaptive management involves changing management policies when it seems appropriate to do so based on information obtained from experiments designed to better define the impacts of particular management policies or practices and routine monitoring.

Shared vision modeling is the practice of involving all interested stakeholders in the development and use of models that address issues of concern to them. It is a way of getting stakeholders to trust the models being used and to get a consensus on how their water resource system works or functions, and hopefully, if possible, how they would like it to function.

Sustainability pertains to developing and managing our natural resources in ways that provide opportunities for those that succeed us in future generations to meet their needs and objectives as we strive to meet our own needs.

3.4 Distinguish what a manager does from what an analyst (modeler) does.

A manager manages people and the process of making decisions. An analyst provides information that should be useful in the process of making decisions.

3.5 Identify some typical or common water resources planning or management problems that are suitable for analysis using quantitative systems analysis techniques.

- Single and multiple purpose reservoir sizing and operation and impact prediction
- Wastewater treatment location, efficiency, impacts
- Groundwater development, wellfield location, pumping rates, artificial recharge rates
- Flood control and damage reduction, mix of structural and nonstructural measures
- Urban stormwater management
- Interbasin transfers, locations, amounts, canal or pipe and pumping capacities
- Hydropower capacity and production
- Water distribution systems and wastewater collection systems, capacities, pumping rates and capacities, reservoir or tank capacities
- Irrigation design and operation, drainage systems

3.6 Consider the following five alternatives for the production of energy (10^3 kwh/day) and irrigation supplies (10^6 m³/month):

Alternative	Energy Production	Irrigation Supply
A	22	20
B	10	35
C	20	32
D	12	21
E	6	25

Which alternative would be the best in your opinion and why? Why might a decision maker select alternative E even realizing other alternatives exist that can give more hydropower energy and irrigation supply?

The alternatives that are efficient in that you have to give up some benefits to get more of another are easily seen on an energy vs irrigation plot of these five alternatives. Alternatives A, C, and B are efficient. Based on these two types of benefits, the selection would be one of these three alternatives. If either D or E are chosen, clearly other objectives are being considered.

3.7 Define a model similar to Equations 3.1 to 3.3 for finding the dimensions of a cylindrical tank that minimizes the total cost of storing a specified volume of water. What are the unknown decision variables? What are the model parameters? Develop an iterative approach for solving this model.

Minimize Totalcost

Subject to:

$$\text{Totalcost} = \text{Sidecost} + \text{Topcost} + \text{Bottomcost}$$

$$\text{Sidecost} = C_s 2 (\pi r) r h$$

$$\text{Topcost} = C_t (\pi r^2)$$

$$\text{Bottomcost} = C_b (\pi r^2)$$

$$(\pi r^2) h > \text{Volume}$$

Of course the number of variables in this model can be reduced by omitting the first four definitional constraints so there are just r and h as unknown variables. Decision variables are r and h .

Model parameters are π (approximately 22/7), the costs per unit area for side (C_s), for top (C_t) and for base (C_b) and Volume. The values of these parameters have to be defined.

One way to find a good solution is to define an increment of r , Δr , and an increment of h , Δh . Then determine which increment to add to the total r or h already existing that has the greatest marginal volume increase per marginal cost increase, $d\text{Volume}/d\text{Totalcost}$ at the existing values for r and h . Alternatively, one can add incremental values of r and h so as to keep the side cost as close to 2/3rds of the total cost as possible, but this fact is not generally known. Keep adding increments until the volume constraint is satisfied.

The least-cost values of r and h can be determined from the following equations (found using Lagrange multipliers discussed in Chapter 4):

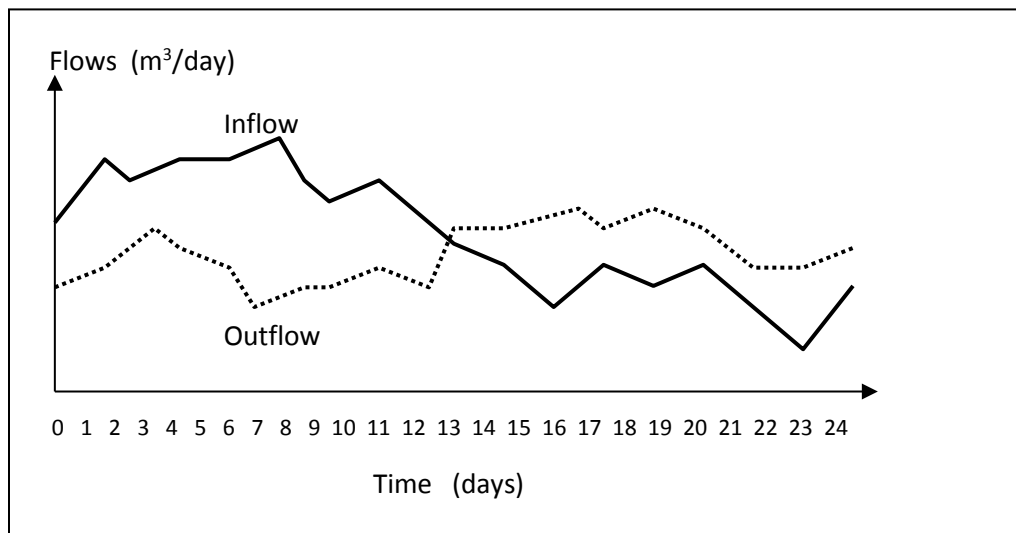
$$r = [(\text{Volume}) (C_s) / ((p_i) (C_b + C_t))]^{1/3}$$

$$h = [(\text{Volume}) (C_b + C_t)^2 / ((p_i) (C_s)^2)]^{1/3}$$

3.8 Briefly distinguish between simulation and optimization.

Simulation is used for determining system performance associated with specified values of all model variables and parameters. Optimization is often used for the preliminary screening of alternatives to determine a set of good decision variable values that can then be simulated to determine more precisely the system performance.

3.9 Consider a tank, a lake or reservoir or an aquifer having inflows and outflows as shown in the graph below.



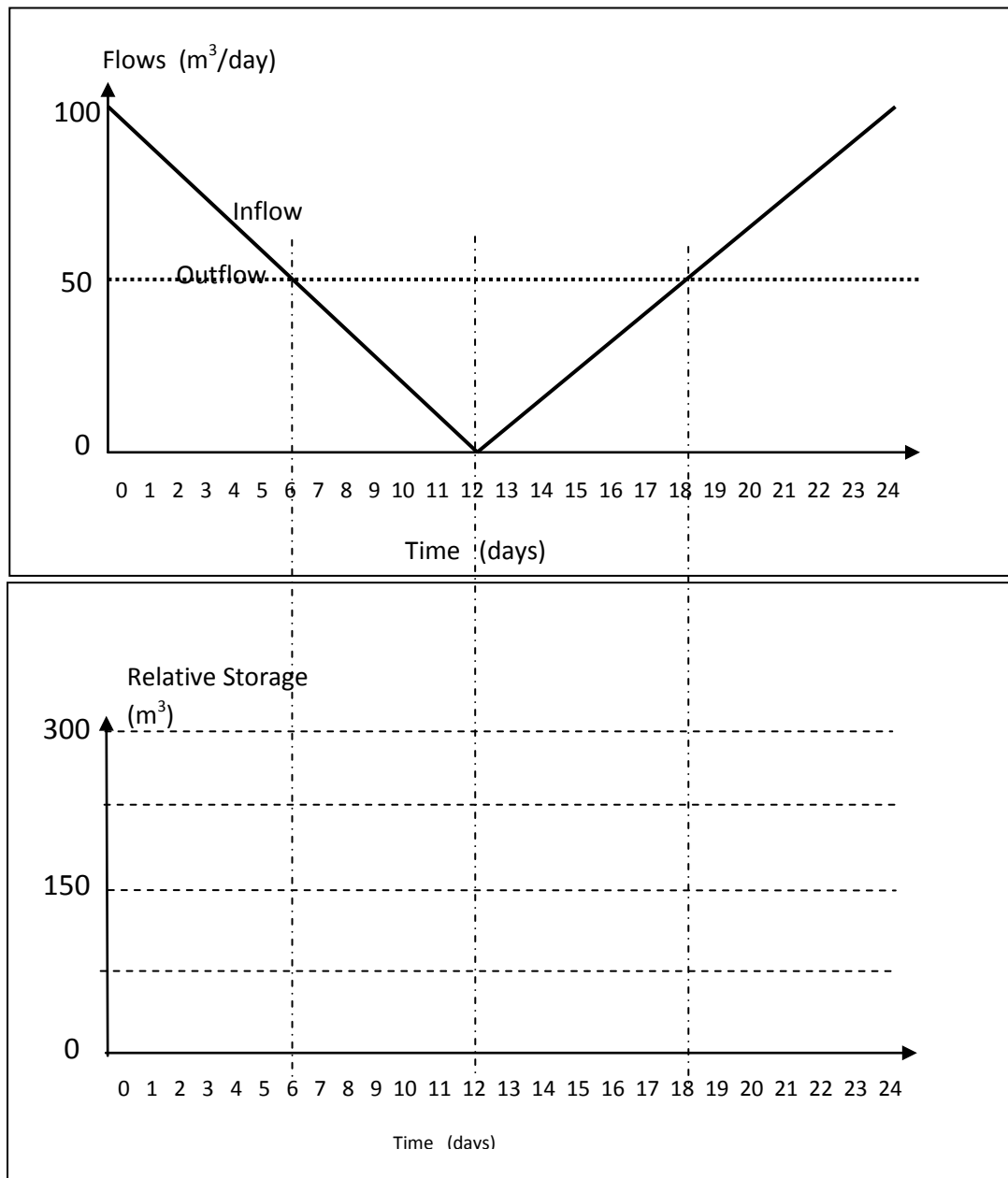
- When was the inflow its maximum and minimum values?
 - When was the outflow its minimum value?
 - When was the storage volume its maximum value?
 - When was the storage volume its minimum value?
 - Write a mass balance equation for the time series of storage volumes assuming constant inflows and outflows during each time period.
- Maximum at about day 7, minimum between days 22 and 23.
 - Minimum at about day 6 or between days 6 and 7.
 - Maximum at about day 12 or between days 12 and 13.
 - Minimum at day 0 since the net inflow is greater than the net outflow just from observing the net inflow and outflow areas between the two graphs.
 - Let S_t represent the initial storage volume in period t . (Note: it cannot represent the storage volume during period t , since that volume will change during the period

anytime the inflow and outflow values differ.) Let Q_t^{in} represent the inflow and Q_t^{out} represent the outflow during period t . The mass balance equation is:

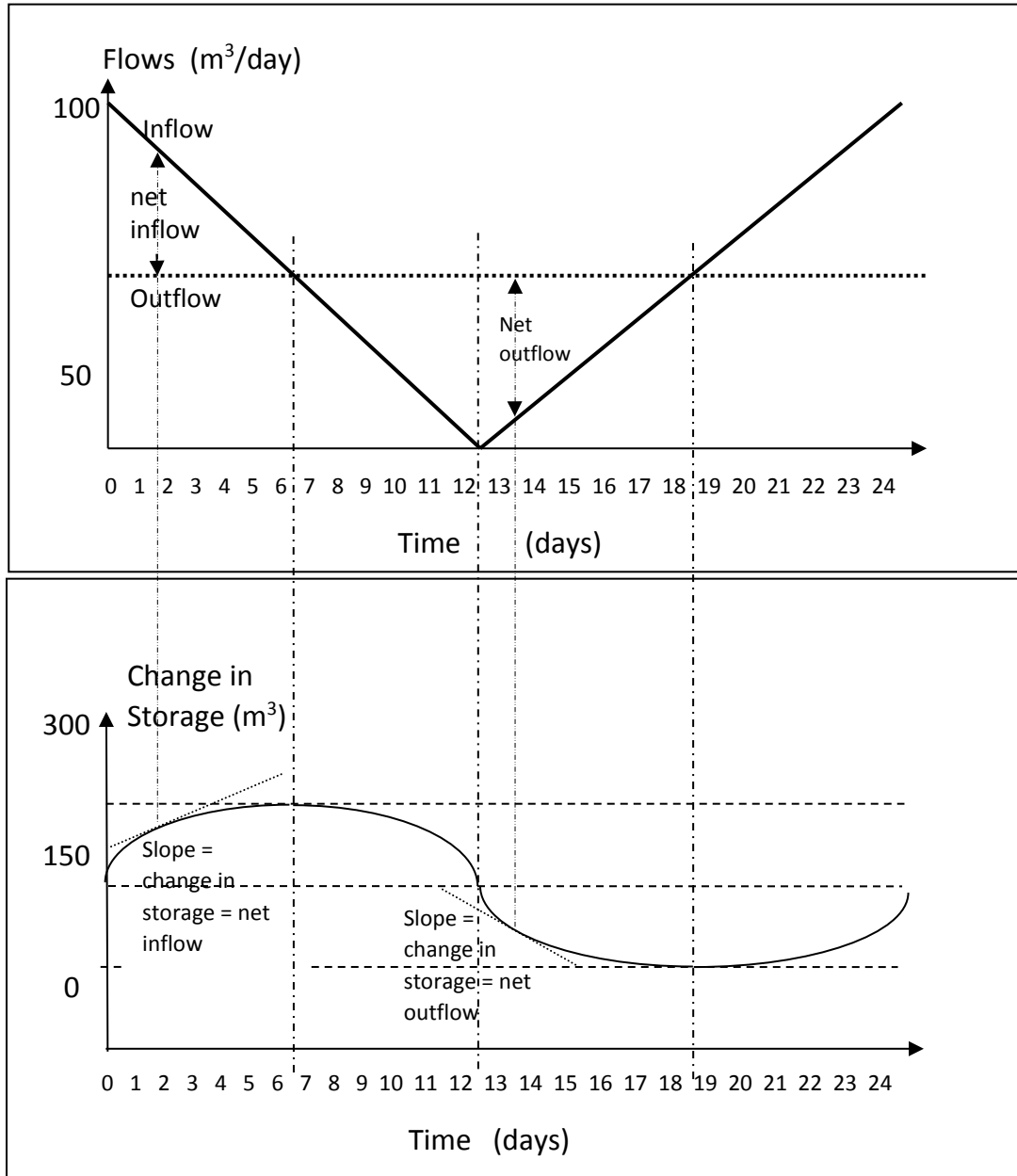
$$S_t + Q_t^{\text{in}} - Q_t^{\text{out}} = S_{t+1}$$

since the final storage in period t equals the initial storage in period $t+1$.

3.10 Given the changing inflows and constant outflow from a tank or reservoir, as shown in the graph below, sketch a plot of the storage volumes over the same period of time, beginning at 150. Show how to determine the value of the slope of the storage volume plot at any time from the inflow and outflow ($= 50 \text{ m}^3/\text{day}$) graph below.



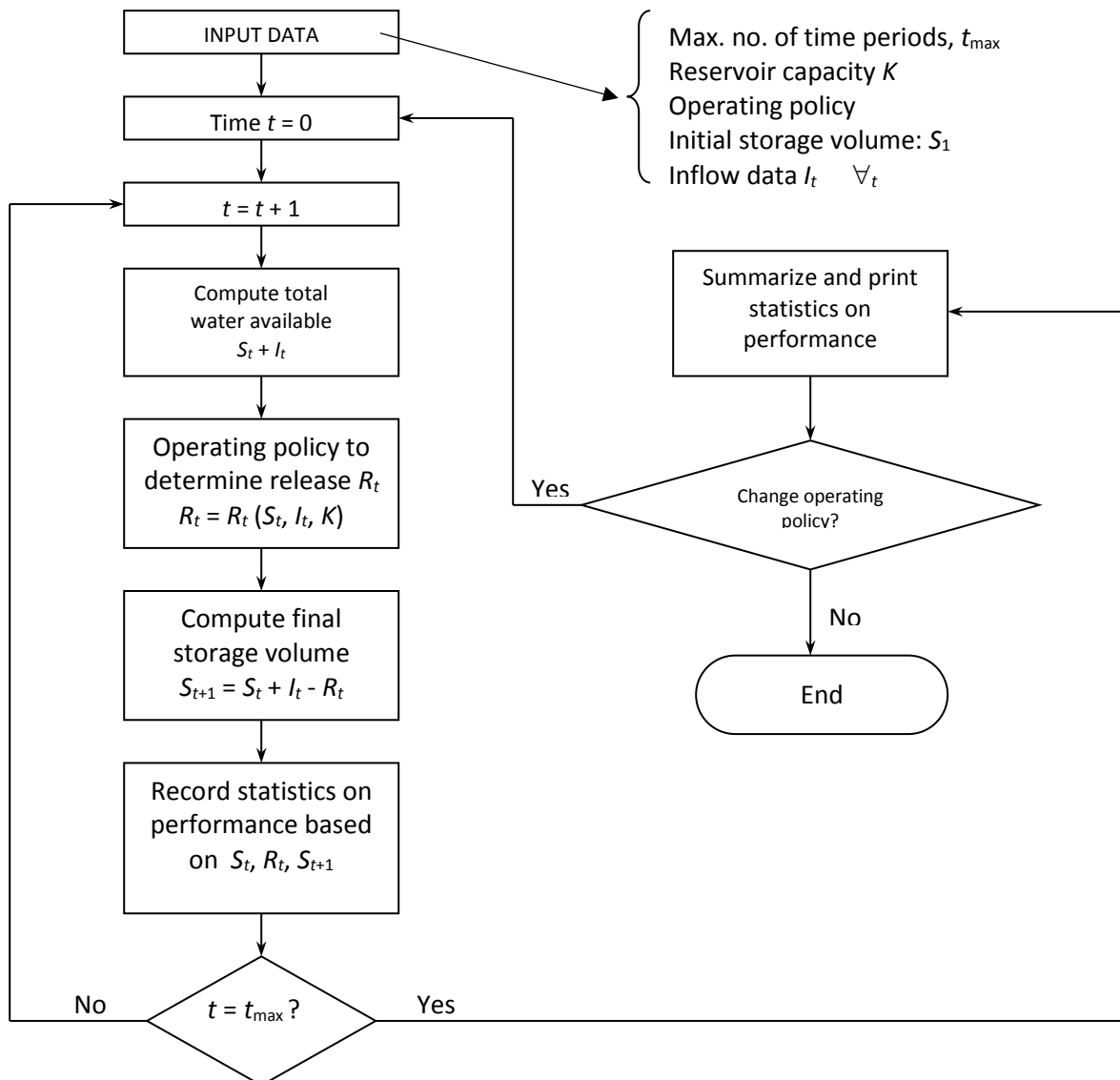
Solution:



Note: During periods 1 through 6 the inflow is greater than the outflow. The net amount entering the tank or reservoir is $(0.5)(100 - 50)(6) = 150$ cubic meters. The outflow exceeds the inflow in the next 12 periods. The net outflow is $(0.5)(50 - 0)(12) = 300$ cubic meters. In periods 19 through 24 the inflow again exceeds the outflow, and a total of 150 cubic meters is added to the storage. The same results can be found using calculus.

3.11 Describe, using words and a flow diagram, how you might simulate the operation of a storage reservoir over time. To simulate a reservoir, what data do you need to have or know?

Simulate over a series of time periods the mass balance that takes place in a reservoir, making sure the storage volume does not exceed the reservoir capacity. This requires knowledge of the inflows in each time period, the reservoir operating policy (how much to release as a function of storage volume and time period) and the performance measures that are functions of storage volumes and/or releases.



3.12 Identify and discuss a water resources planning situation that illustrates the need for a combined optimization-simulation study in order to identify the best plan and its impacts.

Any problem in which there would be a very large number of alternatives that could be simulated. The issue is what to simulate when the simulation of all reasonable alternatives would be too time consuming.

3.13 Write a flow chart/computer simulation program for computing the maximum yield of water that can be obtained given any value of active reservoir storage capacity, K , using.

Year y	Flow Q_y	Year y	Flow Q_y
1	5	9	3
2	7	10	6
3	8	11	8
4	4	12	9
5	3	13	3
6	3	14	4
7	2	15	9
8	1		

Find the values of the storage capacity K required for yields of 2, 3, 3.5, 4, 4.5 and 5.

There are many possible ways of doing this. Perhaps the simplest is to assume a value of K and an initial storage capacity of K , and then just find the maximum discrete value of the yield that can be obtained from a simulation of at least two cycles through the hydrologic record. The release policy will be to release the yield only in each period, unless a spill is necessary and then only that which is necessary, leaving a full reservoir. As K is increased, the simulation can begin with the yield that was just obtained for the smaller value of K .

For a yield of 2, $K = 1$; 3, 3; 3.5, 5.5; 4, 8; 4.5, 11; 5, 14.

3.14 How many different simulations of a water resource system would be required to ensure that there is at least a 95% chance that the best solution obtained is within the better 5% of all possible solutions that could be obtained? What assumptions must be made in order for your answer to be valid? Can any statement be made comparing the value of the best solution obtained from the all the simulations to the value of the truly optimal solution?

Assuming n independent simulations whose decision variables have been selected at random, the probability that all are outside the 5% range would be 0.95^n . The probability that at least one simulation is within the 5% range is $1 - 0.95^n$. Thus n must be such that $1 - 0.95^n \geq 0.95$. $n = 59$.

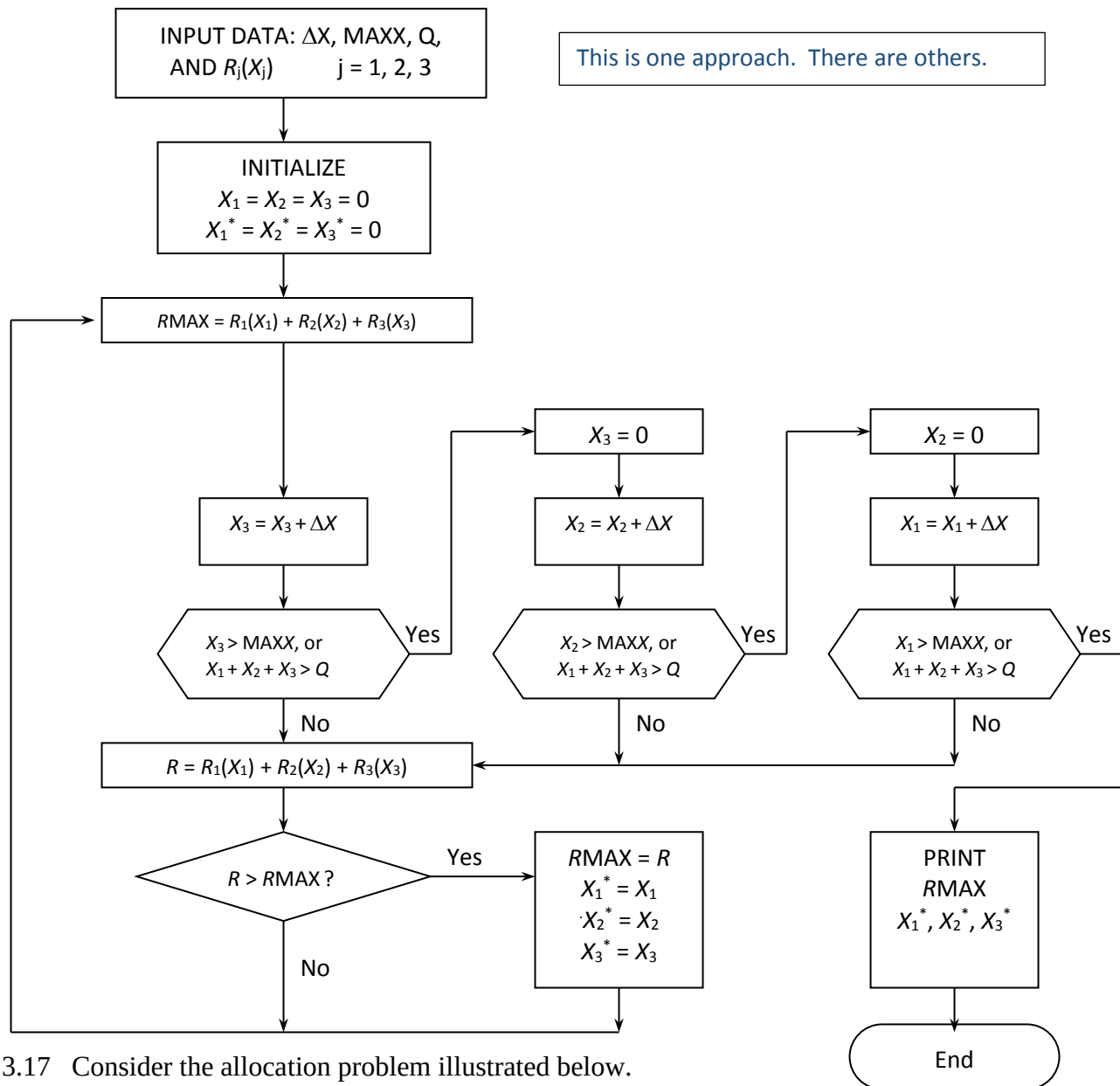
There is no way to determine just how far the best solution is to the true optimal one, unless optimization is used. Note: Creating random feasible values of all decision variables of large simulation problems is not very easy. Hence this approach is rarely used.

3.15 Assume in a particular river basin 20 development projects are being proposed. Assume each project has a fixed capacity and operating policy and it is only a question of which of the 20 projects would maximize the net benefits to the region. Assuming 5 minutes of computer time is

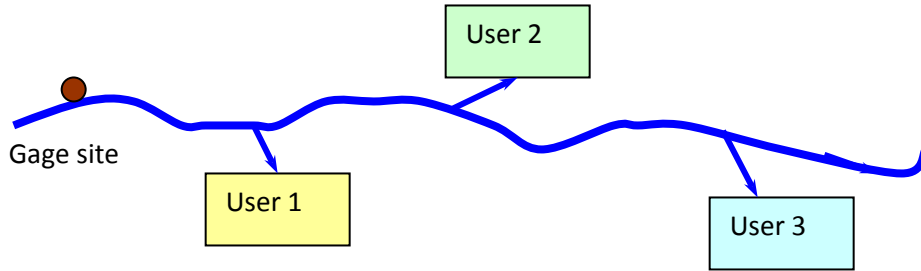
required to simulate and evaluate each combination of projects, show that it would require 36 days of computer time even if 99% of the alternative combinations could be discarded using “good judgment.” What does this suggest about the use of simulation for regional interdependent multi-project water resources planning?

Need some way to narrow down the number of simulations needed. Optimization is one approach for doing this. Chapter IV illustrates an approach to this preliminary screening procedure.

3.16 Assume you wish to determine the allocation of water X_j to three different users j , who obtain benefits $R_j(X_j)$. The total water available is Q . Write a flow chart showing how you can find the allocation to each user that results in the highest total benefits.



3.17 Consider the allocation problem illustrated below.

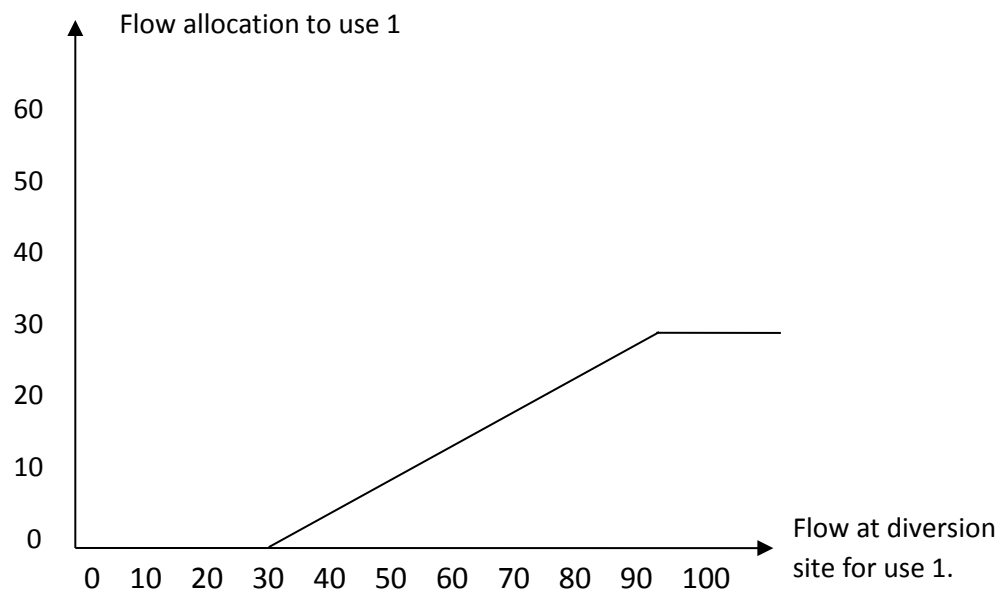


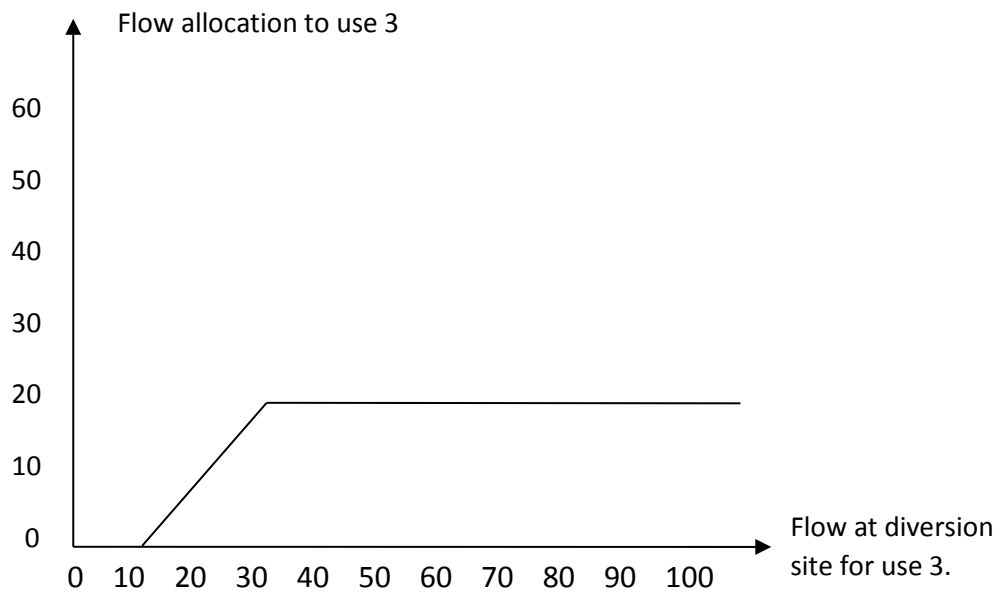
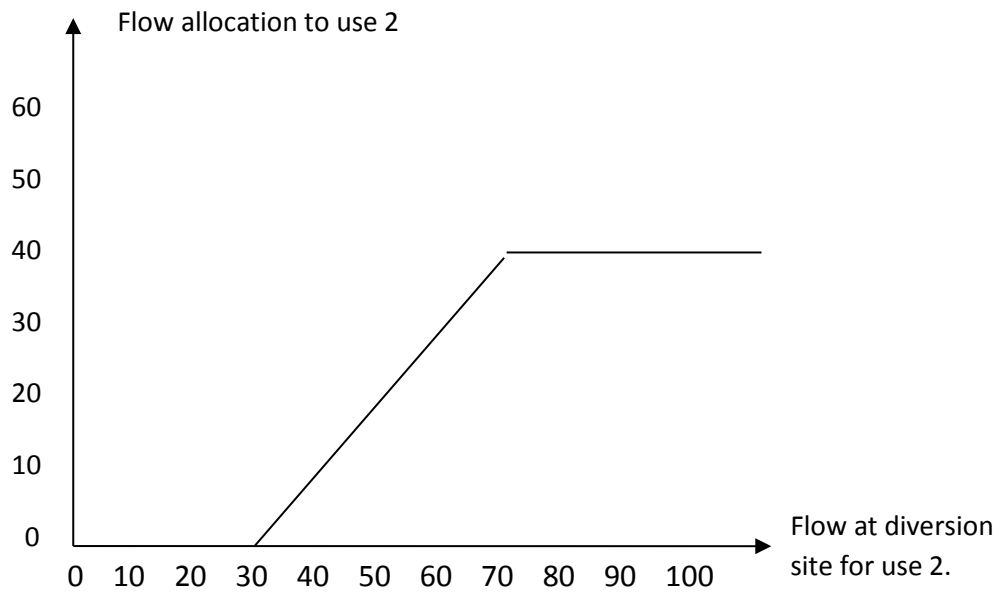
The allocation priority in each simulation period t is:

First 10 units of streamflow at the gage remain in the stream.
 Next 20 units go to User 3.
 Next 60 units are equally shared by Users 1 and 2.
 Next 10 units go to User 2.
 Remainder goes downstream.

- a) Assume no incremental flow along the stream and no return flow from users. Define the allocation policy at each site. This will be a graph of the allocation as a function of the flow at the allocation site.

Solution.





- b) Simulate this allocation policy using any river basin simulation model such as RIBASIM, WEAP, Modsim, or other selected model, including your own, for any specified inflow series ranging from 0 to 130 units.

Chapter 4

Engineering economics:

4.1 Consider two alternative water resource projects, A and B. Project A will cost \$2,533,000 and will return \$1,000,000 at the end of 5 years and \$4,000,000 at the end of 10 years. Project B will cost \$4,000,000 and will return \$2,000,000 at the end of 5 and 15 years, and another \$3,000,000 at the end of 10 years. Project A has a life of 10 years, and B has a life of 15 years. Assuming an interest rate of 0.1 (10%) per year:

- (a) What is the present value of each project?
- (b) What is each project's annual net benefit?
- (c) Would the preferred project differ if the interest rates were 0.05?
- (d) Assuming that each of these projects would be replaced with a similar project having the same time stream of costs and returns, show that by extending each series of projects to a common terminal year (e.g., 30 years), the annual net benefits of each series of projects will be same as found in part (b).

$$\begin{aligned} \text{(a) Present Value of A} &= \{-2.533 + 1.0(1.10)^{-5} + 4.0(1.10)^{-10}\}10^6 \\ &= -369,900 \end{aligned}$$

$$\begin{aligned} \text{Present Value of B} &= \{-4.0 + 2.0[(1.10)^{-5} + (1.10)^{-15}] + 3.0(1.10)^{-10}\}10^6 \\ &= -1,122,700 \end{aligned}$$

$$\text{(b) Net Annual Benefit of A} = (-370,000)\left(\frac{0.1(1.1)^{10}}{(1.1)^{10} - 1}\right) = -60,200$$

$$\text{Net Annual Benefit of B} = (-1,123,000)\left(\frac{0.1(1.1)^{15}}{(1.1)^{15} - 1}\right) = -147,600$$

$$\text{(c) Present Value of A} = \{-2.533 + 1.0(1.05)^{-5} + 4.0(1.05)^{-10}\}10^6 = 706,200$$

$$\begin{aligned} \text{Present Value of B} &= \{-4.0 + 2.0[(1.05)^{-5} + (1.05)^{-15}] + 3.0(1.05)^{-10}\}10^6 \\ &= 370,800 \end{aligned}$$

$$\text{Net Annual Benefit of A} = (706,200)\left(\frac{0.05(1.05)^{10}}{(1.05)^{10} - 1}\right) = 91,500$$

$$\text{Net Annual Benefit of B} = (370,800)\left(\frac{0.05(1.05)^{15}}{(1.05)^{15} - 1}\right) = 35,700$$

Preferred project is A at 5% interest. Neither is preferred at 10% interest, but A is preferable to B.

(d) Payment of 369,900 at beginning of years 1, 11 and 21 for A has a present value of –567,500, which in turn has a 30-year net annual benefit of –60,200 (same as (b) above). Payments of 1,122,700 at the beginning of years 1 and 16 have a present value of –

1,391,500, which in turn has a 30-year net annual benefit of −147,600 (same as (b) above).

4.2 Show that $A \sum_{t=1}^T (1+r)^{-t} = \frac{(1+r)^T - 1}{r(1+r)^T} A$, the present value of a series of equal payments, A , at the end of each year for T years. What is the impact of an increasing interest rate over time on the present value?

$$\text{To show that } \sum_{t=1}^T (1+r)^{-t} = \frac{(1+r)^T - 1}{r(1+r)^T}.$$

$$\sum_{t=1}^T (1+r)^{-t} = \frac{1}{1+r} + \left(\frac{1}{1+r}\right)^2 + \dots + \left(\frac{1}{1+r}\right)^T \quad (\text{a})$$

$$\left(\frac{1}{1+r}\right) \left(\sum_{t=1}^T (1+r)^{-t}\right) = \left(\frac{1}{1+r}\right)^2 + \dots + \left(\frac{1}{1+r}\right)^T + \left(\frac{1}{1+r}\right)^{T+1} \quad (\text{b})$$

subtracting (b) from (a)

$$\sum_{t=1}^T (1+r)^{-t} \left(1 - \frac{1}{1+r}\right) = \frac{1}{1+r} - \left(\frac{1}{1+r}\right)^{T+1} = \frac{1}{1+r} \left(1 - \left(\frac{1}{1+r}\right)^T\right)$$

$$\sum_{t=1}^T (1+r)^{-t} = \frac{1}{1+r} \frac{\left(\frac{(1+r)^T - 1}{(1+r)^T}\right)}{\left(\frac{1+r-1}{1+r}\right)}$$

$$= \frac{(1+r)^T - 1}{r(1+r)^T}$$

Increasing the interest rate over time decreases the present value.

4.3 a) Show that if compounding occurs at the end of m equal length periods within a year in which the nominal interest rate is r , then the *effective* annual interest rate, r' , is equal to

$$r' = \left(1 + \frac{r}{m}\right)^m - 1$$

Consider one year. $1+r' = (1+r/m)(1+r/m)(1+r/m)\dots(1+r/m) =$ the product of m terms. Compounding takes place at the end of each period for m periods at a within-year period interest rate of r/m . Hence $1+r' = (1+r/m)^m$

a) Show that when compounding is continuous (i.e., when the number of periods $m \rightarrow \infty$), the compound interest factor required to convert a present value to a future value in year T is e^{rT} . [Hint: Use the fact that $\lim_{k \rightarrow \infty} (1 + 1/k)^k = e$, the base of natural logarithms.]

The compound interest factor for computing in m periods per year for T years is

$$\left(1 + \frac{r}{m}\right)^{mT}$$

Letting $k = m/r$ and noting that $k \rightarrow \infty$ as $m \rightarrow \infty$, $\lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^{krT} = e^{rT}$

4.4 The term “capitalized cost” refers to the present value PV of an infinite series of end-of-year equal payments, A . Assuming an interest rate of r , show that as the terminal period $T \rightarrow \infty$, $PV = A/r$.

$$\lim_{T \rightarrow \infty} A \left(\frac{(1+r)^T - 1}{r(1+r)^T} \right) = \lim_{T \rightarrow \infty} A \left(\frac{1}{r} - \frac{1}{r(1+r)^T} \right) = \frac{A}{r} = \text{Present value of annual amount paid } A \text{ at the end of every year forever, assuming an interest rate of } r.$$

4.5 The *internal rate of return* of any project or plan is the interest rate that equals the present value of all receipts or income with the present value of all costs. Show that the internal rate of return of projects A and B in Exercise 4.1 are approximately 8% and 6%, respectively. These are the interest rates r , for each project, that essentially satisfy the equation

$$\sum_{t=0}^T (R_t - C_t)(1+r)^{-t} = 0$$

A: Present Value of 0 = $\{-2.533 + 1.0(1+r)^{-5} + 4.0(1+r)^{-10}\}$: $r = 0.08$

B: Present Value of 0 = $\{-4.0 + 2.0[(1+r)^{-5} + (1+r)^{-15}] + 3.0(1+r)^{-10}\}$: $r = 0.06$

4.6 In Exercise 4.1, the maximum annual benefits were used as an economic criterion for plan selection. The maximum benefit-cost ratio, or annual benefits divided by annual costs, is another criterion. Benefit-cost ratios should be no less than one if the annual benefits are to exceed the annual costs. Consider two projects, I and II:

	Project	
	I	II
Annual benefits	20	2
Annual costs	18	1.5
Annual net benefits	2	0.5
Benefit-cost ratio	1.11	1.3

What additional information is needed before one can determine which project is the most economical project?

If there are funds available for the more expensive project, then the return from investing the funds remaining should the cheaper project be selected must be known before either project can be identified as the preferred one. Annual benefit-cost ratios, or net benefits, can be used interchangeably to evaluate alternative investment plans only if the total costs are the same. If the costs are the same, the plan having the largest benefit-cost ratio will also have the largest net benefits.

4.7 Bonds are often sold to raise money for water resources project investments. Each bond is a promise to pay a specified amount of interest, usually semiannually, and to pay the face value of the bond at some specified future date. The selling price of a bond may differ from its face value. Since the interest payments are specified in advance, the current market interest rates dictate the purchase price of the bond.

Consider a bond having a face value of \$10,000, paying \$500 annually for 10 years. The bond or “coupon” interest rate based on its face value is $500/10,000$, or 5%. If the bond is purchased for \$10,000, the actual interest rate paid to the owner will equal the bond or “coupon” rate. But suppose that one can invest money in similar quality (equal risk) bonds or notes and receive 10% interest. As long as this is possible, the \$10,000, 5% bond will not sell in a competitive market. In order to sell it, its purchase price has to be such that the actual interest rate paid to the owner will be 10%. In this case, show that the purchase price will be \$6927.

$$\$6927 = 500 \left[\frac{(1.10)^{10} - 1}{0.10(1.10)^{10}} \right] + \frac{10,000}{(1.10)^{10}}$$

The interest paid by some bonds, especially municipal bonds, may be exempt from state and federal income taxes. If an investor is in the 30% income tax bracket, for example, a 5% municipal tax-exempt bond is equivalent to about a 7 % taxable bond. This tax exemption helps reduce local taxes needed to pay the interest on municipal bonds, as well as providing attractive investment opportunities to individuals in high tax brackets.

Lagrange Multipliers

4.8 What is the meaning of the Lagrange multiplier associated with the following model?

$$\begin{array}{ll} \text{Maximize} & \text{Benefit}(X) - \text{Cost}(X) \\ \text{Subject to:} & X \leq 23 \end{array}$$

The Lagrange multiplier is the change in the objective function [Benefit(X) – Cost(X)] associated with a unit change in 23. If the constraint is binding it is the slope of the objective function [Benefit(X) – Cost(X)] at $X = 23$. Otherwise it is 0. It is also called a dual variable, dual price or shadow price.

4.9 Assume water can be allocated to three users. The allocation, x_j , to each use j provides the following returns: $R(x_1) = (12 x_1 - x_1^2)$, $R(x_2) = (8 x_2 - x_2^2)$ and $R(x_3) = (18 x_3 - 3 x_3^2)$. Assume that the objective is to maximize the total return, $F(X)$, from all three allocations and that the sum of all allocations cannot exceed 10.

- How much would each use like to have?
- Show that at the maximum total return solution the marginal values, $\partial(R(x_j))/\partial x_j$, are each equal to the shadow price or Lagrange multiplier (dual variable) λ associated with the constraint on the amount of water available.

c) Finally, without resolving a Lagrange multiplier problem, what would the solution be if 15 units of water were available to allocate to the three users and what would be the value of the Lagrange multiplier?

a) If each x_j is unrestricted, optimal allocation = 6, 4 and 3 for $j=1, 2$, and 3 respectively.

b) If $\sum_{j=1}^3 x_j \leq 10$, optimal allocations are 33/7, 19/7, 18/7 respectively and

$$\lambda = 18/7 = \frac{dF}{dx_j} \Big|_{x_j, optimal} \text{ where } F \text{ is the original objective function.}$$

c) If $\sum_{j=1}^3 x_j \leq 15$, then the solution is 6, 4, and 3 with $\lambda = 0$ since $6 + 4 + 3 \leq 15$.

This should be obvious from the first part of this exercise so no computational work needed. Excess resource is available.

4.10 In Exercise 4.9, how would the Lagrange multiplier procedure differ if the objective function, $F(\mathbf{X})$, were to be minimized?

There is no difference in the procedure for solving Lagrange multiplier problem. Procedure is identical for maximization and minimization, therefore boundary values and second conditions must be checked to see if the solution is a maximum or minimum.

4.11 Assume that the objective was to minimize the sum of squared deviations of the actual allocations x_j from some desired or known target allocations T_j . Given a supply of water Q less than the sum of all target allocations T_j , structure a planning model and its corresponding Lagrangian. Will a global minimum be obtained from solving the partial differential equations derived from the Lagrangian? Why?

$$\text{Minimize } \sum_{j=1}^3 (T_j - x_j)^2$$

$$\text{Subject to } \sum_{j=1}^3 x_j \leq Q$$

Since $\sum_j T_j > Q$, all of Q will be used, hence

$L = \sum_j (T_j - x_j)^2 - \lambda(\sum_j x_j - Q)$ and a global solution will result since the squared deviation functions are convex and suitable for minimization.

Differentiating the equation with respect to each of the $n+1$ unknowns, and setting the resulting equations to 0 yields:

$$\frac{\partial L}{\partial x_j} = 2(x_j - T_j) - \lambda = 0 \text{ for all } n \text{ } j.$$

$$\frac{\partial L}{\partial \lambda} = \sum_{j=1}^n x_j - Q = 0$$

$$\text{The optimal solution is: } \begin{cases} x_j = \frac{Q + nT_j - \sum_{j=1}^n T_j}{n} \\ \lambda = \frac{2}{n} \left(Q - \sum_{j=1}^n T_j \right) \end{cases} \text{ for } j = 1, 2, \dots, n.$$

The solution will be global since the functions are convex for minimization.

4.12 Using Lagrange multipliers, prove that the least-cost design of a cylindrical storage tank of any volume $V > 0$ has one-third of its cost in its base and top and two-thirds of its cost in its side, regardless of the cost per unit area of its base or side. (It is these types of rules that end up in handbooks in engineering design.)

$$\text{Min total cost} = 2C_s \pi r h + C_b \pi r^2 = C$$

$$\text{Subject to } \pi r^2 h \leq V$$

$$\frac{\partial L}{\partial r} = 0 = 2C_s \pi h + 2C_b \pi r - \lambda 2\pi r h = C_s h + C_b r - \lambda r h \quad (a)$$

$$\frac{\partial L}{\partial h} = 0 = 2C_s \pi r - \lambda \pi r^2 = 2C_s - r \lambda \text{ therefore } \lambda = 2C_s/r \quad (b)$$

$$\frac{\partial L}{\partial \lambda} = 0 = \pi r^2 h - V \quad (c)$$

Combining (a) and (b)

$$0 = C_s h + C_b r - 2C_s h = C_b r - C_s h$$

$$\text{Therefore } r = C_s h / C_b \text{ and } h = C_b r / C_s \quad (d)$$

Substituting either of these relationships into the ratio:

$$\text{Cost of Side/Total Cost} = \frac{C_s 2\pi r h}{C_s 2\pi r h + C_b \pi r^2} = \frac{2}{3}$$

or

$$\text{Cost of Base/Total Cost} = \frac{C_b \pi r^2}{C_s 2\pi r h + C_b \pi r^2} = \frac{1}{3}$$

To obtain an explicit solution, note from (c) and (d) that

$$r^2 = V/\pi h = C_s V / (\pi C_v r)$$

hence

$$r = \sqrt[3]{C_s V / \pi C_b}$$

and

$$h = V/\pi r^2 = C_b^2 V / (\pi C_s^2 h^2)$$

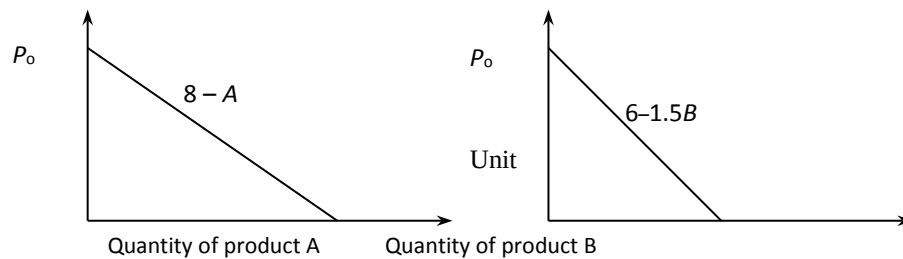
hence

$$h = \sqrt[3]{C_b^2 V / \pi C_s^2}$$

and

$$\lambda = 2C_s / r = 2C_s / \sqrt[3]{C_s V / \pi C_b}$$

4.13 An industrial firm makes two products, A and B. These products require water and other resources. Water is the scarce resource—they have plenty of other needed resources. The products they make are unique, and hence they can set the unit price of each product at any value they want to. However experience tells them that the higher the unit price for a product, the less amount of that product they will sell. The relationship between unit price and quantity that can be sold is given by the following two demand functions. Assume for simplicity that the unit price for product A is $(8 - A)$ and for product B is $(6 - 1.5B)$.



- (a) What are the amounts of A and B, and their unit prices, that maximize the total revenue obtained?
- (b) Suppose the total amount of A and B could not exceed some amount $T^{\max}$. What are the amounts of A and B, and their unit prices, that maximize total revenue, if

i) $T^{\max} = 10$

ii) $T^{\max} = 5$

Water is needed to make each unit of A and B. The production functions relating the amount of water X_A needed to make A, and the amount of water X_B needed to make B, are $A = 0.5 X_A$, and $B = 0.25 X_B$, respectively.

- (c) Find the amounts of A and B and their unit prices that maximize total revenue assuming the total amount of water available is 10 units.
- (d) What is the value of the dual variable, or shadow price, associated with the 10 units of available water?

Solutions:

- (a) Total Revenue Functions:

$$TR_A = (8 - A)A = 8A - A^2$$

$$TR_B = (6 - 1.5B)B = 6B - 1.5B^2$$

Taking the derivatives of equations and setting them equal to zero to maximize revenue:

$$dTR_A/dA = 8 - 2A = 0 \quad \text{or } A = 4;$$

$$dTR_B/dB = 6 - 3B = 0 \quad \text{or } B = 2;$$

The unit prices therefore are:

$$P_A = 8 - A = 4$$

$$P_B = 6 - 1.5B = 2$$

So the total revenue

$$TR = TR_A + TR_B = 22$$

- (b) In this case (i) $A + B \leq T^{\max} = 10$
(ii) $A + B \leq T^{\max} = 5.0$

The solution for A and B would be same as (b) in case of (i) since

$$A + B = 6.0 \leq 10$$

However, assuming we did not know the solution, the Lagrange form of the objective function is:

$$L = 8A - A^2 + 6B - 1.5B^2 - \lambda(A + B + s^2 - T^{\max})$$

$$A, B, \lambda, s^2 \geq 0$$

where s is the slack variable.

Taking partial derivatives with respect to A , B , λ , and s respectively and solving yields:

$$A = 4 - \frac{1}{2}\lambda$$

$$B = 2 - \frac{1}{3}\lambda$$

$$\lambda s = 0 \quad (\text{either } \lambda \text{ or } s \text{ has to be } 0)$$

$$T^{\max} = A + B + s^2$$

Now for $T^{\max} = 10$, solving yields the same solution as (a) with $s^2 = 4$, implying that $\lambda = 0$, which eventually shows that a constraint of 10 does not have an effect on the objective function.

For $T^{\max} = 5$, the solution:

$$A = 17/5; \quad B = 8/5; \quad \lambda = 6/5; \quad s^2 = 0; \quad TR = 21 \frac{2}{5}$$

(c) $X_A + X_B \leq 10$
or
 $2A + 4B \leq 10$ in terms of A and B

$$L = 8A - A^2 + 6B - B^2 - \lambda(2A + 4B + s^2 - 10)$$

Solution:

$$A = 35/11; \quad B = 10/11; \quad \lambda = 9/11; \quad s^2 = 0;$$

$$TR = 19 \frac{6}{11}$$

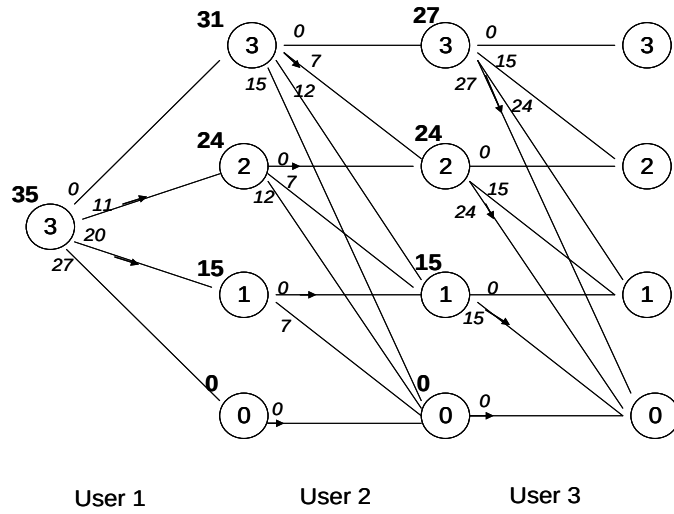
(d) $\lambda = 9/11$, as found in part (c)

Dynamic programming

4.14 Solve for the optimal integer allocations x_1 , x_2 , and x_3 for the problem defined by Exercise 4.9 assuming the total available water is 3 and 4. Also solve for the optimal allocation policy if the total water available is 7 and each x_j must not exceed 4.

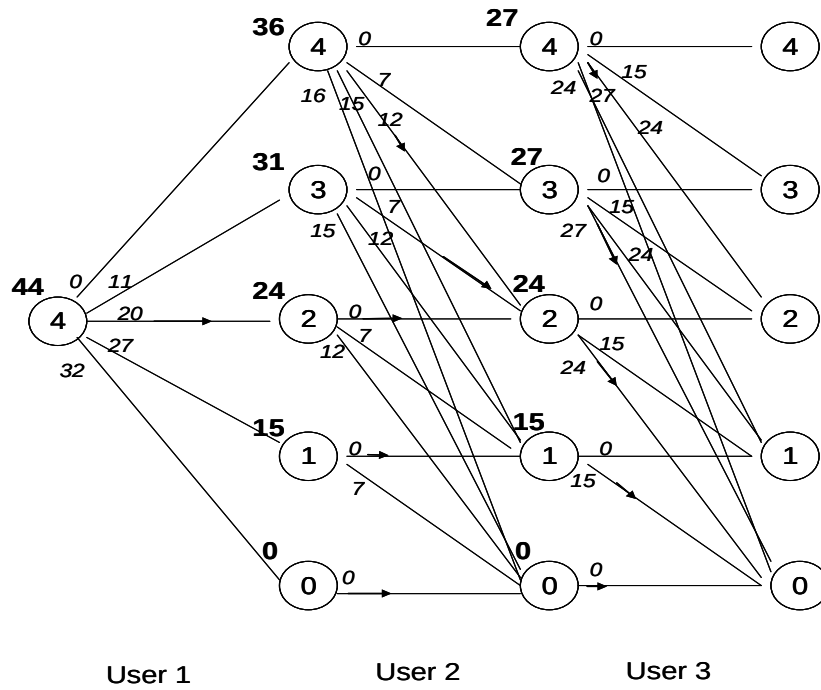
For a total quantity of 3 units, we find 2 optimal allocations:

$$x_1 = 2 \quad x_2 = 0 \quad x_3 = 1 \quad \text{and} \quad x_1 = 1 \quad x_2 = 0 \quad x_3 = 2$$



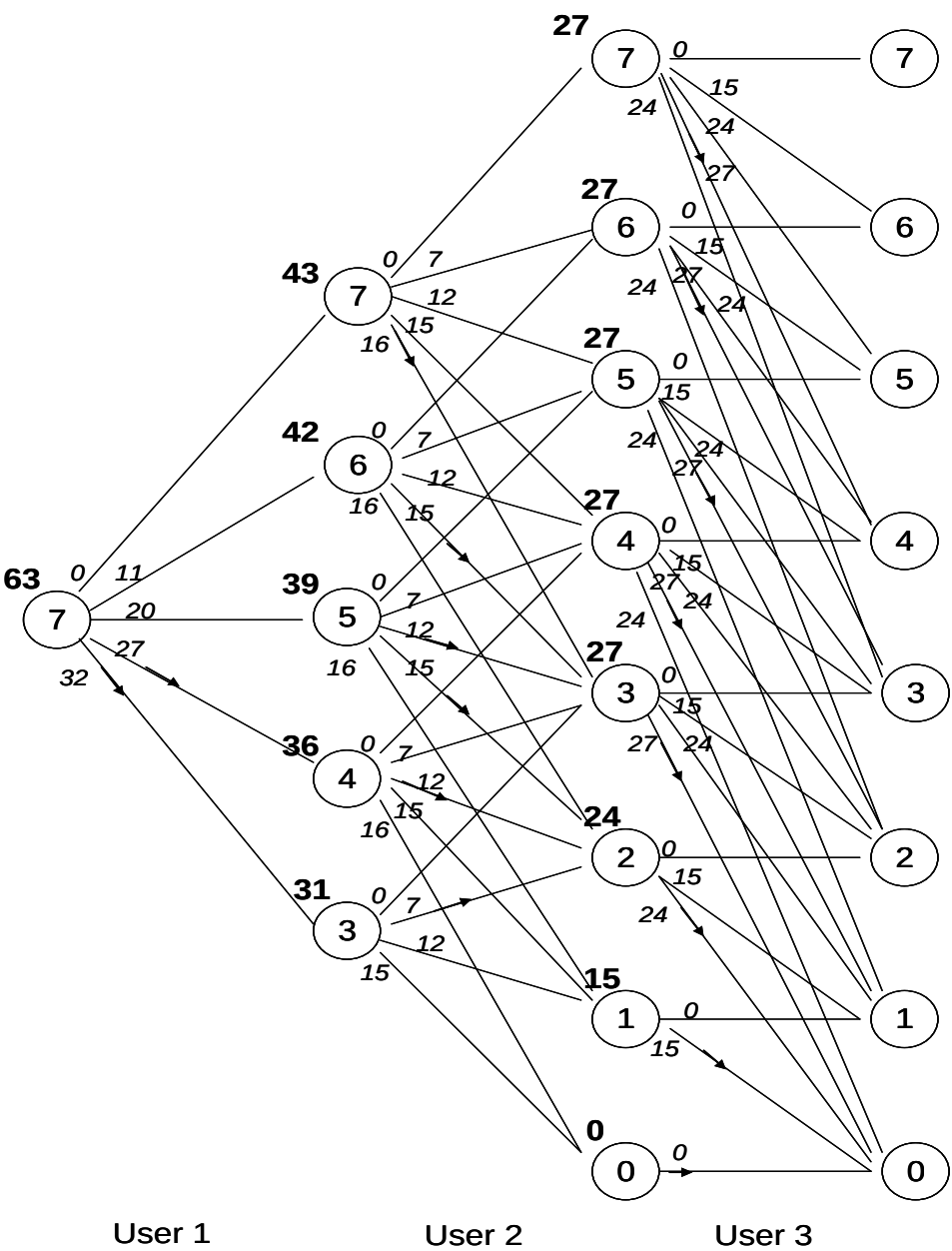
For a total quantity of 4 units, the optimal allocations are:

$$x_1 = 2 \quad x_2 = 0 \quad x_3 = 2$$



For a total quantity of 7 units, we find 2 optimal allocations:

$$x_1 = 3 \quad x_2 = 2 \quad x_3 = 2 \quad \text{and} \quad x_1 = 4 \quad x_2 = 1 \quad x_3 = 2$$



Available Resource	Optimal Allocations			Maximum Objective Function value
Q	x_1	x_2	x_3	
3	1(2)	0	2(1)	35
4	2	0	2	44
7	3(4)	2(1)	2	63

4.15 Consider a three-season reservoir operation problem. The inflows are 10, 50 and 20 in seasons 1, 2 and 3 respectively. Find the operating policy that minimizes the sum of total squared deviations from a constant storage target of 20 and a constant release target of 25 in each of the three seasons. Develop a discrete dynamic programming model that considers only 4 discrete storage values: 0, 10, 20 and 30. Assume the releases cannot be less than 10 or greater than 40. Show how the model's recursive equations change depending on whether the decisions are the releases or the final storage volumes. Verify the optimal operating policy is the same regardless of whether the decision variables are the releases or the final storage volumes in each period. Which model do you think is easier to solve? How would each model change if more importance were given to the desired releases than to the desired storage volumes?

The general recursion equation assuming final storage volume decisions is:

$$f_t^n(s_t) = \text{minimum} \{ (20 - s_t)^2 + (25 - (s_t + i_t - s_{t+1}))^2 + f_{t+1}^{n-1}(s_{t+1}) \} \quad \text{for } 0 \leq s_t \leq K$$

$$0 \leq s_{t+1} \leq K$$

$$s_{t+1} \leq s_t + i_t$$

$$10 \leq s_t + i_t - s_{t+1} \leq 40$$

The general recursion equation assuming release decisions is:

$$f_t^n(s_t) = \text{minimum} \{ (20 - s_t)^2 + (25 - r_t)^2 + f_{t+1}^{n-1}(s_{t+1}) \} \quad \text{for } 0 \leq s_t \leq K$$

$$10 \leq r_t \leq s_t + i_t \leq 40$$

$$r_t \leq s_t + i_t - K$$

$$s_t + i_t - r_t = s_{t+1}$$

Note, the dynamic programming network remains the same in both cases. The nodes are storage values, the links releases. The solution will be the same whether the decision variable is a release or final storage, since knowing one implies knowing the other.

Multiplying the second term of the objective function by a weight greater than one will give more importance to meeting the release targets. Alternatively, the exponent of the second term of objective could be increased if kept an even integer number.

Optimal final storage volumes and releases given initial storage volume S:

S	R ₁	S ₂	R ₂	S ₃	R ₃	S ₁
0	10	0	30	20	10	10
10	10,20	10,0	30	30	10	20
20	20	10	40	30	20	20
30	30	10	--	--	30	20

More detail on the solution at each iteration, using a backward-moving algorithm and final storage (S_{t+1}) as a decision variable is:

$n = 1$			$n = 2$			$n = 3$		
S_3	$f_3^1(S_3)$	S_1^*	S_2	$f_2^2(S_2)$	S_3^*	S_1	$f_1^3(S_1)$	S_2^*
0	425	0	0	450	20	0	1075	0
10	125	0,10	10	250	20	10	575	0,10
20	25	10,20	20	350	30	20	275	10
30	125	20,30	30	-	-	30	375	10

$n = 4$			$n = 5$			$n = 6$		
S_3	$f_3^4(S_3)$	S_1^*	S_2	$f_2^5(S_2)$	S_3^*	S_1	$f_1^6(S)$	S_2^*
0	1200	10	0	725	20	0	1350	0
10	600	20	10	525	30	10	850	0,10
20	300	20	20	625	30	20	550	10
30	400	30	30	-	-	30	650	10

$n = 7$		
S_3	$f_3^7(S_3)$	S_1^*
0	1475	10
10	875	20
20	575	20
30	675	20

Note: Steady-state solution is reached after first iteration

4.16 Show that the constraint limiting a reservoir release, r_t , to be no greater than the initial storage volume, s_t , plus inflow, i_t , is redundant to the continuity equation $s_t + i_t - r_t = s_{t+1}$.

The final storage volume, s_{t+1} , is the surplus variable needed to convert the inequality ($s_t + i_t \geq r_t$) to an equation, and since the continuity constraint equation ($s_t + i_t - r_t = s_{t+1}$) is required, the inequality is redundant.

4.17 Develop a general recursive equation for a forward-moving dynamic programming solution procedure for a single reservoir operating problem. Define all variables and functions used. Why is this not a very useful approach to finding a reservoir operating policy?

$$f_t^n(s_{t+1}) = \text{minimum}\{C_t(s_t, s_{t+1}) + f_{t-1}^{n+1}(s_t)\} \quad 0 \leq s_{t+1} \leq K$$

$$0 \leq s_t \leq K$$

$$s_t \geq s_{t+1} - i_t$$

or

$$f_t^n(s_{t+1}) = \text{minimum}\{C_t(s_{t+1} - i_t + r_t, s_{t+1}) + f_{t-1}^{n+1}(s_{t+1} - i_t + r_t)\} \quad 0 \leq s_{t+1} \leq K$$

$$i_t - s_{t+1} \leq r_t \leq K - s_{t+1} + i_t$$

= minimum objective function value from reservoir releases for n periods up through period t given a final storage volume of s_{t+1} in period t .

This approach to solving DP is not useful for knowing what to release at the beginning of each time period.

4.18 The following table provides estimates for the recent values of the costs of additional wastewater treatment plant capacity needed at the end of each 5-year period for the next 20 years. Find the capacity expansion schedule that minimizes the present values of the total future costs. If there is more than one least cost solution, indicate which one you think is better, and why?

		DISCOUNTED COST OF ADDITIONAL CAPACITY:					Total Required Capacity at End of Period
Period	Years	UNITS OF ADDITIONAL CAPACITY					
		2	4	6	8	10	
1	1-5	12	15	18	23	26	2
2	6-10	8	11	13	15		6
3	11-15	6	8				8
4	16-20	4					10

The cost in each period t must be paid at the beginning of the period. What was the discount factor used to convert the costs at the beginning of each period t to present value costs shown above? In other words how would a cost at the beginning of period t be discounted to the beginning of period 1, given an annual interest rate of r ? (Only the algebraic expression of the discount factor is asked, not the numerical value of r .)

Let S_t = existing capacity at the beginning of period t
 D_t = required capacity at the end of period t
 X_t = capacity addition in period t
 $f_t(S_t)$ = minimum total discounted cost of future capacity expansion through period 4 given S_t at beginning of period t .

Recursive equations:

$$f_5(10) = 0$$

$$f_t(S_t) = \text{minimum}\{Cost_t(X_t) + f_{t+1}(S_t + X_t)\} \quad D_{t-1} \leq S_t \leq 10$$

$$X_t$$

$$0 \leq X_t \leq D_t - S_t$$

Solutions:

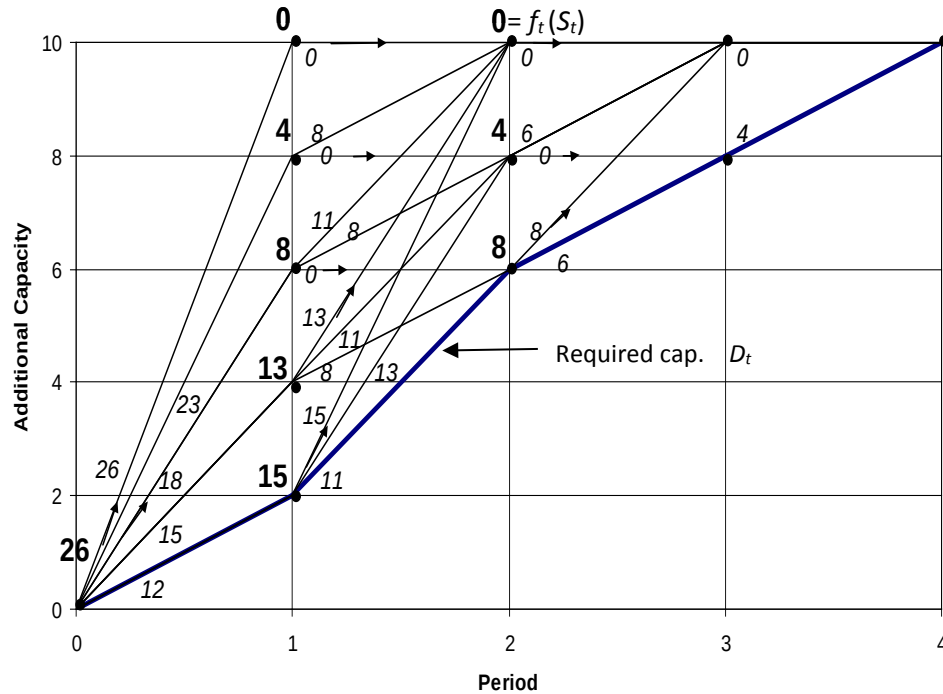
Stage T	State S_t	$Cost_t(X_t) + f_{t+1}(S_t + X_t)$							$f_t(S_t)$	X_t
		X_t :	0	2	4	6	8	10		
4	8	-		4					4	2
	10	0							0	0
3	6	-		10	8				8	4
	8	4		6					4	0
	10	0							0	0
2	2	-	-	19	17	15			15	8
	4	-		16	15	13			13	6
	6	8		12	11				8	0
	8	4		8					4	0
	10	0							0	0
1	0	-		27	28	26	27	26	26	6,10

Hence two solutions costing 26 are

	X_1	X_2	X_3	X_4
Solution A	6	0	4	0
Solution B	10	0	0	0

Solution A permits a greater opportunity to adjust to changing (especially decreasing) future demands for capacity and costs of capacity expansion. The discount factor for each of the four periods t is at $= (1 + r)^{-5(t-1)}$, where r is the annual interest rate expressed as a fraction.

This problem can be easily solved by drawing the appropriate network of links (feasible decisions) from each feasible state (possible existing capacity).



4.19 Consider a wastewater treatment plant in which it is possible to include five different treatment processes in series. These treatment processes must together remove at least 90% of the 100 units of influent waste. Assuming the R_i is the amount of waste removed by process i , the following conditions must hold:

$$20 \leq R_1 \leq 30$$

$$0 \leq R_2 \leq 30$$

$$0 \leq R_3 \leq 10$$

$$0 \leq R_4 \leq 20$$

$$0 \leq R_5 \leq 30$$

- (a) Write the constrained optimization-planning model for finding the least-cost combination of the removals R_i that together will remove 90% of the influent waste. The cost of the various discrete sizes of each unit process i depend upon the waste entering the process i as well as the amount of waste removed, as indicated in the table below.

PROCESS i :		1	2	3	4	5
Influent, I_i	Removal, R_i	Annual Cost = $C_i(I_i, R_i)$				
100	20	5				
100	30	10				
80	10		3	3	1	
80	20		9		2	
80	30		13			
70	10		4	5	2	
70	20		10		3	
70	30		15			
60	10			6	2	3
60	20				4	6
60	30					9
50	10			7	3	4
50	20				5	8
50	30					10
40	10			8	5	5
40	20				7	12
40	30					18
30	10				8	8
30	20				10	12
20	10					8

- (b) Draw the dynamic programming network and solve this problem by dynamic programming. Indicate on the network the calculations required to find the least-cost path from state 100 at stage 1 to state 10 at stage 6 using both forward- and backward-moving dynamic programming solution procedures.
- (c) Could the following conditions be included in the original dynamic programming model and still be solved without requiring R_4 to be 0 in the first case and R_3 to be 0 in the second case?
- (i) $R_4 = 0$ if $R_3 = 0$, or
 - (ii) $R_3 = 0$ if $R_2 \leq 20$.

Let I_i be the influent waste to process i .

$$(a) \text{ Min. } \sum_{i=1}^5 C_i(I_i, R_i)$$

Subject to:

$$\begin{aligned} 100 - R_1 &= I_2 & 20 \leq R_1 \leq 30 \\ I_2 - R_2 &= I_3 & 0 \leq R_2 \leq 30 \end{aligned}$$

$$\begin{array}{ll}
 I_3 - R_3 = I_4 & 0 \leq R_3 \leq 10 \\
 I_4 - R_4 = I_5 & 0 \leq R_4 \leq 20 \\
 I_5 - R_5 = 10 & 0 \leq R_5 \leq 30
 \end{array}$$

$$(b) f_i(I_i) = \min_{R_i} \{C_i(I_i, R_i) + f_{i+1}(I_i - R_i)\} \quad \text{for } 10 \leq I_i \leq 100$$

Bounds on R_i as defined above in (a)
as appropriate for each stage i

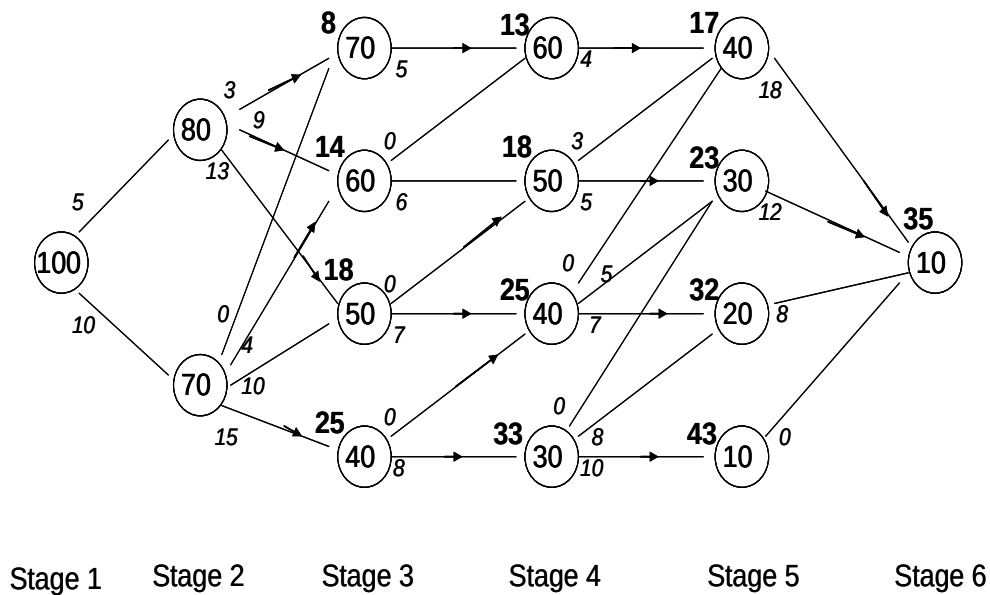
Solutions:	(A)	(B)	both yield min. cost of 35
R_1	= 20	20	
R_2	= 10	30	
R_3	= 10	0	
R_4	= 20	20	
R_5	= 30	20	

(c) Both of these conditional constraints can be incorporated into the model as shown by redrawing the network.

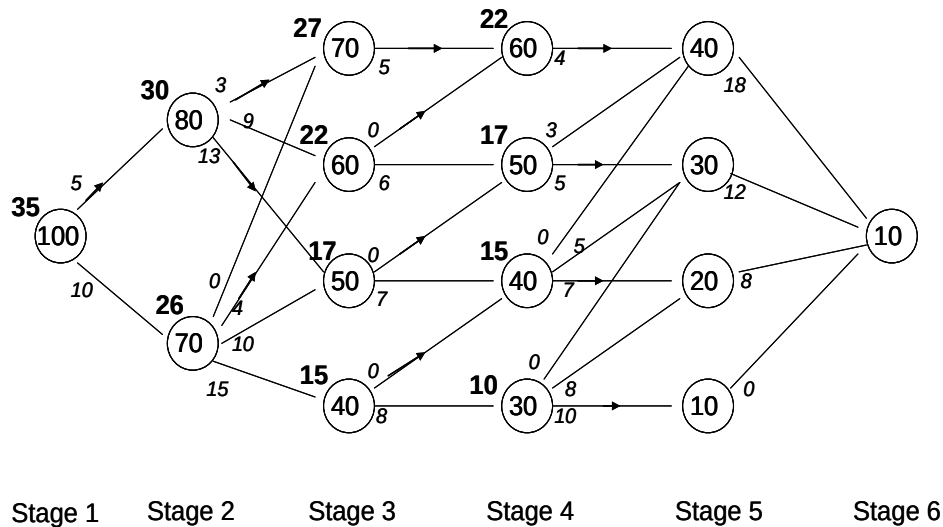
Network Notation:

Only feasible decisions (links) are shown. The functions $f_i(I_i)$ are the minimum costs of removing $I_i - 10$ units of water in the i^{th} and following process (based on a backward moving algorithm) and the functions $g_i(I_i)$ are the minimum costs of removing $I_i - 10$ units of waste in the preceding processes, given I_i units of influent into the i^{th} process.

Forward-moving dynamic programming:



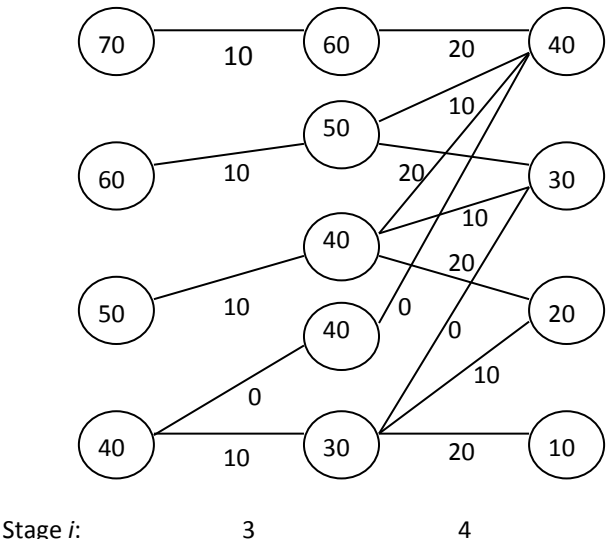
Backward-moving dynamic programming:



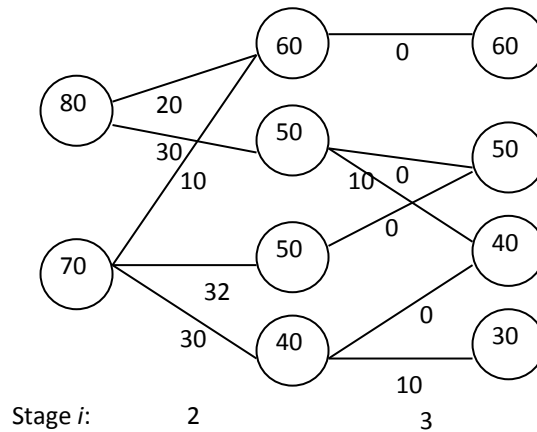
Both methods give the same two least-cost paths:

<i>Waste Removed</i>	<i>Solution # 1</i>	<i>Solution # 2</i>
Stage 1	20	20
Stage 2	10	30
Stage 3	10	0
Stage 4	20	20
Stage 5	30	20

If $R_4 = 0$ when $R_3 = 0$



If $R_3 = 0$ when $R_2 \leq 20$



4.20 The city of Eutro Falls is under a court order to reduce the amount of phosphorus that which it discharges in its sewage to Lake Algae. The city presently has three wastewater treatment plants. Each plant i currently discharges P_i kg/day of phosphorus into the lake. Some or all plants must reduce their discharges so that the *total* for the three plants does not exceed P kg/day.

Let X_i be the fraction or percent of the phosphorus removed by additional treatment at plant i , and the $C_i(X_i)$ the cost of such treatment (\$/year) at each plant i .

- Structure a planning model to determine the least cost (i.e., a cost effective) treatment plant for the city.
- Restructure the model for the solution by dynamic programming. Define the stages, states, decision variables, and the recursive equation for each stage.

a) Objective function: Min: $\sum_{i=1}^3 C_i(X_i)$

Subject to:

$$\sum_{i=1}^3 P_i(1 - X_i) \leq P$$

$$X_1, X_2, X_3 \leq 1$$

all variables are nonnegative

b)

For backward moving DP model:

Stage is the individual plant i .

State is the amount of pollutant in the lake, discrete values (S_i) from 0 to P .

Decision is the amount to treat, X_i .

$$F_4(S_4) = 0 \quad \text{for all } S_4$$

$$F_i(S_i) = \min_{X_i} \{ C_i(X_i) + F_{i+1}(S_{i+1} = S_i + P_i(1-X_i)) \} \quad 0 \leq S_i \leq P$$

$$0 \leq X_i \leq 1$$

$$S_i + P_i(1-X_i) \leq P$$

- c) Now assume $P_1 = 20$; $P_2 = 15$; $P_3 = 25$; and $P = 20$. Make up some cost data and check the model if it works.

4.21 Find (draw) a *rule curve* for operating a single reservoir that maximizes the sum of the benefits for flood control, recreation, water supply and hydropower. Assume the average inflows in four seasons of a year are 40, 80, 60, 20, and the active reservoir capacity is 100. For an average storage S and for a release of R in a season, the hydropower benefits are 2 times the square root of the product of S and R , $2(SR)^{0.5}$, and the water supply benefits are $3R^{0.7}$ in each season. The recreation benefits are $40 - (70 - S)^2$ in the third season. The flood control benefits are $20 - (40 - S)^2$ in the second season. Specify the dynamic programming recursion equations you are using to solve the problem.

This is just the normal backward moving reservoir operation model but with a changing objective in some of the periods. Each objective is a function of the state variable, the initial storage, and the decision variable, either the release or the final storage volume.

General Model:

$$\begin{aligned} \text{Maximize} \quad & \sum_{t=1}^4 B_t[S_t, R_t] \\ \text{Subject to} \quad & S_{t+1} = S_t + Q_t - R_t \\ & S_t \leq K = 100 \end{aligned}$$

DP Model:

$$\begin{aligned} F_t^n(S_t) &= \max \{ B_t(S_t, R_t) + F_{t+1}^{n-1}(S_{t+1}) \} \quad t = 1, 2, 3, 4 \\ \max[0, S_t + Q_t - K] &\leq R_t \leq S_t + Q_t \\ S_{t+1} &= S_t + Q_t - R_t \end{aligned}$$

where

$$B_1 = [(S_1 R_1)^{0.5} + (S_2 R_1)^{0.5}] + 3R_1^{0.7}$$

$$B_2 = [(S_2 R_2)^{0.5} + (S_3 R_2)^{0.5}] + 3R_2^{0.7} + 20 - [(40 - S_2)^2 + (40 - S_3)^2] / 2$$

$$B_3 = [(S_3 R_3)^{0.5} + (S_4 R_3)^{0.5}] + 3R_3^{0.7} + 40 - [(70 - S_3)^2 + (70 - S_4)^2] / 2$$

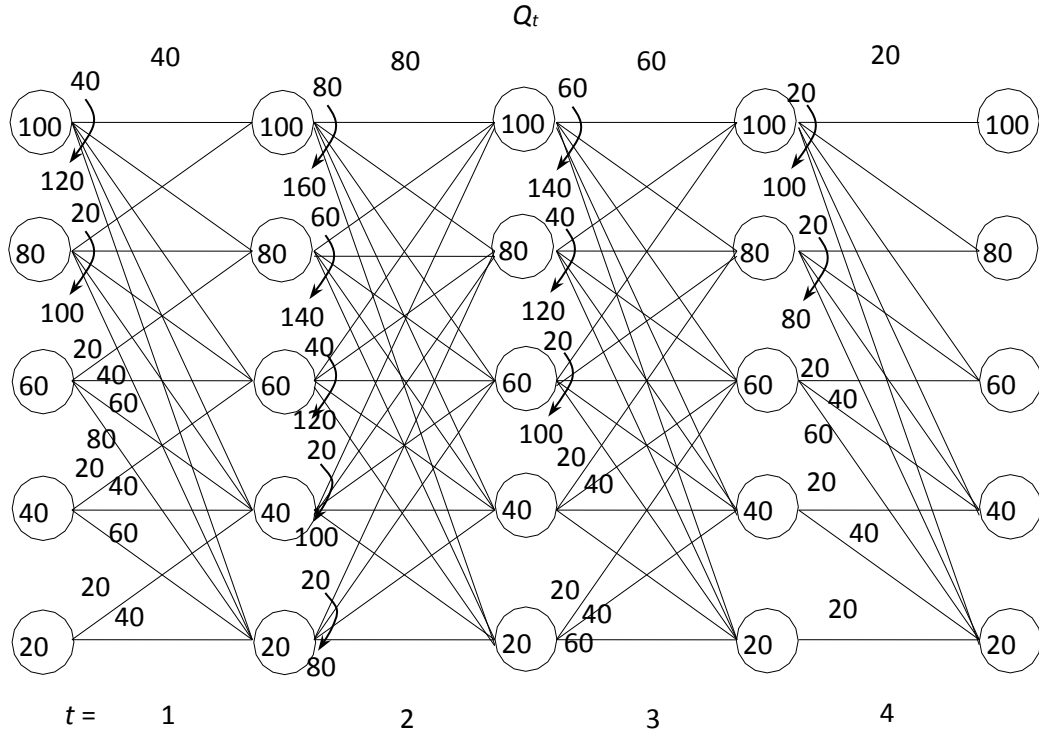
$$B_4 = [(S_4 R_4)^{0.5} + (S_1 R_4)^{0.5}] + 3R_4^{0.7}$$

For simplicity of the DP network, the following storage and release ranges have been assumed.

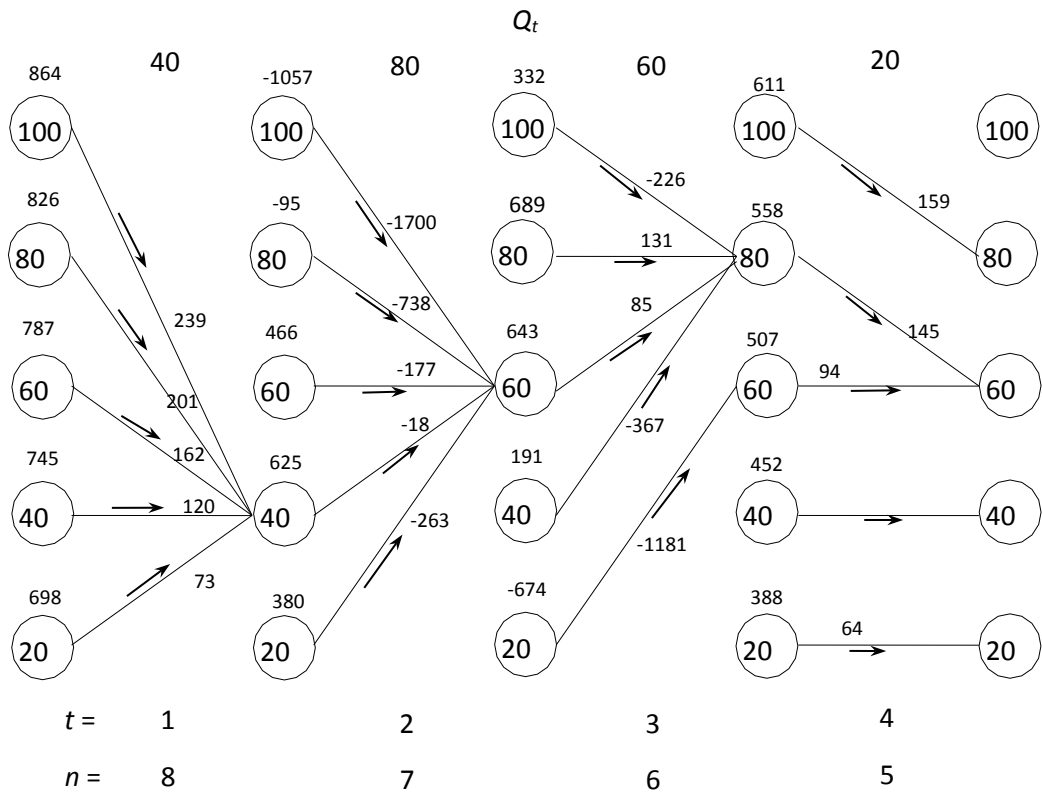
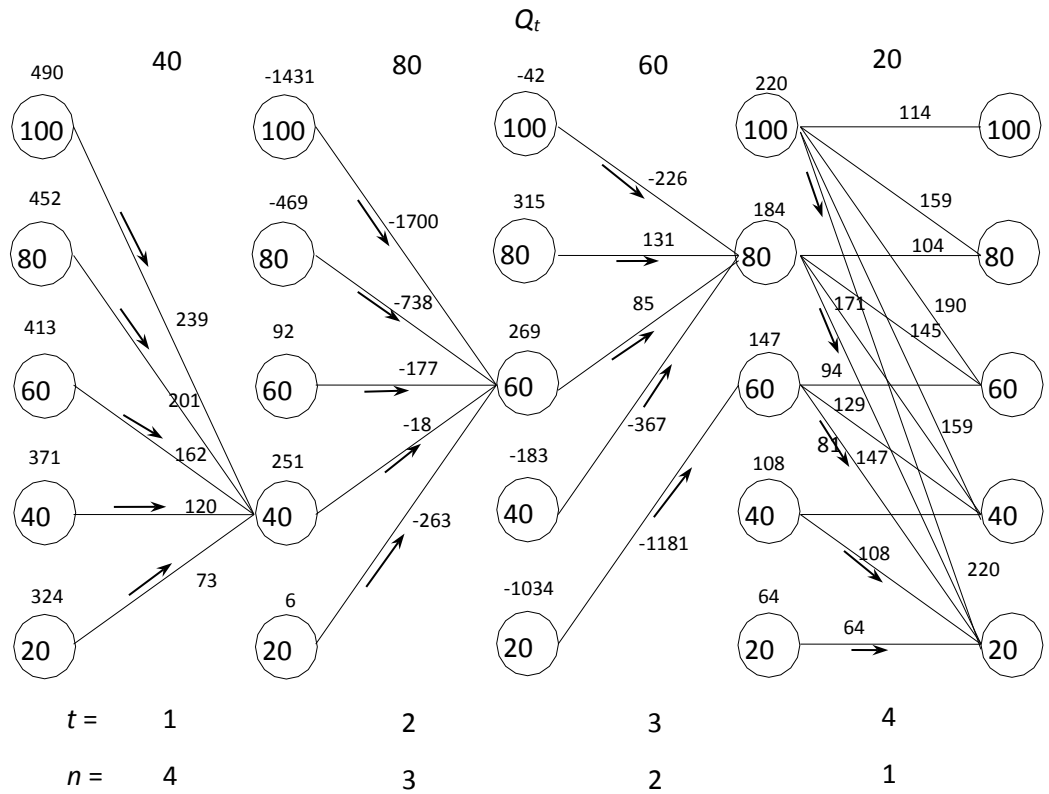
$$S_t = \{20, 40, 60, 80, 100\}$$

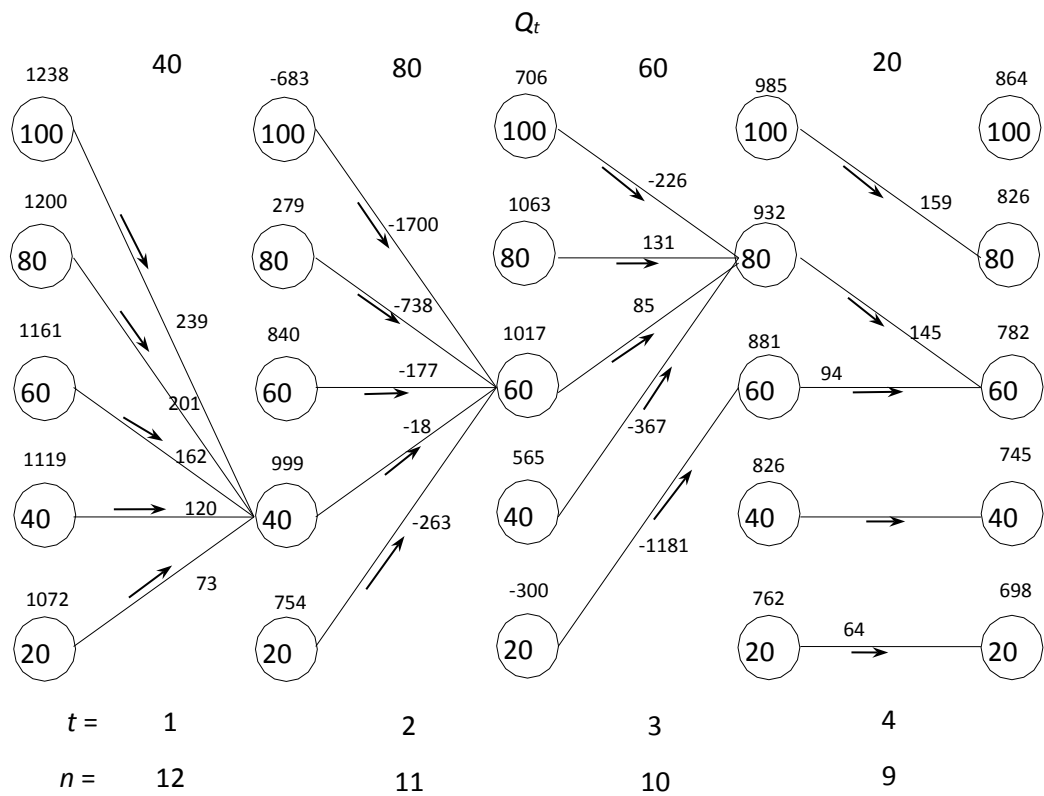
$$R_t = \{20, 40, 60, 80, 100, 120, 140, 160\}$$

The DP model is given below. Nodes show S_t in the beginning and at the end of each season t , and links show the possible releases in each period given the inflow Q_t .



To find a steady state solution, the DP model is solved for $n = 12$ periods.





Note that negative benefits in the relevant seasons haven't been neglected and solution might turn out different in case they are neglected.

The Reservoir Rule Curve

S_t	Release R_t			
	$t = 1$	2	3	4
100	100	120	80	40
80	80	100	60	
60	60	80	40	
40	40	60	20	20
20	20	40		

4.22 How would the model defined in Exercise 4.21 change if there were a water user upstream of this reservoir and you were to find the best water allocation policy for that user, assuming known benefits associated with these allocations that are to be included in the overall maximum benefits objective function?

Now there are two decision variables, the allocation from the inflow and the release from the reservoir. The allocation benefits should be included in the objective function in each period t . For a more detailed solution to such a problem see Chapter XI.

4.23 Suppose there are four water users along a river who benefit from receiving water from the river. Each has a water target, i.e., each expects and plans for a specified amount. These known water targets are $W(1)$, $W(2)$, $W(3)$, and $W(4)$ for the four users respectively. Show how dynamic programming can be used to find two allocation policies. One is to be based on minimizing the maximum deficit deviation from any target allocation. The other is to be based on minimizing the maximum percentage deficit from any target allocation.

First of all the general model can be written:

Unknown variables

D = deficit to any user

$X(i)$ = allocation to each user i

Known variables

$W(i)$ = target water of user i ; $i = 1, 2, 3, 4$

Q = river flow

For LP using LINGO:

Minimize the maximum deficit deviation from any target allocation.

$$\begin{aligned} \text{MIN} &= D; \\ W(i) - X(i) &\leq D \quad \text{for all } i. \\ X(1) + X(2) + X(3) + X(4) &\leq Q; \end{aligned}$$

Minimize the maximum percentage deficit from any target allocation

$$\begin{aligned} \text{MIN} &= PD; \\ (W(i) - X(i)) / W(i) &\leq PD; \\ X(1) + X(2) + X(3) + X(4) &\leq Q; \end{aligned}$$

For DP:

Set up the dynamic programming network for solving this problem, using discrete values for the state variable (in the nodes) which will be the water available to allocate to the remaining uses.

Backward dynamic programming solution procedure

$F_i(S_i)$ = maximum deficit (or percentage deficit) associated current and further allocations given water available to allocate, S_i to use i .

S_i = available water for allocation at state i

X_i = water allocated to user i at state i .

D_i = deficit deviation of water to user i .

W_i = water target for user i

Q = a particular river flow

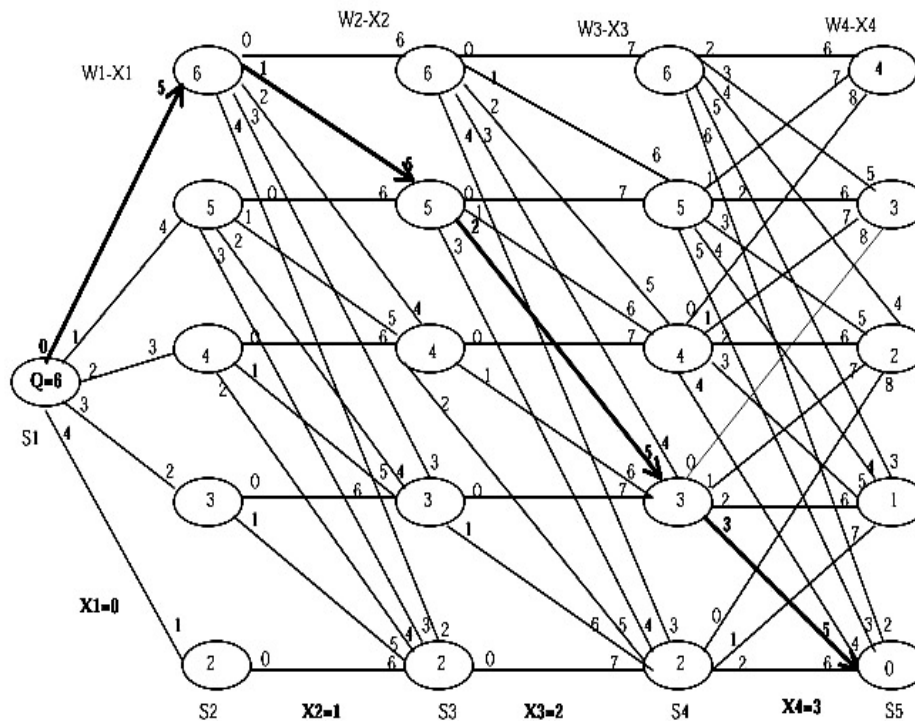
$$F_i(S_i) = \min\{\max[F_i(D_i = W_i - X_i), F_{i+1}(S_{i+1})]\} \text{ for all } S_i \text{ from } 0 \text{ to } Q.$$

$$X_i \leq S_i$$

Assumption:

a particular river flow, Q , = 6 unit of water

W_1, W_2, W_3, W_4 = 5, 6, 7, and 8 units, respectively



Solution: At a particular $Q = 6$, the optimal solution for allocation is to allocate zero unit of water to X_1 , 1 unit of water to X_2 , 2 units of water to X_3 and 3 units of water to X_4 , and the deficit to each user is 5 units of water.

Gradient “Hill-climbing” methods

4.24 Solve Exercise 4.13(b) using hill-climbing techniques and assuming discrete integer values and $T^{\max} = 5$. For example, which product would you produce if you could make only 1 unit of either A or B ? If you could make another unit of A or B , which would you make? Continue this process up to 5 units of products A and/or B .

$$TR_A = 8A - A^2 \quad TR_B = 6B - 1.5B^2$$

Finding the marginal increase in revenue, ΔTR , associated with a unit increase in allocation:

Allocation A	TR_A	ΔTR_A	Allocation B	TR_B	ΔTR_B
0	0		0	0	
		7			4.5
1	7		1	4.5	
		5			1.5
2	12		2	6	
		3			-1.5
3	15		3	4.5	
		1			
4	16				

Thus

Allocate first two to A ($TR = 12$), next one to B ($TR = 16.5$), next one to A ($TR = 19.5$), last one to B ($TR = 21$).

4.25 Under what conditions will hill-climbing methods for maximization or minimization not work?

When there are multiple places where the slope of the objective function changes sign. i.e, when there are multiple values of the function where the slopes are 0.

Linear and non-linear programming

4.26 Consider the industrial firm that makes two products A and B as described in Exercise 4.13(b). Using *Lingo* (or any other program you wish):

- Find the amounts of A and B and their unit prices that maximize total revenue assuming the total amount of water available is 10 units.
- What is the value of the dual variable, or shadow price, associated with the 10 units of available water?
- Suppose the demand functions are not really certain. How sensitive are the allocations of water to changes in the parameter values of those functions? How sensitive are the allocations to the parameter values in the production functions?

The Lingo-model is given below, allowing alternative amounts for the total amount of water, Wmax.

Objective: Maximize the Total Revenue;

```

max = TR;
TR = PA*A + PB*B;
PA = 8 - A;
PB = 6 - 1.5*B;
A = 0.5*XA;
B = 0.25*XB;
[water] XA + XB < Wmax;
Data:
Wmax = ?;
Enddata

```

Local optimal solution found at step: 9

Objective value: 19.54545

Variable	Value	Reduced Cost
TR	19.54545	0.000000
PA	4.818182	0.000000
A	3.181818	0.000000
PB	4.636364	0.000000
B	0.9090909	0.000000
XA	6.363636	0.000000
XB	3.636364	0.000000
WMAX	10.00000	0.000000

Row	Slack or Surplus	Dual Price
1	19.54545	1.000000
2	0.000000	-1.000000
3	0.000000	-3.181818
4	0.000000	-0.9090909
5	0.000000	-1.636364
6	0.000000	-3.272727
WATER	0.000000	-0.8181818

Solution:

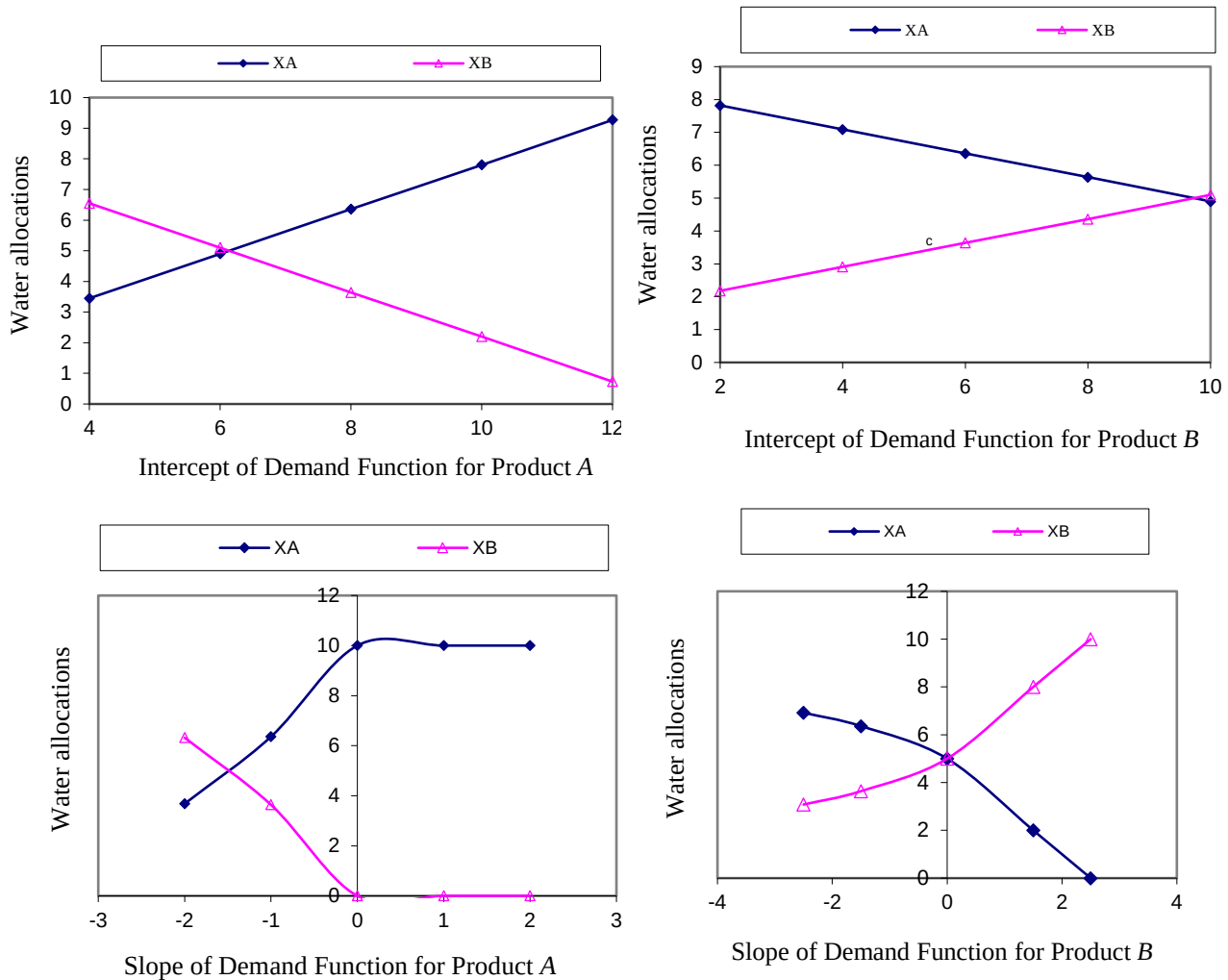
$X_A = 6 \frac{9}{25}$; $X_B = 3 \frac{16}{25}$; $TR = 19 \frac{6}{11}$

$A = 35/11$; $B = 10/11$;

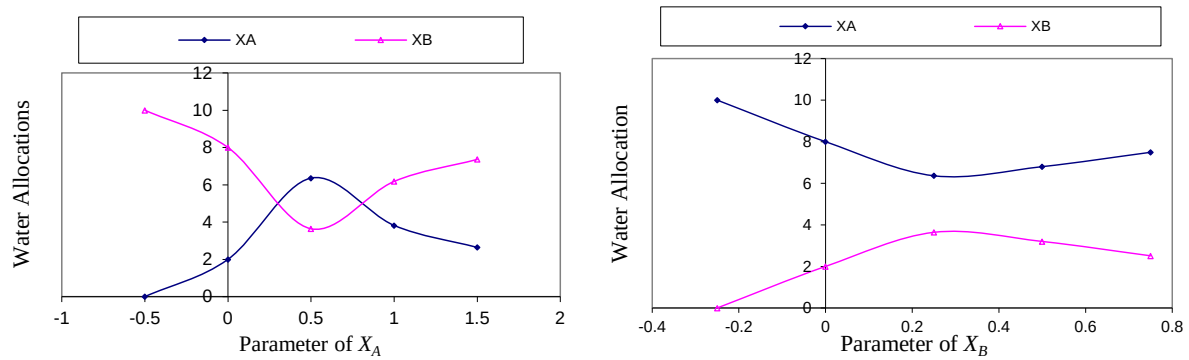
$\lambda = 9/11 = \text{dual variable of } X_A + X_B \leq W_{\max} = 10$ the water constraint;

If another unit of water were available the total income would increase by 0.818 (the 'dual price' of row 7) assuming the objective function were linear. Since it is not, it is safer to say that if Wmax increased by 0.1, the income would increase by about 0.0818.

Sensitivity of Demand Functions Parameters to Water Allocations



Sensitivity of Water Allocation Parameters to Water Allocations



4.27 Assume that there are m industries or municipalities that discharge their wastes into a river. Denote the discharge sites by the subscript i and let W_i be the kg of waste discharged into the river each day at those sites i . To improve the river water quality downstream, wastewater treatment plants may be required at each site i . Let x_i be the fraction of waste removed by treatment at each site i . Develop a model for estimating how much waste removal is required at each site to maintain acceptable water quality in the river at a minimum total cost. Use the following additional notation:

a_{ij} = decrease in quality at site j per unit of waste discharged at site i

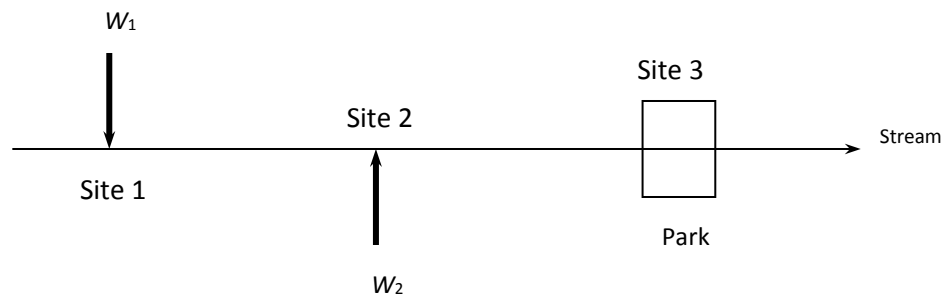
q_j = quality at site j that would result if all controlled upstream discharges were eliminated (i.e., $W_1 = W_2 = 0$)

Q_j = minimum acceptable quality at site j

C_i = cost per unit (fraction) of waste removed at site i

$$\begin{aligned} &\text{Minimize} && \sum_{i=1}^m C_i (x_i) \\ &\text{subject to:} && q_j - \sum_{i=1}^m a_{ij} w_i (1 - x_i) \geq Q_j && \forall j \\ &&& 0 \leq x_i \leq 1 && \forall i \end{aligned}$$

4.28 Assume that there are two sites along a stream, $i = 1, 2$, at which waste (BOD) is discharged. Currently, without any wastewater treatment, the quality (DO), q_2 and q_3 , at each of sites 2 and 3 is less than the minimum desired, Q_2 and Q_3 , respectively. For each unit of waste removed at site i upstream of site j , the quality improves by A_{ij} . How much treatment is required at sites 1 and 2 that meets the standards at a minimum total cost?



Following are the necessary data:

C_i = cost per unit fraction of waste treatment at site i (both C_1 and C_2 are unknown but for the same amount of treatment, whatever that amount, $C_1 > C_2$)

R_i = decision variables, unknown waste removal fractions at sites $i = 1, 2$

$$\begin{array}{lll}
 A_{12} = 1/20 & W_1 = 100 & Q_2 = 6 \\
 A_{13} = 1/40 & W_2 = 75 & Q_3 = 4 \\
 A_{23} = 1/30 & q_2 = 3 & q_3 = 1
 \end{array}$$

$$\begin{array}{ll}
 \text{Minimize} & \sum_{i=1}^2 C_i(R_i) \\
 \text{subject to:} & q_j + \sum_{i=1}^2 A_{ij} W_i R_i \geq Q_j \quad j = 2, 3 \\
 & 0 \leq R_i \leq 1 \quad i = 1, 2
 \end{array}$$

or

$$\text{Minimize} \quad C_1 R_1 + C_2 R_2$$

Subject to

(at $j = 2$)

$$3 + (1/20)(100) R_1 \geq 6 \quad \text{or} \quad R_1 \geq 3/5 = 0.6$$

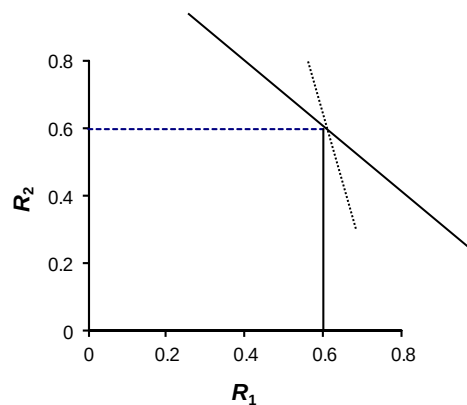
(at $j = 3$)

$$1 + (1/40)(100) R_1 + (1/30)(75) R_2 \geq 4 \quad \text{or} \\
 R_1 + R_2 \geq 3/2.5 = 1.2$$

$$0 \leq R_1 \leq 1$$

$$0 \leq R_2 \leq 1$$

Hence if $C_1 > C_2$, $R_1 = R_2 = 0.6$



4.29 Define a linear programming model for finding the tradeoff between active storage capacity and the maximum percentage deviation from a known target storage volume and a known target release in each period. How could the solution of the model be used to define a reservoir policy?

$$\begin{array}{ll}
\text{Minimize} & D \\
\text{Subject to} & D \geq (T^S - S_t)/T^S \quad \forall_t \\
& D \leq (T^S - S_t)/T^S \quad \forall_t \\
& D \geq (T^R - R_t)/T^R \quad \forall_t \\
& D \leq (T^R - R_t)/T^R \quad \forall_t \\
- & S_t + I_t - R_t = S_{t+1} \quad \forall_t \text{ last } t + 1 = 1 \\
& S_t \leq K \quad (K \text{ is known}) \quad \forall_t \\
& \text{all variables} \geq 0
\end{array}$$

The solution of this or other equivalent models will give some indication of appropriate deviations from storage and release targets to be used as a guide for reservoir operation. However, the deterministic nature of the model limits its usefulness when future inflows I_t are uncertain.

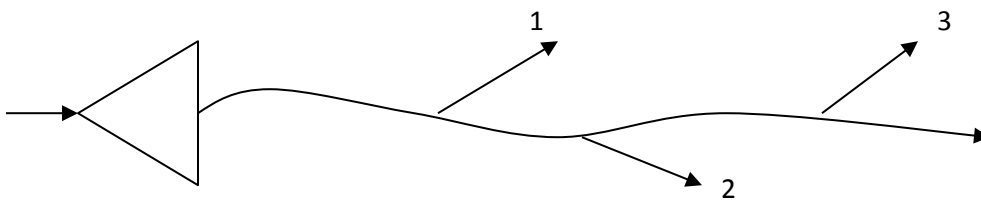
If we want to find the capacity K that minimizes the maximum percentage deviation (D) from target storage (T^S) and target release (T^R) in each period :

$$\text{Objective: } \min \{D + 0.0001 \cdot K\}$$

Subject to the same constraints as above.

This model will give the smallest storage capacity K that enables to minimize the maximum percentage deviation. No larger capacity can improve the result. On the contrary, a capacity smaller will result in an increase of the deviations. But even if the reservoir is already built and we cannot change its capacity, this model is interesting to give the optimal storage volume and optimal release to be implemented at each period.

4.30 Consider the possibility of building a reservoir upstream of three demand sites along a river.



The net benefits derived from each use depend on the reliable amounts of water allocated to each use. Letting x_{it} be the allocation to use i in period t , the net benefits for each period t equal

1. $6x_{1t} - x_{1t}^2$
2. $7x_{2t} - 1.5 x_{2t}^2$
3. $8x_{3t} - 0.5 x_{3t}^2$

Assume the average inflows to the reservoir in each of four seasons of the year equal 10, 2, 8, 12.

- a) Find the tradeoff between the yield (the expected release that can be guaranteed in each season) and the reservoir capacity.

You can do this by solving the following model for various values of capacity K .

$$\begin{aligned} & \text{Maximize } Y \\ & \text{Subject to:} \\ & S1 + 10 - Y \geq S2 \\ & S2 + 2 - Y \geq S3 \\ & S3 + 8 - Y \geq S4 \\ & S4 + 12 - Y \geq S1 \\ & S1 \leq K \\ & S2 \leq K \\ & S3 \leq K \\ & S4 \leq K \end{aligned}$$

- b) Find the tradeoff between the yield and the maximum total net benefits that can be obtained from allocating that yield among the three users.

You can do this by solving the following model for various values of yield Y .

$$\begin{aligned} & \text{Maximize } (6x_1 - x_1^2) + (7x_2 - 1.5x_2^2) + (8x_3 - 0.5x_3^2) \\ & \text{Subject to:} \\ & x_1 + x_2 + x_3 \leq Y. \end{aligned}$$

- c) Find the tradeoff between the reservoir capacity and the total net benefits one can obtain from allocating the total releases, not just the reliable yield, to the downstream users.

You can find this by solving the following model for various values of capacity K .

$$\begin{aligned} & \text{Maximize } \sum_t [(6x_1(t) - x_1(t)^2) + (7x_2(t) - 1.5x_2(t)^2) + \\ & (8x_3(t) - 0.5x_3(t)^2)] \\ & \text{Subject to:} \\ & S(1) + 10 - r(1) = S(2) \\ & S(2) + 2 - r(2) = S(3) \\ & S(3) + 8 - r(3) = S(4) \\ & S(4) + 12 - r(4) = S(1) \\ & S(t) \leq K \quad \text{for } t = 1, 2, 3, 4. \end{aligned}$$

$$x_1(t) + x_2(t) + x_3(t) \leq r(t) \quad \text{for } t = 1, 2, 3, 4.$$

- d) Assuming a reservoir capacity of 7, and dividing the release into integer increments of 2 (i.e., 2, 4, 6 and 8), using linear programming, find the optimal operating policy. Assume the maximum release cannot exceed 8, and the minimum release cannot be

less than 2. How does this solution differ from that obtained using dynamic programming?

You can find this by solving the following model.

$$\text{Maximize } \sum_{t=1}^4 [(6 x_1(t) - x_1(t)^2) + 7 x_2(t) - 1.5 x_2(t)^2] + (8 x_3(t) - 0.5 x_3(t)^2)]$$

Subject to:

$$S(1) + 10 - r(1) = S(2)$$

$$S(2) + 2 - r(2) = S(3)$$

$$S(3) + 8 - r(3) = S(4)$$

$$S(4) + 12 - r(4) = S(1)$$

$$S(t) \leq 15 \quad \text{for } t = 1, 2, 3, 4.$$

$$x_1(t) + x_2(t) + x_3(t) \leq r(t) \quad \text{for } t = 1, 2, 3, 4.$$

$$2 \leq r(t) \leq 8 \quad \text{and } r(t)/2 \text{ integer for all } t.$$

DP would require dividing the storage volume into discrete values, say 0, 2, 4, and 6. It would give us a sequential policy indicating what to release for each discrete storage volume. Thus it is much easier to derive a policy using DP than LP. LP just gives us the best storage volumes and releases for the given sequence of inflows.

- e) If you were maximizing the total net benefit obtained from the three users and if the water available to allocate to the three users were 15 in a particular time period, what would be the value of the Lagrange multiplier or dual variable associated with the constraint that you cannot allocate more than 15 to the three uses?

The value of the dual variable or Lagrange multiplier would be 0 because the most all the users want is $3 + 7/3 + 8 = 13.33$.

- f) There is the possibility of obtaining recreational benefits in seasons 2 and 3 from reservoir storage. No recreational benefits can occur in seasons 1 and 4. To obtain these benefits facilities must be built, and the question is at what elevation (storage volume) should they be built. This is called the recreational storage volume target. Recreational benefits in each recreation season equal 8 per unit of storage target if the actual storage equals the storage target. If the actual storage is less than the target the losses are 12 per unit deficit – the difference between the target and actual storage volumes. If the actual storage volume is greater than the target volume the losses are 4 per unit excess. What is the reservoir capacity and recreation storage target that maximizes the annual total net benefits obtained from downstream allocations and recreation in the reservoir less the annual cost of the reservoir, $3K^{1.5}$, where K is the reservoir capacity?

Find these variable values by solving the following model:

$$\text{Maximize Allocation_benefits} + \text{Recreation_benefits} - \\ \text{Recreation_losses} - \text{Res_cost}$$

Subject to:

$$\text{Allocation_benefits} = \sum_{t=1}^4 [(6 x_1(t) - x_1(t)^2) + \\ (7 x_2(t) - 1.5 x_2(t)^2) + (8 x_3(t) - 0.5 x_3(t)^2)]$$

$$\text{Recreation_benefits} = 2 (8 T_s)$$

$$\text{Recreation_losses} = 0.5 [12 D(2) + 4 E(2) + 12 D(3) + 4 E(3)] + 0.5 [12 \\ D(3) + 4 E(3) + 12 D(4) + 4 E(4)]$$

$$\text{Res_cost} = 4 K^{1.4}$$

$$S(1) + 10 - r(1) = S(2)$$

$$S(2) + 2 - r(2) = S(3)$$

$$S(3) + 8 - r(3) = S(4)$$

$$S(4) + 12 - r(4) = S(1)$$

$$S(t) \leq K \quad \text{for } t = 1, 2, 3, 4.$$

$$x_1(t) + x_2(t) + x_3(t) \leq r(t) \quad \text{for } t = 1, 2, 3, 4.$$

$$S(t) = T_s - D(t) + E(t) \quad \text{for } t = 2, 3, 4.$$

g) In (f) above, suppose the allocation benefits and net recreation benefits were given weights indicating their relative importance. What happens to the relationship between capacity K and recreation target as the total allocation benefits are given a greater weight in comparison to recreation net benefits?

As the weight given to Allocation_benefits increases, the recreation target decreases in relation to K which remains constant.

4.31 Using the network representation of the wastewater treatment plant design problem defined in Exercise 4.19, write a linear programming model for defining the least-cost sequence of unit treatment process (i.e., the least-cost path through the network). [Hint: Let each decision variable x_{ij} indicate whether or not the link between nodes (or states) i and j connecting two successive stages is on the least-cost or optimal path. The constraints for each node must ensure that what enters the node must also leave the node.]

$$\text{Minimize} \quad \sum_i \sum_j C_{ij} X_{ij}$$

$$\text{subject to:} \quad \sum_j X_{1j} = 1 \quad \text{or} \quad \sum_i X_{i6} = 1$$

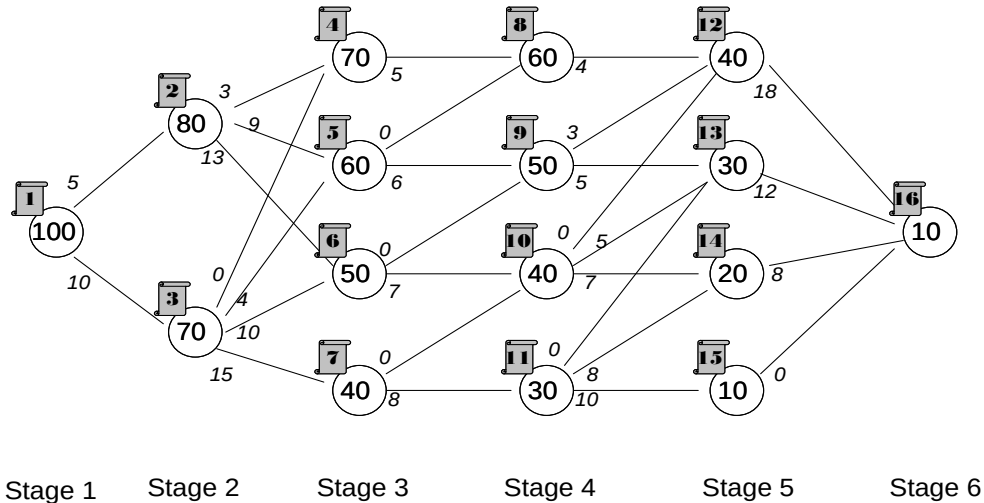
where $i = 1$ is the beginning state node (100 units of waste available) and the sum over j includes only those nodes j representing the discrete effluents of process 1 or influents to process 2. Alternatively, $j = 6$ is that node representing the ending state node (10 units of waste remaining) and the sum over i includes only nodes i representing the discrete

effluents of process 5. The other constraints equate the sum of the X_{ik} 's on those links into a node k to the sum of the X_{kj} 's on those links out of node k for all nodes k .

$$\sum_i X_{ik} = \sum_j X_{kj}$$

Without forcing them to be so, the X_{ij} 's will be either 0 or 1 (unless multiple optimal solutions exist in which case they may be any fraction between 0 and 1).

For the wastewater treatment plant problem, the DP network is:



The model in Lingo is:

```
! Objective: minimize total cost:
;
min = 5*x12 + 10*x13 + 3*x24 + 9*x25 + 13*x26 + 4*x35 + 10*x36 +
15*x37 + 5*x48 + 6*x59 + 7*x610 + 8*x711 + 4*x812 + 3*x912 + 5*x913 +
5*x1013 + 7*x1014 + 8*x1114 + 10*x1115 + 18*x1216 + 12*x1316 +
8*x1416;

! Constraints: what goes in goes out;;
1 = x12 + x13;
x12 = x24 + x25 + x26;
x13 = x34 + x35 + x36 + x37;
x24 + x34 = x48;
x25 + x35 = x59;
x26 + x36 = x69 + x610;
x37 = x710 + x711;
x48 + x58 = x812;
x59 + x69 = x912 + x913;
x610 + x710 = x1012 + x1013 + x1014;
x711 = x1113 + x1114 + x1115;
x812 + x912 + x1012 = x1216;
x913 + x1013 + x1113 = x1316;
x1014 + x1114 = x1416;
x1115 = x1516;
```

Lingo finds one of the two optimal solutions:

Global optimal solution found at iteration: 6
Objective value: 35.00000

Non Zero Variable	Value
X12	1.000000
X26	1.000000
X69	1.000000
X913	1.000000
X1316	1.000000

4.32 Two types of crops can be grown in particular irrigation area each year. Each unit quantity of crop A can be sold for a price P_A and requires W_A units of water, L_A units of land, F_A units of fertilizer, and H_A units of labor. Similarly, crop B can be sold at a unit price of P_B and requires W_B , L_B , F_B and H_B units of water, land, fertilizer, and labor, respectively, per unit of crop. Assume that the available quantities of water, land, fertilizer, and labor are known, and equal W , L , F , and H , respectively.

- (a) Structure a linear programming model for estimating the quantities of each of the two crops that should be produced in order to maximize total income.

Let X_A = units of crop A produced, and X_B = units of crop B produced.

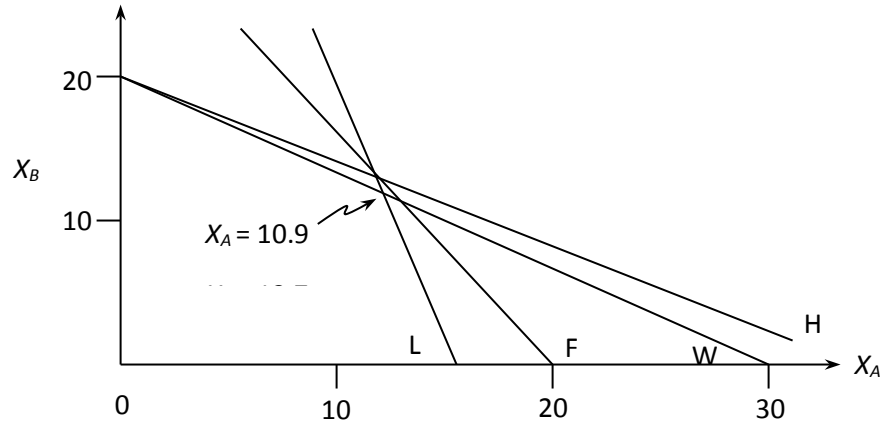
(a) Maximize $P_A X_A + P_B X_B = Y = 30X_A + 25X_B$

Subject to

$$\begin{aligned} W_A X_A + W_B X_B &\leq W & 2X_A + 3X_B &\leq 60 \\ L_A X_A + L_B X_B &\leq L & 5X_A + 2X_B &\leq 80 \\ F_A X_A + F_B X_B &\leq F & 3X_A + 2X_B &\leq 60 \\ H_A X_A + H_B X_B &\leq H & X_A + 2X_B &\leq 40 \\ X_A, X_B &\geq 0 \end{aligned}$$

- (b) Solve the problem graphically, using the following data:

Resource	REQUIREMENTS PER UNIT OF:		Maximum Available Resources
	Crop A	Crop B	
Water	2	3	60
Land	5	2	80
Fertilizer	3	2	60
Labor	1	2	40
Unit Price	30	25	



(c) Define the meaning of the dual variables, and their values, associated with each constraint.

Dual variables

$$(1) \quad \frac{\partial Y}{\partial W} = 5.9 = U_1$$

$$(2) \quad \frac{\partial Y}{\partial L} = 3.6 = U_2$$

all others = 0

(d) Write the dual model of this problem and interpret its objective and constraints.

Minimize

$$WU_1 + LU_2 + FU_3 + HU_4 = 60U_1 + 80U_2 + 60U_3 + 40U_4$$

subject to

$$W_A U_1 + L_A U_2 + F_A U_3 + H_A U_4 \geq P_A$$

$$W_B U_1 + L_B U_2 + F_B U_3 + H_B U_4 \geq P_B$$

$$U_i \geq 0 \quad \forall_i$$

Objective:

Minimize additional income that can be obtained where for each crop the
Total Additional Income/unit crop $\geq$ Income Obtained /Unit Crop

(e) Solve the primal and dual models using an existing computer program, and indicate the meaning of all output data.

Solutions of primal and dual are given in (b) and (c). Other output data will depend on particular computer code used.

(f) Assume that one could purchase additional water, land, fertilizer, and labor with capital that could be borrowed from a bank at an annual interest rate r . How would this opportunity alter the linear programming model? The objective continues to be a maximization of net income. Assume there is a maximum limit on the amount of money that can be borrowed from the bank.

Let $\hat{W}$, $\hat{L}$, $\hat{F}$ and $\hat{H}$ be the additional resources at a cost of C_W , C_L , C_F and C_H per unit resource. Objective:

$$\begin{aligned} &\text{Maximize} && P_A X_A + P_B X_B - B(1 + r) \\ &\text{subject to same constraints as before except right-handed side will be} \\ &&& W + \hat{W}, L + \hat{L}, \text{ etc. and} \\ &&& B = C_W \hat{W} + C_L \hat{L} + C_F \hat{F} + C_H \hat{H} \\ &&& B \leq \text{maximum that can be borrowed.} \end{aligned}$$

(g) Assume that the unit price P_j of crop j is a decreasing linear function ($P_j^0 - b_j x_j$) of the quantity, x_j , produced. How could the linear model be restructured also as to identify not only how much of each crop to produce, but also the unit price at which each crop should be sold in order to maximize total income?

The $P_j X_j$ terms in the objective function are replaced by $(P_j^0 - b_j X_j) X_j$ which is a separable concave function that can be made piece-wise linear for inclusion in an LP model. Once the X_j 's are known, the optimal unit price can be determined as $P_j^0 - b_j X_j$.

Alternative constraints:

- $W_A X_A + W_B X_B \leq W \rightarrow X_B \leq 20 - \frac{2}{3} X_A$
- $L_A X_A + L_B X_B \leq L \rightarrow X_B \leq 40 - \frac{5}{2} X_A$
- $F_A X_A + F_B X_B \leq F \rightarrow X_B \leq 30 - \frac{3}{2} X_A$
- $H_A X_A + H_B X_B \leq H \rightarrow X_B \leq 20 - \frac{1}{2} X_A$

The optimal quantities are 120/11 units of crop A and 140/11 units of crop B for a total income of 7100/11. If we cannot do smaller than 1 unit, we should choose to produce 11 units of crop A and 12 units of crop B for a total income of 630.

4.33 Using linear programming model, derive an annual storage-yield function for a reservoir at a site having the following record of annual flows:

Year y	Flow Q_y	Year y	Flow Q_y
1	5	9	3
2	7	10	6
3	8	11	8
4	4	12	9
5	3	13	3
6	3	14	4
7	2	15	9
8	1		

- a) Find the values of the storage capacity required for yields of 2, 3, 3.5, 4, 4.5, and 5.

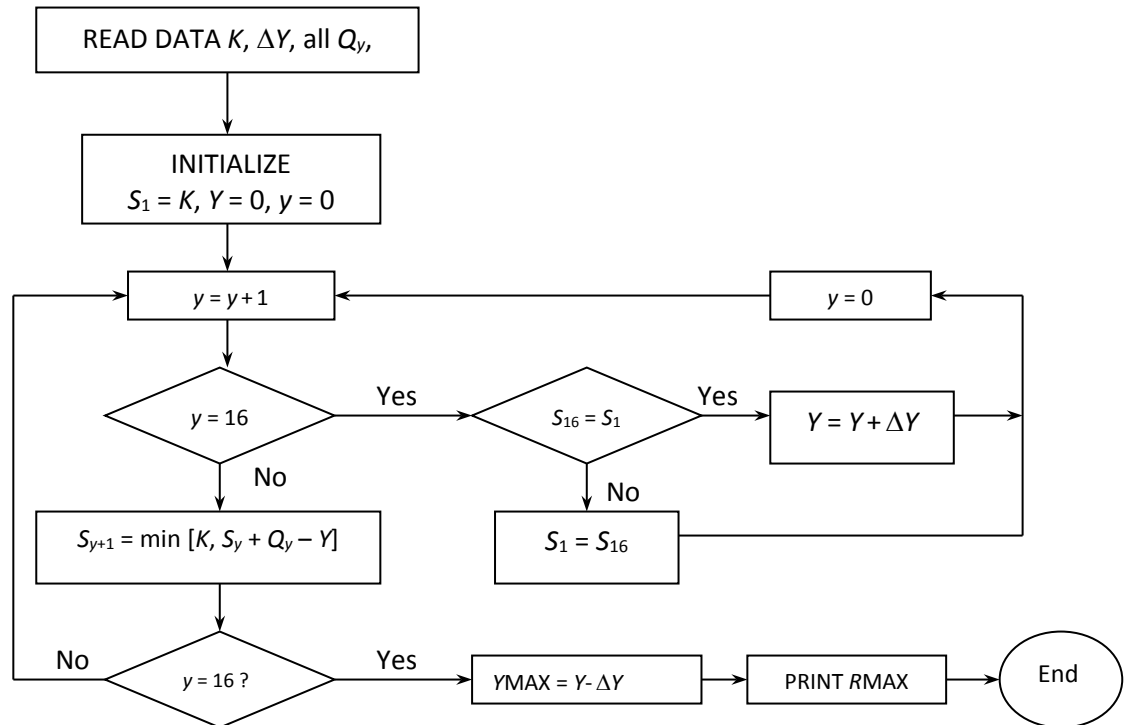
The LP model for finding the storage capacity K required for various known yield Y defined by

$$\begin{aligned} &\text{Minimize} && K \\ &\text{Subject to:} && S_{y+1} = S_y + Q_y - Y - R_y \\ & && S_y \leq K \end{aligned}$$

Alternatively, one can maximize Y subject to various values of K .

For yields of 2, 3, 3.5, 4, 4.5 and 5, the storage K required is 1, 3, 5.5, 8, 11, and 14, respectively.

- b) Develop a flow chart defining a procedure for finding the yields for various increasing values of K .



4.34 Water resources planning usually involves a set of separate tasks. Let the index i denote each task, and H_i the set of tasks that immediately precede task i . The duration of each task i is estimated to be d_i .

- a) Develop a linear programming model to identify the starting times of tasks that minimizes the time, T , required to complete the total planning project.

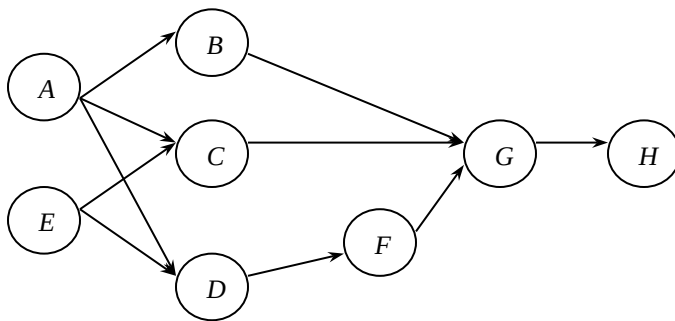
General model:

Minimize T = final project finishing time

Subject to: starting time $t_i \geq t_h + d_h \quad \forall i, h \in H_i$
 $T \geq t_i + d_i \quad \forall \text{ final tasks } i$

b) Apply the general model to the following planning project:

- Task A: Determine planning objectives and stakeholder interests. Duration: 4 months
- Task B: Determine structural and non-structural alternatives that will influence objectives. Duration: 1 month.
- Task C: Develop an optimization model for preliminary screening of alternatives and for estimating tradeoffs among objectives. Duration: 1 month.
- Task D: Identify data requirements and collect data. Duration: 2 months.
- Task E: Develop a data management system for the project. Duration: 3 months.
- Task F: Develop an interactive shared vision simulation model with the stakeholders. Duration: 2 Months.
- Task G: Work with stakeholders in an effort to come to a consensus (a shared vision) of the best plan. Duration: 4 months.
- Task H: Prepare, present and submit a report. Duration: 2 months.



Lingo model for the given planning project example is given below with the solution:

```
min = T;      ! The total Project time;

! Subject to;
TB >= TA + dA; ! Ti is the start time for activity i;
TC >= TA + dA; ! di is the duration of activity i, where;
TC >= TE + dE; ! i = A,...,H;
TD >= TA + dA;
TD >= TE + dE;
TF >= TD + dD;
TG >= TB + dB;
TG >= TC + dC;
TG >= TF + dF;
TH >= TG + dG;
T  >= TH + dH;

Data:
```

```

dA = 4;
dB = 1;
dC = 1;
dD = 2;
dE = 3;
dF = 2;
dG = 4;
dH = 2;
enddata

```

Solution:

Global optimal solution found at step: 6
Objective value: 14.00000

Variable	Value	Reduced Cost
T	14.00000	0.000000
TB	7.000000	0.000000
TA	0.000000	1.000000
DA	4.000000	0.000000
TC	7.000000	0.000000
TE	0.000000	0.000000
DE	3.000000	0.000000
TD	4.000000	0.000000
TF	6.000000	0.000000
DD	2.000000	0.000000
TG	8.000000	0.000000
DB	1.000000	0.000000
DC	1.000000	0.000000
DF	2.000000	0.000000
TH	12.00000	0.000000
DG	4.000000	0.000000
DH	2.000000	0.000000

Row	Slack or Surplus	Dual Price
1	14.00000	1.000000
2	3.000000	0.000000
3	3.000000	0.000000
4	4.000000	0.000000
5	0.000000	-1.000000
6	1.000000	0.000000
7	0.000000	-1.000000
8	0.000000	0.000000
9	0.000000	0.000000
10	0.000000	-1.000000
11	0.000000	-1.000000
12	0.000000	-1.000000

4.35 In Exercise 4.34 suppose the project is penalized if its completion time exceeds a target T . The difference between 14 months and T months is Δ , and the penalty is $P(\Delta)$. You could reduce the time it takes to complete task E by one month at a cost of \$200, and by two months at a cost of \$500. Similarly, suppose the cost of task A could be reduced by a month at a cost of \$600 and two months at a cost of \$1400. Construct a model to find the most economical project completion time. Next modify the linear programming model to find the minimum total added cost if the total project time is to be reduced by 1 or 2 months. What is that added cost and for which tasks?

Given known target project completion time, T , and task durations, d_i , and costs $C_i(\delta_i)$ and $P(\Delta)$ for unknown values of δ_i and Δ , the problem is to

$$\begin{array}{ll}
 \text{Minimize} & P(\Delta) + \sum_i C_i(\delta_i) \\
 \\
 \text{Subject to} & t_i \geq t_h + d_h - \delta_h \quad \forall i, h \in H_i \\
 & \delta_i \leq d_i \text{ or some other limit} < d_i \quad \forall i \\
 & T + \Delta \geq t_i + d_i - \delta_i \quad \forall \text{ final tasks } i \\
 & t_i, \delta_i, \Delta \geq 0 \quad \forall i
 \end{array}$$

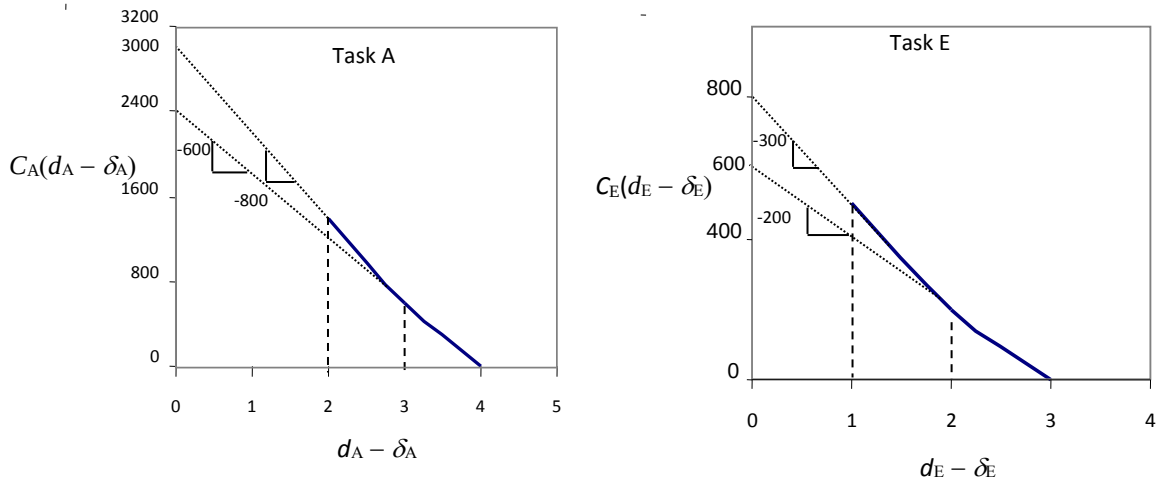
The objective function components may have to be made piece-wise linear for solution by LP

Alternatively, the minimum cost to shorten the project can be determined. This added cost can then be compared to the penalty, $P(\Delta)$.

$$\begin{array}{ll}
 \text{Minimize} & \sum_i C_i(\delta_i) \\
 \\
 \text{Subject to} & t_i \geq t_h + d_h - \delta_h \quad \forall i, h \in H_i \\
 & \delta_i \leq d_i \text{ or some other limit} < d_i \quad \forall i \\
 & T \geq t_i + d_i - \delta_i \quad \forall \text{ final tasks } i \\
 & t_i, \delta_i \geq 0 \quad \forall i
 \end{array}$$

Again, the objective function components may have to be made piece-wise linear for solution by LP

The Cost functions for task A and E can be linearized. The objective is to minimize the added cost of reducing the project time. The shortened project duration, T , of the project can be set at any value less than or equal to 14, which is the normal completion time.



Using the cost functions above, the Lingo model for the given data is following:

```
!Objective;
min = C;

C = CA + CE;          !costs for reduction in durations of A and E;

! Subject to;
3000 - 800*(dA - delA) <= CA; ! These are the linearized cost functions;
2400 - 600*(dA - delA) <= CA; ! as shown on the graphs;
800 - 300*(dE - delE) <= CE;
600 - 200*(dE - delE) <= CE;
TD >= TE + dE - delE; ! delE is the possible reduction in duration of task E;
TD >= TA + dA - delA;  ! delA is the possible reduction in duration of
task A;
TC >= TE + dE - delE; TC >= TA + dA - delA; TB >= TA + dA - delA;
TH >= TG + dG; TG >= TF + dF; TG >= TC + dC; TG >= TB + dB; TF >= TD +
dD;
T >= TH + dH;

Data:
T = ?; dA = 4; dB = 1; dC = 1; dD = 2; dE = 3; dF = 2; dG = 4; dH = 2;
enddata
```

Solution; $T = 14$ (No Reduction in Duration):

Global optimal solution found at step: 7		
Objective value: 0.0000000E+00		
Variable	Value	Reduced Cost
C	0.0000000	0.0000000
CA	0.0000000	1.0000000
CE	0.0000000	1.0000000
DA	4.0000000	0.0000000
DELA	0.0000000	0.0000000
DE	3.0000000	0.0000000
DELE	0.0000000	0.0000000
TD	4.0000000	0.0000000
TE	0.0000000	0.0000000
TA	0.0000000	0.0000000
TC	7.0000000	0.0000000
TB	7.0000000	0.0000000
TH	12.0000000	0.0000000
TG	8.0000000	0.0000000
DG	4.0000000	0.0000000
TF	6.0000000	0.0000000
DF	2.0000000	0.0000000
DC	1.0000000	0.0000000
DB	1.0000000	0.0000000
DD	2.0000000	0.0000000
T	14.0000000	0.0000000
DH	2.0000000	0.0000000

Solution; $T = 13$ [Note: Upon reducing the total project time by 1 month, task A reduces by one month. This task is on the critical path. The available float (slack in the solution) of 1 day for task E is lost and it too is now on the critical path]:

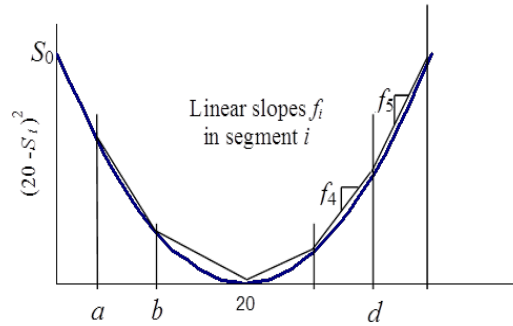
Global optimal solution found at step: 9		
Objective value:		
600.0000		
Variable	Value	Reduced Cost
C	600.0000	0.0000000
CA	600.0000	0.0000000
CE	0.0000000	1.0000000
DA	4.000000	0.0000000
DELA	1.000000	0.0000000
DE	3.000000	0.0000000
DELE	0.0000000	0.0000000
TD	3.000000	0.0000000
TE	0.0000000	0.0000000
TA	0.0000000	600.0000
TC	6.000000	0.0000000
TB	6.000000	0.0000000
TH	11.00000	0.0000000
TG	7.000000	0.0000000
DG	4.000000	0.0000000
TF	5.000000	0.0000000
DF	2.000000	0.0000000
DC	1.000000	0.0000000
DB	1.000000	0.0000000
DD	2.000000	0.0000000
T	13.00000	0.0000000
DH	2.000000	0.0000000

Solution; $T = 12$: [Both tasks incur costs, duration of task A reduces by 2 months and the duration of task E reduces by one month.]

Global optimal solution found at step:		12
Objective value:		1600.000
Variable	Value	Reduced Cost
C	1600.000	0.0000000
CA	1400.000	0.0000000
CE	200.0000	0.0000000
DA	4.000000	0.0000000
DELA	2.000000	0.0000000
DE	3.000000	0.0000000
DELE	1.000000	0.0000000
TD	2.000000	0.0000000
TE	0.0000000	200.0000
TA	0.0000000	800.0000
TC	5.000000	0.0000000
TB	5.000000	0.0000000
TH	10.00000	0.0000000
TG	6.000000	0.0000000
DG	4.000000	0.0000000
TF	4.000000	0.0000000
DF	2.000000	0.0000000
DC	1.000000	0.0000000
DB	1.000000	0.0000000
DD	2.000000	0.0000000
T	12.00000	0.0000000
DH	2.000000	0.0000000

4.36 Solve the reservoir operation problem described in Exercise 4.15 using linear programming. If the reservoir capacity is unknown, show how a cost function (that includes fixed costs and economies of scale) for the reservoir capacity could be included in the linear programming model.

$$\begin{aligned}
 &\text{Minimize} && \sum_{t=1}^T (20 - S_t)^2 + (25 - R_t)^2 \\
 &\text{Subject to} && S_t + I_t - R_t = S_{t+1} && \forall_t \text{ if } t = 3, t+1 = 1. \\
 &&& S_t \leq K && \forall_t \\
 &&& (20 - S_t)^2 \cong S_O + f_1 S_{1t} + f_2 S_{2t} + f_3 S_{3t} + f_4 S_{4t} + \dots \\
 &&& \text{and } S_{it} \leq \text{width of segment } i && \forall_i \\
 &&& S_t = \sum_i S_{it}
 \end{aligned}$$



or

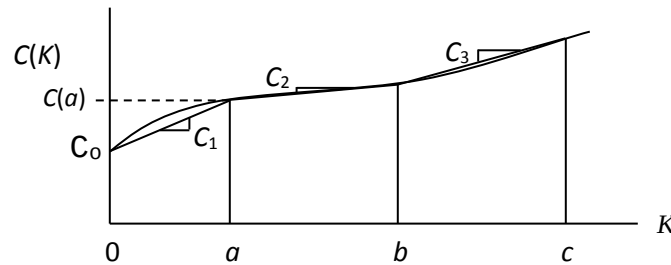
$$(20 - S_t)^2 \cong S_0 w_1 + (20 - a)^2 w_2 + (20 - b)^2 w_3 + 0w_4 + (20 - c)^2 w_5 + (20 - d)^2 w_6$$

where $\sum_i w_i = 1$ and $S_t = aw_2 + bw_3 + 20w_4 + cw_5 + dw_6$

A similar procedure can be used to define $(25 - R_t)^2$ and the separate linear functions and their associated constraints to replace the original separable nonlinear objective function.

Added to the objective function would be the fixed cost times an integer 0,1 variable z_1 plus the linearized variable cost.

For example:



$$C(K) = C_0 z_1 + C_1 K_1 + C(a) z_2 + C_2 K_2 + C_3 K_3$$

where $K_1 \leq a z_1$

$$K_2 \leq (b - a) z_2$$

$$K_3 \leq (c - b) z_2$$

$$K = K_1 + a z_2 + K_2 + K_3$$

$z_1 + z_2 \leq 1$ and z_1, z_2 are 0,1 binary variables.

4.37 An upstream reservoir could be built to serve two downstream users. Each user has a constant water demand target. The first user's target is 30; the second user's target is 50. These targets apply to each of 6 within-year seasons. Find the tradeoff between the required reservoir capacity and maximum deficit to any user at any time, for an average year. The average flows into the reservoir in each of the six successive seasons are: 40, 80, 100, 130, 70, 50.

X_1 = the first user's target allocation in each of six within-year seasons
 X_2 = the second user's target allocation in each of 6 within-year seasons
 Q_t = the average flow into the reservoir in season t , $t = 1, 2, 3, \dots, 6$
 K = storage capacity of the reservoir

Lingo Program

```

MIN = D;

30 - X1 < D; ! Deficit to user 1;
50 - X2 < D; ! Deficit to user 2;
S1 + Q1 - X1 - X2 > S2; S1 < K;
S2 + Q2 - X1 - X2 > S3; S2 < K;
S3 + Q3 - X1 - X2 > S4; S3 < K;
S4 + Q4 - X1 - X2 > S5; S4 < K;
S5 + Q5 - X1 - X2 > S6; S5 < K;
S6 + Q6 - X1 - X2 > S1; S6 < K;

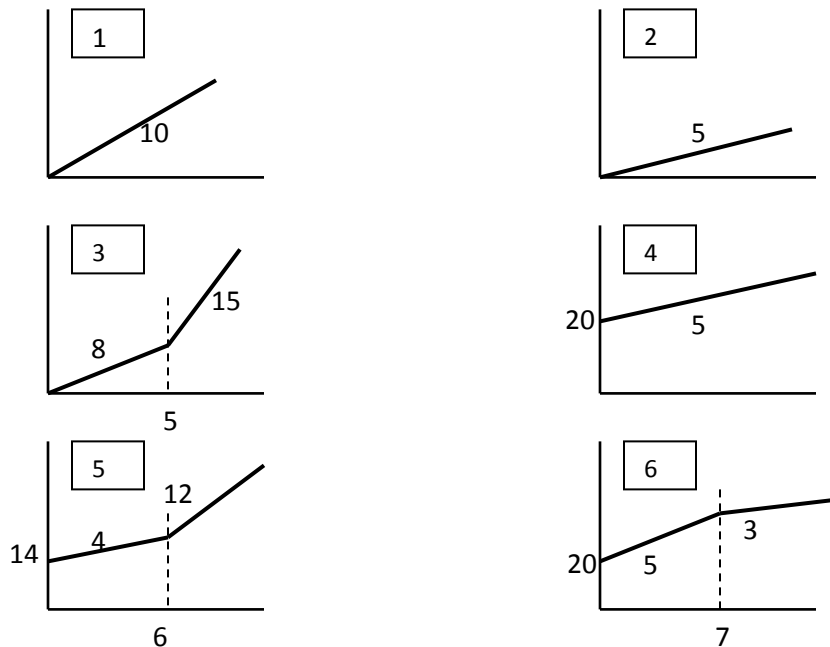
DATA: Q1 = 40; Q2 = 80; Q3 = 100; Q4 = 130; Q5 = 70;
Q6 = 50; K = ?; ENDDATA
  
```

K	D	X1	X2	S1	Q1	S2	Q2	S3	Q3	S4	Q4	S5	Q5	S6	Q6
0	20	10	30	0	40	0	80	0	100	0	130	0	70	0	50
10	15	15	35	10	40	0	80	0	100	0	130	10	70	10	50
20	12.5	17.5	37.5	15	40	0	80	0	100	0	130	5	70	20	50
30	10	20	40	20	40	0	80	0	100	0	130	20	70	30	50
40	7.5	22.5	42.5	25	40	0	80	0	100	0	130	35	70	40	50
50	5	25	45	30	40	0	80	0	100	0	130	50	70	50	50
60	3.33	26.67	46.67	33.33	40	0	80	0	100	3.33	130	60	70	56.67	50
70	1.67	28.33	48.33	36.67	40	0	80	0	100	16.67	130	70	70	63.33	50
74	1	29	49	38	40	0	80	0	100	22	130	74	70	66	50
75	0.83	29.17	49.17	38.33	40	0	80	1.67	100	23.33	130	75	70	66.67	50
80	0.83	29.17	49.17	38.33	40	0	80	1.67	100	23.33	130	75	70	66.67	50
100	0.83	29.17	49.17	38.33	40	0	80	1.67	100	23.33	130	75	70	66.67	50

When $K = 0$, the maximum deficit to each user is 20, and the minimum deficit to each user is 0.83 when K greater or equal to 75.

4.38 Two groundwater well fields can be used to meet the water demands of a single user. The maximum capacity of the A well field is 15 units of water per period, and the maximum capacity of the B well field is 10 units of water per period. The annual cost of building and operating each well field, each period, is a function of the amount of water pumped and transported from that well field. Three sets of cost functions are shown below: Construct a LP model and use it to define and then plot the total least-cost function and the associated individual well field capacities required to meet demands from 0 to 25, assuming cost functions 1 and 2 apply to well fields A and B respectively. Next define another least-cost function and associated capacities assuming cost functions 3 and 4 apply to A and B respectively. Finally define a least-cost function and associated capacities assuming well field cost functions 5 and 6 apply. You can check your model

results just using common sense – the least-cost functions should be obvious, even without using optimization.



Define:

W_a = unit of water from A well field
 W_b = unit of water from B well field
 K_a = max. capacity of A well field
 K_b = max. capacity of B well field
 D_w = demand of the user = 0, 1, 2, ..., 25.

For cost functions 1 and 2:

C_a = cost per unit of water getting from A well field
 C_b = cost per unit of water getting from B well field

Lingo model:

```

MIN = CostA + CostB;

CostA      = (Ca*Wa);
CostB      = (Cb*Wb);

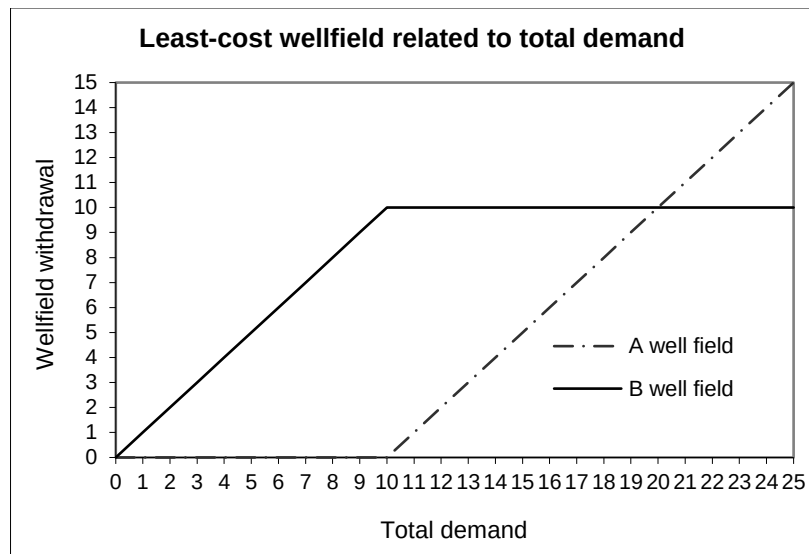
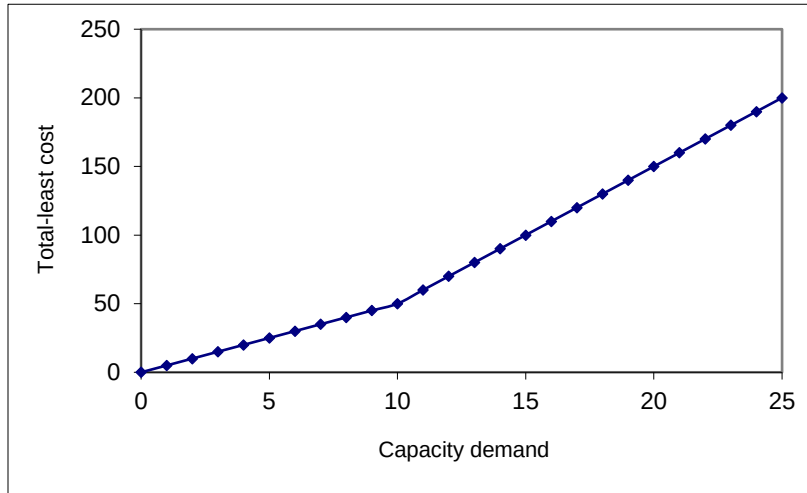
Wa          < Ka;
Wb          < Kb;
Wa + Wb     = Dw;

DATA:
Ca = 10;    Cb = 5;   Ka = 15;   Kb = 10;
Dw = ?;
ENDDATA
  
```

Solutions:

Demand	0	1	2	3	4	5	6	7	8	9	10	11	12
Total -least cost	0	5	10	15	20	25	30	35	40	45	50	60	70

Demand	13	14	15	16	17	18	19	20	21	22	23	24	25
Total -least cost	80	90	100	110	120	130	140	150	160	170	180	190	200



Define for cost functions 3 and 4:

Wa1, Wa2 = unit of water from A well field at different cost/unit
 Wb = unit of water getting from B well field
 Ca1, Ca2 = cost per Wa1 and Wa2, respectively, getting from A well field
 Cb = cost per unit of water getting from B well field
 Cob = fixed cost of B well field
 Ib = integer variable, 0, 1

Lingo program

```

MIN = CostA + CostB;

CostA = (Ca1*Wa1 + Ca2*Wa2);
CostB = (Cob*Ib + Cb*Wb);

Wa1      < 5;
Wa2      < Ka-5;
Wa       = Wa1 + Wa2;
Wb       < Kb*Ib;
Wa + Wb  = Dw;
@BIN(Ib);

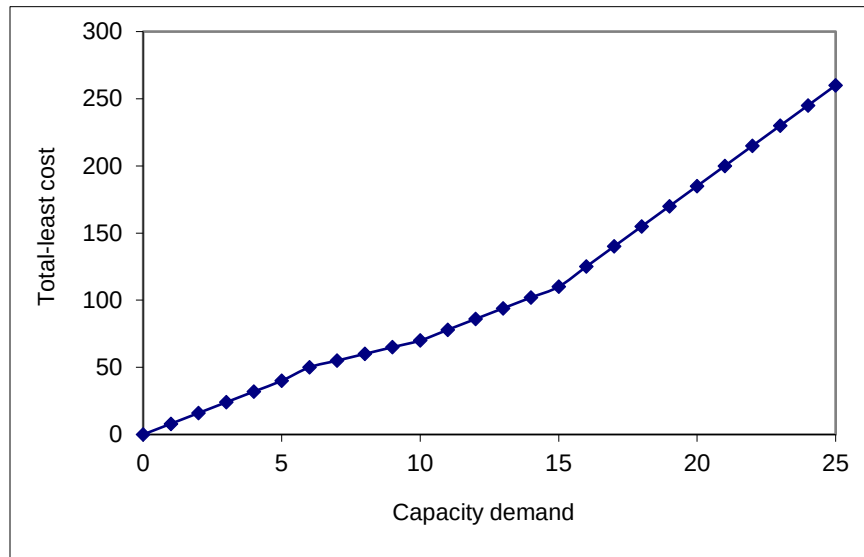
DATA:
    Ca1 = 8; Ca2 = 15; Cb = 5; Cob = 20;
    Ka = 15; Kb = 10;
    Dw = ?;
ENDDATA

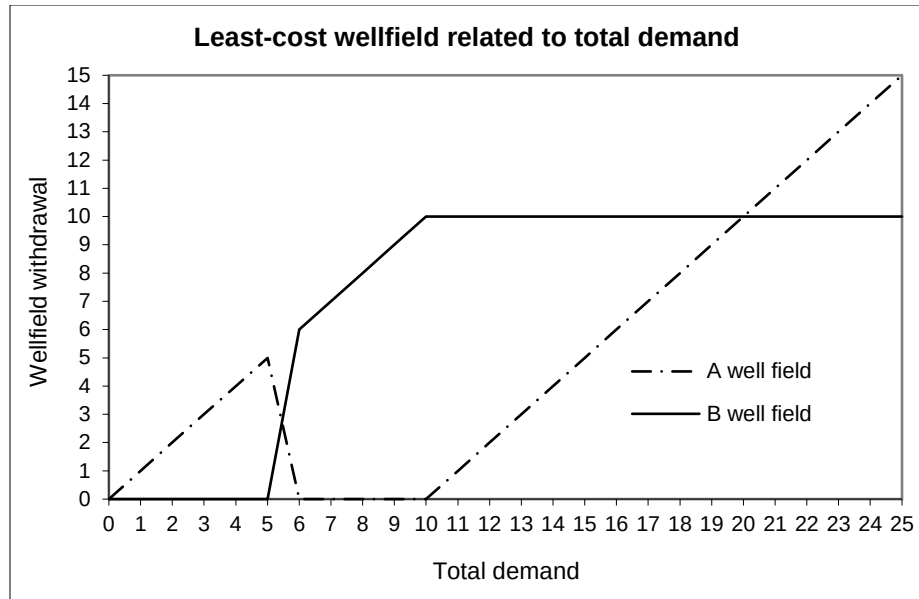
```

Solutions

Demand	0	1	2	3	4	5	6	7	8	9	10	11	12
Total -least cost	0	8	16	24	32	40	50	55	60	65	70	78	86

Demand	13	14	15	16	17	18	19	20	21	22	23	24	25
Total -least cost	94	102	110	125	140	155	170	185	200	215	230	245	260





Note: the actual transition between A and B is between 5.5 and 5.6.

Cost functions 5 and 6

Lingo program

```

MIN = CostA + CostB;

CostA      = (Coa*Ia + Ca1*Wa1 + Ca2*Wa2);
CostB      = (Cob1*Ib1 + Cb1*Wb1 +
              Cob2*Ib2 + Cb2*Wb2);

Wa1        < 6*Ia;
Wa2        < (Ka-6)*Ia;
Wa         = Wa1 + Wa2;
Wb1        < 7*Ib1;
Wb2        < (Kb-7)*Ib2;
Wb         = Wb1 + Wb2;
Dw         = Wa + Wb;
@BIN(Ia);  @BIN(Ib1);  @BIN(Ib2);

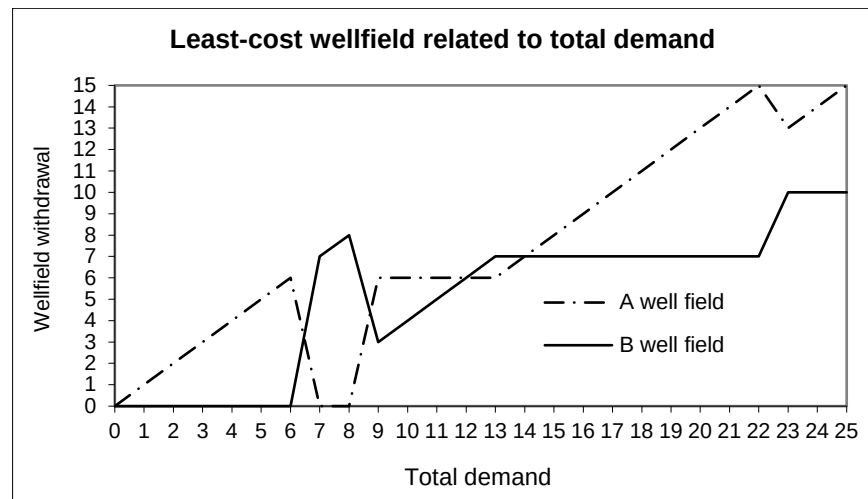
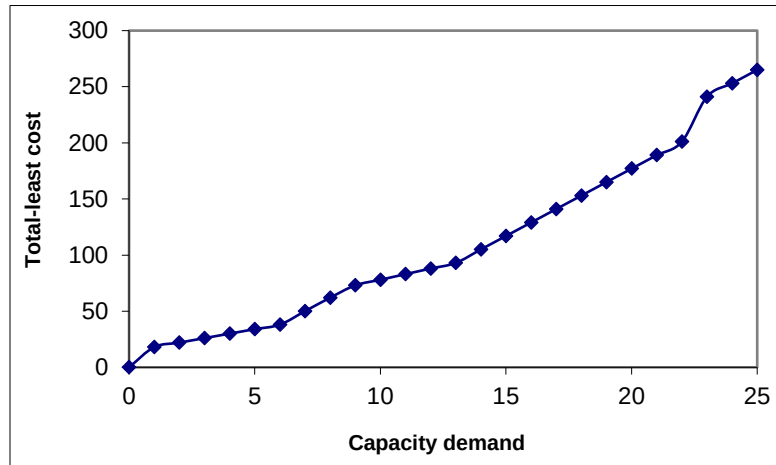
DATA:
Ka      = 15;  Kb      = 10;
Coa     = 14;  Cob1    = 20;  Cob2    = 55;
Ca1     = 4;   Ca2     = 12;  Cb1     = 5;
Cb2     = 3;   Dw      = ?;
ENDDATA

```

Solutions

Demand	0	1	2	3	4	5	6	7	8	9	10	11	12
Total -least cost	0	18	22	26	30	34	38	50	62	73	78	83	88

Demand	13	14	15	16	17	18	19	20	21	22	23	24	25
Total -least cost	93	105	117	129	141	153	165	177	189	201	241	253	265



Note: More accurate plots would show vertical lines when A goes to 0 and B enters solution, or when A becomes constant and B increases.

4.39 Referring to Exercise 4.38 above, assume cost functions 5 and 6 represent the cost of adding additional capacity to well fields A and B respectively in any of the next 5 five-year construction periods, i.e., in the next 25 years. Identify and plot the least-cost capacity expansion schedule (one that minimizes the total present value of current and future expansions, assuming demands of 5, 10, 15, 20 and 25 are to be met at the end of years 5, 10, 15, 20 and 25 respectively. Costs, including fixed costs, of capacity expansion in each construction period have to be paid at the beginning of the construction period. Determine the sensitivity of your solution to the interest rate used to compute present value.

Lingo model shown on next page.

In this problem for $r = 0.00$ to 0.10 the least-cost capacity expansion schedule is to add 5 units of capacity to well field A at the 1st, 2nd and 3rd periods, and add 5 units of capacity to well field B at the 3rd and 5th periods to obtain the present value of total least-cost of

adding a total 25 units of capacity within 25-year period. Obviously the present value of the total cost decreases as r increases.

The optimal capacity to meet the total demand of 25 units of capacity at the end of year 25 is insensitive to the discount rate, r , in this problem but this is not always the case.

Lingo program

```

MIN = C1 + C2 + C3 + C4 + C5;

C1 = (Coa*Ia1 + Ca1*A11 + Ca2*A12 + Cob1*Ib11 + Cb1*B11 + Cob2*Ib12 +
      Cb2*B12)*(1/(1+r))^(5*(t1-1));
C2 = (Coa*Ia2 + Ca1*A21 + Ca2*A22 + Cob1*Ib21 + Cb1*B21 + Cob2*Ib22 +
      Cb2*B22)*(1/(1+r))^(5*(t2-1));
C3 = (Coa*Ia3 + Ca1*A31 + Ca2*A32 + Cob1*Ib31 + Cb1*B31 + Cob2*Ib32 +
      Cb2*B32)*(1/(1+r))^(5*(t3-1));
C4 = (Coa*Ia4 + Ca1*A41 + Ca2*A42 + Cob1*Ib41 + Cb1*B41 + Cob2*Ib42 +
      Cb2*B42)*(1/(1+r))^(5*(t4-1));
C5 = (Coa*Ia5 + Ca1*A51 + Ca2*A52 + Cob1*Ib51 + Cb1*B51 + Cob2*Ib52 +
      Cb2*B52)*(1/(1+r))^(5*(t5-1));

A11 < 6*Ia1; A21 < 6*Ia2; A31 < 6*Ia3; A41 < 6*Ia4; A51 < 6*Ia5;
A12 < (Ka-6)*Ia1; A22 < (Ka-6)*Ia2; A32 < (Ka-6)*Ia3;
A42 < (Ka-6)*Ia4; A52 < (Ka-6)*Ia5;

@BIN(Ia1); @BIN(Ia2); @BIN(Ia3); @BIN(Ia4); @BIN(Ia5);
A1 = A11 + A12; A2 = A21 + A22; A3 = A31 + A32; A4 = A41 + A42;
A5 = A51 + A52;
A1 + A2 + A3 + A4 + A5 < Ka;
B11 < 7*Ib11; B21 < 7*Ib21; B31 < 7*Ib31; B41 < 7*Ib41; B51 < 7*Ib51;
B12 < (Kb-7)*Ib12; B22 < (Kb-7)*Ib22; B32 < (Kb-7)*Ib32;
B42 < (Kb-7)*Ib42; B52 < (Kb-7)*Ib52;

@BIN(Ib11); @BIN(Ib12); @BIN(Ib21); @BIN(Ib22); @BIN(Ib31);
@BIN(Ib32); @BIN(Ib41); @BIN(Ib42); @BIN(Ib51); @BIN(Ib52);

B1 = B11 + B12; B2 = B21 + B22; B3 = B31 + B32; B4 = B41 + B42;
B5 = B51 + B52; B1 + B2 + B3 + B4 + B5 < Kb;

A1 + B1 > 5; A1 + A2 + B1 + B2 > 10; A1 + A2 + A3 + B1 + B2 + B3 > 15;
A1 + A2 + A3 + A4 + B1 + B2 + B3 + B4 > 20; A1 + A2 + A3 + A4 + A5 +
B1 + B2 + B3 + B4 + B5 > 25;

DATA
Ka = 15; Kb = 10; Coa = 14; Cob1 = 20; Cob2 = 55; Ca1 = 4; Ca2 = 12;
Cb1 = 5; Cb2 = 3; t1 = 1; t2 = 2; t3 = 3; t4 = 4; t5 = 5; r = ?;
ENDDATA

```

4.40 Consider a crop production problem involving three types of crops. How many hectares of each crop should be planted to maximize total income?

<u>Resources:</u>	<u>Max Limits</u>	<u>Resource requirements</u>			
		<u>Crops:</u>	<u>Corn</u>	<u>Wheat</u>	<u>Oats</u>
Water	1000/week		3.0	1.0	1.5 units/week/ha
Labor	300/week		0.8	0.2	0.3 person hrs/week/ha
Land	625 hectares				
<u>Yield</u> \$/ha			400	200	250

Show a graph that identifies the tradeoffs among crops that can be made without reducing the total income.

X_i = hectares of planting crop i ; i = corn, wheat and oats, respectively
 W_i = water needed by growing each unit of crop i ; per week per ha
 H_i = labor needed for growing each unit of crop i ; per week per ha
 L = total land available = 625 ha
 Y_i = yield/ha of growing crop i

Lingo Program

```

MAX    = Corn*C + Wheat*W + Oats*O;

3*Corn  +   Wheat   + 1.5*Oats < 1000; !water constraint;
0.8*Corn + 0.2*Wheat + 0.3*Oats < 300; !labor constraint;
Corn    + Wheat    + Oats     < 625; !land constraint;

DATA:
C = 400; W = 200; O = 250;
ENDDATA

```

Results from the model

Objective value: 162500.0

Yields:

Income:

Corn = 41.67

16666.67

Wheat = 0.00

0.00

Oats = 583.33

145833.3

or

Corn = 187.5

75000.00

Wheat = 437.5

87500.00

Oates = 0.0

0.0

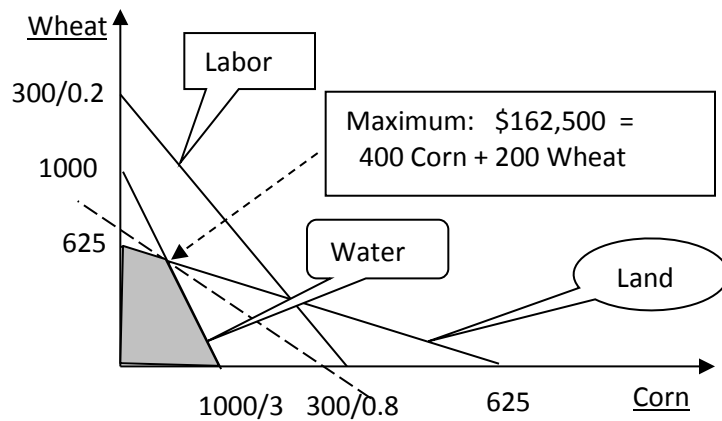
The maximum total income obtaining from the model is 162500. Oats can range from 0 to 583.33 and while corn can range from 187.5 to 41.667 and wheat from 437.5 to 0. The tradeoffs are linear.

Solution is \$ 162,500.

Corn = 187.5 acres; Wheat = 437.5 acres; Oats = 0 acres.

Dual variables: Water = 100; Labor = 0; Land = 100

Graphical Solution. Consider only Corn and Wheat as the unknowns

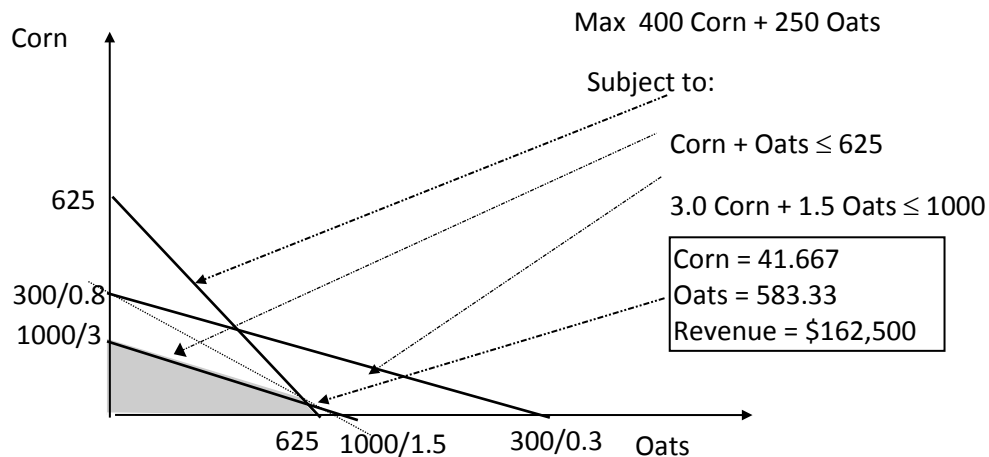


Ranges over which Corn and Wheat will both be in solution, and for which Water and Land will be the binding constraints and for which the dual variables will apply:

Variable:	Current obj. coef.	Max Increase	Max Decrease
Corn	400	200	200
Wheat	200	200	66.667
Row:	Current RHS		
Water	1000	208.3	375
Labor	300	inf.	62.5
Land	625	375	291.667

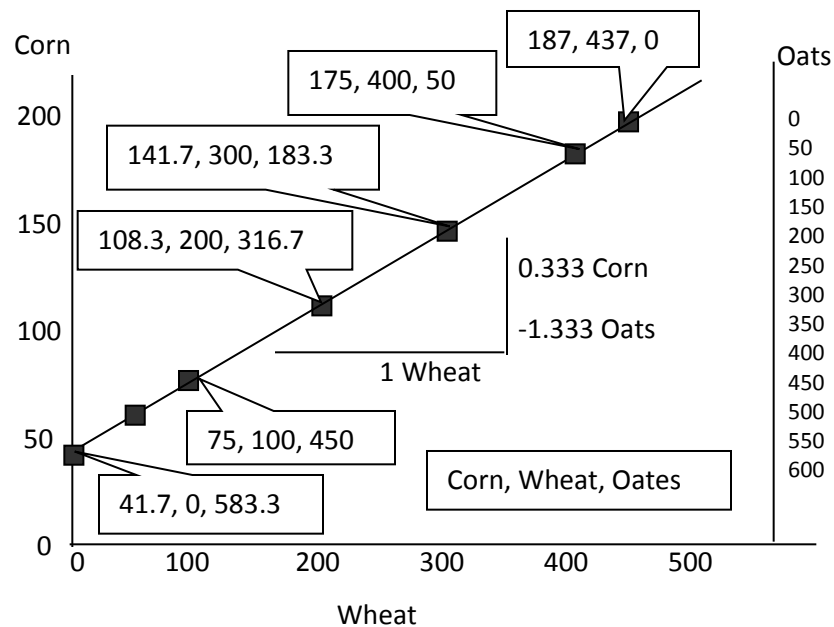
Relative range of objective function coefficients -
 $3 \text{ Corn} + 1 \text{ Wheat} \rightarrow 1 \text{ Corn} + 1 \text{ Wheat}$

But another optimal solution when Wheat = 0



So, there must be a tradeoff between Corn, Wheat and Oats that you can't see in 2 dimensions.

Combinations of Corn, Wheat and Oats produce same profit, by maximizing say Corn, or Wheat, or Oats.



4.41 Releases from a reservoir are used for water supply or for hydropower. The benefit per unit of water allocated to hydropower is BH and the benefit per unit of water allocated to water supply is BW . For any given release the difference between the allocations to the two uses cannot exceed 50% of the total amount of water available. Show graphically how to determine the most profitable allocation of the water for some assumed values of BH and BW . From the graph identify which constraints are binding and what their “dual prices” mean (in words).

- Q = amount of available water, 100 percent units.
- R = amount (%) of water that can be released to uses
- X_w = % water allocating for water supply
- X_h = % water allocating for hydropower
- B_w = benefit per percent of water allocated to water supply, assumed to be 10
- B_h = benefit per percent of water allocated to hydropower, assumed to be 15

Max: $B_w \cdot X_w + B_h \cdot X_h$

S.T:

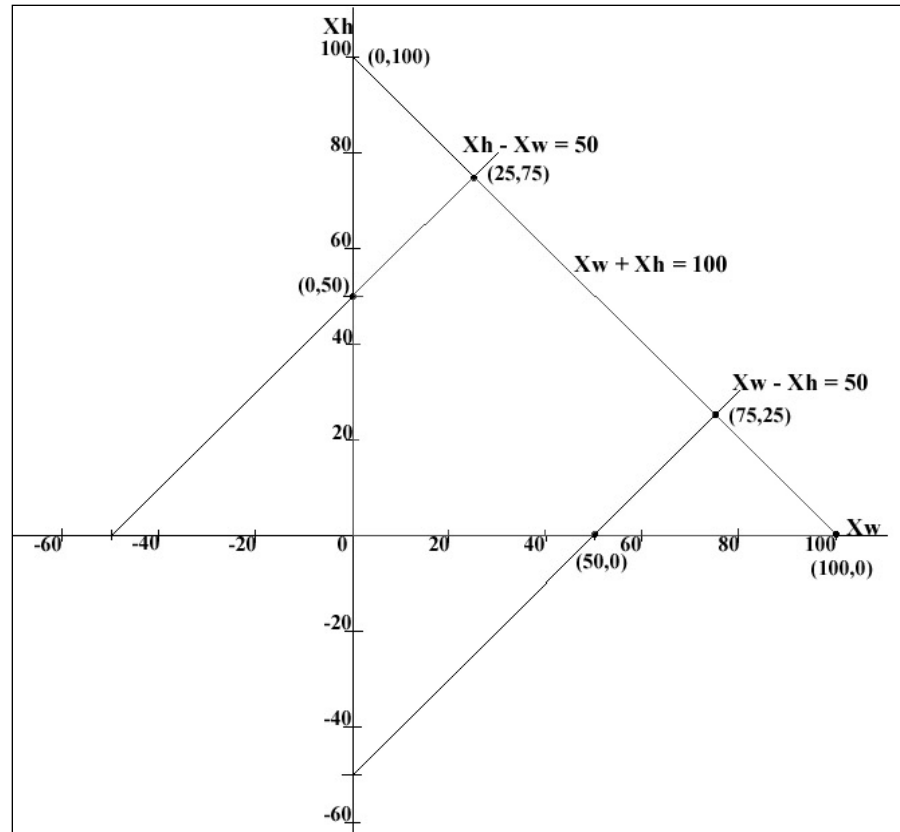
$$X_w - X_h \leq 0.5 \cdot Q$$

$$X_h - X_w \leq 0.5 \cdot Q$$

$$X_w + X_h = R$$

$$R \leq Q$$

All variables are nonnegative



According to the graph above, when the benefit per unit of water allocated to water supply is 10 and that of water allocated to hydro power is 15, the most profitable allocation is to allocate 75 percent of water for hydro power and 25 percent of water for water supply to obtain the optimal objective function value of $(25 \times 10 + 75 \times 15) = 1375$.

All constraints except the first one are binding meaning that all 100 units of available water are released from the reservoir for the both uses. For this maximization problem, the dual price is the amount that the optimal objective function will increase if the value of the right hand side (RHS) of any binding constraint increases one more unit, assuming all coefficient values of the remaining constraints are constant.

4.42 Suppose there are four water users along a river who benefit from receiving water. Each has a known water target, i.e., each expects and plans for a specified amount. These known water targets are W_1 , W_2 , W_3 , and W_4 for the four users respectively. Find two allocation policies. One is to be based on minimizing the maximum deficit deviation from any target allocation. The other is to be based on minimizing the maximum percentage deficit from any target allocation.

Deficit allocations are allocations that are less than the target allocation. For example if a target allocation is 30 and the actual allocation is 20, the deficit is 10. Water in excess of the targets can remain in the river. The policies are to indicate what the allocations should be for any particular river flow Q . The policies can be expressed on a graph showing the amount of Q on the horizontal axis, and each user's allocation on the vertical axis.

Create the two optimization models that can be used to find the two policies and indicate how they would be used to define the policies. What are the unknown variables and what are the known variables? Specify the model in words as well as mathematically.

Unknown variables

D = maximum deficit to any user

X_i = actual allocation to each user i

R = amount of water release

Known variables

W_i = target water of user i ; $i = 1,2,3,4$

Q = river flow

S = water exceeded and remained in the river

Minimize the maximum deficit deviation D from any target allocation W .

```

MIN = D;
      W1 - X1 < D;
      W2 - X2 < D;
      W3 - X3 < D;
      W4 - X4 < D;
      X1 + X2 + X3 + X4 < R;
      R + S = Q;

DATA:
W1 = 10;   W2 = 15;   W3 = 20; W4 = 25; Q=?;
ENDDATA

```

To find the percentage deficit that will be equal for all users:

```

Max = 100*PD;
PD < X1/W1;
PD < X2/W2;
PD < X3/W3;
PD < X4/W4;
PD < 1 ;
X1 + X2 + X3 + X4 < R;
R + S = Q;

DATA:
W1 = 10;   W2 = 15;   W3 = 20; W4 = 25;
Q = ?;
ENDDATA

```

4.43 In Indonesia there exists a wet season followed by a dry season each year. In one area of Indonesia all farmers within an irrigation district plant and grow rice during the wet season. This crop brings the farmer the largest income per hectare; thus they would all prefer to continue growing rice during the dry season. However, there is insufficient water during the dry season to irrigate all 5000 hectares of available irrigable land for rice production. Assume an available irrigation water supply of $32 \times 10^6 \text{ m}^3$ at the beginning of each dry season, and a minimum requirement of $7000 \text{ m}^3/\text{ha}$ for rice and $1800 \text{ m}^3/\text{ha}$ for the second crop.

- (a) What proportion of the 5000 hectares should the irrigation district manager allocate for rice during the dry season each year, provided that all available hectares must be given sufficient water for rice or the second crop?
- (b) Suppose that crop production functions are available for the two crops, indicating the increase in yield per hectare per m^3 of additional water, up to $10,000 \text{ m}^3/\text{ha}$ for the second crop. Develop a model in which the water allocation per hectare, as well as the hectares allocated to each crop, is to be determined, assuming a specified price or return per unit of yield of each crop. Under what conditions would the solution of this model be the same as in part (a)?

a) Let X_R be the hectares of rice to be grown during the dry season and $5000 - X_R$ be the hectares of the second crop. The total amount of water required is

$$7000 X_R + 1800(5000 - X_R) = 32 \times 10^6$$

$$\text{Hence } X_R = [(32 \times 10^6) - ((1.8)(5) \times 10^6)] / [(7 - 1.8) \times 10^3]$$

$$X_R = 4435 \text{ ha of rice}$$

(b) Let the known parameters:

P_R, P_S equal the price per unit yield of rice and the second crop

Y_R, Y_S equal the yield per hectare given the original allocations of water (7000 and $1800 \text{ m}^3/\text{ha}$)

y_R, y_S equal the additional yield per hectare per m^3 of additional water.

Define as the unknown variables

X_R, X_S = the hectares of rice and second crop

W_R, W_S = the additional water allocations per ha

Assuming the objective is to maximize total income, the model can be written:

$$\text{Maximize} \quad P_R(Y_R + y_R W_R)X_R + P_S(Y_S + y_S W_S)X_S$$

Subject to:

Water availability

$$X_R (7000 + W_R) + X_S (1800 + W_S) \leq 32 \times 10^6$$

Land availability

$$X_R + X_S \leq 5000$$

Bounds

$$W_S \leq 10,000$$

The solution of this model will be the same as (a) if it does not pay to increase the water allocations to the crops, and the price times yield per ha of rice exceeds the price times yield per ha of the second crop.

4.44 Along the Nile River in Egypt, irrigation farming is practiced for the production of cotton, maize, rice, sorghum, full and short berseem for animal production, wheat, barley, horsebeans, and winter and summer tomatoes. Cattle and buffalo are also produced, and together with the crops that require labor, water, Fertilizer, and land area (feddans). Farm types or management practices are fairly uniform, and hence in any analysis of irrigation policies in this region this distinction need not be made. Given the accompanying data develop a model for determining the tons of crops and numbers of animals to be grown that will maximize (a) net economic benefits based on Egyptian prices, and (b) net economic benefits based on international prices. Identify all variables used in the model.

Known parameters:

C_i	= miscellaneous cost of land preparation per feddan
P_i^E	= Egyptian price per 1000 tons of crop i
P_i^I	= international price per 1000 tons of crop i
v	= value of meat and dairy production per animal
g	= annual labor cost per worker
f^P	= cost of P fertilizer per ton
f^N	= cost of N fertilizer per ton
Y_i	= yield of crop i , tons/feddan
α	= feddans serviced per animal
β	= tons straw equivalent per ton of berseem carryover from winter to summer
r^w	= berseem requirements per animal in winter
s^{wh}	= straw yield from wheat, tons per feddan
s^{ba}	= straw yield from barley, tons per feddan
r^s	= straw requirements per animal in summer
μ_i^N	= N fertilizer required per feddan of crop i
μ_i^P	= P fertilizer required per feddan of crop i
l_{im}	= labor requirements per feddan in month m , man-days
w_{im}	= water requirements per feddan in month m , 1000 m ³
h_{im}	= land requirements per month, fraction (1 = full month)

Required Constraints. (assume known resource limitations for labor, water, and land):

- Summer and winter fodder (berseem) requirements for the animals.
- Monthly labor limitations.
- Monthly water limitations.
- Land availability each month.
- Minimum number of animals required for cultivation.
- Upper bounds on summer and winter tomatoes (assume these are known).
- Lower bounds on cotton areas (assume this is known).

Other possible constraints:

- Crop balances.

- (b) Fertilizer balances.
- (c) Labor balance.
- (d) Land balance.

This model is a simplification of one developed by Dr. G. Kutcher and his colleagues at the World Bank and in Egypt for studying the agricultural economy of an area of the Nile valley.

Maximize total net income based on Egyptian or International prices (P_i^E or P_i^I)

$$\text{Maximize } \sum_i P_i Q_i \quad + \quad vA \quad - \quad TC$$

Crop income + animal total
Income cost

where Q_i is tons of crop i produced and A is the number of animals produced.

Subject to:

Definition of total cost

$$TC = \sum_i C_i X_i + gL + f^P F^P + f^N F^N$$

Total cost	Land Preparation cost	labor cost	Fertilizer Cost
------------	-----------------------	------------	-----------------

where X_i is the feddans of land in crop i .
 L is the total labor required.
 F^P, F^N is the tons of fertilizer required.

Crop Production

$y_i X_i = Q_i$, the tons of crop i produced, $\forall i$ except berseem

$$y_F X_F + y_S X_S = r^w A + QB$$

Quantity of full and short berseem produced	Animal requirements for berseem in winter	Straw Requirement For animals
---	---	-------------------------------

summer fodder (berseem) requirements

$$s^{wh} X_W + s^{ba} X_B \geq -\beta QB + r^s A$$

Straw yield from wheat	Straw yield from barley	equivalent straw carryover	Straw Requirement for animals
------------------------	-------------------------	----------------------------	-------------------------------

Fertilizer balances

$$\sum_f \mu_i^N X_i = F^N$$

$$\sum_f \mu_i^P X_i = F^P$$

Labor Constraints

$$\sum_i l_{im} X_i \leq L_m \quad \forall m$$

$$\left[\begin{array}{c} \text{Labor requirements} \\ \text{Per month} \end{array} \right] \leq \left[\begin{array}{c} \text{labor} \\ \text{supply} \end{array} \right]$$

$$\sum_i \sum_m l_{im} X_i = 240 L$$

$$\left[\begin{array}{c} \text{Labor requirements} \\ \text{in year} \end{array} \right] = \left[\begin{array}{c} \text{Man-days} \\ \text{Per year} \end{array} \right] \left[\begin{array}{c} \text{Number of} \\ \text{workers} \end{array} \right]$$

Water Availability

$$\sum_i w_{im} X_i \leq w_m \quad \forall m$$

$$\left[\begin{array}{c} \text{Water} \\ \text{requirements} \end{array} \right] \leq \left[\begin{array}{c} \text{water} \\ \text{supply} \end{array} \right]$$

Land Availability

$$\sum_i h_{im} X_i \leq Land \quad \forall m$$

$$\left[\begin{array}{c} \text{Land requirements} \\ \text{in month } m \end{array} \right] \leq \left[\begin{array}{c} \text{Land} \\ \text{available} \end{array} \right]$$

Animal Requirements

$$A \leq \frac{1}{\alpha} \sum_i X_i$$

$$\left[\begin{array}{c} \text{Number of} \\ \text{animals} \end{array} \right] \leq \left[\begin{array}{c} \text{Animal needs} \\ \text{Per feddan} \end{array} \right] \left[\begin{array}{c} \text{Number of} \\ \text{feddans} \end{array} \right]$$

Bounds

$$X_i^{\min} \leq X_i \quad \text{as required for cotton}$$

$$y_i X_i \leq y_i X_i^{\min} \quad \text{as required for tomatoes}$$

4.45 In Algeria there are two distinct cropping intensities, depending upon the availability of water. Consider a single crop that can be grown under intensive rotation or extensive rotation on a total of A hectares. Assume that the annual water requirements for the intensive rotation policy are 16000 m^3 per hectare, and for the extensive rotation policy they 4000 m^3 per hectare. The annual net production returns are 4000 and 2000 dinars, respectively. If the total water available is $320,000 \text{ m}^3$, show that as the available land area A increases, the rotation policy that maximizes total net income changes from one that is totally intensive to one that is increasingly extensive.

Would the same conclusion hold if instead of fixed net incomes of 4000 and 2000 dinars per hectares of intensive and extensive rotation, the net income depended on the quantity of crop produced? Assuming that intensive rotation produces twice as much produced by extensive rotation, and that the net income per unit of crop Y is defined by the simple linear function $5 - 0.05Y$, develop and solve a linear programming model to determine the optimal rotation policies if A equals 20, 50, and 80. Need this net income or price function be linear to be included in a linear programming model?

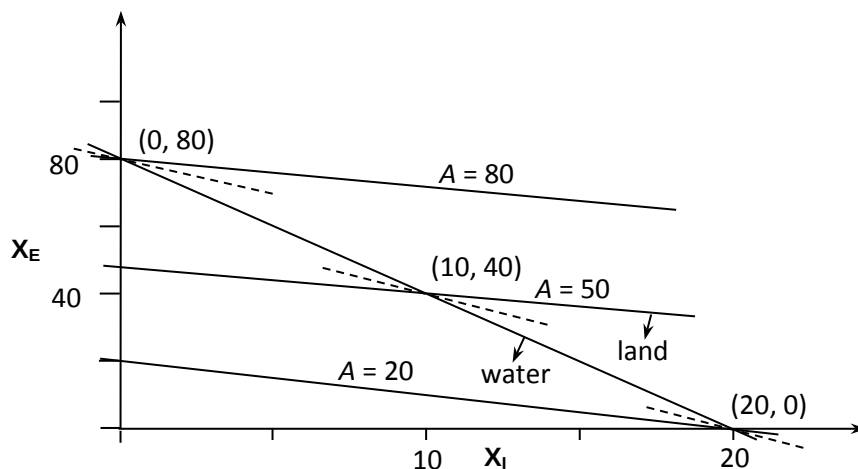
Solution:

Let X_I be the hectares of intensive cropping intensity and X_E be the hectares of extensive cropping intensity. Maximizing net annual production:

$$\text{Maximize} \quad 4000 X_I + 2000 X_E$$

$$\text{Subject to} \quad \begin{aligned} 16 X_I + 4 X_E &\leq 320 \\ X_I + X_E &\leq A \\ X_I, X_E &\geq 0 \end{aligned}$$

Solving graphically:



This graph shows that as A increases, the optimum value of X_I decreases and X_E increases.

If the price times yield per hectare ratio $P_I Y_I / P_E Y_E$ were greater than 4, then for any value of $A > 0$ all available land would be allocated to intensive irrigation up to the limit possible with the available water (in this case up to 20 hectares).

Assuming the yield per hectare of extensive rotation is Y_E then to maximize net income, one would

Maximize	$(5 - 0.05Y) Y$	
Subject to	$Y = Y_E(2X_I + X_E)$	which defines total yield Y
	$16 X_I + 4 X_E \leq 320$	for water
	$X_I + X_E \leq A = \{20, 50, 80\}$	for land
	$X_I, X_E \geq 0$	

For any value of $Y_E > 0$ and $Y \leq 50$, the value of X_I and X_E will be determined by maximizing Y subject to the water and land and non-negativity constraints. Since net income decreases for $Y > 50$, $2X_I + X_E$ will not exceed $50/Y_E$. If Y can exceed 50 it will pay to keep some land idle and given the data available, it is not clear which of all feasible solutions would be preferred.

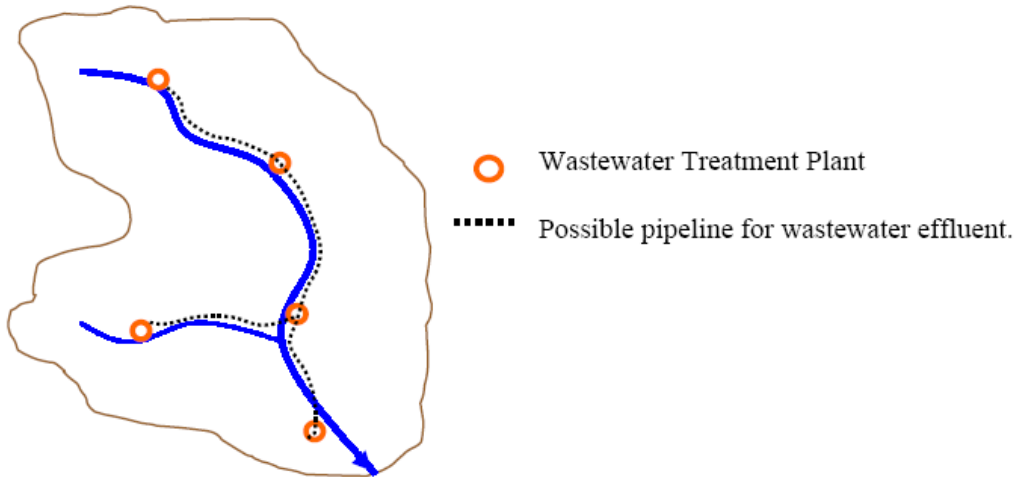
Assuming Y_E is such that Y will not exceed 50, the optimum X_I and X_E can be found by the following model:

Maximize	$2X_I + X_E$
Subject to	$4 X_I + X_E \leq 80$
	$X_I + X_E \leq A = \{20, 50, 80\}$
	$X_I, X_E \geq 0$

This objective has the same slope as the previous model graphed above hence the solutions (X_I, X_E) for $A = \{20, 50, 80\}$, and $(20, 0)$, $(10, 40)$, $(0, 80)$, respectively.

4.46 Current stream quality is below desired minimum levels throughout the stream in spite of treatment at each of the treatment plant and discharge sites shown below. Currently effluent standards are not being met, and minimum desired streamflow concentrations can be met by meeting effluent standards. All current wastewater discharges must undergo additional treatment. The issue is where additional treatment is to occur and how much.

Develop a model to identify cost-effective options for meeting effluent standards where ever wastewater is discharged into the stream. The decisions variables include the amount of wastewater to treat at each site and then release to the river. Any wastewater at any site that is not undergoing additional treatment can be piped to other sites. Identify other issues that could affect the eventual decision.



Assume known current wastewater flows at site $i = q_i$.

Additional treatment to meet effluent standards cost $= a_i + b_i(D_i)^{c_i}$ where D_i is the total wastewater flow undergoing additional treatment at site i and $c_i < 1$.

Pipeline and pumping for each pipeline segment costs approximately $\alpha_{ij} + \beta(q_{ij})^\gamma$ where q_{ij} is pipeline flow between adjacent sites i and j and $\gamma < 1$.

Minimize sum over i $(a_i + b_i(D_i)^{c_i})$ + sum over appropriate ij $(\alpha_{ij} + \beta(q_{ij})^\gamma)$

Subject to:

$$q_1 = D_1 + q_{12} - q_{21}$$

$$q_2 = D_2 - q_{12} + q_{21} + q_{23}$$

$$q_3 = D_3 - q_{23} - q_{43} - q_{53} + q_{32} + q_{34} + q_{35}$$

$$q_4 = D_4 - q_{34} + q_{43}$$

$$q_5 = D_5 - q_{35} + q_{53}$$

need 0,1 variables associated with fixed costs as well, and constraints to make them work.

Chapter 5

5.1 An upstream reservoir serves as a recreation site for swimmers, wind surfers and boaters. It also serves as a flood storage reservoir in the second of four seasons or time periods in a year. The reservoir's releases can be diverted to an irrigation area. A wetland area further downstream receives the unallocated portion of the reservoir release plus the return flow from the irrigation area. The irrigation return flow contains a salinity concentration that can damage the ecosystem.

a) Assume there exist recreation lake level targets, irrigation allocation targets, and wetland flow and salinity targets. The challenge is to determine the reservoir releases and irrigation allocations so as to 'best' meet these targets. This is the crisp' problem.

Data:

Reservoir storage capacity: 30 mcm;

During period 2 the flood storage capacity is 5 mcm.

Irrigation return flow fraction: 0.3 (i.e., 30% of that diverted for irrigation);

Salinity concentration of reservoir water: 1 ppt;

Salinity concentration of irrigation return flow: 20 ppt;

Reservoir average inflows for four seasons, respectively: 5, 50, 20, 10 mcm;

Targets for part a):

Target maximum salinity concentration in wetland: 3 ppt;

Target storage target for all four seasons: 20 mcm;

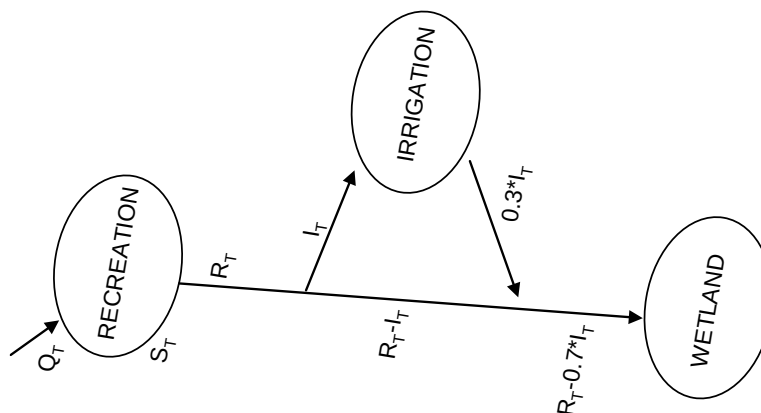
Minimum flow target in wetland in each season, respectively: 10, 20, 15, 15 mcm;

Maximum flow target in wetland in each season, respectively: 20, 30, 25, 25 mcm;

Target irrigation allocations: 0, 20, 15, 5 mcm;

Solution:

The problem's objective needs to be defined. There are a number of reasonable objectives. One could be to minimize the maximum deviation from any target at any place and at any time. Another could be to minimize the sum of maximum deviations for each use over all seasons. The solutions below simply illustrate some candidate objectives.



Crisp problem:

Consider the minimization of maximum storage and irrigation target deviations subject to constraints on the salinity and flow ranges in the wetland.

Objective: $\min\{d_s + d_I\}$: Sum of maximum % deviations from storage and irrigation targets.

Subject to:

- Conservation of mass for the reservoir:
 - o $S_1 + 5 - R_1 = S_2$
 - o $S_2 + 50 - R_2 = S_3$
 - o $S_3 + 20 - R_3 = S_4$
 - o $S_4 + 10 - R_4 = S_1$
- Storage requirements:
 - o $S_1 \leq 30$
 - o $S_2 \leq 25$
 - o $S_3 \leq 25$
 - o $S_4 \leq 30$
- Wetland flow requirements:
 - o $10 \leq (R_1 - 0.7 \cdot I_1) \leq 20$
 - o $20 \leq (R_2 - 0.7 \cdot I_2) \leq 30$
 - o $15 \leq (R_3 - 0.7 \cdot I_3) \leq 25$
 - o $15 \leq (R_4 - 0.7 \cdot I_4) \leq 25$
- Wetland salinity requirements:
 - o $[1 \times (R_1 - I_1) + 20 \times 0.3 \cdot I_1] \leq 3 \times (R_1 - 0.7 \cdot I_1)$
 - o $[1 \times (R_2 - I_2) + 20 \times 0.3 \cdot I_2] \leq 3 \times (R_2 - 0.7 \cdot I_2)$
 - o $[1 \times (R_3 - I_3) + 20 \times 0.3 \cdot I_3] \leq 3 \times (R_3 - 0.7 \cdot I_3)$
 - o $[1 \times (R_4 - I_4) + 20 \times 0.3 \cdot I_4] \leq 3 \times (R_4 - 0.7 \cdot I_4)$
- Storage targets (percentage deviations):
 - o $1 - \frac{S_1}{20} \leq d_s \quad \frac{S_1}{20} - 1 \leq d_s$
 - o $1 - \frac{S_2}{20} \leq d_s \quad \frac{S_2}{20} - 1 \leq d_s$
 - o $1 - \frac{S_3}{20} \leq d_s \quad \frac{S_3}{20} - 1 \leq d_s$
 - o $1 - \frac{S_4}{20} \leq d_s \quad \frac{S_4}{20} - 1 \leq d_s$

- Irrigation allocations targets (percentage deviations):

$$\begin{aligned} & \circ 1 - \frac{I_2}{20} \leq d_I \\ & \circ 1 - \frac{I_3}{15} \leq d_I \\ & \circ 1 - \frac{I_4}{5} \leq d_I \end{aligned}$$

Note: There is no need to specify that the irrigation allocations should be less than the releases ($I_T \leq R_T$). The model will automatically allocate flows as low as possible for irrigation, because of the wetland salinity requirements. No allocation is needed in period 1.

Lingo finds a solution where the maximum percentage deviation from target storage is 38% ($(d_s)_{\min} = 0.382$) and the maximum percentage deviation from target irrigation allocations is 60% ($(d_I)_{\min} = 0.601$):

Policy	Season 1	Season 2	Season 3	Season 4
Storage (<i>target</i>)	17.4 (20)	12.4 (20)	25.0 (20)	23.8 (20)
Release	10.0	37.4	21.2	16.4
Irrigation (<i>target</i>)	0.0 (0)	10.5 (20)	6.0 (15)	2.0 (5)
Wetland flow	10.0	30.0	17.0	15.0
Wetland salinity	1.0	3.0	3.0	1.8

An alternative crisp problem minimizes a weighted maximum deviation where and when it occurs plus the sum of all deviations from all targets.

Lingo model: (Note: New notation, but it should be obvious.)

```
SETS:
PERIODS /1..4/: I, R, DS, DA, DQ, DEQ, A, TQmin, TA, TQmax, Q, PS, wq;
NUMBERS /1..5/: S;
ENDSETS

min = w*D + @sum(Periods(t): DS(t) + DA(t) + DQ(t) + DEQ(t) );

@FOR (PERIODS(t):
    S(t+1) = S(t) + I(t) - R(t); S(1) = S(TN + 1);
    S(t) < K; S(t) - TS < DS(t); TS - S(t) < DS(t);
    TA(t) - A(t) < DA(t); PS(t) - PSmax < DEQ(t); R(t) - A(t) = Q(t);
    Q(t) + (AC)*A(t) = wq(t); TQmin(t) - wq(t) < DQ(t);
    wq(t) - TQmax(t) < DQ(t);
    (Q(t)*RC + (AC)*A(t)*IC) = PS(t)*(Q(t) + (AC)*A(t));
    (Q(t)*RC + (AC)*A(t)*IC) = PS(t)*(Q(t) + (AC)*A(t));
    DS(t) < D; DA(t) < D; DQ(t) < D; DEQ(t) < D;
);
! Note: Flood control constraints are redundant and hence omitted here. ;
DATA:
    TN = 4; K = 30; AC = 0.3; RC = 1; IC = 20; PSmax = 3; TS = 20;
    I = 5, 50, 20, 10; TQmin = 10, 20, 15, 15; TQmax = 20, 30, 25, 25;
    TA = 0, 20, 15, 5; w = ?;
ENDDATA
```

Giving priority to the maximum deviation, D, the value of D is 3.2

Summarizing: Previous crisp solution (This crisp solution)

Policy	Season 1	Season 2	Season 3	Season 4
Storage (target 20)	17.4 (18.7)	12.4 (16.8)	25.0 (23.2)	23.8 (21.8)
Release	10.0 (6.8)	37.4 (45.7)	21.2 (21.4)	16.4 (13.1)
Irrigation	0.0 (0)	10.5 (20.0)	6.0 (11.8)	2.0 (1.8)
Wetland flow	10.0 (6.8)	30.0 (29.7)	17.0 (13.1)	15.0 (11.8)
Wetland salinity	1.0 (1.0)	3.0 (4.8)	3.0 (6.2)	1.8 (1.9)

Alternatively, consider converting the above example to percent deviations.

```

SETS:
PERIODS /1..4/: I, R, DS, DA, DQ, DEQ, A, TQmin, TA, TQmax, Q, PS, wq;
NUMBERS /1..5/: S;
ENDSETS

min = w*D + @sum(Periods(t): DS(t) + DA(t) + DQ(t) + DEQ(t) );

@FOR (PERIODS(t):
    S(t+1) = S(t) + I(t) - R(t); S(1) = S(TN + 1);
    S(t) < K; (S(t) - TS)/TS < DS(t); (TS - S(t))/TS < DS(t);
    (TA(t) - A(t))/TA(t) < DA(t); (PS(t) - PSmax)/PSmax < DEQ(t);
    R(t) - A(t) = Q(t);
    Q(t) + (AC)*A(t) = wq(t); (TQmin(t) - wq(t))/TQmin(t) < DQ(t);
    (wq(t) - TQmax(t))/TQmax(t) < DQ(t);
    (Q(t)*RC + (AC)*A(t)*IC) = PS(t)*(Q(t) + (AC)*A(t) );
    (Q(t)*RC + (AC)*A(t)*IC) = PS(t)*(Q(t) + (AC)*A(t) );
    DS(t) < D; DA(t) < D; DQ(t) < D; DEQ(t) < D;
);
! Note: Flood control constraints are redundant and hence omitted here. ;
DATA:
    TN = 4; K = 30; AC = 0.3; RC = 1; IC = 20; PSmax = 3; TS = 20;
    I = 5, 50, 20, 10; TQmin = 10, 20, 15, 15; TQmax = 20, 30, 25, 25;
    TA = 0.001, 20, 15, 5; w = ?;
ENDDATA

```

Solution: The minimum maximum percent deviation is 34.

Summarizing: First crisp solution (Second crisp solution) [This crisp sol.]

Policy	Season 1	Season 2	Season 3	Season 4
Storage (target 20)	17.4 (18.7) [18.5]	12.4 (16.8) [13.5]	25.0 (23.2) [26.5]	23.8 (21.8) [20.8]
Release	10.0 (6.8) [10.0]	37.4 (45.7) [37.0]	21.2 (21.4) [25.7]	16.4 (13.1) [12.3]
Irrigation	0.0 (0.0) [0.0]	10.5 (20.0) [13.2]	6.0 (11.8) [9.9]	2.0 (1.8) [3.5]
Wetland flow	10.0 (6.8) [10]	30.0 (29.7) [27.7]	17.0 (13.1) [18.7]	15.0 (11.8) [9.9]
Wetland salinity	1.0 (1.0) [1.0]	3.0 (4.8) [3.7]	3.0 (6.2) [4.0]	1.8 (1.9) [3.0]

b) Next create fuzzy membership functions to replace the targets and solve the problem. Assume that the targets used in a) above are expressed in qualitative terms as membership functions. The membership functions indicate the degree of satisfaction for these targets. Solve for the 'best' reservoir release and allocation policy that maximizes the minimum membership function value. Each membership function defines the relative level of satisfaction, where a value of 1 indicates complete stakeholder satisfaction. This is the qualitative problem.

Fuzzy problem:

This is now a maximization problem: we want to maximize the intersection of the membership functions, which is the minimum of all of them.

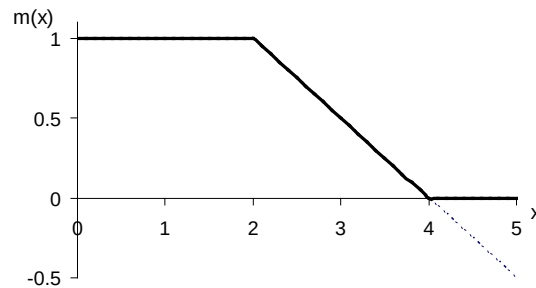
Target maximum wetland salinity concentration:

$$m(x) = Z + (1 - Z) \cdot (1 - 0.5(x - 2))$$

$$x \leq 2Z + 10(1 - Z)$$

$$x \geq 2(1 - Z)$$

Z is a binary 0,1 variable



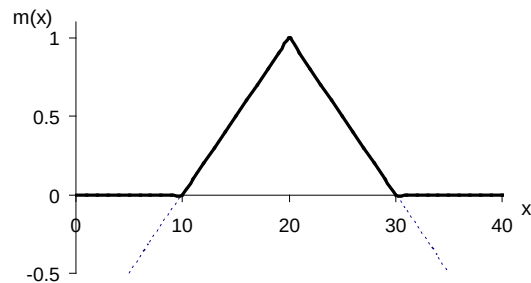
To avoid introducing a second binary variable, we allow the membership function to be negative when it should be zero.

Storage target:

$$m(x) = 0.1 \cdot (x_1 - 10 - x_2)$$

$$x_1 + x_2 = x$$

$$x_1 \leq 20$$



No need to introduce binary variables if we allow the membership function to be negative. As a result, we have a concave function to be maximized, and we can use the segments method.

Target irrigation allocations:

We use the same membership function. No constraint is needed for season 1, because no allocation is desired. For season 2, we take the same threshold values as shown previously (10, 20, 30). For season 3, we take (5, 15, 25) and for season 4 (0, 5, 10).

Flow target in wetland:

$$m(x) = Z_1 \cdot 0.5 \cdot (x - 9) + Z_2 + (1 - Z_1 - Z_2) \cdot (1 - 0.5(x - 19))$$

$$x \leq 11Z_1 + 100(1 - Z_1)$$

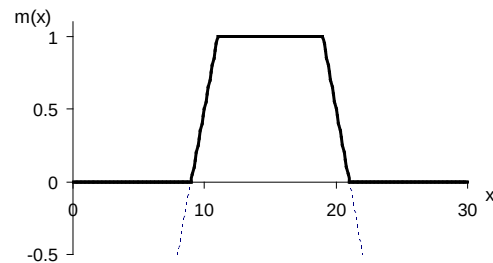
$$x \geq 11Z_2$$

$$x \leq 19Z_2 + 100(1 - Z_2)$$

$$x \geq 19(1 - Z_1 - Z_2)$$

$$Z_1 + Z_2 \leq 1$$

Z_1 and Z_2 are binary 0,1 variables



Once again, allowing the membership function to be negative permits to save the cost of introducing two more binary variables. For season 2, the bounds are 19, 21, 29 and 31. For seasons 3 and 4, the bounds are 14, 16, 24 and 26.

Lingo finds a solution where the minimum membership function has a value of 0.25. The ‘best’ reservoir release and allocation policy is similar to the ones obtained from our crisp models:

Policy	Season 1	Season 2	Season 3	Season 4
Storage	17.1	12.5	25.0	22.5
Release	9.5	37.5	22.5	15.4
Irrigation	0.0	12.6	7.5	1.3
Wetland flow	9.5	28.7	17.3	14.5
Wetland salinity	1.0	3.5	3.5	1.3

Crisp model solutions for comparison.

Policy	Season 1	Season 2	Season 3	Season 4
Storage (target 20)	17.4 (18.7) [18.5]	12.4 (16.8) [13.5]	25.0 (23.2) [26.5]	23.8 (21.8) [20.8]
Release	10.0 (6.8) [10.0]	37.4 (45.7) [37.0]	21.2 (21.4) [25.7]	16.4 (13.1) [12.3]
Irrigation	0.0 (0.0) [0.0]	10.5 (20.0) [13.2]	6.0 (11.8) [9.9]	2.0 (1.8) [3.5]
Wetland flow	10.0 (6.8) [10]	30.0 (29.7) [27.7]	17.0 (13.1) [18.7]	15.0 (11.8) [9.9]
Wetland salinity	1.0 (1.0) [1.0]	3.0 (4.8) [3.7]	3.0 (6.2) [4.0]	1.8 (1.9) [3.0]

Fuzzy Lingo model:

```
! Objective: maximize the minimum membership function;

max = M ;

! Conservation of mass;

5 + s1 - r1 = s2;
50 + s2 - r2 = s3;
20 + s3 - r3 = s4;
10 + s4 - r4 = s1;

! Storage requirements;

s1 < 30;
s2 < 25;
s3 < 25;
s4 < 30;

! Definition of flow in wetland;

w1 = r1 - 0.7*i1;
w2 = r2 - 0.7*i2;
w3 = r3 - 0.7*i3;
w4 = r4 - 0.7*i4;

! Definition of salinity in wetland;

x1 = (r1 + 5*i1)/(r1 - 0.7*i1);
x2 = (r2 + 5*i2)/(r2 - 0.7*i2);
x3 = (r3 + 5*i3)/(r3 - 0.7*i3);
x4 = (r4 + 5*i4)/(r4 - 0.7*i4);

! Target storages;

! Season 1;
0.1*(s1s1-10-s2s1) > M;
s1s1 + s2s1 = s1;
s1s1 < 20;
! Season 2;
0.1*(s1s2-10-s2s2) > M;
s1s2 + s2s2 = s2;
s1s2 < 20;
! Season 3;
0.1*(s1s3-10-s2s3) > M;
s1s3 + s2s3 = s3;
s1s3 < 20;
! Season 4;
0.1*(s1s4-10-s2s4) > M;
s1s4 + s2s4 = s4;
s1s4 < 20;
```

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```

! Target irrigation allocations;

! Season 2;
0.1*(i1i2-10-i2i2) > M;
i1i2 + i2i2 = i2;
i1i2 < 20;
! Season 3;
0.1*(i1i3-5-i2i3) > M;
i1i3 + i2i3 = i3;
i1i3 < 15;
! Season 4;
0.2*(i1i4-i2i4) > M;
i1i4 + i2i4 = i4;
i1i4 < 5;

! Target salinity;

! Season 1;
z1x1 + (1-0.5*(x1-2))*(1-z1x1) > M;
x1 > 2*(1-z1x1);
x1 < 2*z1x1 + 10*(1-z1x1);
@BIN(z1x1);
! Season 2;
z1x2 + (1-0.5*(x2-2))*(1-z1x2) > M;
x2 > 2*(1-z1x2);
x2 < 2*z1x2 + 10*(1-z1x2);
@BIN(z1x2);
! Season 3;
z1x3 + (1-0.5*(x3-2))*(1-z1x3) > M;
x3 > 2*(1-z1x3);
x3 < 2*z1x3 + 10*(1-z1x3);
@BIN(z1x3);
! Season 4;
z1x4 + (1-0.5*(x4-2))*(1-z1x4) > M;
x4 > 2*(1-z1x4);
x4 < 2*z1x4 + 10*(1-z1x4);
@BIN(z1x4);

! Flow target in wetland;

! Season 1;
0.5*(w1-9)*z1w1 + z2w1 +
(1-0.5*(w1-19))*(1-z1w1-z2w1) > M;
w1 < 11*z1w1 + 100*(1-z1w1);
w1 > 11*z2w1;
w1 < 19*z2w1 + 100*(1-z2w1);
w1 > 19*(1-z1w1-z2w1);
z1w1 + z2w1 < 1;
@BIN(z1w1); @BIN(z2w1);
! Season 2;
0.5*(w2-19)*z1w2 + z2w2 + (1-0.5*(w2-29))*
(1-z1w2-z2w2) > M;
w2 < 21*z1w2 + 100*(1-z1w2);
w2 > 21*z2w2;
w2 < 29*z2w2 + 100*(1-z2w2);
w2 > 29*(1-z1w2-z2w2);
z1w2 + z2w2 < 1;
@BIN(z1w2); @BIN(z2w2);

```

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```

! Season 3;
0.5*(w3-14)*z1w3 + z2w3 + (1-0.5*(w3-24))*
    (1-z1w3-z2w3) > M;
w3 < 16*z1w3 + 100*(1-z1w3);
w3 > 16*z2w3;
w3 < 24*z2w3 + 100*(1-z2w3);
w3 > 24*(1-z1w3-z2w3);
z1w3 + z2w3 < 1;
@BIN(z1w3); @BIN(z2w3);
! Season 4;
0.5*(w4-14)*z1w4 + z2w4 + (1-0.5*(w4-24))*
    (1-z1w4-z2w4) > M;
w4 < 16*z1w4 + 100*(1-z1w4);
w4 > 16*z2w4;
w4 < 24*z2w4 + 100*(1-z2w4);
w4 > 24*(1-z1w4-z2w4);
z1w4 + z2w4 < 1;

```

This model could be written using vectors for time periods.

5.2 Develop a flow chart showing how you would apply genetic algorithms to finding the parameters, a_{ij} , of a water quality prediction model, such as the one we have used to find the concentration downstream of an upstream discharge site i . This will be based on observed values of mass inputs, W_i , and concentrations, C_j , and flows, Q_j , at a downstream site j .

$$C_j = \sum_i W_i a_{ij} / Q_j$$

The objective to be used for fitness is to minimize the sum of the differences between the observed C_j and the computed C_j . To convert this to a maximization objective you could use something like the following:

$$\text{Max } 1 / (1 + D)$$

Where $D \geq (C_j \text{ obs} - C_j \text{ calculated})$

$D \geq (C_j \text{ calculated} - C_j \text{ obs.})$

There are many ways to approach this problem. Using numbers to the base 10, each parameter a_{ij} can be expressed using 3 digits (e.g. 213 means that $a_{ij}=0.213$). If we consider n sites ($i=1\dots n-1, j=2\dots n$), we are looking for $n(n-1)/2$ parameters, and thus each string will have $3n(n-1)/2$ digits. The order will be $a_{12}, a_{13}, \dots, a_{1(n-1)}, a_{23}, \dots, a_{(n-1)n}$.

We can define the population size (typically 1,000) and define the probabilities for crossover and mutation. A mutation reduces the value of the parameter by 1, or if the original parameter is 000, mutation changes it to 999.

Beginning with a randomly generated initial population of strings, the process can proceed:

- Pair individuals and determine if a crossover is to be performed on each pair, and where (we can assume an equal probability of a crossover between any two parameters).

- Determine if any parameter in the resulting individual strings is to be mutated.
- Evaluate the fitness associated with each individual string (child) by calculating:

$$F = \max \left\{ \frac{1}{1 + D} \right\}$$

where:

$$D \geq C_j obs - \sum_{i=1}^{j-1} \frac{W_i a_{ij}}{Q_j} \quad \forall j = 2..n$$

$$D \geq \sum_{i=1}^{j-1} \frac{W_i a_{ij}}{Q_j} - C_j obs \quad \forall j = 2..n$$

Record the best individual string.

- The fitness associated with the population will be the sum of individual fitness. The probability that a child will be selected for the new population is simply its share in the overall fitness.
- Randomly generate the new population from the children.

Repeat the process (typically 1,000 times) until there is no significant improvement in the best solution found from all past iterations (e.g. less than 1% of change in its associated fitness).

5.3 Use a genetic algorithm program to predict the parameter values asked for in problem 5.2, and then an artificial neural network ANN to obtain a predictor of downstream water quality based on the values of these parameters. You may use the model and data presented in Section 5.2 of Chapter 4 if you wish.

5.4 Using a genetic algorithm program to find the allocations X_i that maximize the total benefits to the three water users i along a stream, whose individual benefits are:

Use 1: $6 X_1 - X_1^2$

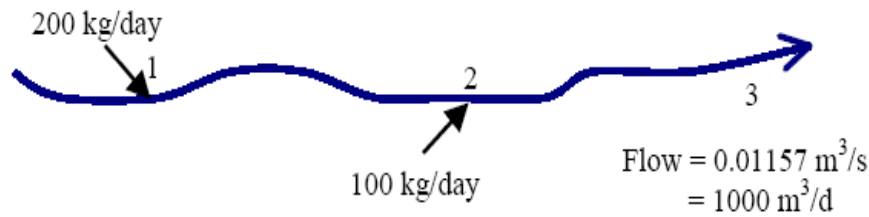
Use 2: $7 X_2 - X_2^2$

Use 3: $8 X_3 - X_3^2$

Assume the available stream flow is some known value (ranging from 0 to 20).

Determine the effect of different genetic algorithm parameter values on the ability to find the best solution.

5.5 Consider the wastewater treatment problem illustrated in the drawing below.



The initial stream concentration just upstream of site 1 is 32. The maximum concentration of the pollutant just upstream of site 2 is 20 mg/l (g/m^3), and at site 3 it is 25 mg/l. Assume the marginal cost per fraction (or percentage) of the waste load removed at site 1 is no less than that cost at site 2, regardless of the amount removed.

Using a suitable genetic algorithm program, solve for the least cost wastewater treatment at sites 1 and 2 that will satisfy the quality constraints at sites 2 and 3 respectively.

Discuss the sensitivity of the GA parameter values in finding the best solution. You can get the exact solution using LINGO as discussed in Section 5.3 in Chapter 4.

5.6 Develop an artificial neural network for flow routing given the following two sets of upstream and downstream flows. Use one set of 5-periods for training (finding the unknown weights and other variables) and the other set for validation of the calculated parameter values (weights and bias constants).

Develop the simplest artificial neural network you can that does an adequate job of prediction.

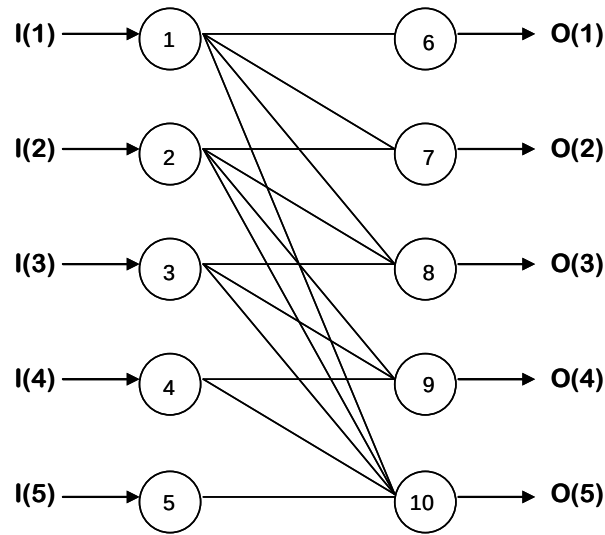
<u>Time period</u>	<u>Upstream flow</u>	<u>Downstream flow</u>
1	450	366
2	685	593
3	830	755
4	580	636
5	200	325
1	550	439
2	255	304
3	830	678
4	680	679
5	470	534

[These outflows come from the following model, assuming an initial storage in period 1 of 50, the detention storage that will remain in the reach even if the inflows go to 0. For each period t :

$$\text{Outflow}(t) = 1.5(-50 + \text{initial storage}(t) + \text{inflow}(t))^{0.9}$$

where the outflow is the downstream flow and inflow is the upstream flow.]

The simplest ANN is a linear model, where the outflows in period t are a linear combination of the inflows plus a constant:



Based on the equation given, we generated 10 sets of 5 periods for training, and 4 sets for validation. The weights of the ANN were determined by minimizing the maximum percentage deviations on the training sets. The model was evaluated by calculating the maximum percentage deviations on the validation sets.

The best linear model is:

$$O(1) = 17.4 + 0.776 \cdot I(1)$$

$$O(2) = 6.4 + 0.198 \cdot I(1) + 0.738 \cdot I(2)$$

$$O(3) = -12.3 + 0.056 \cdot I(1) + 0.223 \cdot I(2) + 0.718 \cdot I(3)$$

$$O(4) = 11.3 + 0.003 \cdot I(1) + 0.070 \cdot I(2) + 0.200 \cdot I(3) + 0.686 \cdot I(4)$$

$$O(5) = -10.7 + 0.016 \cdot I(1) + 0.041 \cdot I(2) + 0.044 \cdot I(3) + 0.185 \cdot I(4) + 0.747 \cdot I(5)$$

As we can see, the latest inflows are given more weights than the previous ones, which makes sense: the outflow in period t is more likely to depend on the inflow during this period, than on the inflow two periods before.

The maximum percentage deviation is 2.3% for the training sets and 4.2% for the validation sets. The average absolute percentage deviation is 1.3% for both the training and validation sets. With these excellent results, we did not need to develop a more complex ANN with hidden layers and sigmoid functions.

SETS:
 SET/1..10/: O1, O2, O3, O4, O5, I1, I2, I3, I4, I5, X1, X2, X3, X4, X5;
 ENDSETS

! Objective: minimize squared percentage deviations;
 min = D;

@FOR(SET(k):

X1(k) = (w11-w12) + (w111-w112)*I1(k);
 X2(k) = (w21-w22) + (w121-w122)*I1(k) + (w221-w222)*I2(k);
 X3(k) = (w31-w32) + (w131-w132)*I1(k) + (w231-w232)*I2(k) + (w331-w332)*I3(k);
 X4(k) = (w41-w42) + (w141-w142)*I1(k) + (w241-w242)*I2(k) + (w341-w342)*I3(k) + (w441-w442)*I4(k);
 X5(k) = (w51-w52) + (w151-w152)*I1(k) + (w251-w252)*I2(k) + (w351-w352)*I3(k) + (w451-w452)*I4(k) + (w551-w552)*I5(k);

((X1(k)-O1(k))/O1(k))^2 < D;
 ((X2(k)-O2(k))/O2(k))^2 < D;
 ((X3(k)-O3(k))/O3(k))^2 < D;
 ((X4(k)-O4(k))/O4(k))^2 < D;
 ((X5(k)-O5(k))/O5(k))^2 < D;);

Data:
 I1 = 450, 550, 650, 600, 500, 400, 230, 439, 87, 159;
 I2 = 685, 255, 380, 609, 325, 380, 320, 309, 590, 458;
 I3 = 830, 830, 325, 843, 599, 625, 85, 580, 324, 480;
 I4 = 580, 680, 780, 480, 380, 780, 235, 569, 450, 689;
 I5 = 200, 470, 256, 143, 387, 586, 324, 589, 129, 489;
 O1 = 366, 439, 510, 475, 403, 330, 200, 358, 83, 144;
 O2 = 593, 304, 417, 569, 346, 367, 292, 322, 470, 383;
 O3 = 755, 678, 350, 757, 528, 551, 130, 509, 365, 453;
 O4 = 636, 679, 655, 567, 423, 709, 214, 551, 427, 619;
 O5 = 325, 534, 373, 260, 397, 625, 298, 578, 204, 528;
 enddata

Training Sets

Time period	Upstream flow	Downstream flow	Estimated flow	Percentage Deviation
1	450	366	366	0.0%
2	685	593	601	1.3%
3	830	755	761	0.8%
4	580	636	625	1.7%
5	200	325	318	2.2%
1	550	439	444	1.1%
2	255	304	304	0.0%
3	830	678	671	1.0%
4	680	679	664	2.2%
5	470	534	522	2.2%
1	650	510	522	2.3%
2	380	417	416	0.2%
3	325	350	342	2.3%
4	780	655	640	2.3%
5	256	373	365	2.1%
1	600	475	483	1.7%
2	609	569	575	1.1%
3	843	757	762	0.7%
4	480	567	554	2.3%
5	143	260	256	1.5%

Time period	Upstream flow	Downstream flow	Estimated flow	Percentage Deviation
1	500	403	405	0.6%
2	325	346	346	0.0%
3	599	528	518	1.9%
4	380	423	416	1.7%
5	387	397	396	0.3%
1	400	330	328	0.7%
2	380	367	366	0.3%
3	625	551	543	1.5%
4	780	709	699	1.4%
5	586	625	621	0.6%
1	230	200	196	2.1%
2	320	292	288	1.4%
3	85	130	133	2.3%
4	235	214	213	0.5%
5	324	298	295	1.0%
1	439	358	356	0.5%
2	309	322	322	0.0%
3	580	509	497	2.3%
4	569	551	541	1.8%
5	589	578	580	0.3%
1	87	83	85	2.3%
2	590	470	459	2.3%
3	324	365	357	2.2%
4	450	427	427	0.0%
5	129	204	209	2.3%
1	159	144	141	2.3%
2	458	383	376	1.8%
3	480	453	443	2.2%
4	689	619	613	1.0%
5	489	528	524	0.8%

Validation Sets:

Time period	Upstream flow	Downstream flow	Estimated flow	Percentage Deviation
1	750	580	599	3.3%
2	325	399	395	1.0%
3	509	478	467	2.3%
4	380	408	399	2.2%
5	873	733	760	3.7%
1	100	95	95	0.0%
2	609	485	476	1.9%
3	743	665	662	0.5%
4	480	537	533	0.7%
5	643	611	617	1.0%
1	200	177	173	2.3%
2	685	551	552	0.2%
3	380	430	424	1.4%
4	580	537	534	0.6%
5	420	454	458	0.9%
1	300	254	250	1.6%
2	255	255	254	0.4%
3	230	236	226	4.2%
4	680	559	543	2.9%
5	730	677	686	1.3%

Chapter 6

6.1 Identify a water resources planning study with which you have some familiarity. Make a list of the basic information used in the study and the methods used transform that information into decisions, recommendations, and conclusions.

- (a) Indicate the major sources of uncertainty and possible error in the basic information and in the transformation of that information into decisions, recommendations, and conclusions.
- (b) In systems studies, sources of error and uncertainty are sometimes grouped into three categories:
 1. Uncertainty due to the natural variability of rainfall, temperature, and stream flows which affect a system's operation.
 2. Uncertainty due to errors made in the estimation of the models' parameters with a limited amount of data.
 3. Uncertainty or errors introduced into the analysis because conceptual and/or mathematical models do not reflect the true nature of the relationships being described.

Indicate, if applicable, into which category each of the sources of error or uncertainty you have identified falls.

Solution depends on example selected by student.

6.2 The following matrix displays the joint probabilities of different weather conditions and of different recreation benefit levels obtained from use of a reservoir in a state park:

<i>Weather</i>	POSSIBLE RECREATION BENEFITS		
	RB_1	RB_2	RB_3
Wet	0.10	0.20	0.10
Dry	0.10	0.30	0.20

- (a) Compute the probabilities of recreation levels RB_1 , RB_2 , RB_3 , and of dry and wet weather.
- (b) Compute the conditional probabilities $P(\text{wet} \mid RB_1)$, $P(RB_3 \mid \text{dry})$, and $P(RB_2 \mid \text{wet})$.

$$\begin{array}{lcl} \text{(a)} & P[RB_1] = 0.10 + 0.10 = 0.20 \\ & P[RB_2] = 0.20 + 0.30 = 0.50 \\ & P[RB_3] = 0.10 + 0.20 = 0.30 \end{array} \left. \vphantom{\begin{array}{l} P[RB_1] \\ P[RB_2] \\ P[RB_3] \end{array}} \right\} \text{Sum to one}$$

$$\left. \begin{array}{l} P [\text{Dry}] = 0.60 \\ P [\text{Wet}] = 0.40 \end{array} \right\} \quad \text{Sum to one}$$

$$(b) \quad P [\text{Wet} | \text{RB}_1] = 0.10/0.20 = 0.50$$

$$P [\text{RB}_3 | \text{Dry}] = 1/3 = 0.333$$

$$P [\text{RB}_2 | \text{Wet}] = 0.50$$

6.3 In flood protection planning, the 100-year flood, which is an estimate of the quantile $x_{0.99}$, is often used as the design flow. Assuming that the floods in different years are independently distributed:

- (a) Show that the probability of at least one 100-year flood in a 5-year period is 0.049.
- (b) What is the probability of at least one 100-year flood in a 100-year period?
- (c) If floods at 1000 different sites occur independently, what is the probability of at least one 100-year flood at some site in any single year?

(a) The probability of at least one flood is 1 minus the probability of no flood:

$$1 - (0.99)^5 = 0.049$$

$$(b) \quad 1 - (0.99)^{100} = 0.63$$

$$(c) \quad 1 - (0.99)^{1000} = 0.999957$$

6.4 The price to be charged for water by an irrigation district has yet to be determined. Currently it appears as if there is a 60% probability that the price will be \$10 per unit of water and a 40% probability that the price will be \$5 per unit. The demand for water is uncertain. The estimated probabilities of different demands given alternative prices are as follows:

Price	QUANTITIES DEMANDED GIVEN PRICE:				
	30	55	80	100	120
\$ 5	0.00	0.15	0.30	0.35	0.20
\$ 10	0.20	0.30	0.40	0.10	0.00

- (a) What is the most likely value of future revenue from water sales?
- (b) What are the mean and variance of future water sales?
- (c) What is the median value and interquartile range of future water sales?
- (d) What price will maximize the revenue from the sale of water?

(a) The probability of the 2 times 5 possible outcomes (given the probabilities of the two prices) are

Quantity	30	55	80	100	120
\$ 5	0.00	0.06	0.12	0.14	0.08
\$ 10	0.12	0.18	0.24	0.06	0.00

The revenue in each case is

150	275	400	500	600
300	550	800	1000	1200

Because no two different events produce the same revenue, the most likely revenue corresponds to the most likely event. The most likely event is \$10 price and 80 units sold for \$800 revenue.

(b) The mean of future water sales is

$$30[0.4(0) + 0.6(0.2)] + 55[0.4(0.15) + 0.6(0.3)] \\ + \dots + 120[0.4(0.2) + 0.6(0)] = 75.20$$

The mean of future revenues is 569.5.

The variance of future water sales is

$$(30 - 75.2)^2 (0.120) + \dots + (120 - 75.2)^2 (0.08) = 634.96 = (25.2)^2$$

The variance of future revenue is $42,057 = (205.1)^2$.

(c) To derive the median and interquantile range of the demand which equals the future water sales, note that

$$\begin{array}{ll} P[D \leq 30] = 0.120 & \\ P[D \leq 55] = 0.360 & P[D = 55] = 0.24 \\ P[D \leq 80] = 0.720 & P[D = 80] = 0.36 \\ P[D \leq 100] = 0.920 & P[D = 100] = 0.20 \\ P[D \leq 120] = 1.000 & P[D = 120] = 0.08 \end{array}$$

The median is $D = 80$ because

$$\begin{array}{l} P[D < 80] = 0.36 \\ P[D \leq 80] = 0.72 \end{array}$$

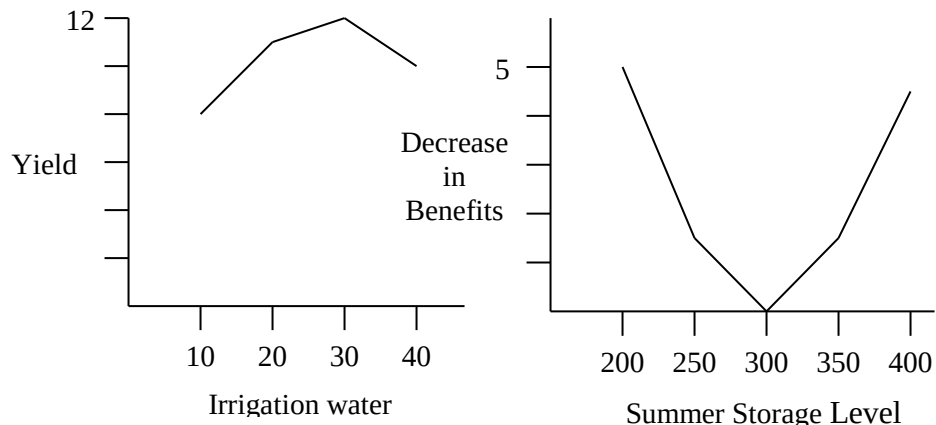
The interquantile range is $55 - 100$.

(d) Price = \$ 10 $\Rightarrow$ expected revenue = \$645.00
Price = \$ 5 $\Rightarrow$ expected revenue = \$456.30
Hence \$10 maximizes the expected revenue.

6.5 Plot the following data on possible recreation losses and irrigated agricultural yields. Show that use of the expected storage level or expected allocation underestimates the expected value of the convex function describing reservoir losses while it overestimates the expected value of the concave function describing crop yield. A concave function $f(x)$ has the property that $f(x) \leq f(x_0) + f'(x_0)(x - x_0)$ for any x_0 ; prove that use of $f(E[X])$ will always overestimate the expected value of a concave function $f(X)$ when X is a random variable.

<i>Irrigation Water Allocation</i>	<i>Crop Yield/Hectare</i>	<i>Probability of Allocation</i>
10	6.5	0.20
20	10.0	0.30
30	12.0	0.30
40	11.0	0.20

<i>Summer Storage Level</i>	<i>Decrease in Recreation Benefits</i>	<i>Probability of Storage level</i>
200	5	0.10
250	2	0.20
300	0	0.40
350	1	0.20
400	4	0.10



Expected irrigation water allocation = 25

Corresponding yield $\cong 11$

Expected yield = 10.1 < 11

Expected storage level = 300
 Corresponding decrease in benefits = 0
 Expected decrease in benefits = 1.5 > 0

Proof of Assertion: By definition of concavity

$$f(x) \leq f(E[x]) + f'(E[x])(x - E[x])$$

Hence

$$E[f(x)] \leq f(E[x]) + f'(E[x])(E[x] - E[x]) = f(E[x])$$

6.6 Complications can be added to the economic evaluation of a project by uncertainty concerning the usefulness life of the project. For example, the time at which the useful life of a reservoir will end due to silting is never known with certainty when the reservoir is being planned. If the discount rate is high and the life is relatively long, the uncertainty may not very important. However, if the life of a reservoir, or of a wastewater treatment facility, or any other such project, relatively short, the practice of using the expected life to calculate present costs or benefits may be misleading.

In this problem, assume that a project resulting in \$1000 of net benefits at the end of each year expected to last between 10 and 30 years, each year within the range of 11 to 30 being equally likely. Given a discount rate of 10%:

- Compute the present value of net benefits NB_0 , assuming a 20-year project life.
- Compare this with the expected present net benefits $E[NB_0]$ taking account of uncertainty in the project lifetime.
- Compute the probability that the actual present net benefits is at least \$1000 less than NB_0 , the benefit estimate based on a 20-year life.
- What is the chance of getting \$1000 more than the original estimate NB_0 ?

(a) Assuming a 20-year life,

$$E[NB] = 1000[(1.1)^{-1} + \dots + (1.1)^{-20}] = 8513.6$$

(b) $E[NB] =$

$$1000[(1.1)^{-1} + \dots + (1.1)^{-10} + (1.1)^{-11} + 0.95(1.1)^{-12} + 0.90(1.1)^{-13} + \dots + 0.05(1.1)^{-30}] = 8358.8$$

(c) $P[NB \geq 8513.6 - 1000] = P[NB \geq 7513.6]$

$NB < 7513.6$ if reservoir fails at the end of year 11, 12, 13, 14, or 15, hence $5/20 = 0.25$ is the probability that $NB < 7513.6$ so that 0.75 is the probability that $NB \geq 7513.6$.

- (d) $P[NB > 8513.6 + 1000] = 0$ because present value if reservoir lasts 30 years is only 9426.9.

6.7 A continuous random variable that could describe the proportion of fish or other animals in different large samples which have some distinctive features is the beta distribution whose density is ($\alpha > 0, \beta > 0$):

$$f_X(x) = \begin{cases} cx^{\alpha-1}(1-x)^{\beta-1} & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Directly calculate the value of c and the mean and variance of X for $\alpha = \beta = 2$.
- (b) In general, $c = \Gamma(\alpha + \beta)/\Gamma(\alpha)\Gamma(\beta)$, where $\Gamma(\alpha)$ is the gamma function equal to $(\alpha - 1)!$ for integer α . Using this information, derive the general expression for the mean and variance of X . To obtain a formula which gives the values of the integrals of interest, note that the expression for c must be such that the integral over $(0, 1)$ of the density function is unity for any α and β .

(a) We have the density function

$$f_x(x) = cx(1-x) \quad \text{when } \alpha = \beta = 2$$

$$1 = \int_0^1 f_x(x) dx = c \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{c}{6} \Rightarrow c = 6$$

$$E[X] = \int_0^1 xf_x(x) dx = 6 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = \frac{6}{12} = \frac{1}{2}$$

$$\text{Var}[X] = \int_0^1 x^2 f_x(x) dx - (E[X])^2 = 6 \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 - \left(\frac{1}{2} \right)^2 = \frac{6}{20} - \frac{1}{4} = \frac{1}{20}$$

(b) For the mean

$$\begin{aligned} E[X] &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x \cdot x^{\alpha-1} (1-x)^{\beta-1} dx \\ &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \left\{ \frac{\Gamma(\alpha + 1)\Gamma(\beta)}{\Gamma(\alpha + \beta + 1)} \right\} = \frac{\alpha}{\alpha + \beta} \quad \text{using } \Gamma(\alpha + 1) = \alpha \Gamma(\alpha). \end{aligned}$$

For the variance one should first calculate

$$E[x^2] = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^2 \cdot x^{\alpha-1} (1-x)^{\beta-1} dx$$

$$= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \left\{ \frac{\Gamma(\alpha + 2)\Gamma(\beta)}{\Gamma(\alpha + \beta + 2)} \right\} = \frac{(\alpha + 1)\alpha}{(\alpha + \beta + 1)(\alpha + \beta)}$$

$$\text{Var}(x) = E[x^2] - (E[x])^2$$

$$= \frac{[\alpha(\alpha + 1)](\alpha + \beta) - \alpha^2(\alpha + \beta + 1)}{(\alpha + \beta + 1)(\alpha + \beta)^2}$$

$$= \frac{\alpha\beta}{(\alpha + \beta + 1)(\alpha + \beta)^2}$$

6.8 The joint probability density of rainfall at two places or rainy days could be described by

$$f_{X,Y}(x, y) \begin{cases} 2/(x + y + 1)^3 & x, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Calculate and graph:

- (a) $F_{XY}(x, y)$, the joint distribution function of X and Y .
- (b) $F_Y(y)$, the marginal cumulative distribution function of Y , and $f_Y(y)$, the density function of Y .
- (c) $f_{Y|X}(y|x)$, the conditional density function of Y given that $X = x$, and $F_{Y|X}(y|x)$, the conditional cumulative distribution function of Y given that $X = x$ (the cumulative distribution function is obtained by integrating the density function).

Show that

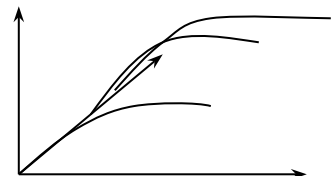
$$F_{Y|X}(y|x=0) > F_Y(y) \quad \text{for } y > 0$$

Find a value of x_0 and y_0 for which

$$F_{Y|X}(y_0|x_0) < F_Y(y_0)$$

(a) For the density function $f_{XY}(x, y) = \frac{2}{(x + y + 1)^3}$

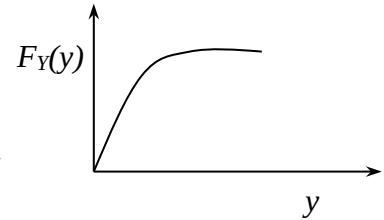
$x, y \geq 0$, one obtains



$$\begin{aligned}
 F_{XY}(x, y) &= \int_0^x d\bar{x} \int_0^y d\bar{y} \left[\frac{2}{(x+y+1)^3} \right] \\
 &= \int_0^x d\bar{x} \left[\frac{-1}{(\bar{x} + \bar{y} + 1)^2} \right]_{\bar{y}=0}^{\bar{y}=y} \\
 &= 1 - \frac{1}{y+1} - \frac{1}{x+1} + \frac{1}{x+y+1} \quad x, y \geq 0
 \end{aligned}$$

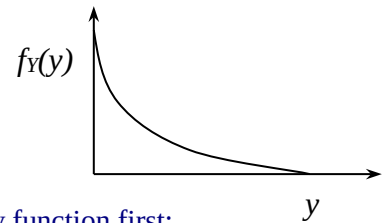
(b) Letting $x \rightarrow \infty$ one obtains

$$\lim_{x \rightarrow \infty} F_{XY}(x, y) = F_Y(y) = 1 - \frac{1}{y+1} = \frac{y}{y+1}$$



hence

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{-(-1)}{(y+1)^2} = \frac{1}{(y+1)^2}$$



(c) Here one should calculate the conditional density function first:

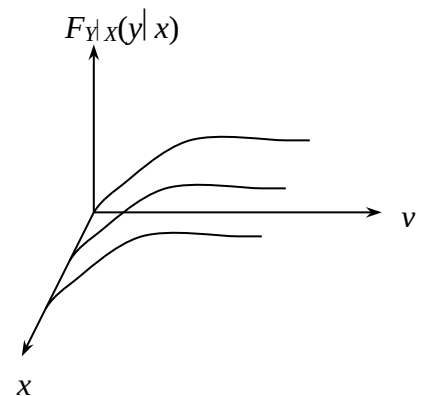
$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{\frac{2}{(x+y+1)^3}}{\frac{1}{(x+1)^2}} = \frac{2(x+y)^2}{(x+y+1)^3}$$

where by symmetry, $f_X(x) = \frac{1}{(x+1)^2}$ given the result in part (b).

Now

$$\begin{aligned}
 F_{Y|X}(y|x) &= \int_0^y \frac{2(x+\bar{y})^2}{(x+\bar{y}+1)^3} d\bar{y} \\
 &= 1 - \frac{(x+1)^2}{(x+y+1)^2}
 \end{aligned}$$

$$\begin{aligned}
 F_{Y|X}(y|x=0) &= \frac{[(y+1)^2 - 1]}{(y+1)^2} \\
 &= \frac{[y(y+1) + y]}{(y+1)^2}
 \end{aligned}$$



$$= \frac{y}{y+1} + \frac{y}{(y+1)^2} > \frac{y}{(y+1)} = F_Y(y) \text{ for } y > 0.$$

To find an (x_0, y_0) so that $F_{Y|X}(y_0, x_0) < F_Y(y_0)$ consider $x_0 = 2.00$

$$F_{Y|X}(y|x_0) = 1 - \frac{(x+1)^2}{(x+y+1)^2} \Big|_{x=2} = 1 - \frac{9}{(y+3)^2}$$

y	$F_{Y X}(y 2)$	$F_Y(y)$	} $F_{Y X}(y 2) < F_Y(y)$
0	0	0	
0.5	0.265	0.333	
1.0	0.438	0.50	
4.0	0.816	0.80	
9.0	0.938	0.90	

6.9 Let X and Y be two continuous independent random variables. Prove that

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

for any two real-valued functions g and h . Then show that $\text{Cov}(X, Y) = 0$ if X and Y are independent.

$$\begin{aligned} E[g(X)h(Y)] &= \int \int g(X)h(Y) f_{XY}(x, y) dx dy \\ &= \int g(X) f_X(x) dx \int h(Y) f_Y(y) dy \\ &= E[g(X)]E[h(Y)] \end{aligned}$$

where $f_{XY}(x, y) = f_X(x) f_Y(y)$ for independent random variables. Proof for covariance follows because $E[xy] = E[x]E[y]$; hence $E[xy] - E[x]E[y] = 0$.

6.10 A frequent problem is that observations (X, Y) are taken on such quantities as flow and concentration and then a derived quantity $g(X, Y)$ such as mass flux is calculated. Given that one has estimates of the standard deviations of the observations X and Y and their correlation, an estimate of the standard deviation of $g(X, Y)$ is needed. Using a second-order Taylor series expansion for the mean of $g(X, Y)$ as a function of its partial derivatives and of the means, variances, covariance of the X and Y . Using a first-order approximation of $g(X, Y)$, obtained an estimates of the variances of $g(X, Y)$ as a function of its partial derivatives and the moments of X and Y . Note, the covariance of X and Y equals

$$E[(X - \mu_X)(Y - \mu_Y)] = \sigma_{XY}^2$$

Using the second-order Taylor series approximation

$$g(x, y) \cong g(\mu_x, \mu_y) + \frac{\partial g}{\partial x}(X - \mu_x) + \frac{\partial g}{\partial y}(Y - \mu_y) + \frac{1}{2} \frac{\partial^2 g}{\partial x^2}(X - \mu_x)^2 + \frac{\partial^2 g}{\partial x \partial y}(X - \mu_x)(Y - \mu_y) + \frac{1}{2} \frac{\partial^2 g}{\partial y^2}(Y - \mu_y)^2$$

One finds that

$$E[g(x, y)] \cong g(\mu_x, \mu_y) + \frac{1}{2} \frac{\partial^2 g}{\partial x^2} V[X] + \frac{\partial^2 g}{\partial x \partial y} \text{Cov}[X, Y] + \frac{1}{2} \frac{\partial^2 g}{\partial y^2} V[Y].$$

Using only the first 3 terms in the Taylor series, the expected value of $g(X, Y)$ would be

$$E[g(X, Y)] = g(\mu_x, \mu_y)$$

Hence an approximation to the variance of $g(X, Y)$ using the first order Taylor series is

$$V[g(x, y)] \cong E[g(X, Y) - g(\mu_x, \mu_y)]^2 = \left(\frac{\partial g}{\partial x}\right)^2 V[X] + 2\left(\frac{\partial g}{\partial x}\right)\left(\frac{\partial g}{\partial y}\right) \text{Cov}[X, Y] + \left(\frac{\partial g}{\partial y}\right)^2 V[Y]$$

Higher-order terms can be obtained.

6.11 A study of the behavior of water waves impinging upon and reflecting off a breakwater located on a sloping beach was conducted in small tank. The height (crest-to-trough) of the waves was measured a short distance from the wave generator and at several points along the beach different distances from the breakwater were measured and their mean and standard error recorded.

<i>Location</i>	<i>Mean Wave Height (cm)</i>	<i>Standard Error of Mean (cm)</i>
Near wave generator	3.32	0.06
1.9 cm from breakwater	4.42	0.09
1.9 cm from breakwater	2.59	0.09
1.9 cm from breakwater	3.26	0.06

a) At which points were the wave heights significantly different from the height near wave generator assuming that errors were independent?

Of interest to the experimenter is the ratio of the wave heights near the breakwater to the initial wave heights near the breakwater to the initial wave heights in the deep water. Using the results in Exercise 6.10, estimate the standard error of this ratio at the three points assuming that errors made in measuring the height of waves at the three points and near the wave generator are independent.

b) At which point does the ratio appear to be significantly different from 1.00?

Using the results of Exercise 6.10, show that the ratio of the mean wave heights is probably a biased estimate of the actual ratio. Does this bias appear to be important?

(a) Determination of points where wave heights x_i ($i = 1, 2, 3$) differ for height x_o at wave generator

Location	Difference $\Delta = \bar{x}_i - \bar{x}_o$	Standard Deviation $s_\Delta = \sqrt{s_i^2 - s_o^2}$	t-ratio $\frac{\Delta}{s_\Delta}$	Difference $\Pr[\tilde{t} \geq t]$
1. (1.9 cm)	1.10	0.108	10.2	< 0.001
2. (5.0 cm)	-0.73	0.108	-6.75	< 0.001
3. (10.0 cm)	-0.06	0.085	-0.707	0.48

Hence wave height at locations 1 and 2 are significantly different from that at x_o .

(b) First a formula for the variance of $\frac{\bar{x}_i}{\bar{x}_o}$ is required.

$$\text{Var}\left[\frac{\bar{x}_i}{\bar{x}_o}\right] = \text{Var}[g(\bar{x}_i, \bar{x}_o)] \cong \left(\frac{\partial g}{\partial \bar{x}_i}\right)^2 \text{Var}[\bar{x}_i] + 2 \frac{\partial g}{\partial \bar{x}_i} \frac{\partial g}{\partial \bar{x}_o} \text{Cov}[\bar{x}_i, \bar{x}_o] + \left(\frac{\partial g}{\partial \bar{x}_o}\right)^2 \text{Var}[\bar{x}_o]$$

where the covariance is zero if x_i and x_o are independently distributed. Thus

$$\begin{aligned} \text{Var}\left[\frac{\bar{x}_i}{\bar{x}_o}\right] &= \left(\frac{1}{\bar{x}_o}\right)^2 \text{Var}[\bar{x}_i] + \left(-\frac{\bar{x}_i}{(\bar{x}_o)^2}\right)^2 \text{Var}[\bar{x}_o] \\ &= \frac{(\bar{x}_o)^2 \text{Var}(\bar{x}_i) + (\bar{x}_i)^2 \text{Var}(\bar{x}_o)}{(\bar{x}_o)^2} \end{aligned}$$

Analysis of the ratio yields:

Location	Ratio	S. D.	t-ratio Ratio-1/SD	Significance $\Pr[\tilde{t} \geq t]$
1. (1.9 cm)	1.33	0.036	9.17	< 0.001
2. (5.0 cm)	0.780	0.030	7.23	< 0.001
3. (10.0 cm)	0.982	0.025	0.71	0.48

One obtains the same result as before.

Is ratio of means an unbiased estimator?

$$\begin{aligned}
 E[g(\bar{x}_i, \bar{x}_o)] &\cong \\
 g(\mu_{\bar{x}_i}, \mu_{\bar{x}_o}) &+ \frac{1}{2} \frac{\partial^2 g}{\partial \bar{x}_i^2} \text{Var}[\bar{x}_i] + \frac{1}{2} \frac{\partial^2 g}{\partial \bar{x}_o^2} \text{Var}[\bar{x}_o] + \frac{\partial^2 g}{\partial \bar{x}_i \partial \bar{x}_o} \text{Cov}[\bar{x}_i, \bar{x}_o] \\
 &= \frac{\mu_{\bar{x}_i}}{\mu_{\bar{x}_o}} + \frac{1}{2} (0) \text{Var}[\bar{x}_i] + \frac{1}{2} \left[2 \frac{\bar{x}_i}{(\bar{x}_o)^3} \right] \text{Var}[\bar{x}_o] + 0
 \end{aligned}$$

$$= \text{Ratio} + \text{Ratio} \frac{\text{Var}(\bar{x}_o)}{(\bar{x}_o)^2}$$

$$= \text{Ratio} \left[1 + \frac{\text{Var}(\bar{x}_o)}{(\bar{x}_o)^2} \right]$$

Location	$\frac{\text{Var}(\bar{x}_o)}{(\bar{x}_o)^2}$	
1. (1.9 cm)	0.00041	} Bias does not appear to exceed 0.12% of actual value
2. (5.0 cm)	0.0012	
3. (10.0 cm)	0.00033	

6.12 Derive Kirby's bound, Equation 6.45, on the estimate of the coefficient of skewness by computing the sample estimates of the skewness of the most skewed sample it would be possible to observe. Derive also the upper bound $(n - 1)^{1/2}$ for the estimate of the population coefficient of variation

$$\frac{v_x}{\bar{x}}$$

when all the observations must be nonnegative.

To derive the bound on sample estimates of the skew coefficient, one need not restrict the observations to be positive. For simplicity and without loss of generality, assume for sample $\{x_1, \dots, x_n\}$ that $\bar{x} = 0$.

The most skewed sample which can be observed is then

$$x_i = a \quad \text{for one } i$$

$$x_j = -a/(n-1) \quad \text{for all other } j \neq i$$

Note $\bar{x} = \frac{1}{n} \{a + (n-1)[-a/(n-1)]\} = 0$. The sample estimate of the skew coefficient is then

$$\begin{aligned} \hat{\gamma} &= \frac{\frac{1}{n} \sum (x_k - \bar{x})^3}{[\frac{1}{n} \sum (x_k - \bar{x})^2]^{3/2}} \\ &= \frac{\frac{1}{n} [a^3 + (n-1)(\frac{-a}{n-1})^3]}{\{\frac{1}{n} [a^2 + (n-1)(\frac{-a}{n-1})^2]\}^{3/2}} \\ &= \frac{\frac{1}{n} [1 - \frac{1}{(n-1)^2}]}{\{\frac{1}{n} [1 + (\frac{1}{n-1})]\}^{3/2}} = \frac{\frac{1}{(n-1)^2 n} [n^2 - 2n + 1 - 1]}{\{\frac{n-1+1}{n(n-1)}\}^{3/2}} \\ &= \frac{\frac{n-2}{(n-1)^2}}{(\frac{1}{n-1})^{3/2}} = \frac{n-2}{\sqrt{n-1}} \end{aligned}$$

To derive the meaningful bound on sample estimates of the coefficient of variation we restrict the observations to be positive. Without loss of generality take $\bar{x} = 1$. Subject to this constraint, one wishes to maximise the variance. This maximum is achieved by the distribution

$$x_1, \dots, x_{n-1} = 0$$

$$x_n = n$$

$$nv_x^2 = \sum_{i=1}^n (x_i - 1)^2 = (n-1)(-1)^2 + 1(n-1)^2 = n(n-1)$$

so that

$$\frac{v_x}{\bar{x}} = \sqrt{n-1}$$

To see that the assumed distribution is at least a local maximum, let

$$\left. \begin{array}{l} x_1 = \varepsilon \\ x_2, \dots, x_{n-1} = 0 \\ x_n = n - \varepsilon \end{array} \right\} \bar{x} = \frac{\sum x_i}{n} = 1$$

$$nv_x^2 = (\varepsilon - 1)^2 + (n - 2)(-1)^2 + (n - \varepsilon - 1)^2$$

$$\left. \frac{d(nv_x^2)}{d\varepsilon} \right|_{\varepsilon=0} = 2(\varepsilon - 1) - 2(n - \varepsilon - 1) \Big|_{\varepsilon=0}$$

$$= -2 - 2(n - 1) = -2n.$$

One should decrease the value of x_i below zero if possible. This range is prohibited by the constraint $x_i \geq 0$ hence this is at least a local maximum.

6.13 The errors in the predictions of water quality models are sometimes described by the double exponential distribution whose density is

$$f(x) = \frac{\alpha}{2} \exp(-\alpha|x - \beta|) \quad -\infty < x < +\infty$$

What are the maximum likelihood estimates of α and β . Note that

$$\frac{d}{d\beta}|x - \beta| = \begin{cases} -1 & x > \beta \\ +1 & x < \beta \end{cases}$$

Is there always a unique solution for β ?

To derive maximum likelihood parameter estimates one can use the log-likelihood function:

$$L = \ln f(x_1, \dots, x_n | \alpha, \beta)$$

$$\begin{aligned}
&= \ln \prod_{i=1}^n f(x_i | \alpha, \beta) = \sum_{i=1}^n \ln \left[\frac{\alpha}{2} \exp(-\alpha |x - \beta|) \right] \\
&= n \ln \frac{1}{2} + n \ln \alpha - \alpha \sum_{i=1}^n |x - \beta|
\end{aligned}$$

Taking derivatives with respect to α and β one obtains

$$0 = \frac{\partial L}{\partial \alpha} = \frac{n}{\alpha} - \sum_{i=1}^n \frac{1}{|x - \beta|} \quad \text{or} \quad \frac{1}{\alpha} = \frac{1}{n} \sum_{i=1}^n \frac{1}{|x - \beta|}$$

$$0 = \frac{\partial L}{\partial \beta} = -\alpha \{(\text{the number of } x\text{'s} < \beta) - (\text{the number of } x\text{'s} > \beta)\}$$

The number in braces is zero only if β equals a sample median, i.e., $\frac{1}{2}$ of the observations are greater than β and $\frac{1}{2}$ are less. This number is certain to be unique only if the number of observations is odd. For $\beta = \text{median}$, $1/\alpha$ equals what is called the average deviation. It is an estimate of scale just like the standard deviation s_x .

6.14 Derive the equations that one would need to solve to obtain maximum likelihood estimates of the two parameters α and β of the gamma distribution. Note an analytical expression for $d\Gamma(\alpha)/d\alpha$ is not available so that a closed form expression for maximum likelihood estimate of α is not available. What is the maximum likelihood estimate of β as a function of the maximum likelihood estimates of α ?

The density function for the gamma distribution is

$$f(x | \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\beta x) \quad x > 0$$

Hence the log-likelihood function for a sample $x_1, \dots, x_n$ is

$$\begin{aligned}
L &= \ln \Pi f(x | \alpha, \beta) = \sum \ln f(x | \alpha, \beta) \\
&= \sum [-\ln \Gamma(\alpha) + \alpha \ln \beta + (\alpha - 1) \ln x_i - \beta x_i] \\
&= -n \ln \Gamma(\alpha) + n\alpha \ln \beta + (\alpha - 1) \sum \ln x_i - n\beta \bar{x}
\end{aligned}$$

Taking derivatives with respect to α and β yields

$$0 = \frac{\partial L}{\partial \alpha} = \frac{-n}{\Gamma(\alpha)} \frac{d\Gamma(\alpha)}{d\alpha} + n \ln \beta + \sum \ln x_i$$

$$0 = \frac{\partial L}{\partial \beta} = \frac{n\alpha}{\beta} - n\bar{x}$$

The second relationship yields $\hat{\beta} = \frac{\hat{\alpha}}{\bar{x}}$. This insures that

$$\bar{x} = \frac{\hat{\alpha}}{\hat{\beta}}$$

so that the maximum likelihood estimates of α and β preserve the sample average. Substituting this expression into the other equation yields

$$-\frac{1}{\Gamma(\hat{\alpha})} \frac{d\Gamma(\alpha)}{d\alpha} \Big|_{\hat{\alpha}} + \ln \hat{\alpha} = \ln \bar{x} - \frac{\sum \ln x_i}{n}$$

This equation for α depends on only the sample average $\bar{x}$ and the geometric mean

$$x_g = \exp\left[\frac{\sum \ln x_i}{n}\right] = \left(\prod_{i=1}^n x_i\right)^{1/n}$$

These are called “sufficient statistics.” No closed form expression for α as a function of $\bar{x}$ and x_g is available.

6.15 The log-Pearson Type-III distribution is often used to model flood flows. If X has a log-Pearson Type-III distribution, then

$$Y = \ln(X) - m$$

has a two parameter gamma distribution where e^m is the lower bound of X if $\beta > 0$ and e^m is the upper bound of X if $\beta < 0$. The density function of Y can be written

$$f_Y(y)dy = \frac{(\beta y)^{\alpha-1}}{\Gamma(\alpha)} \exp(-\beta y) d(\beta y) \quad 0 < \beta y < +\infty$$

Calculate the mean and variance of X in terms of α , β and m . Note that

$$E[X^r] = E[(\exp(Y + m))^r] = \exp(rm) E[\exp(rY)]$$

To evaluate the required integrals remember that the constant terms in the definition of $f_Y(y)$ ensure that the integral of this density function over the range of y must be unity for any values of α and β so long as $\alpha > 0$ and $\beta y > 0$. For what values of r and β does the mean of X fail to exist? How do the values of m , α and β affect the shape and scale of the distribution of X ?

A relationship for the expectation of x^r may be derived using the relationship:

$$\begin{aligned} & \int_{\beta y=0}^{+\infty} (\beta y)^{\alpha-1} e^{-(\beta y)} d(\beta y) \\ &= \int_{x=0}^{+\infty} (x)^{\alpha-1} e^{-(\beta x)} dx = \Gamma(\alpha) \quad \text{where } x = \beta y \end{aligned}$$

Now

$$\begin{aligned} E[x^r] &= \int_{\beta y=0}^{+\infty} (e^{y+m})^r \frac{(\beta y)^{\alpha-1}}{\Gamma(\alpha)} \exp[-\beta y] d(\beta y) \\ &= \frac{e^{rm}}{\Gamma(\alpha)} \int_{\beta y=0}^{+\infty} (\beta y)^{\alpha-1} \exp[-(\beta-r)y] d(\beta y) \end{aligned}$$

If $(\beta-r)$ has the same sign as β , this can be transformed to

$$\begin{aligned} &= \frac{e^{rm}}{\Gamma(\alpha)} \left[\frac{\beta}{\beta-r} \right]^\alpha \int_{(\beta-r)y=0}^{+\infty} [(\beta-r)y]^{\alpha-1} \exp[-(\beta-r)y] d[(\beta-r)y] \\ &= e^{rm} \left[\frac{\beta}{\beta-r} \right]^\alpha \end{aligned}$$

For $r = 1$

$$E[X] = e^m \left[\frac{\beta}{\beta-1} \right]^\alpha$$

Using the results for $r = 1$ and 2

$$\begin{aligned} \text{Var}[X] &= E[X^2] - (E[X])^2 \\ &= e^{2m} \left[\left(\frac{\beta}{\beta-2} \right)^\alpha - \left(\frac{\beta}{\beta-1} \right)^{2\alpha} \right] \end{aligned}$$

assuming $(\beta-2)$ and β have the same signs, i.e., $\beta > 2$ or $\beta < 0$.

What happens when $(\beta-r)$ and β have opposite signs?

Then when $(\beta y) \rightarrow +\infty$, $(\beta-r)y \rightarrow -\infty$ and the term $\exp[-(\beta-r)y]$ will not go to zero at the upper bound of the integration and hence $E[X^r]$ will be infinite. Thus, $E[X^r]$ only exists if β and $(\beta-r)$ have the same signs, which means either $\beta > r$ (which is the more common case corresponding to interval $y \geq 0$, $X \geq e^m$) or else $\beta < 0$ (which means $0 \leq x \leq e^m$, because $y < 0$).

The sign of β determines whether or not X has an upper bound. For $\beta > 0$, X has no upper bound and $E[X^r]$ fails to exist for $r > \beta$. Thus β also determines the shape of X 's distribution.

Within the bound set by β , α also determines the shape and moments of the distribution of X .

The parameter m only affects the scale of X . In particular, the distribution of X/e^m depends only on α and β .

6.16 When plotting observations to compare the empirical and fitted distributions of streamflows, or other variables, it is necessary to assign a cumulative probability to each observation. These are called *plotting positions*. As noted in the text, for the i^{th} largest observation X_i ,

$$E[F_X(X_i)] = i/(n + 1)$$

Thus the *Weibull plotting position* $i/(n + 1)$ is one logical choice. Other commonly used plotting positions are the *Hazen plotting position* $(i - 3/8)/(n + 1/4)$. The plotting position $(i - 3/8)/(n + 1/4)$ is a reasonable choice because its use provides a good approximation to the expected value of X_i . In particular for standard normal variables

$$E[X_i] \cong \Phi^{-1}[(i - 3/8)/(n + 1/4)]$$

where $\Phi(\cdot)$ is the cumulative distribution function of a standard normal variable. While much debate centers on the appropriate plotting position to use to estimate $p_i = F_X(X_i)$, often people fail to realize how imprecise all such estimates must be. Noting that

$$\text{Var}(p_i) = \frac{i(n - i - 1)}{(n + 1)^2 (n + 2)},$$

contrast the difference between the estimates $\hat{p}_i$ of p_i provided by these three plotting positions and the standard deviation of p_i . Provide a numerical example. What do you conclude?

The mean and variance of the actual value of $p_i = F_X(x_i)$, where x_i is the i^{th} largest observation in a sample of size n , are given in the problem:

$$E[p_i] = \frac{i}{n + 1}$$

$$\text{Var}[p_i] = \frac{i(n - i + 1)}{(n + 1)^2 (n + 2)}$$

For $i = 1$ or n , the variance of p_i becomes

$$\frac{n}{(n+1)^2(n+2)} \cong \frac{1}{n^2}$$

hence the standard deviation of p_i will be about n^{-1} .

For $i \approx n/2$,

$$\text{Var}[p_i] = \frac{\left(\frac{n}{2}\right)^2}{(n+1)^2(n+2)} \cong \frac{1}{4n}$$

so that the standard deviation of $\text{Var}[p_i]$ is about $\frac{0.5}{\sqrt{n}}$, which for $n > 6$ is much

larger than the standard deviation of p_i obtained for $i = 1$ or n . These two sets of extreme results provide bounds for what in the center and at the extreme of the distribution.

Now the differences between the 3 estimators of p_i depend only on i and n . Hence for fixed i and n these differences are constant. Thus, to compare these difference we examine the fixed bias of the estimators for different values of i .

ESTIMATOR			
	$\frac{i}{n+1}$	$\frac{i - \frac{1}{2}}{n}$	$\frac{i - 0.375}{n + 0.250}$
a. Bias	none	$\frac{2i - n + 1}{2n(n+1)}$	$\frac{0.75i - 0.375n - 0.375}{(n + 0.250)(n+1)}$
b. Bias for large n and $i = 1$	none	$-\frac{1}{2n}$	$-\frac{3}{8n}$
c. Bias for large n and $i = n/2$	none	0	0

Now for the standard deviation of p_i :

$$\left[\frac{i(n-i+1)}{(n+1)^2(n+2)} \right]^{1/2} = \begin{cases} \frac{1}{n} & \text{For } i=1 \text{ and large } n \\ \frac{1}{2\sqrt{n}} & \text{For } i= n/2 \text{ and large } n \end{cases}$$

not that for large n and $i = 1$, the standard deviation of p_i is $1/n$ while the bias is $-1/(2n)$ and $-3/(8n)$. Hence the standard deviation is twice as large and will be the dominant error. For $i \cong n/2$ and large n , the bias goes to zero and the variance $0.25/n$ constitutes the only error. A numerical example with small n shows that these large sample results hold in that situation. Let $n = 9$.

ESTIMATOR			
	$\frac{i}{n+1}$	$\frac{i - \frac{1}{2}}{n}$	$\frac{i - 0.375}{n + 0.250}$
$i = 1$			
$\hat{p}_i =$	0.1000	0.0555	0.0676
Bias =	0.0000	-0.0444	-0.0324
Bias ² + Var(p_i) =	0.00818	0.01016	0.00923
$i = 3$			
$\hat{p}_i =$	0.300	0.277	0.284
Bias =	0.0000	-0.023	-0.016
Bias ² + Var(p_i) =	0.0191	0.0196	0.0194

Of course $i/(n + 1)$ has no bias while the other estimators do. However, as can be seen this bias adds very little to their mean square errors [Bias² + Var(p_i)]. On the other hand we must remember that these plotting positions are often used to calculate $\hat{x}_i = F_X^{-1}(\hat{p}_i)$ which are compared to the observed values x_i . Which estimate will minimize $E[(x_i - \hat{x}_i)^2]$ depends upon the true distribution of the X .

6.17 The following data represent a sequence of annual flood flows, the maximum flow rate observed each year, for the Sebou river at the Azib Soltane gaging station in Morocco.

<i>Maximum Discharge</i>		<i>Maximum Discharge</i>	
<i>Date</i>	<i>(m³/s)</i>	<i>Date</i>	<i>(m³/s)</i>
03/26/33	445	03/13/54	750
12/11/33	1410	02/27/55	603
11/17/34	475	04/08/56	880
03/13/36	978	01/03/57	485
12/18/36	461	12/15/58	812
12/15/37	362	12/23/59	1420
04/08/39	530	01/16/60	4090
02/04/40	350	01/26/61	376
02/21/41	1100	03/24/62	904
02/25/42	980	01/07/63	4120
12/20/42	575	12/21/63	1740
02/29/44	694	03/02/65	973
12/21/44	612	02/23/66	378
12/24/45	540	10/11/66	827
05/15/47	381	04/01/68	626
05/11/48	334	02/28/69	3170
05/11/49	670	01/13/70	2790
01/01/50	769	04/04/71	1130
12/30/50	1570	01/18/72	437
01/26/52	512	02/16/73	312
01/20/53	613		

- Construct a histogram of the Sebou flood flow data to see what the flow distribution looks like.
- Calculate the mean, variance, and sample skew. Based on Table 6.3, does the sample skew appear to be significantly different from zero?
- Fit a normal distribution to the data and use the Kolmogorov-Smirnov test to determine if the fit is adequate. Draw a quantile-quantile plot of the fitted quantiles $F^{-1}[(i - 3/8)/(n + 1/4)]$ versus the observed quantiles x_i and include on the graph the Kolmogorov-Smirnov bounds on each x_i , as shown in Figures 6.2a and 6.2b.
- Repeat part (c) using a two-parameter lognormal distribution.
- Repeat part (c) using a three-parameter lognormal distribution. The Kolmogorov-Smirnov test is now approximate if applied to $\log_e[X_i - \tau]$, where τ is calculated using Equation 6.81 or some other method of your choice.
- Repeat part (c) for two- and three- parameter versions of the gamma distribution. Again, the Kolmogorov-Smirnov test is approximate.
- A powerful test of normality is provided by the correlation test. As described by Filliben (1975), one should approximate $p_i = F_X(x_i)$ by

$$\hat{p}_i = \begin{cases} 1 - (0.5)^{1/n} & i = 1 \\ (i - 0.3175)/(n + 0.365) & i = 2, \dots, n - 1 \\ (0.5)^{1/n} & i = n \end{cases}$$

Then one obtains a test for normality by calculation of the correlation r between the ordered observations X_i and m_i the median value of the i^{th} largest observation in a sample of n standard normal random variables so that

$$m_i = \Phi^{-1}(\hat{p}_i)$$

where $\Phi(x)$ is the cumulative distribution function of the standard normal distribution. The value of r is then

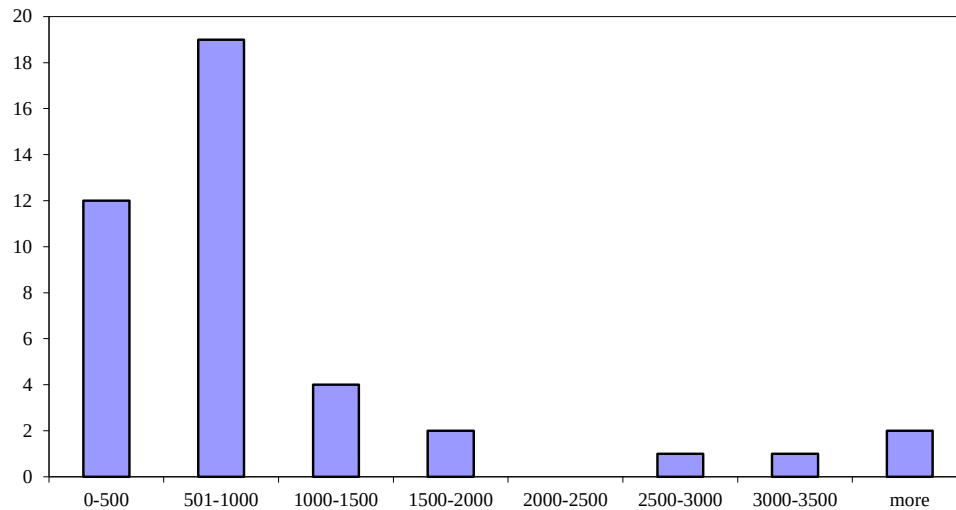
$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})^2 (m_i - \bar{m})^2}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (m_i - \bar{m})^2}}$$

Some significance levels for the value of r are (Filliben 1975)

n	SIGNIFICANCE LEVEL		
	1%	5%	10%
10	0.876	0.917	0.934
20	0.925	0.950	0.960
30	0.947	0.964	0.970
40	0.958	0.972	0.977
50	0.965	0.977	0.981
60	0.970	0.980	0.983

The probability of observing a value of r less than the given value, were the observations actually drawn from a normal distribution, equals the specified probability. Use this test to determine whether a normal or two-parameter lognormal distribution provides an adequate model for these flows.

(a) The construction of a histogram is a simple and easy way of seeing what data look like. Here the observations range from 300-4100.



Clearly the flood flows on this river are highly skewed. Thirty-one of the observations are less than 1000 m³/s while two are greater than 4000 m³/s.

(b) $\bar{x} = 1004.49$

$$v_x^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = 919.42$$

$$\hat{\gamma} = 2.282$$

If the observations were normally distributed then $g = 0$, $E[\hat{\gamma}] = 0$, and

$$S.D.(\hat{\gamma}) \cong \frac{41 - 20}{50 - 20}(0.33) + \frac{50 - 41}{50 - 20}(0.47)$$

$$= 0.37$$

Now $t = \frac{2.282 - 0}{0.37} = 6.162$

Hence it appears to be next to impossible for such a sample skew to have arisen from a sample from a normal distribution.

(c) – (g)

Distribution	Kolmogrov-Smirnov Statistic	Filliben r Statistic
Normal	0.27**	0.813**
2-P lognormal	0.117	0.959*
3-P lognormal	0.062	0.995
2-p gamma	0.169**	0.919**
3-p gamma	0.142*	0.944**

* Significant at 10% level (= 0.126 for K-S test and ~ 0.977 for r)

** Significant at 1% level (= 0.161 for K-S test and ~ 0.959 for r)

Using the K-S test, only the two lognormal distributions appear to fit the data. The Filliben r -test concurs except it indicates that the 2-parameter lognormal distribution is unsatisfactory given that the calculated value of r is right at the 1% significance level.

The quantile-quantile plots (shown below) bear out these conclusions. The quantile-quantile plot for the 2-parameter lognormal distribution has a distinct curvature when the fitted distribution failed to reproduce the high skew exhibited by the data. The observations plot right on a straight line with the 3-parameter lognormal distribution. Its lower bound of 276.0 was calculated using Equation 6.81.

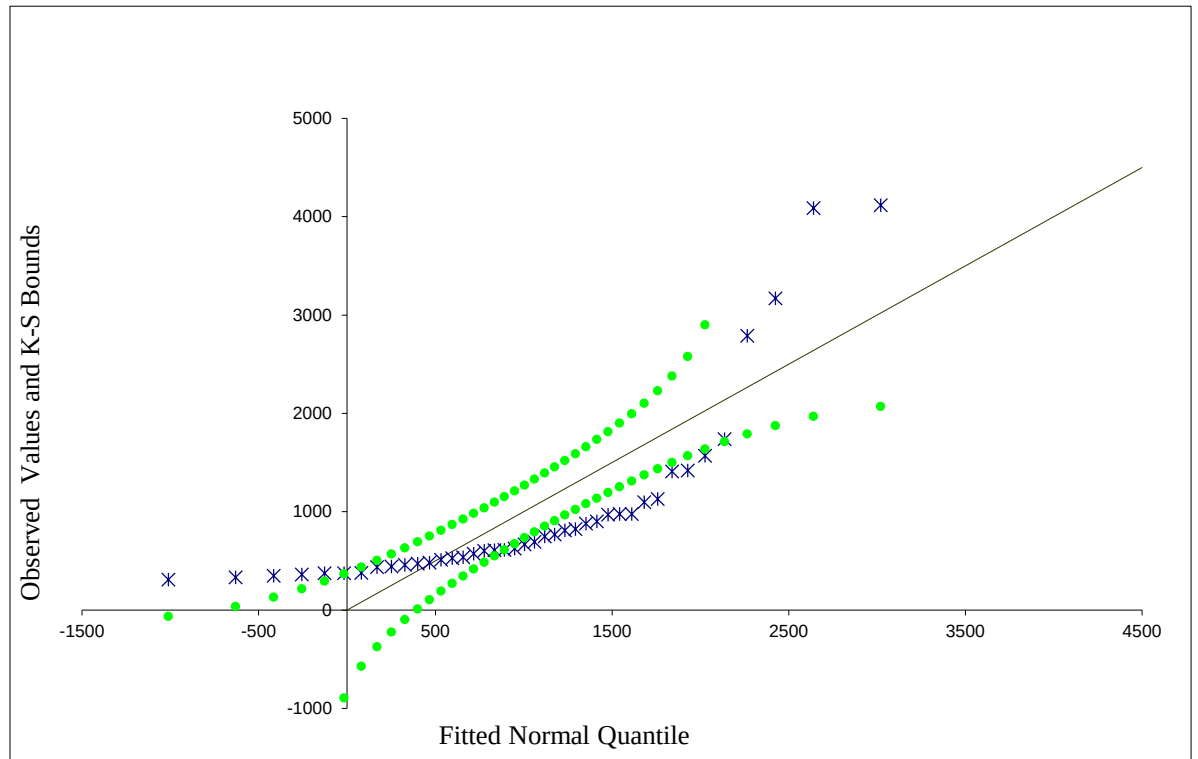
$$\hat{\tau} = \frac{x_1 \times x_n - \hat{x}_{0.50}^2}{x_1 + x_n - 2\hat{x}_{0.50}}$$

Note, the statistically inefficient moment method was used to fit the reported gamma distributions. Had a better fitting procedure been used, these distributions should have fit the data better.

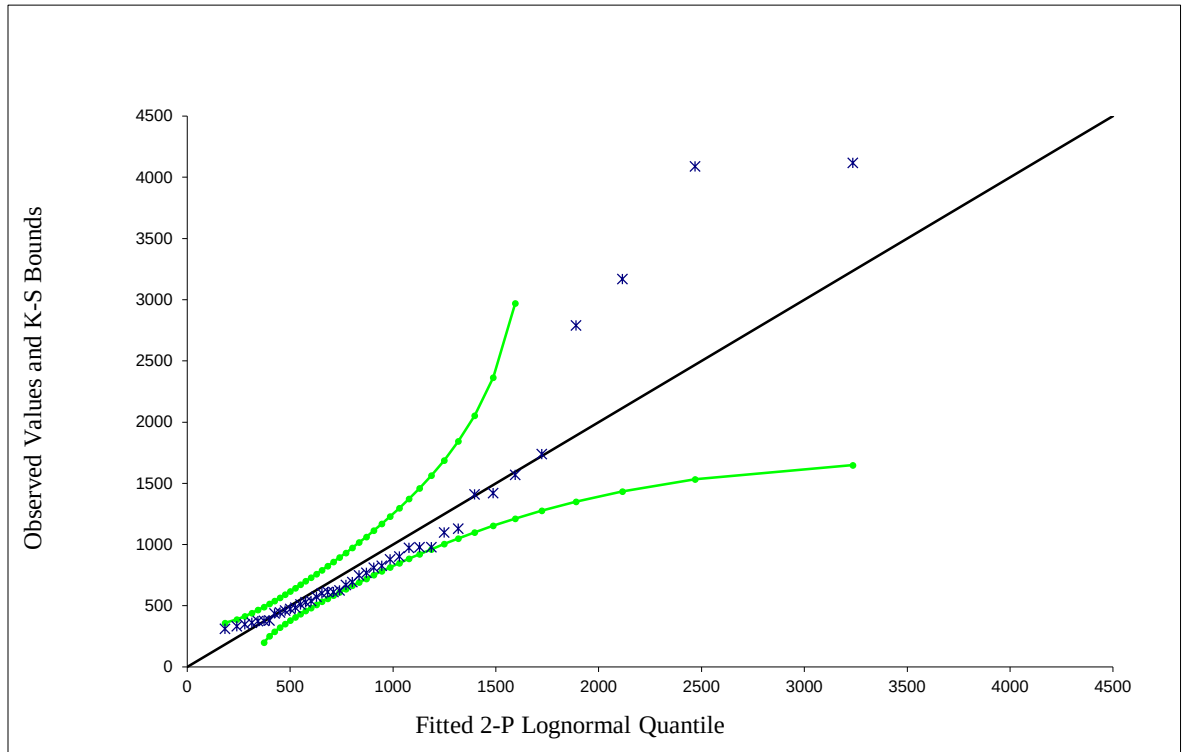
Note, the statistically inefficient moment method was used to fit the reported gamma distributions. Had a better fitting procedure been used, these distributions should have fit the data better.

In the three plots below, the observations were transformed to have a normal distribution by taking logarithms and were plotted against the corresponding fitted quantiles.

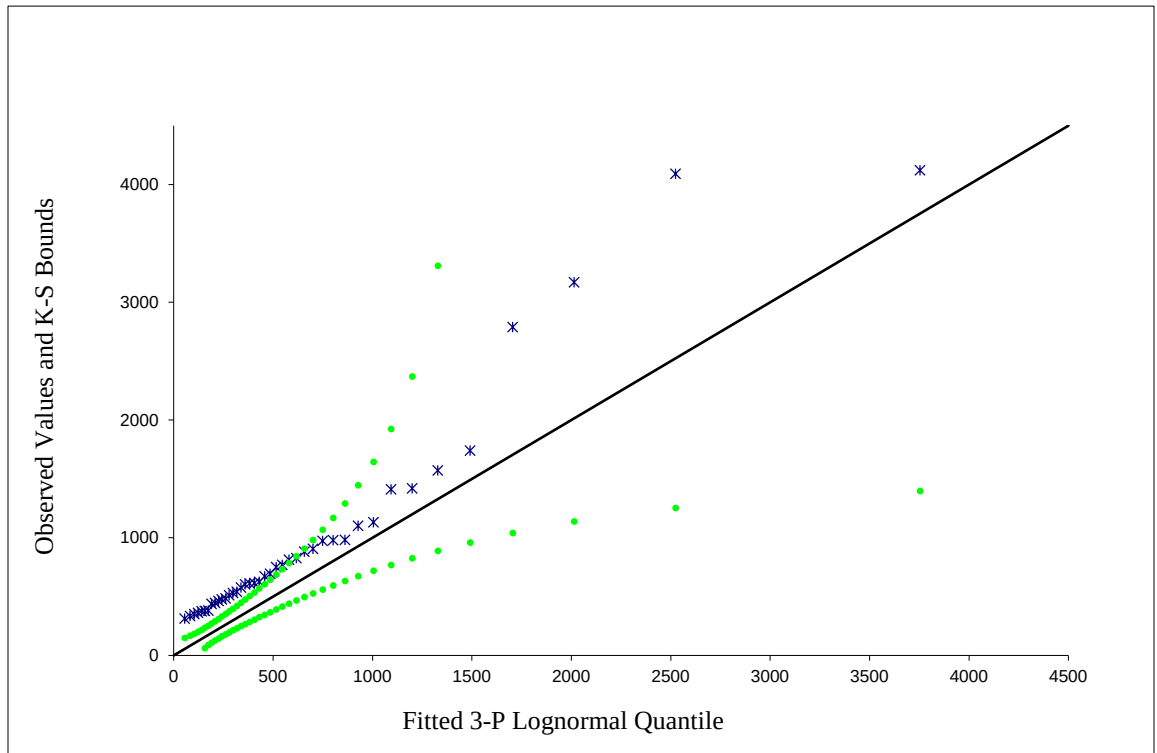
Quantile-Quantile Plot of Normal Distribution Fitted to Observations



Quantile-Quantile Plot of 2-P Lognormal Distribution Fitted to Observations



Quantile-Quantile Plot of 3-P Lognormal Distribution Fitted to Observations



6.18 A small community is considering the immediate expansion of its wastewater treatment facilities so that the expanded facility can meet the current deficit of 0.25 MGD and the anticipated growth in demand over the next 25 years. Future growth is expected to result in the need of an additional 0.75 MGD. The expected demand for capacity as a function of time is

$$\text{Demand} = 0.25 \text{ MGD} + G(1 - e^{-0.23t})$$

where t is the time in years and $G = 0.75$ MGD. The initial capital costs and maintenance and operating costs related to capital are $\$1.2 \times 10^6 C^{0.70}$ where C is the plant capacity (MGD). Calculate the loss of economic efficiency LEE and the misrepresentation of minimal costs (MMC) that would result if a designer incorrectly assigned G a value of 0.563 or 0.938 ($\pm 25\%$) when determining the required capacity of the treatment plant. [Note: When evaluating the true cost of a non-optimal design which provides insufficient capacity to meet demand over a 25-year period, include the cost of building a second treatment plant; use an interest rate of 7% per year to calculate the present value of any subsequent expansions.] In this problem, how important is an error in G compared to an error in the elasticity of costs equal to 0.70? One MGD, a million gallons per day, is equivalent to $0.0438 \text{ m}^3/\text{s}$.

Analysis of the cost data and population projections indicates:

G	Target Capacity	Initial Cost (\$)	Present value of Subsequent Cost (\$) ¹	Total cost (\$)
0.563	0.813	1.038×10^6	0.247×10^6	1.285×10^6
0.750	1.00	1.200×10^6	0	1.200×10^6
0.938	1.188	1.353×10^6	0	1.353×10^6

SENSITIVITY

G	LEE	MMC
0.563	7.1%	-13.5%
0.750	0	0
0.938	12.8%	12.8%

¹ If capacity 0.813 is provided initially then $1.00 - 0.813 = 0.186$ MGD of capacity must be provided at a later date. To minimize cost, capacity is built as late as possible when

$$0.813 = 0.25 + 0.75[1 - e^{-0.23t}]$$

$$t = \frac{-1}{0.23} \ln[1 - \frac{0.563}{0.750}] = 6.03 \text{ years}$$

The subsequent cost is

$$(1.07)^{-6.03} [1.2 \times 10^6 (0.187)^{0.7}] = 0.247 \times 10^6 \$$$

6.19 A municipal water utility is planning the expansion of their water acquisition system over the next 50 years. The demand for water is expected to grow and is given by

$$D = 10t(1 - 0.006t)$$

where t is the time in years. It is expected that two pipelines will be installed along an acquired right-of-way to bring water to the city from a distant reservoir. One pipe will be installed immediately and then a second pipe when the demand just equals the capacity C in year t is

$$PV = (\alpha + \beta C^\gamma) e^{-rt}$$

where

$$\alpha = 29.5$$

$$\beta = 5.2$$

$$\gamma = 0.5$$

$$r = 0.07/\text{year}$$

Using a 50-year planning horizon, what is the capacity of the first pipe which minimizes the total present value of the construction of the two pipelines? When is the second pipe built? If a $\pm 25\%$ error is made in estimating γ or r , what are the losses of economic efficiency (LEE) and the misrepresentation of minimal costs (MMC)? When finding the optimal decision with each set of parameters, find the time of the second expansion to the nearest year; a computer program that finds the total present value of costs as a function of the time of the second expansion t for $t = 1, \dots, 50$ would be helpful. (A second pipe need not be built.)

Note that if a second pipe is built, then the fixed cost α need not be paid. The results are:

Case	γ	r	$\hat{t}$	Capacity First Pipe	$C(\hat{t} w)$	$C(\hat{t} \hat{w})$	LEE	MMC
0	0.5	0.07	27	226.6	120.91	-	0	0
1	0.375	0.07	50	350	126.78	76.28	4.9%	-37%
2	0.625	0.07	20	176	122.68	200.67	1.5%	66%
3	0.5	0.0525	50	350	126.78	126.78	4.9%	4.9%
4	0.5	0.0875	23	198.26	121.42	115.22	0.4%	-4.7%

Some students will still charge the fixed cost even when the second pipe is not built. In that case, the optimal times for the second expansion for cases 1 and 3 becomes 39 and 35 years where:

Case	$\hat{t}$	Capacity First Pipe	$C(\hat{t} w)$	$C(\hat{t} \hat{w})$	LEE	MMC
1	39	298.7	124.10	76.99	2.6%	-36%
3	35	276.5	122.36	127.76	1.2%	5.7%

6.20 A national planning agency for a small country must decide how to develop the water resources of a region. Three development plans have been proposed, which are denoted d_1 , d_2 , and d_3 . Their respective costs are $200f$, $100f$, and $100f$ where f is a million farths, the national currency. The national benefits which are derived from the chosen development plan depend, in part, on the international market for the goods and agricultural commodities that would be produced. Consider three possible international market outcomes, m_1 , m_2 , and m_3 . The national benefits if development plan 1 selected would be, receptively, 400, 290, 250. The national benefits from selection of plan 2 would be 350, 160, 120, while the benefits from selection of plan 3 would be 250, 200, 160.

- Is any plan inferior or dominated?
- If one felt that probabilities could not be assigned to m_1 , m_2 , and m_3 but wished to avoid poor outcomes, what would be an appropriate decision criterion, and why? Which decisions would be selected using this criterion?
- If $\Pr[m_1] = 0.50$ and $\Pr[m_2] = \Pr[m_3] = 0.25$, how would each of the expected net benefits and expected regret criteria rank the decisions?

The first step is to construct the net benefit matrix for this problem. One obtains:

Decision	Net Benefits: Possible Outcomes		
	m_1	m_2	m_3
d_1	200	90	50
d_2	250	60	20
d_3	150	100	60

- (a) No plan is inferior.
 (b) If one felt probabilities could not be assigned to m_1 , m_2 , and m_3 , it would be reasonable to maximize the minimum possible benefits

$$\begin{aligned} & \text{Max}_{d_i} [\min_m \text{NB}(d_i | m)] \\ & \max_{d_i} \left\{ \begin{array}{l} 50 \text{ for } d_1 \\ 20 \text{ for } d_2 \\ 60 \text{ for } d_3 \end{array} \right\} = 60 \text{ with } d_3 \end{aligned}$$

- (c) Using the given probabilities

$$\begin{aligned} E[\text{NB}(d_1 | m)] &= 135 \\ E[\text{NB}(d_2 | m)] &= 145 \\ E[\text{NB}(d_3 | m)] &= 115 \end{aligned}$$

The ranking produced by these probabilities is

$$d_2 > d_1 > d_3$$

and d_2 is the most preferred one.

6.21 Show that if one has a choice between two water management plans yielding benefits X and Y , where X is stochastically smaller than Y , then for any reasonable utility function, plan Y is preferred to X .

The result of this exercise allows one to rank some plans without actually knowing the utility function for a situation. Assume that X is stochastically smaller than Y which means

$$F_Y(z) \geq F_X(z) \quad \text{for all } z$$

Or equivalently

$$F_X^{-1}(p) = x(p) \leq y(p) = F_Y^{-1}(p) \quad \text{for all } p.$$

Thus if someone assumes that some utility function $U(z)$ is a monotonically increasing function of benefits z ,

$$U[x(p)] \leq U[y(p)] \quad \text{for all } p.$$

Hence

$$\begin{aligned} 0 &\geq E[U(x(p)) - U(y(p))] = E[U(x(p))] - E[U(y(p))] \\ &= E[U]_X - E[U]_Y \end{aligned}$$

So that the plan corresponding to X cannot be preferred to the plan corresponding to Y .

6.22 A reservoir system was simulated for 100 years and the average annual benefits and their variance was found to be

$$\bar{B} = 4.93$$

$$s_B^2 = 3.23$$

The correlation of annual benefits was also calculated and is:

k	r_k
0	1.000
1	0.389
2	0.250
3	0.062
4	0.079
5	0.041

- (a) Assume that $\rho(l) = 0$ for $l > k$, compute (using Equation 6.137) the standard error of the calculated average benefits for $k = 0, 1, 2, 3, 4$, and 5. Also calculate the standard error of the calculated benefits, assuming that annual benefits may be thought of as a stochastic process with a correlation structure $\rho_B(k) = [\rho_B(1)]^k$. What is the effect of the correlation structure among the observed benefits on the standard error of their average?
- (b) At the 90% and 95% levels, which of the r_k are significantly different from zero, assuming that $\rho_B(l) = 0$ for $l > k$?

This exercise illustrates the use of formula reported in Section 7.3.

(a) This part depends upon Equation 6.127,

$$\begin{aligned} \text{Var}[\bar{B}] &= \sigma_B^2 [1 + 2 \sum_{l=1}^{n-1} (1 - \frac{k}{n}) \rho(k)] \\ &\cong \frac{s_B^2}{n} [1 + 2 \sum_{l=1}^{n-1} (1 - \frac{k}{n}) r_k] \end{aligned}$$

The reported value of s_B^2 is 3.23, where $n = 100$. Using these values one obtains:

k	$\text{Var}[\bar{x}]$	Standard Error $\bar{x}$
0	0.032	0.180
1	0.057	0.239
2	0.073	0.270
3	0.077	0.277
4	0.082	0.286
5	0.084	0.290

If one assumes that $\rho(k) = (0.389)^k$, Equation 6.127 becomes Equation 6.129:

$$\begin{aligned} \text{Var}[\bar{x}] &= \frac{\sigma^2}{n} \left\{ 1 + \frac{2\rho}{n} \left[\frac{n(1-\rho) - (1-\rho^n)}{(1-\rho)^2} \right] \right\} \\ &\cong \frac{\sigma^2}{n} \left[1 + \frac{2\rho}{(1-\rho)} \right] \\ &\cong \frac{s^2}{n} \left[1 + \frac{2(\hat{r}_1)}{(1-\hat{r}_1)} \right] = 0.0734 = (0.271)^2 \end{aligned}$$

Clearly correlation among the observations increase the standard error of $\bar{x}$.

(b) This problem makes use of Equation 6.137:

$$\text{Var}[r_k] = \frac{1}{n} [1 + 2 \sum_{l=1}^q \rho^2(l)] \quad \text{for } k > q$$

where one assumes that $\rho(l) = 0$ for $l > q$. Letting $q = k - 1$, one obtains:

$$\begin{aligned} k = 1, \quad q = 0 \quad \text{Var}[r_1] &= \frac{1}{n} = 0.010 = (0.10)^2 \\ k = 2, \quad q = 1 \quad \text{Var}[r_2] &= \frac{1}{n} [1 + 2\rho^2(1)] \cong \frac{1}{100} [1 + 2(0.389)] \\ &= 0.0130 = (0.114)^2 \\ k = 3, \quad q = 2 \quad \text{Var}[r_3] &= \frac{1}{n} [1 + 2\rho^2(1) + 2\rho^2(2)] \cong \frac{1}{100} [1.428] \\ &= 0.0143 = (0.119)^2 \end{aligned}$$

and so forth. The tabulated results are

k	Observed r_k	S.D.(r_k)	$t = \frac{r_k}{S.D.(r_k)}$	$\Pr[\hat{t} \geq t]$
1	0.389	0.100	3.89	< 1%
2	0.250	0.114	2.19	~ 2.8%
3	0.062	0.119	0.52	~ 60%
4	0.079	0.120	0.66	~ 51%
5	0.041	0.120	0.34	

Clearly r_k for $k = 1$ and 2 are significant at the 5% level. The r_k 's for $k = 3, 4,$ and 5 are not significant.

6.23 Replicated reservoir simulations using two operating policies produced the following results:

<i>Replicate</i>	BENEFITS	
	<i>Policy 1</i>	<i>Policy 2</i>
1	6.27	4.20
2	3.95	2.58
3	4.49	3.87
4	5.10	5.70
5	5.31	4.02
6	7.15	6.75
7	6.90	4.21
8	6.03	4.13
9	6.35	3.68
10	6.95	7.45
11	7.96	6.86
Mean, $\bar{X}_i$	6.042	4.859
Standard deviation of values, s_{xi}	1.217	1.570

- Construct a 90% confidence limits for each of the two means $\bar{X}_i$.
- With what confidence interval can you state that Policy 1 produces higher benefits than Policy 2 using the sign test and using the t -test?
- If the corresponding replicate with each policy were independent, estimate with what confidence one could have concluded that Policy 1 produces higher benefits with the t -test.

First we complete the lower part of the table.

	Variable		
	X1	X1 - X2	X2
Mean	6.042	1.183	4.859
s	1.217	1.130	1.570

- (a) To construct 90% confidence intervals for the two means we assume that the benefits from each replicate are normally distributed.

Then

$$t = \frac{\bar{x}_i - \mu_i}{s_i / \sqrt{n}}$$

has Student's t-distribution with $n - 1 = 10$ degrees of freedom.

Hence

$$\Pr\left[\left| \frac{\bar{x}_i - \mu_i}{s_i / \sqrt{n}} \right| \leq t_\alpha \right] = 1 - \alpha$$

or

$$\Pr\left[\bar{x}_i - \frac{s_i}{\sqrt{n}} t_\alpha \leq \mu_i \leq \bar{x}_i + \frac{s_i}{\sqrt{n}} t_\alpha \right] = 1 - \alpha$$

for $\alpha = 0.10$, $t_\alpha = 1.812$ (from table of percentage points of Student's t-distribution $\Pr(|t| \leq t_\alpha) = 1 - 2\alpha$) so that

$$i = 1: \quad 6.042 - 0.665 \leq \mu_1 \leq 6.042 + 0.665 \\ 5.38 \leq \mu_1 \leq 6.71$$

$$i = 2: \quad 4.859 - 0.858 \leq \mu_2 \leq 4.859 + 0.858 \\ 4.00 \leq \mu_2 \leq 5.72$$

- (b) Using the sign test, one counts the number of times Policy 1 was better than Policy 2. This happened in 9 of 11 simulations. The probability of such strong evidence (9, 10, or 11 of outcomes in favor of Policy 1) were it equally likely that Policy 1 would be better or worse than Policy 2, is

$$\begin{aligned} \Pr[k \geq 9] &= \binom{11}{11} \left(\frac{1}{2}\right)^{11} \left(\frac{1}{2}\right)^0 + \binom{11}{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^1 + \binom{11}{9} \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^2 \\ &= \left(\frac{1}{2}\right)^{11} \left[1 + 11 + \frac{11 \cdot 10}{1.2}\right] = 0.033 \end{aligned}$$

where $\binom{n}{k} = \frac{n!}{[(n-k)!k!]}$ is a combinatorial.

Thus one can reject the hypothesis of no difference between the policies in favor of Policy 1 is better at the 3.3% level. Using the t test:

$$t = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_{x_1 - x_2} / \sqrt{n}} = \frac{1.183}{1.130 / \sqrt{11}} = 3.471$$

$$\Pr[t \geq 3.471] \leq 0.005.$$

(c) If the replicates with each policy had been independent, the sign test would have no power. The variance one would use in the t -test would be (approximately)

$$s_1^2 + s = (1.271)^2 + (1.570)^2 = 3.946 = (1.96)^2$$

$$t = \frac{1.183}{1.96 / \sqrt{11}} = 2.00$$

$$\Pr[t \geq 2.00] \cong 0.04$$

for 10 degrees of freedom. Clearly Policy 1 is still better than Policy 2 but the results are not as significant as those obtained by applying the t -test to $x_1 - x_2$.

6.24 Assume that annual streamflow at a gaging site have been grouped into three categories or states. State 1 is 5 to 15 m³/s, state 2 is 15 to 25 m³/s, and state 3 is 25 to 35 m³/s, and these grouping contain all the flows on records. The following transition probabilities have been computed from record:

P_{ij}	$j =$		
	1	2	3
1	0.5	0.3	0.2
2	0.3	0.3	0.4
3	0.1	0.5	0.4

- (a) If the flow for the current year is between 15 and 25 m³/s, what is the probability that the annual flow 2 years from now will be in the range 25 to 35 m³/s?

- (b) What is the probability of a dry, an average, and a wet year many years from now?

This is a simple problem illustrating the use of Markov chains. The transition matrix is

$$P = \begin{pmatrix} 0.50 & 0.30 & 0.20 \\ 0.30 & 0.30 & 0.40 \\ 0.10 & 0.50 & 0.40 \end{pmatrix}$$

- (a) If the current river stage is between 15 m³/s and 25 m³/s, then the system is in state 2. The probability vector for the subsequent year (containing the probabilities the flow level is in states 1, 2, 3, respectively) is $(0,1,0)P = (0.30, 0.30, 0.40)$. The probability vector for the next year is

$$(0.30, 0.30, 0.40)P = (0.28, 0.38, 0.34)$$

Thus the probability of a flow in the range 25 – 35 m³/s (the third state) equals 0.34 or 34%.

- (b) The steady-state probabilities describing the likely state of the system in the distant future are obtained by solving

$$(p_1, p_2, p_3)P = (p_1, p_2, p_3)$$

$$p_1 + p_2 + p_3 = 1$$

The solution is (0.290, 0.368, 0.342).

6.25 A Markov chain model for the streamflows in two different seasons has the following transition probabilities

STREAMFLOW next Season 2			
STREAMFLOW IN SEASON 1	0-3 m ³ /s	3-6 m ³ /s	≥ 6 m ³ /s
0-10 m ³ /s	0.25	0.50	0.25
≥ 10 m ³ /s	0.05	0.55	0.40

STREAMFLOW next Season 1		
STREAMFLOW IN SEASON 2	0-10 m ³ /s	≥ 10 m ³ /s
0-3 m ³ /s	0.70	0.30
3-6 m ³ /s	0.50	0.50
≥ 6 m ³ /s	0.40	0.60

Calculate the steady-state probabilities of the flows in each interval in each season.

This problem can be solved in several steps. First one can determine the year-to-year transition probabilities for flows in some season to the rows in that same season the next year.

Using season 1,

$$\begin{aligned}
 P &= \begin{pmatrix} 0.25 & 0.50 & 0.25 \\ 0.05 & 0.55 & 0.40 \end{pmatrix} \bullet \begin{pmatrix} 0.70 & 0.30 \\ 0.50 & 0.50 \\ 0.40 & 0.60 \end{pmatrix} \\
 &= \begin{pmatrix} 0.525 & 0.475 \\ 0.470 & 0.530 \end{pmatrix} \\
 &= \left(\sum_k p_{ik}^1 p_{kj}^2 \right)
 \end{aligned}$$

Here p_{ik}^1 is the probability of going from state i in season 1 to state k in season 2; p_{kj}^2 is the probability of going from state k in season 2 to state j in the following season 1. Summing over all k yields the probability of going from state i in season 1 to state j in the following season 1. This transition matrix can be used to obtain the steady state probabilities for season 1. Solving

$$\begin{aligned}
 0.525p_1^1 + 0.470p_2^1 &= p_1^1 && \text{or} \\
 0.475p_1^1 + 0.530p_2^1 &= p_2^1 && \text{and} \\
 p_1 + p_2 &= 1
 \end{aligned}$$

one obtains

$$\begin{aligned}
 p_1^1 &= 0.4974 \\
 p_2^1 &= 0.5026
 \end{aligned}$$

The steady-state probabilities for season 2 are then

$$\begin{aligned}
 (p_1^2, p_2^2, p_3^2) &= (p_1^1, p_2^1) \begin{pmatrix} 0.25 & 0.50 & 0.25 \\ 0.05 & 0.55 & 0.40 \end{pmatrix} \\
 &= (0.1495, 0.5251, 0.3254)
 \end{aligned}$$

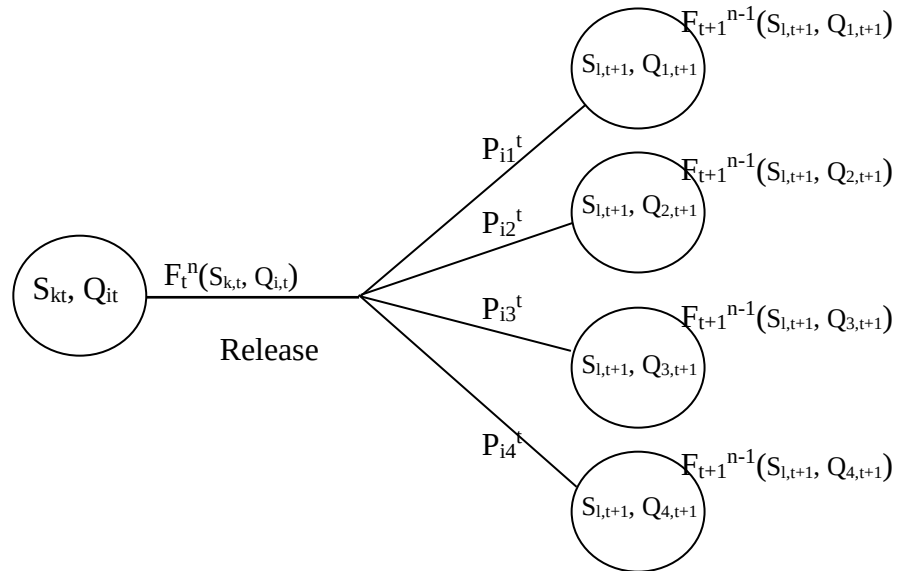
6.26 Can you modify the deterministic discrete DP reservoir-operating model to include the uncertainty, expressed as P_{ij}^t , of the inflows, as in Exercise 6.25?

(Hints: The operating policy would define the release (or final storage) in each season as a function of not only the initial storage but also the inflow. If the inflow changes, so might the release or final storage volume. Hence you need to discretize the inflows as well as the storage volumes. Both storage and inflow are state variables. Assume for this model you can predict with certainty the inflow in each period at the beginning of the period. So, each node of the network represents a known initial storage and inflow value. You cannot predict with certainty the following period's flows, only their probabilities. What does the network look like now?

Using the final storage as the decision variable (from which the release can be computed):

$$F_t^n(S_{kt}, Q_{it}) = \max_{\substack{S_{l,t+1} \leq K \\ S_{l,t+1} \leq S_{kt} + Q_{it}}} \text{or min} \{ \text{Obj}_t(S_{kt}, Q_{it}, S_{l,t+1}) + \sum_j P_{ij}^t F_{t+1}^{n-1}(S_{l,t+1}, Q_{j,t+1}) \}$$

Portion of network for a particular combination of initial and final storage volumes: The final storage volumes are the same, but the inflow, $Q_{j,t+1}$, in period $t+1$ changes. Each of those inflows has a conditional probability P_{ij}^t .



For each release link from a (S_{kt}, Q_{it}) node, a possible $F_t^n(S_{k,t}, Q_{i,t})$ value = $P_{i1}^t F_{t+1}^{n-1}(S_{l,t+1}, Q_{1,t+1}) + \dots + P_{i4}^t F_{t+1}^{n-1}(S_{l,t+1}, Q_{4,t+1})$.

The F values are expected values of the objective being maximized or minimized associated with a season t and number of remaining periods n .

6.27 Assume that there exist two possible discrete flows Q_{it} into a small reservoir in each of two periods t each year having probabilities P_{it} . Find the steady-state operating policy (release as a function of initial reservoir volumes and current period's inflow) for the reservoir that minimizes the expected sum of squared deviations from storage and release targets. Limit the storage volumes to integer values that vary from 3 to 5. Assume a storage volume target of 4 and a release target of 2 in each period t . (Assume only integer values of all states and decision variables and that each period's inflow is known at the beginning of the period.) Find the annual expected sum of squared deviations from the storage and release targets.

Period, t	FLOWS, Q_{it}		PROBABILITIES, P_{it}	
	$i = 1$	$i = 2$	$i = 1$	$i = 2$
1	1	2	0.17	0.83
2	3	4	0.29	0.71

This is an application of Exercise 6.26 except the flow probabilities are independent of the previous flow.

First it is advantageous to calculate the loss functions for each season.

S_1, Q_1	R_1			S_2			ΣDev^2		
3, 1	0	1		4	3		5	2	
3, 2	0	1	2	5	4	3	5	2	1
4, 1	0	1	2	5	4	3	4	1	0
4, 2	1	2	3	5	4	3	1	0	1
5, 1	1	2	3	5	4	3	2	1	2
5, 2	2	3	4	5	4	3	1	2	5

S_2, Q_2	R_2			S_1			ΣDev^2		
3, 3	1	2	3	5	4	3	2	1	2
3, 4	2	3	4	5	4	3	1	2	5
4, 3	2	3	4	5	4	3	0	1	4
4, 4	3	4	5	5	4	3	1	4	9
5, 3	3	4	5	5	4	3	2	5	10
5, 4	4	5	6	5	4	3	5	10	17

Now, let $f_t^n(S_t, Q_t)$ = expected sum of squared deviations from storage and release targets in remaining n periods when start with storage S_t in period $t = 1, 2$;
Let all $f_t^n(S_t, Q_t) = 0$ for $n = 0$.

Last stage: $n = 1, t = 2$

$$f_2^1(3,3) = \min \{ 2 + 0, 1 + 0, 2 + 0 \} = 1 \text{ with } R_2 = 2.$$

$$f_2^1(3,4) = \min \{ 1 + 0, 2 + 0, 5 + 0 \} = 1 \text{ with } R_2 = 2.$$

$$f_2^1(4,3) = \min \{ 0 + 0, 1 + 0, 4 + 0 \} = 0 \text{ with } R_2 = 2.$$

$$f_2^1(4,4) = \min \{ 1 + 0, 4 + 0, 9 + 0 \} = 1 \text{ with } R_2 = 3.$$

$$f_2^1(5,3) = \min \{ 2 + 0, 5 + 0, 10 + 0 \} = 2 \text{ with } R_2 = 3.$$

$$f_2^1(5,4) = \min \{ 5 + 0, 10 + 0, 17 + 0 \} = 5 \text{ with } R_2 = 4.$$

Next to last stage: $n = 2, t = 1$

$$f_1^2(3,1) = \min \{ 5 + 0(0.29) + 1(0.71), 2 + 1(0.29) + 1(0.71) \} = 3 \\ \text{with } R_1 = 1.$$

$$f_1^2(3,2) = \min \{ 5 + 2(0.29) + 5(0.71), 2 + 0(0.29) + 1(0.71), \\ 1 + 1(0.29) + 1(0.71) \} = 2 \text{ with } R_1 = 2.$$

$$f_1^2(4,1) = \min \{ 4 + 2(0.29) + 5(0.71), 1 + 0(0.29) + 1(0.71), \\ 0 + 1(0.29) + 1(0.71) \} = 1 \text{ with } R_1 = 2.$$

$$f_1^2(4,2) = \min \{ 1 + 2(0.29) + 5(0.71), 0 + 0(0.29) + 1(0.71), \\ 1 + 1(0.29) + 1(0.71) \} = 0.71 \text{ with } R_1 = 2.$$

$$f_1^2(5,1) = \min \{ 2 + 2(0.29) + 5(0.71), 1 + 0(0.29) + 1(0.71), \\ 2 + 1(0.29) + 1(0.71) \} = 1.71 \text{ with } R_1 = 2.$$

$$f_1^2(5,2) = \min \{ 1 + 2(0.29) + 5(0.71), 2 + 0(0.29) + 1(0.71), \\ 5 + 1(0.29) + 1(0.71) \} = 2.71 \text{ with } R_1 = 3.$$

Continuing... $n = 3, t = 2$

$$f_2^3(3,3) = \min \{ 2 + 1.71(0.17) + 2.71(0.83), 1 + 1(0.17) + \\ 0.71(0.83), 2 + 3(0.17) + 2(0.83) \} = 1.76 \text{ with } R_2 = 2.$$

$$f_2^3(3,4) = \min \{ 1 + 1.71(0.17) + 2.71(0.83), 2 + 1(0.17) + \\ 0.71(0.83), 5 + 3(0.17) + 2(0.83) \} = 2.76 \text{ with } R_2 = 3.$$

$$f_2^3(4,3) = \min \{ 0 + 1.71(0.17) + 2.71(0.83), 1 + 1(0.17) + \\ 0.71(0.83), 4 + 3(0.17) + 2(0.83) \} = 1.76 \text{ with } R_2 = 3.$$

$$f_2^3(4,4) = \min \{ 1 + 1.71(0.17) + 2.71(0.83), 4 + 1(0.17) + \\ 0.71(0.83), 9 + 3(0.17) + 2(0.83) \} = 3.54 \text{ with } R_2 = 3.$$

$$f_2^3(5,3) = \min \{ 2 + 1.71(0.17) + 2.71(0.83), 5 + 1(0.17) + \\ 0.71(0.83), 10 + 3(0.17) + 2(0.83) \} = 4.54 \text{ with } R_2 = 3.$$

$$f_2^3(5,4) = \min \{ 5 + 1.71(0.17) + 2.71(0.83), 10 + 1(0.17) + \\ 0.71(0.83), 17 + 3(0.17) + 2(0.83) \} = 7.54 \text{ with } R_2 = 4.$$

Continuing... n = 4, t = 1

$$\begin{aligned}
 f_1^4(3,1) &= \min \{ 5 + 1.76(0.29) + 3.54(0.71), 2 + 1.76(0.29) + 2.76(0.71) \} = 4.47 \text{ with } R_1 = 1. \\
 f_1^4(3,2) &= \min \{ 5 + 4.54(0.29) + 7.54(0.71), 2 + 1.76(0.29) + 3.54(0.71), 1 + 1.76(0.29) + 2.76(0.71) \} = 3.47 \text{ with } R_1 = 2. \\
 f_1^4(4,1) &= \min \{ 4 + 4.54(0.29) + 7.54(0.71), 1 + 1.76(0.29) + 3.54(0.71), 0 + 1.76(0.29) + 2.76(0.71) \} = 2.47 \text{ with } R_1 = 2. \\
 f_1^4(4,2) &= \min \{ 1 + 4.54(0.29) + 7.54(0.71), 0 + 1.76(0.29) + 3.54(0.71), 1 + 1.76(0.29) + 2.76(0.71) \} = 3.02 \text{ with } R_1 = 2. \\
 f_1^4(5,1) &= \min \{ 2 + 4.54(0.29) + 7.54(0.71), 1 + 1.76(0.29) + 3.54(0.71), 2 + 1.76(0.29) + 2.76(0.71) \} = 4.02 \text{ with } R_1 = 2. \\
 f_1^4(5,2) &= \min \{ 1 + 4.54(0.29) + 7.54(0.71), 2 + 1.76(0.29) + 3.54(0.71), 5 + 1.76(0.29) + 2.76(0.71) \} = 5.02 \text{ with } R_1 = 3.
 \end{aligned}$$

Now this process should continue until $f_t^{n+2}(S_t, Q_t) - f_t^n(S_t, Q_t)$ is a constant representing annual expected squared deviations from targets. Based on results so far the policy appears to be

	Storage	Inflow	Release	Sq. Deviations
Period 1	3	1	1	2
	3	2	2	1
	4	1	2	0
	4	2	2	0
	5	1	2	1
	5	2	3	2
Period 2	3	3	2	1
	3	4	3	2
	4	3	3	1
	4	4	3	1
	5	3	3	2
	5	4	4	5

Note: Once this policy is implemented S_1 will never be 3 and S_2 will never be 5.

Given this policy the joint probabilities $P_t(S, Q)$ can be calculated:

$$\begin{aligned}
 P_2(3, 3) &= [P_1(4,1)] 0.29 = 0.01625952 \\
 P_2(3, 4) &= [P_1(4,1)] 0.71 = 0.03980780 \\
 P_2(4, 3) &= [P_1(4,2) + P_1(5,1) + P_1(5,2)] 0.29 = 0.2737405 \\
 P_2(4, 4) &= [P_1(4,2) + P_1(5,1) + P_1(5,2)] 0.71 = 0.6701922
 \end{aligned}$$

$$P_1(4, 1) = [P_2(3,3) + P_2(3,4) + P_2(4,3)] 0.17 = 0.05606733$$

$$P_1(4, 2) = [P_2(3,3) + P_2(3,4) + P_2(4,3)] 0.83 = 0.2737405$$

$$P_1(5, 1) = [P_2(4,4)] 0.17 = 0.1139327$$

$$P_1(5, 2) = [P_2(4,4)] 0.83 = 0.5562595$$

$$P_1(4,1) + P_1(4,2) + P_1(5,1) + P_1(5,2) = 1$$

Summing the products of these joint probabilities times the sum of squared deviations yields 2.266260.

6.28 Assume that the streamflow Q at a particular site has cumulative distribution function $F_Q(q) = q/(1 + q)$ for $q \geq 0$. Show how to compute the mean streamflow, and the probability that any specified value of streamflow, q , will be exceeded.

The mean streamflow is the area under the probability of exceedance function. The probability of exceedance function is $1 - \text{cumulative distribution function } F_Q(q)$.

$$1 - q/(1 + q) = 1/(1+q)$$

Integrating this exceedance function from $q = 0$ to ∞ yields $\ln(1+\infty) - \ln(1) = \infty$, so clearly the mean does not exist. The probability p that any specified value of q being exceeded is $1/(1+q)$.

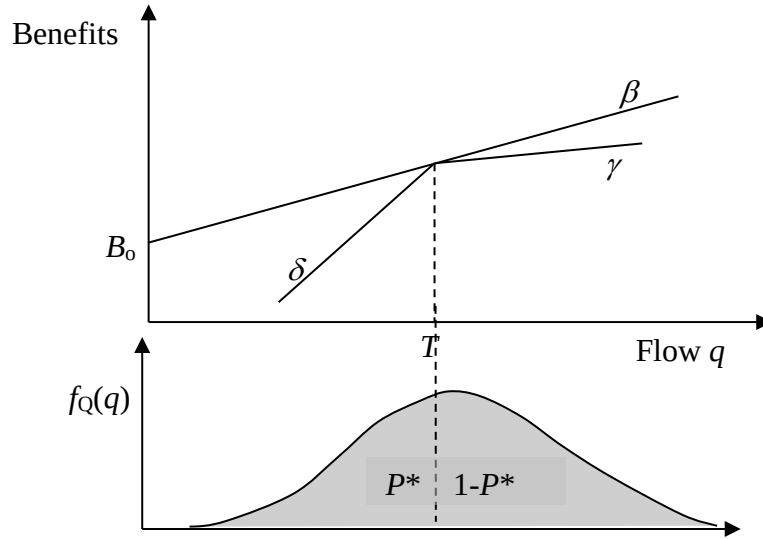
6.29 Assume that a potential water user can withdraw water from an unregulated stream, and that the probability distribution function $F_Q()$ of the available streamflow Q is known. Calculate the value of the withdrawal target T that will maximize the expected net benefits from the water's use given the two short-run benefit functions specified below.

(a) The benefits from streamflow Q when the target is T are

$$B(Q | T) = \begin{cases} B_0 + \beta T + \gamma(Q - T) & Q \geq T \\ B_0 + \beta T + \delta(Q - T) & Q < T \end{cases}$$

where $\delta > \beta > \gamma$. In this case, the optimal target T^* can be expressed as a function of $P^* = F_Q(T) = \Pr\{Q \leq T\}$, the probability that the random streamflow Q will be less than or equal to T . Prove that

$$P^* = (\beta - \gamma) / (\delta - \gamma).$$



(b) The benefits from streamflow Q when the target is T are

$$B(Q | T) = B_0 + \beta T - \delta(Q - T)^2$$

Solution:

Here one wishes to maximize the expected benefits given by

$$EB(T) = \int_0^{\infty} B(Q | T) f(Q) dQ$$

(a) To maximize $EB(T)$ differentiate the above expression with respect to T and use the general result that

$$\frac{d}{dx} \int_0^x g(s, x) ds = g(x, x) + \int_0^x \frac{\partial g(s, x)}{\partial x} ds$$

when $g(s, x)$ is continuously differentiable.

Thus to find the T that maximizes the expected benefits:

$$\begin{aligned} \frac{d}{dT} EB(T) &= B(T | T) f(T) + \int_0^T \frac{\partial}{\partial T} B(Q | T) f(Q) dQ - B(T | T) f(T) \\ &\quad + \int_T^{\infty} \frac{\partial}{\partial T} B(Q | T) f(Q) dQ = 0 \\ &= \int_0^T \frac{\partial}{\partial T} [B_0 + \beta T + \delta(Q - T)] f(Q) dQ \end{aligned}$$

$$\begin{aligned}
& + \int_T^{\infty} \frac{\partial}{\partial T} [B_o + \beta T + \gamma(Q - T)] f(Q) dQ \\
& = \int_0^T [\beta - \delta] f(Q) dQ + \int_T^{\infty} [\beta - \gamma] f(Q) dQ \\
& = (\beta - \delta)F(T) + (\beta - \gamma)[1 - F(T)]
\end{aligned}$$

Hence

$$F(T) = \frac{-(\beta - \gamma)}{(\beta - \delta) - (\beta - \gamma)} = \frac{\beta - \gamma}{\delta - \gamma}$$

Note

$$\begin{aligned}
\frac{d^2}{dT^2} &= \frac{d}{dT} [(\beta - \delta)F(T) + (\beta - \gamma)(1 - F(T))] \\
&= (\beta - \delta)f(T) - (\beta - \gamma)f(T) = (\gamma - \delta)f(T) < 0
\end{aligned}$$

when $\delta > \beta > \gamma$, so that a maximum has indeed been found.

(b) When

$$B(Q | T) = B_o + \beta T + \delta(Q - T)^2$$

we obtain

$$\begin{aligned}
\frac{d}{dT} EB(T) &= \int_0^{\infty} \frac{\partial}{\partial T} [B_o + \beta T + \delta(Q - T)^2] f(Q) dQ = 0 \\
&= \beta \int_0^{\infty} f(Q) dQ + 2\delta \int_0^{\infty} (Q - T) f(Q) dQ \\
&= \beta + 2\delta(E[Q] - T)
\end{aligned}$$

so that

$$T = E[Q] + \frac{\beta}{2\delta} \quad \text{where} \quad \frac{d^2}{dT^2} EB(T) = -2\delta < 0$$

Thus a maximum has indeed been obtained.

6.30 If a random variable is discrete, what effect does this have on the specified confidence of a confidence interval for the median or any other quantile? Give an example.

For given p there may be no x for which $\Pr[X \leq x] = p$

Then the p^{th} quantile will be the x_p for which

$$p_1 = \Pr[X \leq x_p] < p < \Pr[X \leq x_p] = p_2$$

Then

$$\begin{aligned} \Pr[X_r \leq x_p \leq X_s] &= \sum_{k=r}^{s-1} \binom{n}{k} \Pr[k \text{ values} \leq x_p] \Pr[n - k \text{ values} \geq x_p] \\ &= \sum_{k=r}^{s-1} \binom{n}{k} \frac{n!}{(n-k)!k!} p_2^k (1-p_1)^{n-k} > \sum_{k=r}^{s-1} p^k (1-p)^{n-k} \end{aligned}$$

because $p_2 > p$ and also $(1-p_1) > (1-p)$

Hence if a random variable is discrete, the confidence level attached to a confidence interval may be an underestimate.

6.31 (a) Use Wilcoxon test for unpaired samples to test the hypothesis that the distribution of the total shortage TS in Table 6.14 is stochastically less than the total shortage TS reported in Table 6.15. Use only the data from the second 10 simulations reported in the table. Use the fact that observations are paired (i.e., simulation j for $11 \leq j \leq 20$ in both tables were obtained with the same streamflow sequence) to perform the analysis with the sign test.

(b) Use the sign test to demonstrate that the average deficit with Policy 1 (Table 6.14) is stochastically smaller than with Policy 2 (Table 6.15); use all simulations.

(a) The data are displayed in the following table.

<i>Simulation</i>	<i>Total Shortage Table 7.14</i>	<i>Rank</i>	<i>Total Shortage Table 7.15</i>	<i>Rank</i>	<i>Difference T7.14 – T7.15</i>
11	1.80	11.5	3.88	19	-2.08
12	1.28	9	1.92	13	-0.64
13	0.53	5	1.50	10	-0.97
14	0.88	8	2.52	16	-1.64
15	1.99	14	3.76	18	-1.77
16	0.23	4	1.80	11.5	-1.57
17	2.68	17	5.10	20	-2.42
18	0.76	6.5	2.40	15	-1.64
19	0.00	2	0.00	2	0.00
20	0.00	2	0.76	6.5	-0.76
Sum of Ranks $W =$		79			

To perform the Wilcoxon test for unpaired data one must order all the observations. When ties occur, the average of the ranks are used. The resulting ranks are shown above. In this case one has 10 observations of each variable so the distribution of W , the sum of the ranks of variables 1, is essentially normal with mean and variance given by

$$E[W] = \frac{10(10+1)}{2} + \frac{1}{2}(10)(10) = 105$$

$$Var[W] = \frac{1}{12}(10)(10)(10+10+1) = 175$$

$$t = \frac{79-105}{\sqrt{175}} = -1.97$$

$$\Pr[\tilde{t} \leq -1.97] = 0.024$$

Hence one can conclude that the total shortage with Policy 1 are less than those obtained with Policy 2 at the 2.4% level.

If one uses the fact that the observations are paired, the sign test provides an easy test of whether shortages are less severe with Policy 1 than Policy 2. In all cases where the shortages differ, they are more severe with Policy 2.

If we test the hypothesis that there is no difference versus the hypothesis that Policy 1 produces less severe shortages than Policy 2, then the significance of this result is $(0.5)^9 = 0.002$.

(b) Comparing the two columns for average deficits one finds that those in Table 6.14 are smaller 20 times, there are two ties, and those in Table 6.15 are smaller 3 times. The probability of this or a more lopsided result in favor of the first policy were it equally likely that either policy would achieve the smaller average deficit is

$$\left(\frac{1}{2}\right)^{20+3} \left[\binom{23}{0} + \binom{23}{1} + \binom{23}{2} + \binom{23}{3} \right] = 0.00024 = 0.024\%$$

Note, the ties are ignored in this test, thus the result is conditional on observing 23 differences.

6.32 The accompanying table provides an example of the use of non-parametric statistics for examining the adequacy of synthetic streamflow generators. Here the maximum yield that can be supplied with a given size reservoir is considered. The following table gives the rank of the maximum yield obtainable with the historic flows among the set consisting of the historic yield and the maximum yield achievable with 1000 synthetic sequences of 25 different rivers in North America.

- (a) Plot the histogram of the ranks for reservoir sizes $S/\mu_Q = 0.85, 1.35, 2.00$. (Hint: Use the intervals 0-100, 101-200, 201-300, etc.) Do the ranks look uniformly distributed?

**Rank of the Maximum Historic Yield
among 1000 Synthetic Yields**

River Number	NORMALIZED ACTIVE STORAGE, S/μ_Q			
	0.35	0.85	1.35	2.00
1	47	136	128	235
2	296	207	183	156
3	402	146	120	84
4	367	273	141	191
5	453	442	413	502
6	76	92	56	54
7	413	365	273	279
8	274	191	86	51
9	362	121	50	29
10	240	190	188	141
11	266	66	60	118
12	35	433	562	738
13	47	145	647	379
14	570	452	380	359
15	286	392	424	421
16	43	232	112	97
17	22	102	173	266
18	271	172	260	456
19	295	162	272	291
20	307	444	532	410
21	7	624	418	332
22	618	811	801	679
23	1	78	608	778
24	263	902	878	737
25	82	127	758	910

Source: A. I. McLeod and K. W. Hipel, Critical Drought Revisited,
Paper presented at the international Symposium on Risk and Reliability
in Water Resources, Waterloo, Ont., June 26-28, 1978

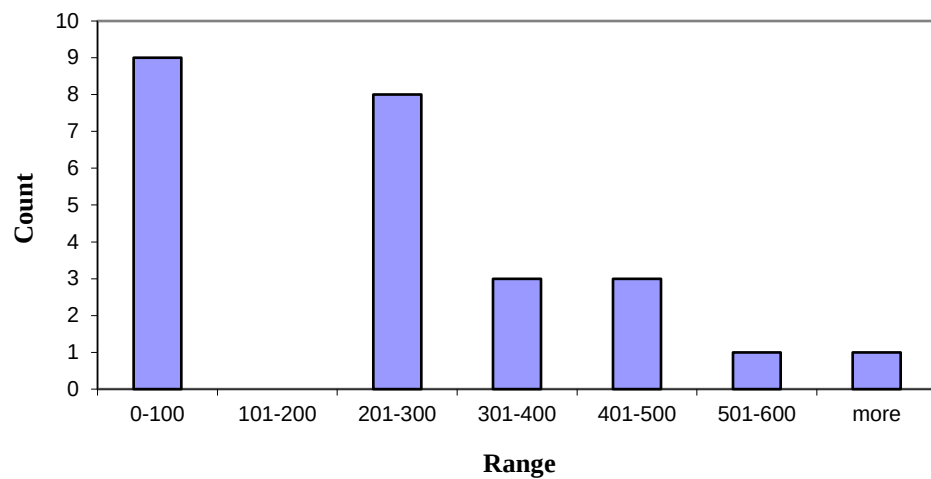
- (b) Do you think this streamflow generation model produces streamflows which are consistent with the historic flows when one uses as a criterion the maximum possible yield? Construct a statistical test to support your conclusion and show that it does support your conclusion. (Idea: You might want to consider if it is equally likely that the rank of the historical yield is

500 and below 501 and above. You could then use the binomial distribution to determine the significance of the results.)

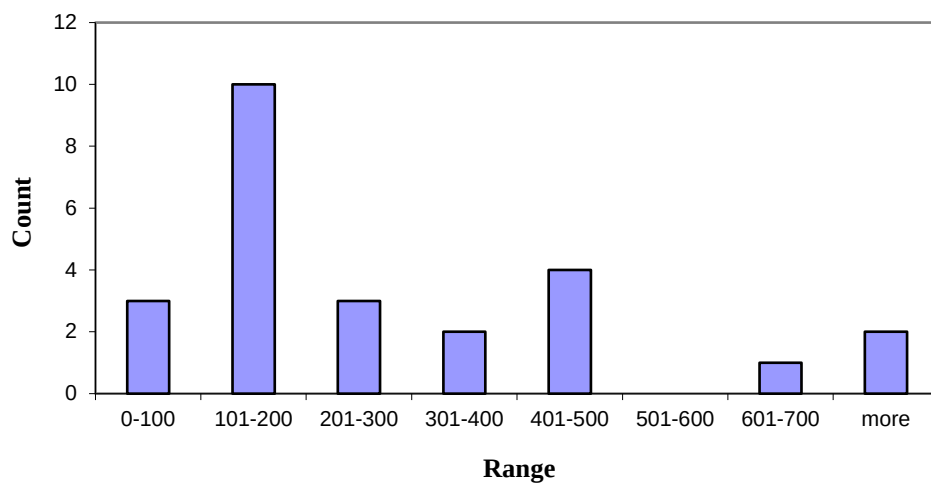
- (c) Use the Kolmogorov-Smirnov test to check if the distribution of the yields obtainable with storage $S/\mu_Q = 1.35$ is significantly different from uniform $F_U(u) = u$ for $0 \leq u \leq 1$. How important do you feel this result is?

(a) Histogram provides an easy way to see what the distribution look like.
We have

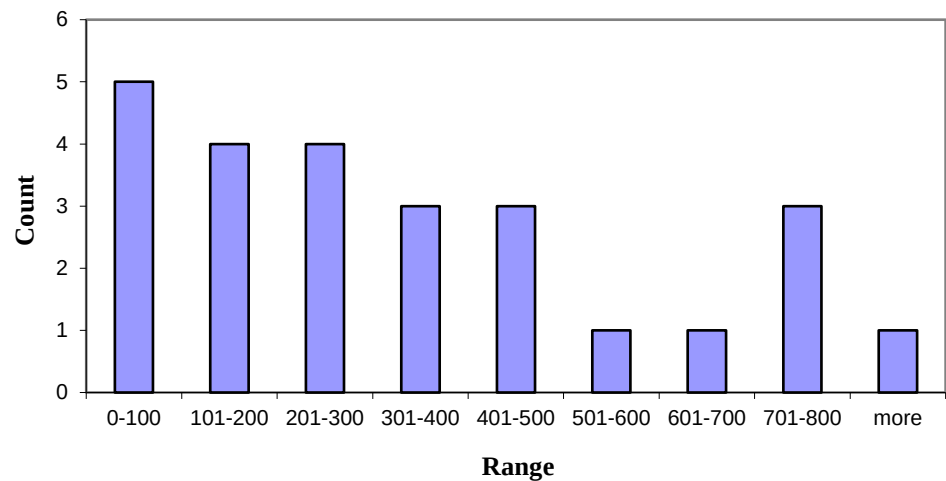
$$S/\mu_Q = 0.35$$



$$S/\mu_Q = 0.85$$



$$S/\mu_Q = 2.00$$



(b) Particularly for low S/μ_Q the ranks tend to always be less than 500. Because there should be a 50-50 chance that the ranks are above or below 500, one can use the binomial distribution (sign test) to see if droughts in the synthetic records tend to be less severe than in the historical record:

S/μ_Q :	0.35	0.85	1.35	2.00
k = Number				
of Ranks > 500 =	2	3	7	6
$\Pr[\hat{k} \leq k] = 0.024$	<0.01%	0.01%	2.2%	0.73%

Clearly, the streamflow model is not producing severe droughts (which determine the value of S) with the frequency actually observed.

i	$(i-1)/n$	Observations=Rank/100	i/n	Maximum Deviation
1	0	0.049	0.04	0.049
2	0.04	0.055	0.08	0.025
3	0.08	0.059	0.12	0.061
4	0.12	0.085	0.16	0.075
5	0.16	0.111	0.20	0.089
6	0.20	0.119	0.24	0.121
7	0.24	0.127	0.28	0.153
8	0.28	0.140	0.32	0.180
9	0.32	0.172	0.36	0.188
10	0.36	0.182	0.40	0.218
11	0.40	0.187	0.44	0.253
12	0.44	0.259	0.48	0.221
13	0.48	0.271	0.52	0.249
14	0.52	0.272	0.56	0.288
15	0.56	0.379	0.60	0.221
16	0.60	0.412	0.64	0.228
17	0.64	0.417	0.68	0.263
18	0.68	0.423	0.72	0.297
19	0.72	0.530	0.76	0.230
20	0.76	0.560	0.80	0.240
21	0.80	0.606	0.84	0.234
22	0.84	0.645	0.88	0.235
23	0.88	0.756	0.92	0.164
24	0.92	0.799	0.96	0.161
25	0.96	0.876	1.00	0.124

Maximum deviation is 0.297. This is significant at the 1% level!

It is important that the ranks are not uniform for one might think that they should be. However, because the streamflow models were actually fit using the historical flows, one might also think that the ranks of the historical S's should cluster around 500. Thus the results obtained in part (b) are more profound than those obtained here.

6.33 Section 7.3 dismisses the bias in ν_x^2 for correlated X 's as unimportant to its variance.

- Calculate the approximate bias in ν_x^2 for the cases corresponding to Table 6.10 and determine if this assertion is justified.
- By numerically evaluating the bias and variance of ν_x^2 , when $n = 25$, determine if the same result holds if $\rho_x(k) = 0.5(0.9)^k$, which is the autocorrelation function of an ARMA (1, 1) process sometimes used to describe annual streamflow series.

The purpose of this exercise is to provide an appreciation of the size of the bias in v_x^2 as compared to its standard error.

From Equation 6.131 one has

$$E[v_x^2] = \sigma_x^2 \left[1 - \frac{1}{n} - \frac{2}{n} \sum_{k=1}^{n-1} \left(1 - \frac{k}{n} \right) \rho_x(k) \right]$$

Comparing Equations 6.127 and 6.129, the equation above can be written

$$E[v_x^2] = \sigma_x^2 \left[1 - \frac{1}{n} - \frac{2\rho[n(1-\rho) - (1-\rho^n)]}{n^2(1-\rho)^2} \right]$$

when $\rho(k) = \rho^k$. For modest size n this is essentially

$$E[v_x^2] \cong \sigma_x^2 \left[1 - \frac{1}{n} - \frac{2\rho}{n(1-\rho)} \right]$$

The following tables were worked out using the exact formula.

		$E[\nu_x^2]/\sigma_x^2$		
$\rho =$		0.0	0.3	0.6
$n =$	25	0.9600	0.9277	0.8520
	50	0.9800	0.9633	0.9230
	100	0.9900	0.9816	0.9608

		Bias = $E[v^2_X]/\sigma^2_X$			
		$\rho =$	0.0	0.3	0.6
$n =$	25		-0.04	-0.0723	-0.01480
	50		-0.02	-0.0367	-0.0770
	100		-0.01	-0.0184	-0.0393

For $r = 0.0$ and 0.3 , the bias is less than 10% of one standard error; for $r = 0.6$, the bias can equal or even exceed one standard error. Because the year-to-year correlation of annual and monthly flows in the same month is generally 0.4 or less, the variance is the dominant factor in determining the mean square error.

(c) For $\rho(k) = 0.5(0.9)^k$ one obtains

$$\begin{aligned}
& \left[1 - \frac{1}{n} - \frac{2}{n} \sum_{k=1}^{n-1} \left(1 - \frac{k}{n} \right) (0.5)(0.9)^k \right] \\
& = \left\{ 1 - \frac{1}{n} - \frac{\rho[n(1-0.9) - (1-0.9^n)]}{n^2(1-0.9)^2} \right\} = 0.7337 \quad \text{for } n = 25
\end{aligned}$$

Thus the estimate of the variance can have a substantial bias if a series' autocorrelations die very slowly.

Now for the variance of v_x^2 in this case.

$$\begin{aligned}
\text{Var}[v_x^2] &= \frac{2\sigma^4}{n} \left[1 + \sum_{k=1}^{\infty} \rho_x^2(k) \right] \\
&= \frac{2\sigma^4}{n} \left\{ 1 + \frac{2[(0.5)(0.9)]^2}{[1 - (0.9)^2]} \right\} \\
&= \frac{2\sigma^4}{n} (3.13) = \frac{(2.5)^2 \sigma^4}{n}
\end{aligned}$$

so that the coefficient of variation of v_x^2 is $2.5/n^{1/2} = 0.50$ for $n = 25$. Thus the bias $(1 - 0.7337)$ to standard error ratio is 53%.

Note for this series $\rho(1) = 0.45$, which is high for most hydrologic series.

6.34 Consider the crop irrigation problem in Exercise 4.31. For the given prices 30 and 25 for crop A and B, the demand for each crop varies over time. Records of demands show for crop A the demand ranges from 0 to 10 uniformly. There is an equal probability of that the demand will be any value between 0 and 10. For crop B the demand ranges from 5 units to 15 units, and the most likely demand is 10. At least 5 units and no more than 15 units of crop B will be demanded. The demand for crop B can be defined by a triangular density function, beginning with 5, having a mean of 10 and an upper limit of 15. Develop and solve a model for finding the maximum expected net revenue from both crops, assuming the costs of additional resources are 2/unit of water, 4/unit of land, 8/unit of fertilizer, and 5/unit of labor. The cost of borrowed money, i.e., the borrowing interest rate, is 8 percent per growing season. How does the solution change if the resource costs are $1/10^{\text{th}}$ of those specified above?

Let:

X_i	= quantity of crop i ; $i = A, B$
P_i	= price/unit of crop i
Pr_{ij}	= probability of each demanded crop i at integer segment j ; $j = 1, 2, 3, 4, \text{ and } 5$.
r	= interest rate = 0.08 per growing season
W, L, F, H	= amounts of water, land, fertilizer and labor purchased.

Lingo model:

```
MAX = revenue - cost;

revenue      = revenueA + revenueB;
revenueA     = PA*(PXA1 + PXA2 + PXA3 + PXA4 + PXA5 + PXA6 +
                PXA7 + PXA8 + PXA9 + PXA10);
revenueB     = PB*(PXB0 + PXB1 + PXB2 + PXB3 + PXB4 + PXB5 +
                PXB6 + PXB7 + PXB8 + PXB9 + PXB10);
cost         = (2*W + 4*L + 8*F + 5*H)*(1+r);

PXA1 = XA1*0.95 ; ! 0.95 = 0.5*0.1 + 0.9 = (1 - 0.5*0.1);
PXA2 = XA2*0.85 ; ! 0.85 = 1 - 0.1 - 0.5*0.1) ;
PXA3 = XA3*0.75 ;
PXA4 = XA4*0.65 ;
PXA5 = XA5*0.55 ;
PXA6 = XA6*0.45 ;
PXA7 = XA7*0.35 ;
PXA8 = XA8*0.25 ;
PXA9 = XA9*0.15 ;
PXA10 = XA10*0.05 ;

XA      = XA1 + XA2 + XA3 + XA4 + XA5 + XA6 + XA7 + XA8 + XA9
+ XA10 ;
XA1 < 1 ; XA2 < 1 ; XA3 < 1 ; XA4 < 1 ; XA5 < 1 ; XA6 < 1 ;
XA7 < 1 ; XA8 < 1 ; XA9 < 1 ; XA10 < 1 ;

PXB0 = XB0*1.0 ;
PXB1 = XB1*(1 - 0.5*PRB1) ;
PXB2 = XB2*(1 - PRB1 - 0.5*PRB2) ;
PXB3 = XB3*(1 - PRB1 - PRB2 - 0.5*PRB3) ;
PXB4 = XB4*(1 - PRB1 - PRB2 - PRB3 - 0.5*PRB4) ;
PXB5 = XB5*(1 - PRB1 - PRB2 - PRB3 - PRB4 - 0.5*PRB5) ;
PXB6 = XA6*(0.5 - 0.5*PRB5) ;
PXB7 = XA7*(0.5 - PRB5 - 0.5*PRB4) ;
PXB8 = XA8*(0.5 - PRB5 - PRB4 - 0.5*PRB3) ;
PXB9 = XA9*(0.5 - PRB5 - PRB4 - PRB3 - 0.5*PRB2) ;
PXB10 = XA10*(0.5 - PRB5 - PRB4 - PRB3 - PRB2 - 0.5*PRB1) ;

XB      = XB0 + XB1 + XB2 + XB3 + XB4 + XB5 + XB6 + XB7 + XB8
+ XB9 + XB10 ;
XB0 < 5 ; XB1 < 1 ; XB2 < 1 ; XB3 < 1 ; XB4 < 1 ; XB5 < 1 ;
XB6 < 1 ; XB7 < 1 ; XB8 < 1 ; XB9 < 1 ; XB10 < 1 ;

2*XA + 3*XB < 60 + W;          !water constraint;
5*XA + 2*XB < 80 + L;          !land constraint;
3*XA + 2*XB < 60 + F;          !fertilizer constraint;
1*XA + 2*XB < 40 + H;          !labor constraint;

DATA:
    PA = 30; PB = 25; r = 0.08;    PRB1 = 0.02; PRB2 =
    0.06; PRB3 = 0.1; PRB4 = 0.14; PRB5 = 0.18;  ENDDATA
```

6.35 In Section 9.2 generated synthetic streamflows sequences were used to simulate a reservoir's operation. In the example, a Thomas-Fiering model was used to generate $\ln Q_{1y}$ and $\ln Q_{2y}$, the logarithms of the flows in the two seasons of each year y , so as to preserve the season-to-season correlation of the untransformed flows. Noting that the annual flow is the sum of the untransformed seasonal flows Q_{1y} and Q_{2y} , calculate the correlation of annual flows produced by this model. The required data are given in Table 6.13. (*Hint*: You need to first calculate the covariance of $\ln Q_{1y}$ and $\ln Q_{1,y+1}$ and then of Q_{1y} and $Q_{2,y+1}$).

To calculate the correlation of the annual flows equal to $Q_{1y} + Q_{2y}$, where Q_{1y} and Q_{2y} are the seasonal flows in year y , consider

$$\begin{aligned}
 & Cov[Q_{\text{annual}, y+1}, Q_{\text{annual}, y}] \\
 &= E[(Q_{1,y+1} + Q_{2,y+1} - \mu_1 - \mu_2)(Q_{1y} + Q_{2y} - \mu_1 - \mu_2)] \\
 &= E\{[(Q_{1,y+1} - \mu_1) + (Q_{2,y+1} - \mu_2)][(Q_{1y} - \mu_1) + (Q_{2y} - \mu_2)]\} \\
 &= E[(Q_{1,y+1} - \mu_1)(Q_{1y} - \mu_1)] + E[(Q_{2,y+1} - \mu_2)(Q_{2y} - \mu_2)] \\
 &\quad + E[(Q_{1,y+1} - \mu_1)(Q_{2y} - \mu_2)] + E[(Q_{2,y+1} - \mu_2)(Q_{1y} - \mu_1)]
 \end{aligned}$$

where μ_1 and μ_2 are the means of the seasonal flows. From Table 6.14:

mean of Q_{1y} is	4.0	
mean of Q_{2y} is	2.5	
standard deviation of Q_{1y} is		1.5
standard deviation of Q_{2y} is		1.0
standard deviation of the annual flows is	2.3	
correlation Q_{1y} with Q_{2y} is		0.65
correlation Q_{2y} with $Q_{1,y+1}$ is		0.60

The corresponding correlations for $X_{1y} = \ln Q_{1y}$ and $X_{2y} = \ln Q_{2y}$ are obtained by inverting the equation for the correlation between Q_{1y} and Q_{2y} (Equation 6.160):

$$\text{corr}(Q_{1y}, Q_{2y}) = \frac{\exp[\text{cov}(X_{1y}, X_{2y})] - 1}{[\exp(\sigma_1^2) - 1]^{1/2} [\exp(\sigma_2^2) - 1]^{1/2}}$$

where σ_1^2 and σ_2^2 are the variances of X_{1y} and X_{2y} .

$$\sigma_1^2 = \ln\left[1 + \left(\frac{\sigma_{Q_1}}{\mu_{Q_1}}\right)^2\right] = \ln\left[1 + \left(\frac{1.5}{4.0}\right)^2\right] = 0.132$$

$$\sigma_2^2 = \ln\left[1 + \left(\frac{\sigma_{Q_2}}{\mu_{Q_2}}\right)^2\right] = 0.148$$

Note $[\exp(\sigma_i^2) - 1]^{1/2} = \sigma_{Q_i} / \mu_{Q_i} = \text{coefficient of variation of } Q_i.$

Now:

$$\begin{aligned}
 \text{Cov}(X_{1y}, X_{2y}) &= \ln\{1 + \text{corr}(Q_{1y}, Q_{2y})[\exp(\sigma_1^2) - 1]^{1/2}[\exp(\sigma_2^2) - 1]^{1/2}\} \\
 &= \ln[1 + \text{corr}(Q_{1y}, Q_{2y})\left(\frac{\sigma_{Q_1}}{\mu_{Q_1}}\right)\left(\frac{\sigma_{Q_2}}{\mu_{Q_2}}\right)] \\
 &= \ln[1 + 0.65\left(\frac{1.5}{4.0}\right)\left(\frac{1.0}{2.5}\right)] \\
 &= 0.0930
 \end{aligned}$$

$$\text{Hence } \text{Corr}(X_{1y}, X_{2y}) = \frac{0.093}{[0.132]^{1/2}[0.148]^{1/2}} = 0.666$$

For X_{2y} and the following variate $X_{1,y+1}$,

$$\begin{aligned}
 \text{Cov}(X_{2y}, X_{1,y+1}) &= \ln[1 + 0.60\left(\frac{1.0}{2.5}\right)\left(\frac{1.5}{4.0}\right)] \\
 &= 0.0862
 \end{aligned}$$

$$\text{Corr}(X_{2y}, X_{1,y+1}) = \frac{0.0862}{[0.132]^{1/2}[0.148]^{1/2}} = 0.617$$

A Thomas-Fiering model preserves the season-to-season correlations of the X_{iy} 's via the relationship:

$$\left(\frac{X_{i+1,y} - \mu_{i+1}}{\sigma_{i+1}}\right) = \rho_i + 1, i\left(\frac{X_{iy} - \mu_i}{\sigma_i}\right) + v_{iy}$$

Thus the given correlations

$$\text{Corr}(X_{1y}, X_{2y}) = 0.666$$

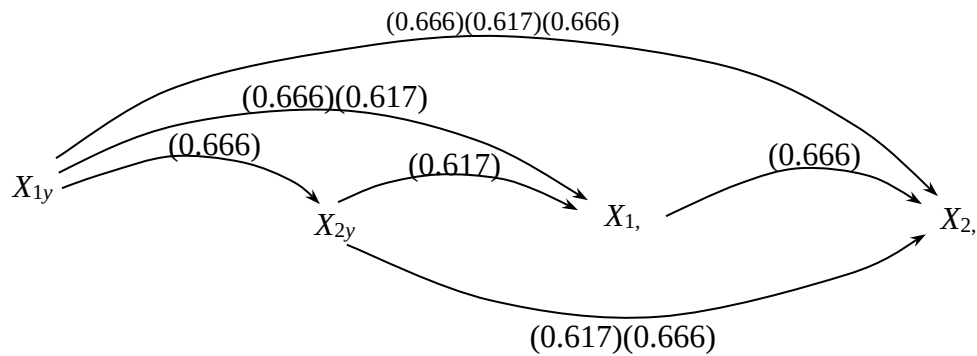
$$\text{Corr}(X_{2y}, X_{1,y+1}) = 0.617$$

then for the generated flows (see following diagrams)

$$\begin{aligned}
 \text{Corr}(X_{1y}, X_{1,y+1}) &= \text{Corr}(X_{1y}, X_{2y}) \text{Corr}(X_{2y}, X_{1,y+1}) \\
 &= (0.666)(0.617) = 0.411
 \end{aligned}$$

$$\begin{aligned}
 \text{Corr}(X_{1y}, X_{2,y+1}) &= \text{Corr}(X_{1y}, X_{1,y+1}) \text{Corr}(X_{1,y+1}, X_{2,y+1}) \\
 &= (0.411)(0.666) = 0.274
 \end{aligned}$$

$$\begin{aligned}
 \text{Corr}(X_{2y}, X_{2,y+1}) &= \text{Corr}(X_{2y}, X_{1,y+1}) \text{Corr}(X_{1,y+1}, X_{2,y+1}) \\
 &= (0.617)(0.666) = 0.411
 \end{aligned}$$



Correlation between X-variables

Now to get the required covariances of the Q_{iy} 's, we are given the values for

$$E[(Q_{2y} - \mu_2)(Q_{1,y+1} - \mu_2)] = (0.60)(1.5)(1.0) = 0.900$$

Now for the others

$$E[(Q_{1y} - \mu_1)(Q_{1,y+1} - \mu_1)] = \left\{ \frac{\exp[(0.411)\sigma_1\sigma_1] - 1}{\left[\frac{\sigma_{Q_1}}{\mu_{Q_1}} \right] \left[\frac{\sigma_{Q_1}}{\mu_{Q_1}} \right]} \right\} \sigma_{Q_1} \sigma_{Q_1}$$

$$= (\mu_{Q_1})^2 \{ \exp[0.411\sigma_1^2] - 1 \}$$

$$= 0.892 \quad (\text{correlation } 0.396)$$

$$E[(Q_{2y} - \mu_2)(Q_{2,y+1} - \mu_2)] = (\mu_{2_1})^2 \{ \exp[0.411\sigma_2^2] - 1 \}$$

$$= 0.392 \quad (\text{correlation } 0.392)$$

$$E[(Q_{1y} - \mu_1)(Q_{2,y+1} - \mu_1)] = (4.0)(2.5) \{ \exp[0.274\sqrt{(0.132)(0.148)}] - 1 \}$$

$$= 0.390 \quad (\text{correlation } 0.260)$$

Finally

$$E[(Q_{1y} + Q_{2y} - \mu_1 - \mu_2)(Q_{1,y+1} + Q_{2,y+1} - \mu_1 - \mu_2)]$$

$$= 0.892 + 0.392 + 0.900 + 0.390$$

$$= 2.574$$

$$\text{Correlation of annual flows} = 2.574/(2.3)^2 = 0.49$$

Rounding will cause answers to vary. Rounding to 2 digits will result in very different answers due to instability of the nonlinear transformations.

6.36 Part of New York City's municipal water supply is drawn from three parallel reservoirs in the upper Delaware River basin. The covariance matrix and lag-1 covariance matrix, as defined in Equations 6.166 and 6.168, were estimated based on the 50-year flow record to be (in m³/sec):

$$S_0 = \begin{bmatrix} 20.002 & 21.436 & 6.618 \\ 21.436 & 25.141 & 6.978 \\ 6.618 & 6.978 & 2.505 \end{bmatrix} = [\text{Cov}(Q_y^i, Q_y^j)]$$

$$S_1 = \begin{bmatrix} 6.487 & 6.818 & 1.638 \\ 7.500 & 7.625 & 1.815 \\ 2.593 & 2.804 & 0.6753 \end{bmatrix} = [\text{Cov}(Q_{y+1}^i, Q_y^j)]$$

Other statistics of the annual flow are:

Site	Reservoir	Mean Flow	Standard Deviation	r_1
1	Pepacton	20.05	4.472	0.3243
2	Cannosville	23.19	5.014	0.3033
3	Neversink	7.12	1.583	0.2696

- (a) Using these data, determine the values of the **A** and **B** matrices of the lag 1 model defined by Equation 6.165. Assume that the flows are adequately modeled by a normal distribution. A lower triangular **B** matrix that satisfies $M = \mathbf{B}\mathbf{B}^T$ may be found by equating the elements of $\mathbf{B}\mathbf{B}^T$ to those of M as follows:

$$M_{11} = b_{11}^2 \rightarrow b_{11} = \sqrt{M_{11}}$$

$$M_{21} = b_{11}b_{21} \rightarrow b_{21} = \frac{M_{21}}{b_{11}} = \frac{M_{21}}{\sqrt{M_{11}}}$$

$$M_{31} = b_{11}b_{31} \rightarrow b_{31} = \frac{M_{31}}{b_{11}} = \frac{M_{31}}{\sqrt{M_{11}}}$$

$$M_{22} = b_{21}^2 + b_{22}^2 \rightarrow b_{22}^2 = \sqrt{M_{22} - b_{21}^2} = \sqrt{M_{22} - M_{21}^2 / M_{11}}$$

and so forth for M_{23} and M_{33} . Note that $b_{ij} = 0$ for $i < j$ and M must be symmetric because $\mathbf{B}\mathbf{B}^T$ is necessarily symmetric.

- (b) Determine $\mathbf{A}$ and $\mathbf{BB}^T$ for the Markov model which would preserve the variances and cross covariances of the flows at each site, but not necessarily the lag 1 cross covariances of the flows. Calculate the lag 1 cross covariances of flows generated with your calculated $\mathbf{A}$ matrix.
- (c) Assume that some model has been built to generate the total annual flow into the three reservoirs. Construct and calculate the parameters of a disaggregation model that, given the total annual inflow to all three reservoirs, will generate annual inflows into each of the reservoirs preserving the variances and cross covariances of the flows. [Hint: The necessary statistics of the total flows can be calculated from those of the individual flows.]

(a) Given the basic formulas

$$S_0 = E[Z_y Z_y^T]$$

$$S_1 = E[Z_{y+1} Z_y^T]$$

the $\mathbf{A}$ matrix of the multivariate lag-1 model is given by

$\mathbf{A} = \mathbf{S}_1 \mathbf{S}_0^{-1}$. One obtains

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} 6.5 & 6.8 & 1.6 \\ 7.5 & 7.6 & 1.8 \\ 2.6 & 2.8 & 0.68 \end{bmatrix} \begin{bmatrix} 6.5 & 6.8 & 1.6 \\ 7.5 & 7.6 & 1.8 \\ 2.6 & 2.8 & 0.68 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 1.03625 & -0.14981 & -1.66649 \\ 1.441188 & -0.307827 & -2.22545 \\ 0.344365 & -0.019372 & -0.586242 \end{bmatrix} \end{aligned}$$

Now $\mathbf{BB}^T = \mathbf{S}_0 - \mathbf{A} \mathbf{S}_0 \mathbf{A}^T = \mathbf{S}_0 - \mathbf{S}_1 \mathbf{A}^T$

$$= \begin{bmatrix} 17.031 & 17.813 & 5.4764 \\ 17.813 & 20.718 & 5.6070 \\ 5.4764 & 5.6070 & 2.06227 \end{bmatrix}$$

To obtain a solution for $\mathbf{B}$, let $\mathbf{B}$ be lower triangular:

$$\mathbf{B} = \begin{bmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$

Then

$$BB^T = \begin{bmatrix} b_{11}^2 & b_{21}b_{11} & b_{31}b_{11} \\ b_{21}b_{11} & b_{21}^2 + b_{22}^2 & b_{31}b_{21} + b_{32}b_{22} \\ b_{31}b_{11} & b_{31}b_{21} + b_{32}b_{22} & b_{31}^2 + b_{32}^2 + b_{33}^2 \end{bmatrix}$$

Solving for each coefficient in turn one obtains

$$b_{11} = \sqrt{17.031} = 4.1269$$

$$b_{21}b_{11} = 17.813 \Rightarrow b_{21} = \frac{17.031}{b_{11}} = 4.3207$$

$$b_{21}^2 + b_{22}^2 = 20.718 \Rightarrow b_{22} = \sqrt{20.718 - b_{21}^2} = 1.4317$$

$$b_{31}b_{11} = 5.4764 \Rightarrow b_{31} = \frac{5.4764}{b_{11}} = 1.3270$$

$$b_{31}b_{21} + b_{32}b_{22} = 5.6070 \Rightarrow b_{32} = \frac{5.6070 - b_{31}b_{21}}{b_{22}} = -0.08852$$

$$b_{31}^2 + b_{32}^2 + b_{33}^2 = 2.06227 \Rightarrow b_{33} = \sqrt{2.06227 - b_{31}^2 - b_{32}^2} = 0.54170$$

hence

$$B = \begin{bmatrix} 4.1269 & 0 & 0 \\ 4.3207 & 1.4317 & 0 \\ 1.3270 & -0.08852 & 0.54170 \end{bmatrix}$$

(b) For the multivariate Markov model, the A matrix is simply

$$A = \begin{bmatrix} 0.3243 & 0 & 0 \\ 0 & 0.3033 & 0 \\ 0 & 0 & 0.54170 \end{bmatrix}$$

and

$$BB^T = S_0 - A S_0 A^T = \begin{bmatrix} 17.898 & 19.326 & 6.0394 \\ 19.326 & 22.828 & 6.4074 \\ 6.0394 & 6.4074 & 2.3229 \end{bmatrix}$$

The lag-1 covariance matrix of flows generated with this model is obtained from $X_{t+1} = AX_t + BV_t$ so that

$$E[X_{t+1}X_t^T] = E[AX_tX_t^T + BV_tX_t^T] = AS_0 = \begin{bmatrix} 6.487 & 6.952 & 2.146 \\ 6.502 & 7.625 & 2.116 \\ 1.784 & 1.881 & 0.6753 \end{bmatrix}$$

Note the diagonal elements of this matrix equal those of S_1 while the off-diagonal elements differ.

(c) To calculate the variance of the total flows, remember that total flow $Z = (1,$

$1, 1)X$ so that $E[ZZ^T] = S_{ZZ} = \sigma_Z^2 = (1,1,1) = S_0 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 117.712$. A disaggregation

model can be used to obtain annual flows at all three sites X given the aggregate inflow Z .

The model is

$$X = AZ + BV$$

The matrix A is 3×1 and satisfies

$$A = S_{XZ} S_{ZZ}^{-1} = \frac{1}{\sigma_Z^2} S_{XZ} = \frac{1}{\sigma^2} S_0 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{bmatrix} 0.40825 \\ 0.45496 \\ 0.13678 \end{bmatrix}$$

The B matrix satisfies

$$BB^T = S_{XX} A S_{ZZ} A^T = S_{XX} - S_{YZ} A^T = \begin{bmatrix} 0.38311 & -0.42786 & 0.044756 \\ -0.42786 & 0.77527 & -0.34741 \\ 0.044756 & -0.34741 & 0.30266 \end{bmatrix}$$

Proceeding as before to find a lower triangular B matrix

$$\begin{aligned} b_{11} &= \sqrt{0.38311} = 0.61896 \\ b_{21} &= \frac{-0.42786}{b_{11}} = -0.69126 \\ b_{22} &= \sqrt{0.77527 - b_{21}^2} = 0.54537 \\ b_{31} &= \frac{0.044756}{b_{11}} = 0.07231 \\ b_{32} &= \frac{-0.34741 - b_{31}b_{21}}{b_{22}} = -0.54536 \\ b_{33} &= \sqrt{0.30266 - b_{31}^2 - b_{32}^2} = 0 \end{aligned}$$

Note $b_{33} = 0$! The third component of V does not affect the values of X . Having obtained the flows at the first two sites, because the total flow at all three sites must sum to the aggregate flow, the flow at the third site is

$$X_3 = Z - X_1 - X_2$$

Rounding errors may make calculated values of $b_{33}^2 < 0$. This makes b_{33} imaginary. This should trouble students. The problem is that Z , X_1 , X_2 , and X_3 are linearly dependent!

6.37 Derive the variance of an ARMA (1, 1) process in terms of ϕ , θ , and σ^2 . [Hint: Multiply both sides of the equation to obtain a second. Be careful to remember which V_t 's are independent of which Z_t 's.]

Consider the ARMA(1, 1) process

$$Z_{t+1} = \phi Z_t + V_t - \theta V_{t-1} \quad (*)$$

Squaring both sides, one obtains

$$\sigma_Z^2 = \phi^2 \sigma_Z^2 + (1 + \theta^2) \sigma_a^2 - 2\phi\theta \text{Cov}(Z_t, V_{t-1})$$

One needs to determine $\text{Cov}(Z_t, V_{t-1})$. Multiply equation (*) by V_t to obtain

$$\text{Cov}(Z_t, V_{t-1}) = \sigma_a^2$$

Hence

$$(1 - \phi^2) \sigma_Z^2 = (1 + \theta^2) \sigma_a^2 - 2\phi\theta \sigma_a^2$$

or

$$\sigma_Z^2 = \left[\frac{1 - 2\phi\theta + \theta^2}{1 - \phi^2} \right] \sigma_a^2$$

6.38 The accompanying table presents a 60-year flow record for the normalized flows of the Gota River near Sjotop-Vannersburg in Sweden.

- Fit an autoregressive Markov model to the annual flow record.
- Using your model, generate a 50-year synthetic flow record. Demonstrate the mean, variance, and correlation of your generated flows deviate from the specified values no more than would be expected as a result of sampling error.
- Calculate the autocorrelations and partial autocovariances of the annual flows for a reasonable number of lags. Calculate the standard errors of the calculated values. Determine reasonable value of p and q for an ARMA (p , q) model of the flows. Determine the parameter values for the selected model.

**Annual Flows, Gota River
near Sjotop-Vannersburg, Sweden**

1898	1.158	1918	0.948	1938	0.892
1899	1.267	1919	0.907	1939	1.020
1900	1.013	1920	0.991	1940	0.869
1901	0.935	1921	0.994	1941	0.772
1902	0.662	1922	0.701	1942	0.606
1903	0.950	1923	0.692	1943	0.739
1904	1.120	1924	1.086	1944	0.813
1905	0.880	1925	1.306	1945	1.173
1906	0.802	1926	0.895	1946	0.916
1907	0.856	1927	1.149	1947	0.880
1908	1.080	1928	1.297	1948	0.601
1909	0.959	1929	1.168	1949	0.720
1910	1.345	1930	1.218	1950	0.955
1911	1.153	1931	1.209	1951	1.186
1912	0.929	1932	0.974	1952	1.140
1913	1.158	1933	0.834	1953	0.992
1914	0.957	1934	0.638	1954	1.048
1915	0.705	1935	0.991	1955	1.123
1916	0.905	1936	1.198	1956	0.774
1917	1.000	1937	1.091	1957	0.769

Source: V. M. Yevdjovich, Fluctuations of Wet and Dry Years,
Part I, Hydrology Paper No. 1, Colorado State University, Fort
Collins, Colo., 1963

- (d) Using the estimated model in (c), generate a 50-year synthetic streamflow record and demonstrate that the mean, variance, and show that first autocorrelations of the synthetic flows deviate from the modeled values by no more than would be expected as a result of sampling error.

(a) A check of the observed annual flows demonstrated that they were well modeled by a normal distribution. To fit a Markov model to the series, one must determine the mean, variance and auto-correlation of the flows. We have

$$\bar{q} = 0.9680$$

$$v_q^2 = \frac{1}{n} \sum (q_t - \bar{q})^2 = (0.1860)^2$$

(dividing by $n - 1$ yields $s_q = 0.18760$)

$$r_1 = \frac{\sum_{t=1}^{n-1} (q_{t+1} - \bar{q})(q_t - \bar{q})}{\sum_{t=1}^n (q_t - \bar{q})^2} = 0.404$$

Thus an autoregressive Markov model for the flow is

$$X_{t+1} = 0.96850 + 0.404(X_t - 0.96850) + 0.1860\sqrt{1 - (0.404)^2} E_t$$

where the E_t 's are $N(0, 1)$ independent random variables. Note that $0.1860[1 - (0.404)^2]^{0.50} = 0.170$.

(b) For a 50-year sequence, the standard deviation of $\bar{x}$, s_x^2 and r_1 are, approximately:

$$S.D.(\bar{x}) \cong \sigma_x \left\{ \frac{1}{n} \left[1 + \frac{2\rho}{(1-\rho)} \right] \right\}^{1/2} = \sigma_x (0.217) = 0.0404$$

$$S.D.(s_x^2) \cong \sigma_x^2 \left[\frac{2(1+\rho^2)}{n(1-\rho)^2} \right]^{1/2} = 0.236\sigma_x^2 = 0.00816$$

$$S.D.(r_1) \cong \left[\frac{1+\rho_1^2}{n} \right]^{1/2} = 0.129$$

It would not be unreasonable for the generated values to be any-where within the true values plus or minus 2 standard deviations (this is a 95% confidence interval so that 1 estimate in 20 on average will fall outside this range.

(c) This problem is best done by use of available computer programs which minimize the unconditional likelihood function series. A good routine is contained in the MINITAB package

The following table gives the estimates of the autocorrelations of the series. The first correlation clearly differs from zero so its standard error is estimated as $[(1-r_1^2)/n]^{1/2}$. The other autocorrelations cannot be distinguished from zero so their standard errors are estimated as if $\rho(1) \neq 0$ while $\rho(k) = 0$ for $k \geq 2$.

k	Autocorrelations of series r_k	Estimate of Standard Error of r_k
1	0.404	$0.118 = \left[\frac{1 - (0.404)^2}{n} \right]^{1/2}$
2	-0.054	$0.149 = \left[\frac{(1 + 2r_1^2)}{n} \right]^{1/2}$
3	-0.098	0.149
4	-0.770	0.149
5	-0.129	0.149
6	-0.028	0.149
7	0.036	0.149
8	0.041	0.149
9	-0.046	0.149
10	-0.055	0.149

k	Partial Autocorrelations $\hat{\phi}_{kk}$	Approximate Standard Error $n^{-1/2}$
1	0.404	0.129
2	-0.259	0.129
3	0.043	0.129
4	-0.076	0.129
5	-0.107	0.129
6	0.082	0.129
7	-0.032	0.129
8	0.030	0.129
9	-0.092	0.129
10	0.002	0.129

Because $r_1 = \hat{\phi}_{11}$ are significant, an AR(1) or ARMA(0, 1) model may be appropriate. Because $\hat{\phi}_{22}$ is also significant, an AR(2) model may be still better. The rapid decrease in the absolute value of the r_k 's rules out an ARMA(1, 1) model of the type often used for series with slowly dying autocorrelation functions.

The following two tables give the values of $\hat{\sigma}_v^2$ and the information criterion $n \ln \hat{\sigma}_v^2 + 2(p + q)$. The specified models were fit using the unconditional maximum likelihood procedure implemented in the MINITAB statistical package. Note, in using the MINITAB package one SHOULD NOT USE the mean square error with backforecasts excluded as an estimate of $\hat{\sigma}_v^2$. Rather, the total SS divided by n is the correct maximum likelihood estimate of $\hat{\sigma}_v^2$.

Values of σ_v^2

	$p =$	0	1	2
$q = 0$		0.03460	0.02876	0.02689
1		0.02725	0.02717	0.02686
2		0.02713	0.02644	0.02601
3		0.02655	0.02588	0.02301

Information Criteria

	$p =$	0	1	2
$q = 0$		-201.83	-210.92	-212.95
1		-214.17	-212.33 ^a	-211.03 ^b
2		-212.42 ^c	-211.98 ^d	-210.96 ^e
3		-211.71 ^f	-211.26 ^g	-216.31 ^h

- a) ϕ_1 parameter obviously not significant ($|t| = 0.33$)
- b) all three parameters not significantly different from zero ($|t| \leq 0.83$)
- c) θ_2 parameter obviously not significant ($|t| \leq 0.51$)
- d) all parameters appear significant ($|t| \geq 2.56$)
- e) ϕ_2 not significant ($|t| < 0.84$) indicating ARMA (1, 2) model preferable, for θ_2 , $|t| = 1.52$
- f) θ_2 and θ_3 not significant ($|t| \leq 1.28$)
- g) θ_1 and θ_3 not significant ($|t| = 1.60$ and 1.06)
- h) ϕ_1 not significant ($|t| = 0.42$) while for $\phi_2 = 0.89$, $|t| = 6.21$.

Minimization of the information criterion indicates that the ARMA(0, 1) model is probably most appropriate. The ARMA(2, 3) model actually has a smaller information criterion value but ϕ_1 is not significant so that this model does not seem reasonable. The AR(2) model's two parameters were both significant and that model yields a smaller value of σ_v^2 than the ARMA(0, 1) model. Still the information criterion indicated that ARMA (0, 1) is a better choice. However, a water resources analyst might want to override this choice if they felt greater persistence exhibited by the AR(2) model was more appropriate for a streamflow model.

The maximum likelihood estimate of θ_1 using the unconditional likelihood function implemented in the MINITAB package is -0.492 . The moment estimate of θ is found by solving

$$0.404 = r_1 \cong \rho_1 = -\theta_1 / (1 + \theta_1^2)$$

which results in

$$\hat{\theta}_1 = -0.508$$

(d) This problem is similar to part (b). For an ARMA (0, 1) model with $\theta_1 = -0.492$

$$\theta_1 = -0.492$$

$$\rho_1 = \frac{0.492}{1 + 0.492^2} = 0.396$$

$$\rho_k = 0 \quad k > 1$$

$$S.D.(\bar{x}) = \sigma_x \left\{ \frac{1}{n} [1 + 2(0.396)] \right\}^{1/2} = 0.0352$$

$$S.D.(v_x^2) = \sigma_x^2 \left[\frac{2\{1 + 2(0.396)^2\}}{n} \right]^{1/2} = 0.007931$$

$$\begin{aligned} S.D.(r_1) &= \left\{ \frac{1}{n} [(\rho_1^2 + 2\rho_1^4) + (1 + \rho_1^2 - 4\rho_1^2 + 2\rho_1^2) + (\rho_1^2 - 4\rho_1^2 + 2\rho_1^4)] \right\}^{1/2} \\ &= \left\{ \frac{1}{n} [(1 - 3\rho_1^2 + 4\rho_1^4)] \right\}^{1/2} = 0.112 \end{aligned}$$

6.39 (a) Assume that one wanted to preserve the covariance matrices $\mathbf{S}_0$ and $\mathbf{S}_1$ of the flows at several site $\mathbf{Z}_y$ by using the multivariate or vector ARMA (0, 1) model

$$\mathbf{Z}_{y+1} = \mathbf{A}\mathbf{V}_y - \mathbf{B}\mathbf{V}_{y-1}$$

where $\mathbf{V}_y$ contains n independent standard normal random variables. What is the relationship between the values of $\mathbf{S}_0$ and $\mathbf{S}_1$ and the matrices $\mathbf{A}$ and $\mathbf{B}$?

(b) Derive estimates of the matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ of the multivariate AR(2) model

$$\mathbf{Z}_{y+1} = \mathbf{A}\mathbf{Z}_y + \mathbf{B}\mathbf{Z}_{y-1} + \mathbf{C}\mathbf{V}_y$$

using the covariance matrices $\mathbf{S}_0$, $\mathbf{S}_1$ and $\mathbf{S}_2$.

(a) Assume that the vector random variable $\underline{Z}_y$ satisfies an ARMA(0, 1) process

$$\underline{Z}_{y+1} = A\underline{V}_y = B\underline{V}_{y-1}$$

Then multiplying both sides by their transpose to obtain

$$S_0 = E[\underline{Z}_{y+1}\underline{Z}_{y+1}^T] = E[(A\underline{V}_y - B\underline{V}_{y-1})(A\underline{V}_y - B\underline{V}_{y-1})^T] = AA^T + BB^T$$

where $E[\underline{V}_y\underline{V}_{y-1}^T] = E[\underline{V}_{y-1}\underline{V}_y^T] = 0$. Multiplying the equation for $\underline{Z}_{y+1}$ by $\underline{Z}_y^T$ and taking expectations yields

$$S_1 = E[\underline{Z}_{y+1}\underline{Z}_{y+1}^T] = E[(A\underline{V}_y - B\underline{V}_{y-1})(A\underline{V}_y - B\underline{V}_{y-1})^T] = -BA^T$$

These are the two equations relating the matrices A and B with S_0 and S_1 . For given S_0 and S_1 , there may be no solution for A and B . For example, in the univariate case when $S_0 = 1$ and $S_1 > 0.50$ there is no real solution. That is

$$S_0 = 1 = a^2 + b^2$$

and

$$S_1 = \rho = -ab > 0.50$$

does not have real roots. This can be proved by noting if

$$-ab > 0.50$$

then

$$-2ab > 1.0$$

Hence $a^2 + b^2 - (-2ab) = (a + b)^2$ must be less than $1 - (2)(0.50) = 0$.

(b) The solution can be obtained by considering the model

$$\underline{Z}_{y+1} = (AB) \begin{bmatrix} \underline{Z}_y \\ \underline{Z}_{y-1} \end{bmatrix} + C\underline{V}_y$$

then

$$[\overline{Z}_{y+1}(\overline{Z}_y^T, \overline{Z}_{y-1}^T)] = (A, B)E\left[\begin{pmatrix} \overline{Z}_y \\ \overline{Z}_{y-1} \end{pmatrix}(\overline{Z}_y^T, \overline{Z}_{y-1}^T)]$$

or

$$(S_1 S_2) = (AB) \begin{bmatrix} S_0 & S_1 \\ S_1^T & S_0 \end{bmatrix}$$

which becomes

$$S_1 = AS_0 + BS_1^T$$

$$S_2 = AS_1 + BS_0$$

solving the first equation for A yields

$$A = S_1 S_0^{-1} - BS_1^T S_0^{-1}$$

substituting into the second equation yields

$$S_2 = S_1 S_0^{-1} S_1 - BS_1^T S_0^{-1} S_1 + BS_0$$

so that

$$B = (S_2 - S_1 S_0^{-1} S_1)(S_0 - S_1^T S_0^{-1} S_1)^{-1}$$

To find C, multiply both sides of the equation defining the model by its own transpose to obtain

$$S_0 = (AB) \begin{bmatrix} S_0 & S_1 \\ S_1^T & S_0 \end{bmatrix} \begin{bmatrix} A^T \\ B^T \end{bmatrix} + CC^T$$

Noting that $(AB) \begin{bmatrix} S_0 & S_1 \\ S_1^T & S_0 \end{bmatrix}$ equals $(S_1 S_2)$, one obtains

$$\begin{aligned} CC^T &= S_0 - (S_1 S_2) \begin{bmatrix} A^T \\ B^T \end{bmatrix} \\ &= S_0 - S_1 A^T - S_2 B^T \end{aligned}$$

6.40 Create a model for the generation of monthly flows. The generated monthly flows should have the same marginal distributions as were fitted to the observed flows of record and should reproduce (i) the month-to-month correlation of the flows, (ii) the month-to-season correlation between each monthly flow and the total flow the previous season, and (iii) the month-to-year correlation between each monthly flow and the total 12-month flow in the previous year. Show how to estimate the model's parameters. How many parameters does your model have? How are the values of the seasonal model? How do you think this model could be improved?

The suggested model may be one reasonable approach to reproducing the relationship or correlation between monthly flows, seasonal flows and annual flows. Let q_t be the flow in month $t = 1, \dots, 12$. If q_t is not normally distributed then perhaps $x_t = \ln(q_t - \tau_t)$ is where τ_t is a lower bound for the 3-parameter lognormal distribution. The proposed model may be written

$$x_{ty} = \beta_0^t + \beta_1^t x_{t-1,y} + \beta_2^t s_{t,y} + \beta_3^t a_{y-1} + v_{ty}$$

where

$\beta_0^t, \beta_1^t, \beta_2^t, \beta_3^t$ are parameters to be estimated
(one set required for each t)

$x_{t-1,y}$ is previous month's transformed flow

$s_{t,y}$ is previous season's flow; note for $t = 1, 2, 3$ the previous season is last season of the previous year

a_{y-1} is annual flow in year $y - 1$, and

v_{ty} is a zero-mean normally distributed random variable with unknown variance σ_t^2 .

Writing the model as

$$x_{ty} = (\beta_0^t, \beta_1^t, \beta_2^t, \beta_3^t) \begin{bmatrix} 1 \\ x_{t-1,y} \\ s_{t,y} \\ a_{y-1} \end{bmatrix} + v_{ty}$$

or more simply

$$Y_{ty} = B_t X_{ty} + v_{ty}$$

The least-squares estimators of B are

$$\hat{B}_t = \left(\sum_{y=1}^n Y_{ty} X_{ty}^T \right) \left(\sum_{y=1}^n X_{ty} X_{ty}^T \right)^{-1}$$

The moment estimates may be written

$$\hat{B}_t = S_{Y_t X_t} \cdot S_{Y_t X_t}^{-1}$$

A good estimator of the variance of v_{ty} in either case is

$$\sigma_t^2 = \frac{1}{n-4} \sum_{y=1}^n (Y_{ty} - \hat{B}_t X_{ty})^2$$

The model has $12 \cdot 5 = 60$ parameters. Forty-eight β parameters and 12 variances. Seasonal and annual values $s_{i,y}$ and a_y are obtained by summing the generated

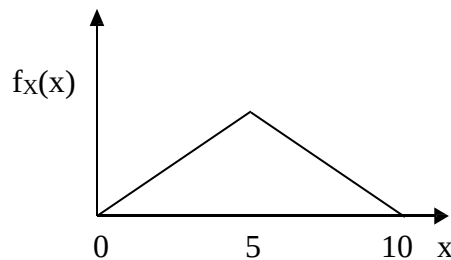
values q_{ty} of the corresponding flows. This may require untransforming the x_{ty} 's if $q_{ty} = \tau_t + \exp(x_{ty})$.

The corresponding generalization of the disaggregation model would be

$$(x_{1,y}, \dots, x_{12,y})^T = A \begin{pmatrix} a_{y-1} \\ x_{12,y-1} \end{pmatrix} + BV + \underline{C}$$

where $\underline{C}$ is a vector of 12 constants. The model has 12 constants in $\underline{C}$, 24 parameters in A and $12(12 + 1)/2 = 78$ parameters in V or 114 parameters. The proposed model by not reproducing the correlation between every month within the same water year by with almost one-half as many parameters as this generalization of the disaggregation approach.

6.41 Consider the symmetric triangular probability density function that ranges from 0 to 10 whose mean, median and most likely value is 5



- a) Generate values of x that come from this distribution.

To do this you need to:

Determine the equations of the cumulative distribution.

Generate uniformly distributed random values of probabilities p

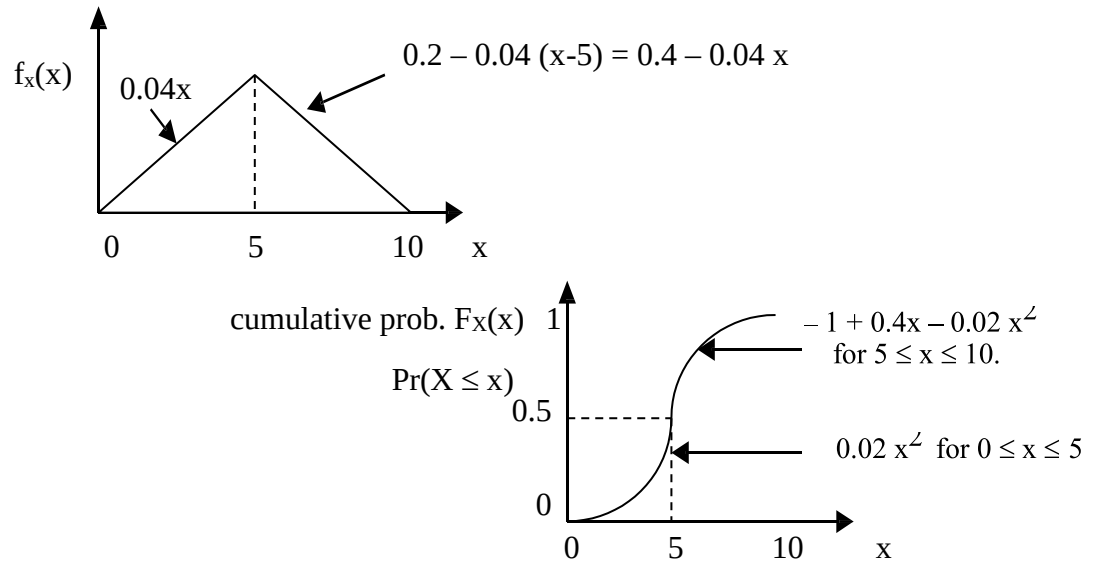
For each p find corresponding value of x , the inverse of the cumulative probability function $F_X(x)$ denoted as $F_X^{-1}(p)$

Using this inverse function, generate a series of 100 random variable values x that would have a probability distribution as shown above.

- b) Calculate the mean, variance and standard deviation of this distribution based on the random values you computed. What is the effect on these statistics of increasing the number of sample values of the random variable, say from 100 to 1000 to 9000?
- c) Calculate and compare with the true mean and variance.

- d) Next, suppose these random values of X are flows entering a reservoir having a capacity of 6. The purpose of the reservoir is to release a target flow of 5 in each time period. Simulate the operation of the reservoir assuming that if there is insufficient water to meet the target release of 5, release what is available, leaving an empty reservoir. Find the mean, variance and standard deviation of reservoir storage and release values.
- e) Finally assume the reservoir releases are to be allocated to three water users whose target allocations are 3, 2.33, and 8. Actual allocations should not exceed these target allocations. Make the allocations such that in each time period the maximum percentage deficit allocation is minimized. Find the mean, variance and standard deviation of each user's percentage deficit allocations.

Solution:



The cumulative distribution function is $0.02 x^2$ for $x = 0$ to 5 and $-1 + 0.4 x - 0.02 x^2$ for x from 5 to 10 .

Inverse functions: cp = cumulative probability ranging from 0 to 1 .

for $0 < cp < 0.5$ $x = (cp/0.02)^{0.5}$

for $0.5 < cp < 1.0$ $x = 10 - ((1-cp)/0.02)^{0.5}$ based on $cp = 1 - 0.02(10-x)^2$

Data:

$n = 100$;

enddata

Sets:

random/1..n/: x, p ;

endsets

$p(1) = .17$;

```

x(1) = @if (p(1) #le# .5, (p(1)/.02)^.5, 10 - ((1-p(1))/0.02)^0.5);
@for(random(t)| t #gt# 1 : p(t) = @rand(p(t-1));
x(t) = @if (p(t) #le# .5, (p(t)/.02)^.5, 10 - ((1-p(t))/0.02)^0.5); );
mean = @sum(random: x)/n ;
var = @sum(random: (x-mean)^2)/n ;
sd = var^0.5;

```

Example of Feasible solution found.

Total solver iterations: 0

Variable	Value
N	100.0000
MEAN	5.016163
VAR	3.318363
SD	1.821638
X(1)	2.915476
X(2)	4.761615
X(3)	6.113916
X(4)	1.555086
X(96)	2.668427
X(97)	6.531466
X(98)	7.781311
X(99)	6.490860
X(100)	6.281422
P(1)	0.1700000
P(2)	0.4534596
P(3)	0.6979671 etc.

Variable	Value
N	1000.000
MEAN	5.091060
VAR	3.991093
SD	1.997772

Variable	Value
N	9000.000
MEAN	5.011419
VAR	4.118654
SD	2.029447

True mean = $\int x \cdot 0.04x \, dx$ from 0 to 5 + $\int x(0.4 - 0.04x) \, dx$ from 5 to 10.
 = $[0.04X^3/3 | X=5] - 0 + [(0.4X^2/2 - 0.04X^3/3) | X=10] - [(0.4X^2/2 - 0.04X^3/3) | X=5]$
 = $(0.04/3)*(5)^3 + 0.2*(10)^2 - (0.04/3)*(10)^3 - 0.2*(5)^2 + (0.04/3)*(5)^3$;
 True Var = $\int (x - \text{mean})^2 \cdot 0.04x \, dx$ from 0 to 5 and $\int (x - \text{mean})^2 (0.4 - 0.04x) \, dx$ from 5 to 10.
 $.02*5*5*5*5 - (.4)*5*5*5 + 3*5*5 + (.8/3)*10*10*10 - 2.5*10*10 - .01*10*10*10*10 + 50$;
 SD = var^{0.5};

Variable	Value
MEAN	5.000000

VAR	4.166667
SD	2.041241

If X are flows entering a reservoir having a capacity of 6 and the purpose of the reservoir is to release a target flow of 5 in each time period, to simulate the operation of the reservoir assuming that if there is insufficient water to meet the target release of 5, release what is available, leaving an empty reservoir one can incorporate the following code in the model presented above. Then it will be easy to find the mean, variance and standard deviation of reservoir storage and release values.

```

Data: RTarget = 5; Kap = 6; enddata
Q(t) = X(t);
R(t) = @smin( RTarget , Q(t)+ S(t) ) ;
FS(t) = Q(t) - R(t) + S(t);
spill(t) = @smax( 0, FS(t) - Kap);
TR(t) = R(t) + spill(t); !Total release ;
S(t+1) = S(t) + q(t) - TR(t);

```

If the reservoir releases are to be allocated to three water users whose target allocations are 3, 2.33, and 8 the actual allocations should not exceed these target allocations and the allocations in each time period are to minimize the maximum percentage deficit allocation is minimized.

Percentage or fraction deficit will be the same for all users and equal to
 $F(t) = @Smax(0, (13.33 - TR(t)) / 13.33$.
 Therefore actual allocation for user = User target allocation times the fraction
 $(1 - F(t))$.
 So, if the reservoir release $TR(t)$ is 5, $F(t) = .625$
 And the three allocations are 1.125, 0.875, 3.0

Now it is easy to find the mean, variance and standard deviation of each user's percentage deficit allocations.

Chapter 7

7.1 Can you modify the deterministic discrete DP reservoir-operating model to include the uncertainty, expressed as P_{ij}^t , of the inflows, as in Exercise 6.25?

(Hints: The operating policy would define the release (or final storage) in each season as a function of not only the initial storage but also the inflow. If the inflow changes, so might the release or final storage volume. Hence you need to discretize the inflows as well as the storage volumes. Both storage and inflow are state variables. Assume for this model you can predict with certainty the inflow in each period at the beginning of the period. So, each node of the network represents a known initial storage and inflow value. You cannot predict with certainty the following period's flows, only their probabilities. What does the network look like now?

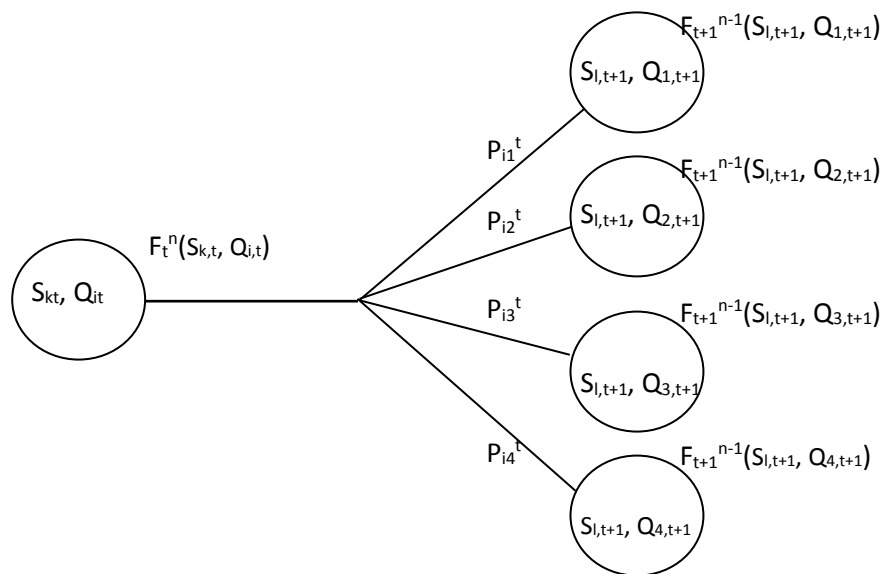
Using the final storage as the decision variable (from which the release can be computed):

$$F_t^n(S_{kt}, Q_{it}) = \max \text{ or } \min \{ \text{Obj}_t(S_{kt}, Q_{it}, S_{l,t+1}) + \sum_j P_{ij}^t F_{t+1}^{n-1}(S_{l,t+1}, Q_{j,t+1}) \}$$

$$S_{l,t+1} \leq K$$

$$S_{l,t+1} \leq S_{kt} + Q_{it}$$

Portion of network for a particular combination of initial and final storage volumes: The final storage volumes are the same, but the inflow, $Q_{j,t+1}$, in period $t+1$ changes. Each of those inflows has a conditional probability P_{ij}^t .



For each release link from a (S_{kt}, Q_{it}) node, a possible $F_t^n(S_{kt}, Q_{it})$ value = $P_{i1}^t F_{t+1}^{n-1}(S_{l,t+1}, Q_{1,t+1}) + \dots + P_{i4}^t F_{t+1}^{n-1}(S_{l,t+1}, Q_{4,t+1})$.

The F values are expected values of the objective being maximized or minimized associated with a season t and number of remaining periods n .

7.2 Assume that there exist two possible discrete flows Q_{it} into a small reservoir in each of two periods t each year having probabilities P_{it} . Find the steady-state operating policy (release as a function of initial reservoir volumes and current period's inflow) for the reservoir that minimizes the expected sum of squared deviations from storage and release targets. Limit the storage volumes to integer values that vary from 3 to 5. Assume a storage volume target of 4 and a release target of 2 in each period t . (Assume only integer values of all states and decision variables and that each period's inflow is known at the beginning of the period.) Find the annual expected sum of squared deviations from the storage and release targets.

Period, t	FLOWS, Q_{it}		PROBABILITIES, P_{it}	
	$i = 1$	$i = 2$	$i = 1$	$i = 2$
1	1	2	0.17	0.83
2	3	4	0.29	0.71

This is an application of Exercise 6.27 except the flow probabilities are independent of the previous flow.

First it is advantageous to calculate the loss functions for each season.

S_1, Q_1	R_1			S_2			ΣDev^2		
3, 1	0	1		4	3		5	2	
3, 2	0	1	2	5	4	3	5	2	1
4, 1	0	1	2	5	4	3	4	1	0
4, 2	1	2	3	5	4	3	1	0	1
5, 1	1	2	3	5	4	3	2	1	2
5, 2	2	3	4	5	4	3	1	2	5

S_2, Q_2	R_2			S_1			ΣDev^2		
3, 3	1	2	3	5	4	3	3	1	2
3, 4	2	3	4	5	4	3	1	2	5
4, 3	2	3	4	5	4	3	0	1	4
4, 4	3	4	5	5	4	3	1	4	9
5, 3	3	4	5	5	4	3	2	5	10
5, 4	4	5	6	5	4	3	5	10	17

Now, let $f_t^n(S_t, Q_t)$ = expected sum of squared deviations from storage and release targets in remaining n periods when start with storage S_t in period $t = 1, 2$; Let all $f_t^n(S_t, Q_t) = 0$ for $n = 0$.

Last stage: $n = 1, t = 2$

$$\begin{aligned} f_2^1(3,3) &= \min \{ 2 + 0, 1 + 0, 2 + 0 \} = 1 \text{ with } R_2 = 2. \\ f_2^1(3,4) &= \min \{ 1 + 0, 2 + 0, 5 + 0 \} = 1 \text{ with } R_2 = 2. \\ f_2^1(4,3) &= \min \{ 0 + 0, 1 + 0, 4 + 0 \} = 0 \text{ with } R_2 = 2. \\ f_2^1(4,4) &= \min \{ 1 + 0, 4 + 0, 9 + 0 \} = 1 \text{ with } R_2 = 3. \\ f_2^1(5,3) &= \min \{ 2 + 0, 5 + 0, 10 + 0 \} = 2 \text{ with } R_2 = 3. \\ f_2^1(5,4) &= \min \{ 5 + 0, 10 + 0, 17 + 0 \} = 5 \text{ with } R_2 = 4. \end{aligned}$$

Next to last stage: $n = 2, t = 1$

$$\begin{aligned} f_1^2(3,1) &= \min \{ 5 + 0(0.29) + 1(0.71), 2 + 1(0.29) + 1(0.71) \} = 3 \text{ with } R_1 = 1. \\ f_1^2(3,2) &= \min \{ 5 + 2(0.29) + 5(0.71), 2 + 0(0.29) + 1(0.71), \\ &\quad 1 + 1(0.29) + 1(0.71) \} = 2 \text{ with } R_1 = 2. \\ f_1^2(4,1) &= \min \{ 4 + 2(0.29) + 5(0.71), 1 + 0(0.29) + 1(0.71), \\ &\quad 0 + 1(0.29) + 1(0.71) \} = 1 \text{ with } R_1 = 2. \\ f_1^2(4,2) &= \min \{ 1 + 2(0.29) + 5(0.71), 0 + 0(0.29) + 1(0.71), \\ &\quad 1 + 1(0.29) + 1(0.71) \} = 0.71 \text{ with } R_1 = 2. \\ f_1^2(5,1) &= \min \{ 2 + 2(0.29) + 5(0.71), 1 + 0(0.29) + 1(0.71), \\ &\quad 2 + 1(0.29) + 1(0.71) \} = 1.71 \text{ with } R_1 = 2. \\ f_1^2(5,2) &= \min \{ 1 + 2(0.29) + 5(0.71), 2 + 0(0.29) + 1(0.71), \\ &\quad 5 + 1(0.29) + 1(0.71) \} = 2.71 \text{ with } R_1 = 3. \end{aligned}$$

Continuing... $n = 3, t = 2$

$$\begin{aligned} f_2^3(3,3) &= \min \{ 2 + 1.71(0.17) + 2.71(0.83), 1 + 1(0.17) + \\ &\quad 0.71(0.83), 2 + 3(0.17) + 2(0.83) \} = 1.76 \text{ with } R_2 = 2. \\ f_2^3(3,4) &= \min \{ 1 + 1.71(0.17) + 2.71(0.83), 2 + 1(0.17) + \\ &\quad 0.71(0.83), 5 + 3(0.17) + 2(0.83) \} = 2.76 \text{ with } R_2 = 3. \\ f_2^3(4,3) &= \min \{ 0 + 1.71(0.17) + 2.71(0.83), 1 + 1(0.17) + \\ &\quad 0.71(0.83), 4 + 3(0.17) + 2(0.83) \} = 1.76 \text{ with } R_2 = 3. \\ f_2^3(4,4) &= \min \{ 1 + 1.71(0.17) + 2.71(0.83), 4 + 1(0.17) + \\ &\quad 0.71(0.83), 9 + 3(0.17) + 2(0.83) \} = 3.54 \text{ with } R_2 = 3. \\ f_2^3(5,3) &= \min \{ 2 + 1.71(0.17) + 2.71(0.83), 5 + 1(0.17) + \\ &\quad 0.71(0.83), 10 + 3(0.17) + 2(0.83) \} = 4.54 \text{ with } R_2 = 3. \\ f_2^3(5,4) &= \min \{ 5 + 1.71(0.17) + 2.71(0.83), 10 + 1(0.17) + \\ &\quad 0.71(0.83), 17 + 3(0.17) + 2(0.83) \} = 7.54 \text{ with } R_2 = 4. \end{aligned}$$

Continuing... n = 4, t = 1

$$f_1^4(3,1) = \min \{ 5 + 1.76(0.29) + 3.54(0.71), 2 + 1.76(0.29) + 2.76(0.71) \} = 4.47 \text{ with } R_1 = 1.$$

$$f_1^4(3,2) = \min \{ 5 + 4.54(0.29) + 7.54(0.71), 2 + 1.76(0.29) + 3.54(0.71), 1 + 1.76(0.29) + 2.76(0.71) \} = 3.47 \text{ with } R_1 = 2.$$

$$f_1^4(4,1) = \min \{ 4 + 4.54(0.29) + 7.54(0.71), 1 + 1.76(0.29) + 3.54(0.71), 0 + 1.76(0.29) + 2.76(0.71) \} = 2.47 \text{ with } R_1 = 2.$$

$$f_1^4(4,2) = \min \{ 1 + 4.54(0.29) + 7.54(0.71), 0 + 1.76(0.29) + 3.54(0.71), 1 + 1.76(0.29) + 2.76(0.71) \} = 3.02 \text{ with } R_1 = 2.$$

$$f_1^4(5,1) = \min \{ 2 + 4.54(0.29) + 7.54(0.71), 1 + 1.76(0.29) + 3.54(0.71), 2 + 1.76(0.29) + 2.76(0.71) \} = 4.02 \text{ with } R_1 = 2.$$

$$f_1^4(5,2) = \min \{ 1 + 4.54(0.29) + 7.54(0.71), 2 + 1.76(0.29) + 3.54(0.71), 5 + 1.76(0.29) + 2.76(0.71) \} = 5.02 \text{ with } R_1 = 3.$$

Now this process should continue until $f_t^{n+2}(S_t, Q_t) - f_t^n(S_t, Q_t)$ is a constant representing annual expected squared deviations from targets. Based on results so far the policy appears to be

	Storage	Inflow	Release	Sq. Deviations
Period 1	3	1	1	2
	3	2	2	1
	4	1	2	0
	4	2	2	0
	5	1	2	1
	5	2	3	2
Period 2	3	3	2	1
	3	4	3	2
	4	3	3	1
	4	4	3	1
	5	3	3	2
	5	4	4	5

Note: Once this policy is implemented S_1 will never be 3 and S_2 will never be 5.

Given this policy the joint probabilities $P_t(S, Q)$ can be calculated:

$$P_2(3, 3) = [P_1(4,1)] 0.29 = 0.01625952$$

$$P_2(3, 4) = [P_1(4,1)] 0.71 = 0.03980780$$

$$P_2(4, 3) = [P_1(4,2) + P_1(5,1) + P_1(5,2)] 0.29 = 0.2737405$$

$$P_2(4, 4) = [P_1(4,2) + P_1(5,1) + P_1(5,2)] 0.71 = 0.6701922$$

$$P_1(4, 1) = [P_2(3,3) + P_2(3,4) + P_2(4,3)] 0.17 = 0.05606733$$

$$P_1(4, 2) = [P_2(3,3) + P_2(3,4) + P_2(4,3)] 0.83 = 0.2737405$$

$$P_1(5, 1) = [P_2(4,4)] 0.17 = 0.1139327$$

$$P_1(5, 2) = [P_2(4,4)] 0.83 = 0.5562595$$

$$P_1(4,1) + P_1(4,2) + P_1(5,1) + P_1(5,2) = 1$$

Summing the products of these joint probabilities times the sum of squared deviations yields 2.266260.

7.3 Develop a linear model for defining the optimal joint probabilities of predefined discrete initial storage volumes, discrete inflows, and discrete final storage volumes in a reservoir in each period t . Let values of the index k represent the different discrete initial storage volumes, S_{kt} . Similarly, let the index i represent the inflows, Q_{it} , and the index l represent the final storage volumes, $S_{l,t+1}$, in period t . Let the index j represent the discrete inflows, $Q_{j,t+1}$, and m represent the discrete final storage volumes, $S_{m,t+2}$, in period $t+1$. Let PR_{kilt} be the unknown joint probability of a discrete initial storage, S_{kt} , an inflow, Q_{it} , and a final storage volume, $S_{l,t+1}$, in period t . It is also the probability of a release associated with a particular combination of k , i , and l in period t . The objective is to maximize the expected net benefits, however measured. The net benefits associated with any combination represented by k , i , and l in period t is B_{kilt} . These net benefits and the conditional inflow probabilities, $P_{ij}^t = \Pr\{Q_{j,t+1}|Q_{it}\}$, are known. Show how the optimal operating policy can be determined once the values of the joint probabilities, PR_{kilt} , are known.

The same policy can be found by DP. Develop a DP model to find the optimal operating policy.

Stochastic LP model:

$$\text{Maximize: } \sum_k \sum_i \sum_l \sum_t B_{kilt} PR_{kilt}$$

Subject to:

$$\sum_k \sum_i PR_{kilt} P_{ij}^t - \sum_m PR_{l,j,m,t+1} = 0 \quad \forall l, j, t \text{ (less one per } t)$$

$$l = l(k, i, t) \quad \text{(the sum is over only those } k \text{ and } i \text{ that are feasible for } l, \text{ i.e., } S_{l,t+1} \leq S_{kt} + Q_{it} \text{ and } S_{l,t+1} \leq \text{capacity})$$

$$\sum_k \sum_i \sum_l PR_{kilt} = 1 \quad \forall t$$

The operating policy can be defined by the conditional probabilities of l given k and i in each period t . These conditional probabilities are: $P\{l|kit\} = PR_{kilt} / \sum_l PR_{kilt}$ for those that are defined. Those that are not indicate the combinations of k and i that will not occur if the policy is followed.

Stochastic DP

$$f_t^n(k, i) = \max_l \{B_{kilt} + \sum_j P_{ij}^t f_{t+1}^{n-1}(l, j)\}$$

where l must be feasible for k and i (i.e., $S_{l,t+1} \leq S_{kt} + Q_{it}$ and $S_{l,t+1} \leq \text{capacity}$)

7.4 Referring to Exercise 7.3, instead of defining a final volume subscript l and m for computing joint probabilities PR_{kilt} , assume that subscripts d and e were used to denote different reservoir release volumes. How would the linear programming model developed be altered to include d and e in place of l and m ? How would the dynamic programming recursion equation be altered?

Stochastic LP

$$\text{Maximize: } \sum_k \sum_i \sum_d \sum_t B_{kidt} PR_{kidt}$$

Subject to:

$$\sum_k \sum_i \sum_d PR_{kidt} P_{ij}^t - \sum_e PR_{l,j,e,t+1} = 0 \quad \forall l, j, t \text{ (less one per } t)$$

$$l = l(k, i, d, t)$$

(selective summation over k, i , and d that yield l in period t)

$$\sum_k \sum_i \sum_d PR_{kidt} = 1$$

$\forall t$

Stochastic DP

$$f_t^n(k, i) = \max_d \{B_{kidt} + \sum_j P_{ij}^t f_{t+1}^{n-1}(l, j)\}$$

where d must be feasible for k and i (i.e., $R_{dt} \leq S_{kt} + Q_{it}$ and $R_{dt} \geq S_{kt} + Q_{it} - \text{capacity}$) and $l = l(k, i, d, t)$ or $S_{l,t+1} = S_{kt} + Q_{it} - R_{dt}$.

7.5 Given joint probabilities PR_{kilt} found from Exercise 7.3, how would one derive the probability distribution of reservoir releases and storage volumes in each period t ?

Find the sums of the joint probabilities PR_{kilt} that result in the same discrete release, $R_{dt} = S_{kt} + Q_{it} - S_{l,t+1}$, and sum PR_{kilt} over all i and l to get probability of a discrete storage k in period t .

7.6 Assume that the streamflow Q at a particular site has cumulative distribution function $F_Q(q) = q/(1 + q)$ for $q \geq 0$. The withdrawal x at that location must satisfy a chance constraint of the form $\Pr[x \geq Q] \leq 1 - \alpha$. Write the deterministic equivalent for each of the following chance constraints:

$$\begin{array}{ll} \Pr[x \leq Q] \geq 0.90 & \Pr[x \geq Q] \leq 0.80 \\ \Pr[x \leq Q] \leq 0.95 & \Pr[x \leq Q] \leq 0.10 \\ \Pr[x \geq Q] \geq 0.75 & \end{array}$$

This exercise involves the concept of chance-constraints and their deterministic equivalent. Given that $F_Q(q) = q/(1 + q)$ for $q > 0$, then $q(F) = F/(1 - F)$. This will define the value, $q^{(1-F)}$, of the random variable Q that is exceeded with probability F . Then for any deterministic decision variable x , the following are equivalent

$$\Pr[x \leq Q] \geq 0.90, \quad x \leq q^{(0.10)} = \frac{1}{9} = 0.111$$

$$\Pr[x \leq Q] \leq 0.95, \quad x \geq q^{(0.05)} = \frac{1}{19} = 0.0526$$

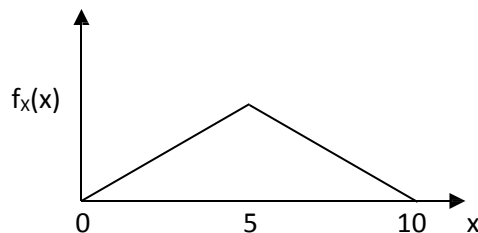
$$\Pr[x \geq Q] \geq 0.75, \quad x \geq q^{(0.75)} = 3$$

$$\Pr[x \geq Q] \leq 0.80, \quad x \leq q^{(0.80)} = 4$$

$$\Pr[x \leq Q] \leq 0.10, \quad x \geq q^{(0.90)} = 9$$

7.7 Monte Carlo Simulation:

Consider the symmetric triangular probability density function that ranges from 0 to 10 whose mean and most likely value is 5



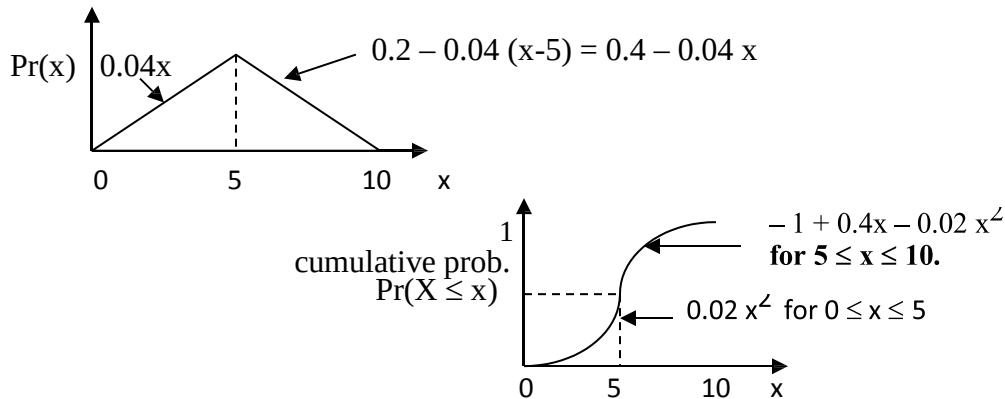
a) Generate values of x that come from this distribution.

To do this you need to:

- Determine the equations of the cumulative distribution.
- Generate uniformly distributed random values of probabilities p
- For each p find corresponding value of x . the inverse of the cumulative probability function $F_X(x)$ denoted as $F_X^{-1}(p)$

- Using this inverse function, generate a series of 100 random variable values x that would have a probability distribution as shown above.
- b) Calculate the mean, variance and standard deviation of this distribution based on the random values you computed. What is the effect on these statistics of increasing the number of sample values of the random variable, say from 100 to 1000 to 9000?
- c) Calculate and compare with the true mean and variance.

Solution:



The cumulative distribution function is $0.02 x^2$ for $x = 0$ to 5 and $-1 + 0.4 x - 0.02 x^2$ for x from 5 to 10 .

Inverse functions: cp = cumulative probability ranging from 0 to 1 .

for $0 < cp < 0.5$ $x = (cp/0.02)^{0.5}$

for $0.5 < cp < 1.0$ $x = 10 - ((1-cp(1))/0.02)^{0.5}$ based on $cp = 1 - 0.02(10-x)^2$

Data:

$n = 100$;

enddata

Sets:

random/1..n/: x, p ;

endsets

$p(1) = .17$;

$x(1) = @if(p(1) \#le\# .5, (p(1)/.02)^{.5}, 10 - ((1-p(1))/0.02)^{0.5})$;

@for(random(t) | t #gt# 1 : $p(t) = @rand(p(t-1))$;

$x(t) = @if(p(t) \#le\# .5, (p(t)/.02)^{.5}, 10 - ((1-p(t))/0.02)^{0.5})$;

mean = @sum(random: x)/ n ;

var = @sum(random: $(x-mean)^2$)/ n ;

sd = var^{0.5};

Example of Feasible solution found.

Total solver iterations: 0

```
Variable Value
N 100.0000
MEAN 5.016163
VAR 3.318363
SD 1.821638
X( 1) 2.915476
X( 2) 4.761615
X( 3) 6.113916
X( 4) 1.555086
.....
X( 96) 2.668427
X( 97) 6.531466
X( 98) 7.781311
X( 99) 6.490860
X(100) 6.281422
P( 1) 0.1700000
P( 2) 0.4534596
P( 3) 0.6979671 etc.
Variable Value
N 1000.000
MEAN 5.091060
VAR 3.991093
SD 1.997772
Variable Value
N 9000.000
MEAN 5.011419
VAR 4.118654
SD 2.029447
```

$$\begin{aligned}\text{True mean} &= \int_0^5 x \cdot 0.04x \, dx + \int_5^{10} x(0.4 - 0.04x) \, dx \\ &= \left[\frac{0.04x^3}{3} \right]_{x=0}^{x=5} - 0 + \left[\frac{0.4x^2}{2} - \frac{0.04x^3}{3} \right]_{x=5}^{x=10} - \left[\frac{0.4x^2}{2} - \frac{0.04x^3}{3} \right]_{x=5} \\ &= (0.04/3) \cdot (5)^3 + 0.2 \cdot (10)^2 - (0.04/3) \cdot (10)^3 - 0.2 \cdot (5)^2 + (0.04/3) \cdot (5)^3;\end{aligned}$$

$$\begin{aligned}\text{True Var} &= \int_0^5 (x - \text{mean})^2 \cdot 0.04x \, dx + \int_5^{10} (x - \text{mean})^2 (0.4 - 0.04x) \, dx \\ &= .02 \cdot 5^5 \cdot 5 - (.4) \cdot 5^5 \cdot 5 + 3 \cdot 5^5 + (.8/3) \cdot 10^4 \cdot 10 - 2.5 \cdot 10^4 \cdot 10 - .01 \cdot 10^4 \cdot 10^2 + 50; \\ \text{SD} &= \text{var}^{0.5};\end{aligned}$$

```
Variable Value
MEAN 5.000000
VAR 4.166667
SD 2.041241
```

- d) Next, suppose these random values of X are flows entering a reservoir having a capacity of 6. The purpose of the reservoir is to release a target flow of 5 in each time period. Simulate the operation of the reservoir assuming that if there is insufficient water to meet the target release of 5, release what is available, leaving an empty reservoir. Find the mean, variance and standard deviation of reservoir storage and release values.

Incorporate the following LINGO code into the flow generation model and then take statistics of storage (S) and release (R , TR , $Spill$) variables.

Data: $R_{Target} = 5$; $Kap = 6$; enddata

$Q(t) = X(t)$;

$R(t) = @smin(R_{Target} , Q(t) + S(t))$;
 $FS(t) = Q(t) - R(t) + S(t)$;
 $spill(t) = @smax(0, FS(t) - Kap)$;
 $TR(t) = R(t) + spill(t)$; !Total release ;
 $S(t+1) = S(t) + q(t) - TR(t)$;

- e) Finally assume the reservoir releases are to be allocated to three water users whose target allocations are 3, 2.33, and 8. Actual allocations should not exceed these target allocations. Make the allocations such that in each time period the maximum percentage deficit allocation is minimized. Find the mean, variance and standard deviation of each user's percentage deficit allocations.

Percentage or fraction deficit will be the same for all users and equal to

$F(t) = @Smax(0, (13.33 - TR(t)) / 13.33$.

Therefore actual allocation for user = User target allocation times the fraction $(1 - F(t))$.

So, if the reservoir release $TR(t)$ is 5, $F(t) = .625$

and the three allocations are 1.125, 0.875, 3.0

Chapter 8

8.1 Distinguish between sensitivity analysis and uncertainty analysis.

A Sensitivity Analysis is the study of how changes in the inputs of a mathematical model or system (numerical or otherwise) affect the values of the outputs. Wikipedia lists the following potential benefits of performing sensitivity analyses.

- Testing the robustness of the results of a model or system in the presence of uncertainty.
- Increased understanding of the relationships between input and output variables in a system or model.
- Uncertainty reduction: identifying model inputs that cause significant uncertainty in the output and should therefore be the focus of attention if the robustness is to be increased (perhaps by further research).
- Searching for errors in the model (by encountering unexpected relationships between inputs and outputs).
- Model simplification – fixing model inputs that have no effect on the output, or identifying and removing redundant parts of the model structure.
- Enhancing communication from modelers to decision makers (e.g. by making recommendations more credible, understandable, compelling or persuasive).
- Finding regions in the space of input factors for which the model output is either maximum or minimum or meets some optimum criterion.
- In case of calibrating models with large number of parameters, a primary sensitivity test can ease the calibration stage by focusing on the sensitive parameters. Not knowing the sensitivity of parameters can result in time being uselessly spent on non-sensitive ones.^[4]

An Uncertainty Analysis attempts to identify and quantify the uncertainty of the output variables of models used in decision making problems.

8.2 Consider the allocation model used in previous chapters involving three water consumers i . Allocations x_i of water can be made from a given total amount Q to the three consumers. The respective benefits are $(6x_1 - x_1^2)$, $(7x_2 - 1.5x_2^2)$ and $(8x_3 - 0.5x_3^2)$. Discuss possible sources of uncertainty in model structure and model output, and identify and display parameter sensitivity.

Uncertainty in model structure: For example the demand functions could be of the form $P_o/(1+SX)$, so the total revenue functions are $P_{oi} x_i / (1+s_i x_i)$. Hence if $Q = 6$, the vector $P_o = \{6, 7, 8\}$ and $S = \{1, 1.5, 0.5\}$ the allocations model and solution would differ from 1, 1, and 4.

Lingo model:

```

Sets:
site/1..3/: X,P,S;
endsets

max = @sum(site: P*x/(1+s*x) );
@sum(site: X ) < Q;

Data:
P = 6 7 8;
S = 1 1.5 0.5;
Q = 6;
enddata

```

Local optimal solution found.

Objective value:		16.58869
Infeasibilities:		0.000000

Variable	Value	Reduced Cost
Q	6.000000	0.000000
X(1)	1.398984	0.000000
X(2)	1.060799	0.000000
X(3)	3.540217	0.000000
P(1)	6.000000	0.000000
P(2)	7.000000	0.000000
P(3)	8.000000	0.000000
S(1)	1.000000	0.000000
S(2)	1.500000	0.000000
S(3)	0.500000	0.000000

Row	Slack or Surplus	Dual Price
1	16.58869	1.000000
2	0.000000	1.042549

Parameter uncertainty: Consider a 10% change in different combinations of the values of each of the parameters P, S, and Q. Assume expected Q = 6.

Parameter Inputs:							Allocation outputs given flow Q												
							0.9Q			Q			1.1Q						
Site:	<u>1</u>	<u>2</u>	<u>3</u>		<u>1</u>	<u>2</u>	<u>3</u>	Site	<u>1</u>	<u>2</u>	<u>3</u>		<u>1</u>	<u>2</u>	<u>3</u>		<u>1</u>	<u>2</u>	<u>3</u>
P:	6	7	8	S:	1	1.5	0.5		.84	.89	3.67		1	1	4		1.2	1.1	4.3
	5.4	6.3	7.2		.9	1.35	.45		.84	.89	3.67		1	1	4		1.2	1.1	4.3
	6.6	7.7	8.8		1.1	1.65	.55		.84	.89	3.67		1	1	4		1.2	1.1	4.3
	6.6	7.7	8.8		.9	1.35	.45		.70	.87	3.83		.86	.98	4.16		1.0	1.1	4.5

and so on for different combinations of parameter P and S values

8.3 Discuss how model output uncertainty is impacted by both model input uncertainty as well as parameter sensitivity.

Parameters are inputs just as are other time series model inputs. They all may have an impact on various model outputs. This is illustrated by the previous exercise. It is clear that a given percent change in one or more parameters could lead to an even bigger change in values of the output variables. What would be useful is to describe just how sensitive specific outputs are to the direction of change of specific parameters and inputs.

8.4 In many water resources studies considerable attention is given to the uncertainty of water supplies (precipitation, streamflows, evaporation, infiltration, etc.) and much less attention is given to the uncertainty of the management objectives, the costs and benefits of infrastructure, the political support associated with alternative possible decisions, and the like. Develop a simple water resources planning model involving the management of water quantity and quality show how these management objective uncertainties may actually dominate the hydrologic uncertainties.

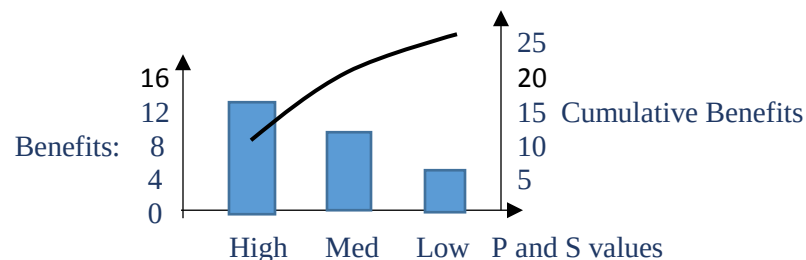
Here the student can select any model involving hydrologic and water quality inputs and do a sensitivity regarding the hydrologic inputs compared to the sensitivity analysis on costs, interest rates, water quality standards, and other policy or economic components of the model.

8.5 Perform a deterministic sensitivity analysis for the consumer 1 in Exercise 8.2. Consider the three parameters, Q , 6 and 1; the latter two numbers are the parameters of the benefit function. Low values of these three parameters are 3, 3, and 0.5 respectively. Most likely values are 6, 6, and 1. High values are 12, 9, and 1.5. Display the results using a Pareto chart, a tornado diagram, and a spider plot.

Using the model: $\max = \text{BENEFIT1} = P \cdot X - s \cdot X \cdot X; X < Q;$
Data: $P = ?; S = ?; Q = ?; \text{enddata}$

Parameter values			Benefit1	Cumulative
$Q = 3, P = 3, S = 0.5$			4.5	27.0
6	6	1	9.0	22.5
12	9	1.5	13.5	13.5

Pareto chart:



Tornado diagram

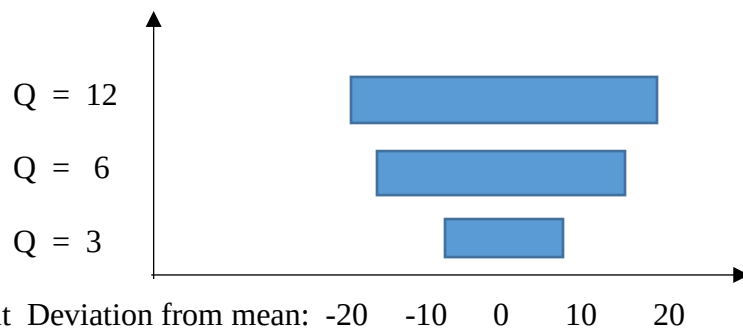
One such diagram can be created for ranges of benefits based on values of Q . Others are possible as well.

		Flow Values Q		
		3	6	12
For $P = 6, S = 1$: Benefits:		9	9	9
3	0.5	4.5	4.5	4.5
9	1.5	13.5	13.5	13.5
3	1.5	1.5	1.5	1.5
9	0.5	22.5	36	40.5
6	0.5	13.5	18	18
6	1.5	6	6	6
9	1	18	20.25	20.25
3	1	2.25	2.25	2.25

For Q = 3: Benefits:	1.5 to 18	Median = $16.5/2 + 1.5 = 9.75$
6	1.5 - 36	$34.5/2 + 1/5 = 18.75$
12	1.5 - 40.5	$39/2 + 1.5 = 21$

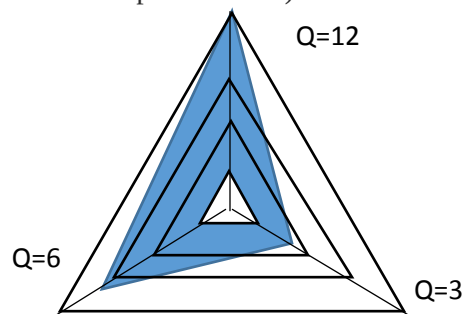
(P,S) Parameter set

P = 9: $0.5 < S < 1.5$	13.5 - 40.5	Median = $17/2 + 13.5 = 22$
P = 6; $0.5 < S < 1.5$	6 - 18	$12/2 + 6 = 12$
P = 3; $0.5 < S < 1.5$	1.5 - 4.5	$3/2 + 1.5 = 3$

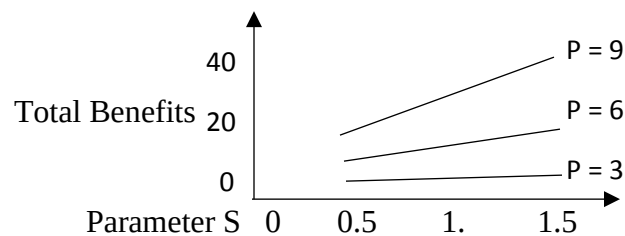


Spider plot:

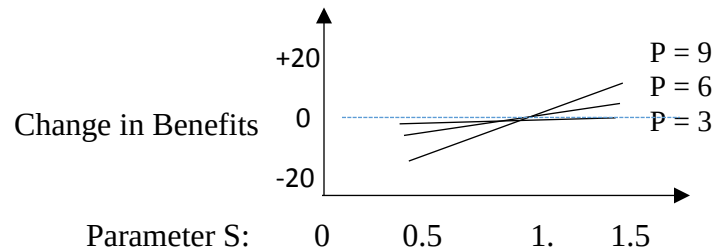
Spider chart plots the values of each category along a separate axis that starts in the center of the chart and ends on the outer ring. A sketch of one for various Q values shown below. (Sketch is not precise here.)



Alternatively, constructing a spider plot similar to Figure 8.14:



Or:



8.6 Referring to water allocation problem defined in Exercise 8.2, assume the available amount of water Q is uncertain. Its cumulative probability distribution is defined by $q/(6+q)$ for non-negative values ($q \geq 0$) of the random variable Q . The expected value of Q does not exist. Perform an uncertainty analysis showing how to define, at least approximately:

- Estimating the mean and standard deviation of the outputs.
- Estimating the probability the performance measure will exceed a specific threshold.
- Assigning a reliability level on a function of the outputs, e.g., the range of function values that is likely to occur with some probability.
- Describing the likelihood of different potential outputs of the system.

Show the application of Monte Carlo sampling and analysis, Latin hypercube sampling, generalized likelihood uncertainty estimation and factorial design methods.

The allocation problem: $\text{Max} = (6x_1 - x_1^2) + (7x_2 - 1.5x_2^2) + (8x_3 - 0.5x_3^2)$

Subject to: $x_1 + x_2 + x_3 \leq q$ when solved for various values of Q will define the maximum benefit allocation policies. Figure 4.4 in Chapter 4 illustrates one such allocation function. Just to show an approach for addressing this particular problem assume $x_1 = 1/4 q$ up to a maximum of 3, $x_2 = 1/4 q$ up to 2.33, and $x_3 = 1/2 q$ up to 8. Also assume the performance measures are the allocations themselves (rather than the derived benefits) and that the allocation thresholds are 2, 2, and 4, respectively. Thus whenever q exceeds 8 all allocation thresholds are met. The probability that q will exceed 8 is $1 - (8/14) = 3/7 = 0.42857$. To simulate this allocation problem we can generate random values p uniformly distributed from 0 to 1 and compute the corresponding value of q , and then the allocations x_1 , x_2 and x_3 . To do this note that if $p = q/(q+6)$ then $q = 6p/(1-p)$ for $p < 1$.

Data:

`n = 200;`

enddata

Sets:

```
time/1..n/: p,q, count ;
site/1..3/: fr, thresh, meanx, varx, sdx ;
joint(site,time): x ;
```

endsets

data:

fr = 0.25, 0.25, 0.5;

thresh = 2, 2, 4;

enddata

p(1) = .4;

q(1) = 6*p(1)/(1-P(1));

count(1) = 0;

@FOR(site(s):

 x(s,1) = @IF(q(1) #ge# 8, thresh(s), fr(s)*q(1));

);

@for(time(t)| t#ge#2:

 p(t) = @rand(p(t-1));

 q(t) = 6*p(t)/(1-p(t));

 count(t) = @IF(q(t)#ge# 8, count(t-1)+1, count(t-1));

 @FOR(site(s):

 x(s,t) = @IF(q(t) #ge# 8, thresh(s), fr(s)*q(t));

);

);

Percent_thresh = 100*count(n)/n;

@For(site(s): meanx(s) = @sum(time(t): x(s,t))/n;

 varx(s) = @sum(time(t): ((x(s,t)-meanx(s))^2)/n);

 sdx(s) = varx(s)^0.5

);

Feasible solution found. Partial output.

Variable	Value
N	200.0000
PERCENT_THRESH	42.50000
P(1)	0.4000000
P(2)	0.6919152E-01
P(3)	0.9593419
P(4)	0.9673175
P(5)	0.7810390
Q(1)	4.000000
Q(2)	0.4460092
Q(3)	141.5722
Q(4)	177.5847
Q(5)	21.40214
COUNT(1)	0.000000
COUNT(2)	0.000000
COUNT(3)	1.000000
COUNT(4)	2.000000
COUNT(5)	3.000000
COUNT(200)	85.00000
FR(1)	0.2500000
FR(2)	0.2500000
FR(3)	0.5000000
THRESH(1)	2.000000
THRESH(2)	2.000000
THRESH(3)	4.000000

MEANX(1)	1.213764
MEANX(2)	1.213764
MEANX(3)	2.427528
VARX(1)	0.6050412
VARX(2)	0.6050412
VARX(3)	2.420165
SDX(1)	0.7778440
SDX(2)	0.7778440
SDX(3)	1.555688

Latin hypercube sampling:

If for some reason one were limited to taking only a few random samples (usually of a more complicated performance map or cumulative distribution than in this problem) perhaps due to computation time or cost, to make sure the samples span the entire distribution, the sampling procedure can be divided into various parts of the distribution. For this example suppose only 8 samples can be taken over the entire distribution. To insure that 2 samples are taken in each of 4 portions of the distribution, $0 \leq p < 0.25$; $0.25 \leq p < 0.5$; $0.5 \leq p < 0.75$; $0.75 \leq p \leq 1$, (each having a probability of 0.25) we can use the following approach:

Data:

n = 2;

enddata

Sets:

segment/1..4/: initial ;
sample/1..n/ ;
samseg(segment, sample): p, q ;

endsets

data:

initial = 0 0.25 0.5 0.75;
seed = ?; !(For example 0.8);

enddata

@For(segment(seg):

p(seg,1) = initial(seg)+ @rand(seed)/4;
q(seg,1) = 6*p(seg,1)/(1-p(seg,1));
p(seg,2) = initial(seg) + @rand(p(seg,1))/4;
q(seg,2) = 6*p(seg,2)/(1-p(seg,2));

);

Variable	Value
N	2.000000
SEED	0.8000000
INITIAL(1)	0.000000
INITIAL(2)	0.2500000
INITIAL(3)	0.5000000
INITIAL(4)	0.7500000
P(1, 1)	0.3459576E-01
P(1, 2)	0.7667304E-01
P(2, 1)	0.2845958
P(2, 2)	0.4107954
P(3, 1)	0.5345958
P(3, 2)	0.7449177
P(4, 1)	0.7845958

P(4, 2)	0.8290401
Q(1, 1)	0.2150131
Q(1, 2)	0.4982398
Q(2, 1)	2.386867
Q(2, 2)	4.183220
Q(3, 1)	6.892018
Q(3, 2)	17.52182
Q(4, 1)	21.85460
Q(4, 2)	29.09595

Note that the values of p and q cover two samples all four segments.

This model can now be extended to include other components of the allocation problem, as desired. But now the allocations in each segment will equal the specified fraction of q in the first two segments (eliminating the need for @IF statements) and only in the third segment beginning with p = 0.5 is the @IF statement needed. The segment beginning with p = 0.75 corresponds to a q = 18, and hence the allocations will always be 2, 2 and 4 respectively.

Generalized likelihood uncertainty estimation:

GLUE is based upon Monte Carlo simulation. Parameter sets may be sampled from any probability distribution. Each parameter set is used to produce model output; the acceptability of each model run is then assessed using a goodness-of-fit criterion that compares the predicted to observed values over some calibration period. The goodness-of-fit function is used to construct a likelihood measure. We will not show an example of this approach based on this exercise problem, but suggest readers refer to [Appraisal of the generalized likelihood uncertainty estimation \(GLUE\) method](#), by Stedinger, et al., 2008, Water Resources Research,

Factorial design:

Assume the allocation policy is to minimize the maximum percent deviation from user 3 target allocations that maximize its total benefits. Consider for brevity, only two parameters, Q and P, each having upper and lower values.

For two discrete values of Q of 4 and 8, and two discrete values of P of 7 and 9 the following % deviations can be calculated.

Allocation Policy	Parameters Q and P	
% deviation*	Q	P
43	4	7
56	4	9
0	8	7
11	8	9

Thus estimate of effect of Q on % deviation = $0.5[(0-43) + (11-56)] = -44$,
and of the effect of P on % deviation = $0.5[(56-43) + (11-0)] = 12$

Clearly these indicators of sensitivity could be improved if three values of each parameter were considered. If three discrete values of Q and P were defined, there would be 9 possible outcomes.

* min= D; $D > (bz - z)/bz$; $bz = P$; $z < Q$;
Data: p=?; Q=?;enddata

Chapter 9

9.1 Distinguish between multiple purposes and multiple objectives and give some examples of complementary and conflicting purposes and objectives of water resources projects.

While various purposes such as water supply, flood control, navigation, recreation, hydropower, etc., could be objectives themselves, they more typically refer to the various types of water uses that bring economic returns or other measures of satisfaction to various users. It is these measures of satisfaction, i.e., economic benefits, income redistribution, regional development, health, employment, etc. that are usually considered the objectives of water resources development.

Conflicting purposes might include water supply and flood control, or flood control and hydropower, or allocation of limited supplies among competing users. Conflicting objectives could be the maximization of total net benefits and equity or income redistribution, or the maximization of water yields, the maximization of yield reliability, and the minimization of costs required to obtain increased water yields and their reliabilities.

Complementary purposes could be increased reservoir releases for water quality improvement, increased recreational benefits and navigation downstream, and for increased flood storage capacity. Two complementary objectives might be the maximization of employment and income redistribution, or increased food production from irrigation and regional development.

9.2 Assume that farmers' demand for water q is a linear function $a - b(p)$ of the price p , where $a, b > 0$. Calculate the farmers' willingness to pay for a quantity of water q . If the cost of delivering a quantity of water q is cq , $c > 0$, how much water should a public agency supply to maximize willingness to pay minus total cost? If the local water district is owned and operated by a private firm whose objective is to maximize profit, how much water would they supply and how much would they earn? The farmers' consumer surplus is their willingness to pay minus what they must pay for the resource. Compare the farmers' consumer surplus in two cases. Do the farmers lose more than the private firm gains by moving from the social optimum to the point that maximizes the firm's profit? Illustrate these relationships with a graph showing the demand curve and the unit cost c of water. Which areas on the graph represent the firm's profits and the farmers' consumer surplus?

Since $q = a - bp$, the unit price $p = \frac{a}{b} - \frac{1}{b}q$.

Willingness to pay for any $q^* = \int_0^{q^*} p dq = \frac{a}{b}q^* - \frac{1}{2b}q^{*2}$

Public Agency's Objective

$$\text{Maximize } \left[\int_0^{q^*} p dq - cq^* \right] = \frac{a}{b} q^* - \frac{1}{2b} q^{*2} - cq^* \quad (1)$$

Setting the derivative of (1) with respect to q^* equal to zero results in $p = c$ (unit price equals unit cost) when q^* maximizes (1) above.

Hence

$$\frac{d(pq^*)}{dq^*} = \frac{a}{b} - \frac{1}{b} q^* = c \quad \text{or} \quad q_{PUB}^* = a - bc$$

Private Company Objective

$$\text{Maximize } [pq^* - cq^*] = \frac{a}{b} q^* - \frac{1}{b} q^{*2} - cq^* \quad (2)$$

Setting the derivative of (2) with respect to q^* equal to zero results in $d(pq^*)/dq^* = c$ (marginal revenue equals unit cost) when q^* maximizes (2) above. Hence

$$\frac{d(pq^*)}{dq^*} = \frac{a}{b} - \frac{2}{b} q^* = c \quad \text{or} \quad q_{PPI}^* = \frac{1}{2}(a - bc)$$

Twice as much water is supplied by a public agency than by the private firm.

Water seller's earnings are pq^* .

If sold by public agency:

$$\left(\frac{a}{b} - \frac{1}{b} q_{PUB}^* \right) (q_{PUB}^*) = c(a - bc)$$

Cost to public agency = $c q_{PRI}^* = c(a - bc)$ thus net income = 0.

If sold by private firm:

$$\left(\frac{a}{b} - \frac{1}{b} q_{PRI}^* \right) (q_{PRI}^*) = \frac{1}{4} \left(\frac{a}{b} + c \right) (a - bc)$$

$$\text{net income: } = \frac{1}{4b} (a - bc)^2$$

Change in income of private firm if quantity of water sold shifts from q_{PUB}^* to q_{PRI}^* :

$$= \frac{1}{4b} (a - bc)^2 \quad (3)$$

Farmer's consumer surplus equals:

$$\int_0^{q^*} p dq - cq^* = \frac{1}{2b} q^{*2}$$

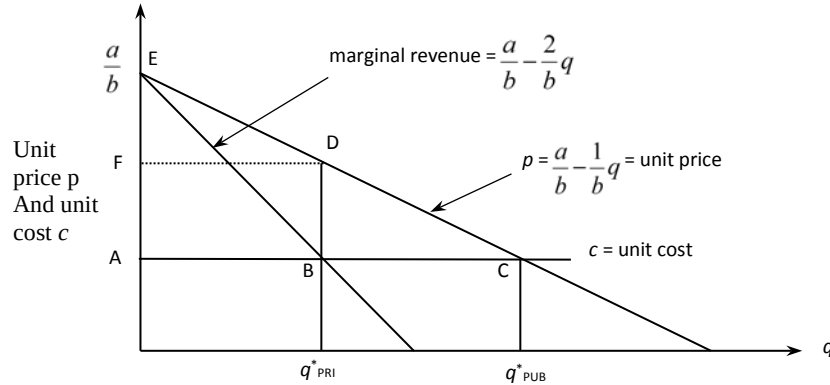
Hence if q^*_{PUB} purchased from a public agency, consumer surplus equals $\frac{1}{2b}(a-bc)^2$, and if purchased from a private firm it equals $\frac{1}{8b}(a-bc)^2$. The decrease in consumer surplus given a shift from q^*_{PUB} to q^*_{PRI} is

$$\frac{3}{8b}(a-bc)^2 \quad (4)$$

Comparing seller's gain (3) with farmer's loss (4) one can prove that the seller's gain is less than the farmer's loss by

$$\left(\frac{3}{8} - \frac{1}{4}\right)(a-bc)^2 = \frac{1}{8}(a-bc)^2$$

If this were not the case, the sellers could pay the farmers their loss and still make the money.



DEF = farmer's consumer surplus if private firm is seller of q^*_{PRI}
ACE = farmer's consumer surplus if public agency is seller of q^*_{PUB}
ABDF = private firm producer's surplus if q^*_{PRI} supplied
BCD = farmer's loss in excess of private firm's gain if q^*_{PRI} supplied rather than q^*_{PUB}

9.3 Consider the water allocation problem used in the earlier chapters of this book. The returns, $B_i(X_i)$ from allocating X_i amount of water to each of three uses i are as follows, along with the optimal allocations from the point of view of each use.

$$\begin{aligned} B_1(X_1) &= 6X_1 - X_1^2 & \rightarrow & X_1^{opt} = 3 & \text{ and } & B_1^{max} = B_1(X_1^{opt}) = 9 \\ B_2(X_2) &= 7X_2 - 1.5X_2^2 & \rightarrow & X_2^{opt} = 7/3 & \text{ and } & B_2^{max} = B_2(X_2^{opt}) = 147/18 \\ B_3(X_3) &= 8X_3 - 0.5X_3^2 & \rightarrow & X_3^{opt} = 8 & \text{ and } & B_3^{max} = B_3(X_3^{opt}) = 32 \end{aligned}$$

Consider this a multi-objective problem. Instead of finding the best overall allocation that maximizes the total return assume the objectives are to maximize the returns from each user.

Show how the weighting, constraint, goal attainment, and goal programming methods can be used to identify the tradeoffs among each of the three objectives for any limiting total amount of water, for example, 6.

The weighting method:

$$\text{Objective: } \max \{w_1 B_1(X_1) + w_2 B_2(X_2) + w_3 B_3(X_3)\}$$

$$\text{Subject to: } X_1 + X_2 + X_3 \leq 6$$

The model is solved for a lot of combinations of weights to generate a set of efficient tradeoffs. To insure efficient tradeoffs, all weights should be > 0 .

Notes:

We can scale each benefit function, and assign new weights (whose sum equals 1) such that the new overall objective function will have a value between 0 and 1. These weights reflect the relative importance given to each objective. If we choose

$$w_1 = \frac{W_1}{9}, \quad w_2 = \frac{W_2}{147/18} \quad \text{and} \quad w_3 = \frac{W_3}{32}, \quad \text{then both overall objective functions are the}$$

$$\text{same, and since } W_1 + W_2 + W_3 = 1, \text{ we have } 9w_1 + \frac{147}{18}w_2 + 32w_3 = 1.$$

$$\text{Objective: } \max \left\{ W_1 \frac{B_1(X_1)}{9} + W_2 \frac{B_2(X_2)}{147/18} + W_3 \frac{B_3(X_3)}{32} \right\}$$

$$\text{Subject to: } X_1 + X_2 + X_3 \leq 6$$

For example, if we weight the first user at 70%, the second at 20% and the third at 10%, we have the following optimal solution:

X_1	X_2	X_3	B_1	B_2	B_3
2.86	2.04	1.10	8.98	8.04	8.19

The constraint method:

$$\text{Objective: } \max \{B_3(X_3)\}$$

$$\text{Subject to: } X_1 + X_2 + X_3 \leq 6$$

$$B_1(X_1) \geq \alpha$$

$$B_2(X_2) \geq \beta$$

The model is solved many times for many values of the lower bounds α and β , where $\alpha \in [0, 9]$ and $\beta \in [0, 147/18]$.

Note: The model can give inefficient values if the constraints applied to the other objectives are not binding.

The goal attainment method:

$$\begin{aligned} \text{Objective: } & \min\{D\} \\ \text{Subject to: } & X_1 + X_2 + X_3 \leq 6 \\ & w_1[9 - B_1(X_1)] \leq D \\ & w_2\left[\frac{147}{18} - B_2(X_2)\right] \leq D \\ & w_3[32 - B_3(X_3)] \leq D \end{aligned}$$

The model is solved for a lot of combinations of weights to generate a set of efficient tradeoffs.

Notes:

- The model will not give an efficient solution when $w_3 = 0$.
- We can assign new weights (whose sum equals 1) that reflect the relative importance of meeting a specific goal compared to meeting the two others.

$$\begin{aligned} \text{Objective: } & \min\{D\} \\ \text{Subject to: } & X_1 + X_2 + X_3 \leq 6 \\ & W_1 \frac{9 - B_1(X_1)}{9} \leq D \\ & W_2 \frac{147/18 - B_2(X_2)}{147/18} \leq D \\ & W_3 \frac{32 - B_3(X_3)}{32} \leq D \end{aligned}$$

For example, if we weight the first user at 70%, the second at 20% and the third at 10%, we have the following optimal solution:

X_1	X_2	X_3	B_1	B_2	B_3
2.23	1.21	2.56	8.41	6.28	17.2

The goal programming method

$$\begin{aligned} \text{Objective: } & \min\{L_1(D_1) + L_2(D_2) + L_3(D_3)\} \\ \text{Subject to: } & X_1 + X_2 + X_3 \leq 6 \\ & 9 - B_1(X_1) \leq D_1 \\ & \frac{147}{18} - B_2(X_2) \leq D_2 \\ & 32 - B_3(X_3) \leq D_3 \end{aligned}$$

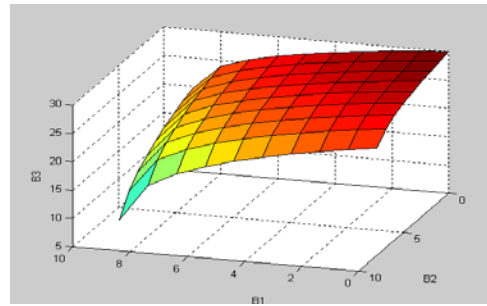
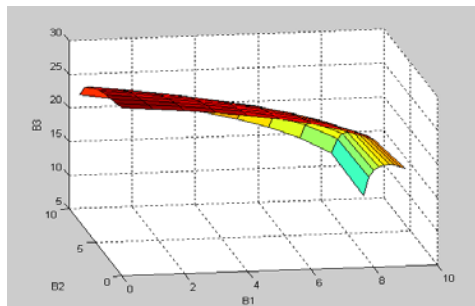
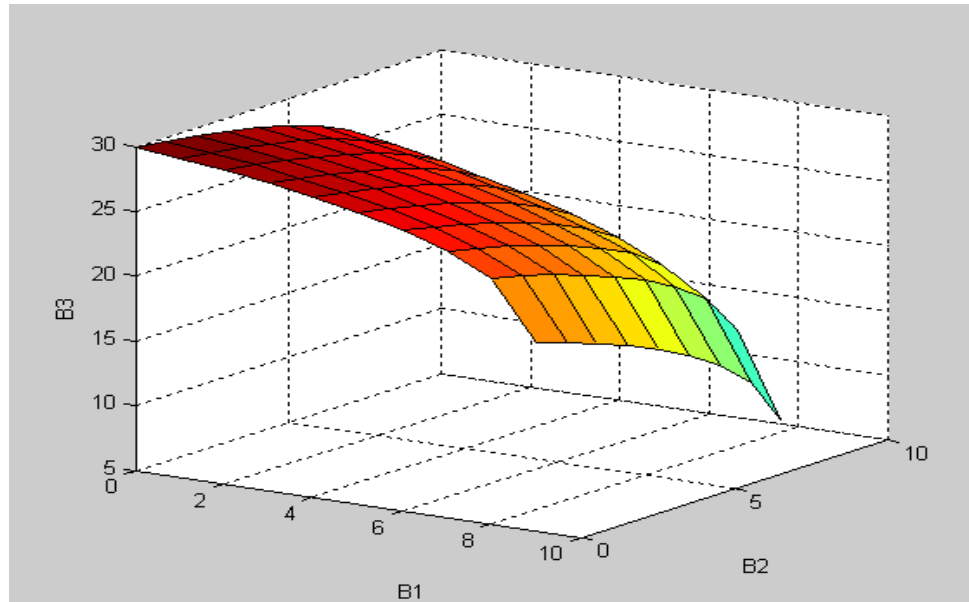
By changing the loss functions, this model can generate a set of efficient tradeoffs.

For example, if we choose $L_1(D_1) = D_1^2$, $L_2(D_2) = \exp(D_2)$, and $L_3(D_3) = D_3$, we have the following optimal solution:

X_1	X_2	X_3	B_1	B_2	B_3
1.89	1.57	2.54	7.77	7.30	17.1

All four methods will give exactly the same efficient surface. When the resource constraint can be expressed as an equality, since the resources available are not enough to satisfy every use, we can find the analytical equation of this surface by expressing X_3 in terms of X_1 and X_2 ($X_3 = 6 - X_1 - X_2$), and thus B_3 in terms of B_1 and B_2 :

$$B_3 = \frac{58}{3} + 2\sqrt{9 - B_1} + \frac{2}{3}\sqrt{49 - 6B_2} - \frac{1}{2}\left(\frac{16}{3} - \sqrt{9 - B_1} - \frac{1}{3}\sqrt{49 - 6B_2}\right)^2$$



x1	x2	x3	b1	b2	b3
0.00	0.00	6.00	0.00	0.00	30.00
0.00	0.15	5.85	0.00	1.00	29.69
0.00	0.31	5.69	0.00	2.00	29.34
0.00	0.48	5.52	0.00	3.00	28.93
0.00	0.67	5.33	0.00	4.00	28.44
0.00	0.88	5.12	0.00	5.00	27.85
0.00	1.13	4.87	0.00	6.00	27.10
0.00	1.45	4.55	0.00	7.00	26.04
0.00	2.00	4.00	0.00	8.00	24.00
0.17	0.00	5.83	1.00	0.00	29.64
0.17	0.15	5.68	1.00	1.00	29.31
0.17	0.31	5.52	1.00	2.00	28.93
0.17	0.48	5.35	1.00	3.00	28.49
0.17	0.67	5.16	1.00	4.00	27.97
0.17	0.88	4.95	1.00	5.00	27.34
0.17	1.13	4.70	1.00	6.00	26.54
0.17	1.45	4.38	1.00	7.00	25.44
0.17	2.00	3.83	1.00	8.00	23.30
0.35	0.00	5.65	2.00	0.00	29.23
0.35	0.15	5.50	2.00	1.00	28.87
0.35	0.31	5.34	2.00	2.00	28.46
0.35	0.48	5.17	2.00	3.00	27.99
0.35	0.67	4.98	2.00	4.00	27.44
0.35	0.88	4.77	2.00	5.00	26.77
0.35	1.13	4.51	2.00	6.00	25.92
0.35	1.45	4.19	2.00	7.00	24.76
0.35	2.00	3.65	2.00	8.00	22.52
0.55	0.00	5.45	3.00	0.00	28.75
0.55	0.15	5.30	3.00	1.00	28.36
0.55	0.31	5.14	3.00	2.00	27.92
0.55	0.48	4.97	3.00	3.00	27.42
0.55	0.67	4.78	3.00	4.00	26.82
0.55	0.88	4.57	3.00	5.00	26.11
0.55	1.13	4.32	3.00	6.00	25.22
0.55	1.45	4.00	3.00	7.00	23.99
0.55	2.00	3.45	3.00	8.00	21.65
0.76	0.00	5.24	4.00	0.00	28.18
0.76	0.15	5.09	4.00	1.00	27.76
0.76	0.31	4.93	4.00	2.00	27.29
0.76	0.48	4.76	4.00	3.00	26.75
0.76	0.67	4.57	4.00	4.00	26.12
0.76	0.88	4.36	4.00	5.00	25.36
0.76	1.13	4.10	4.00	6.00	24.41
0.76	1.45	3.78	4.00	7.00	23.12
0.76	2.00	3.24	4.00	8.00	20.65

x1	x2	x3	b1	b2	b3
1.00	0.00	5.00	5.00	0.00	27.50
1.00	0.15	4.85	5.00	1.00	27.05
1.00	0.31	4.69	5.00	2.00	26.54
1.00	0.48	4.52	5.00	3.00	25.95
1.00	0.67	4.33	5.00	4.00	25.28
1.00	0.88	4.12	5.00	5.00	24.47
1.00	1.13	3.87	5.00	6.00	23.47
1.00	1.45	3.55	5.00	7.00	22.09
1.00	2.00	3.00	5.00	8.00	19.50
1.27	0.00	4.73	6.00	0.00	26.66
1.27	0.15	4.58	6.00	1.00	26.17
1.27	0.31	4.43	6.00	2.00	25.61
1.27	0.48	4.25	6.00	3.00	24.99
1.27	0.67	4.07	6.00	4.00	24.26
1.27	0.88	3.85	6.00	5.00	23.40
1.27	1.13	3.60	6.00	6.00	22.32
1.27	1.45	3.28	6.00	7.00	20.86
1.27	2.00	2.73	6.00	8.00	18.12
1.59	0.00	4.41	7.00	0.00	25.57
1.59	0.15	4.27	7.00	1.00	25.03
1.59	0.31	4.11	7.00	2.00	24.43
1.59	0.48	3.94	7.00	3.00	23.75
1.59	0.67	3.75	7.00	4.00	22.96
1.59	0.88	3.53	7.00	5.00	22.03
1.59	1.13	3.28	7.00	6.00	20.87
1.59	1.45	2.96	7.00	7.00	19.31
1.59	2.00	2.41	7.00	8.00	16.40
2.00	0.00	4.00	8.00	0.00	24.00
2.00	0.15	3.85	8.00	1.00	23.40
2.00	0.31	3.69	8.00	2.00	22.73
2.00	0.48	3.52	8.00	3.00	21.98
2.00	0.67	3.33	8.00	4.00	21.11
2.00	0.88	3.12	8.00	5.00	20.09
2.00	1.13	2.87	8.00	6.00	18.83
2.00	1.45	2.55	8.00	7.00	17.14
2.00	2.00	2.00	8.00	8.00	14.00
3.00	0.00	3.00	9.00	0.00	19.51
3.00	0.15	2.85	9.00	1.00	18.76
3.00	0.31	2.70	9.00	2.00	17.94
3.00	0.48	2.53	9.00	3.00	17.01
3.00	0.67	2.34	9.00	4.00	15.96
3.00	0.88	2.12	9.00	5.00	14.73
3.00	1.13	1.87	9.00	6.00	13.21
3.00	1.45	1.55	9.00	7.00	11.20
3.00	2.00	1.00	9.00	8.00	7.52

9.4 Under what circumstances will the weighting and constraint methods fail to identify efficient solutions?

We consider two objectives to be maximized.

The weighting method:

It will not identify all the efficient values if the efficiency frontier is not strictly concave for maximization or convex for minimization. .

For instance, let's consider the following problem:

$$Z_1(X_1) = X_1^3 \text{ and } Z_2(X_2) = X_2(2 - X_2), \text{ where } X_1 + X_2 = 1.$$

Whatever weights we take, this model will either identify the couples (1,0) or (0,1). But what if we want X_1 and X_2 strictly positive?

The constraint method:

It will not identify all the efficient values if objectives are completely independent of each other, or if the lower bound in the other objective is not binding.

For instance, let's consider the following problem:

$$Z_1(X_1) = 6X_1 - X_1^2 \text{ and } Z_2(X_2) = 8X_2 - 0.5X_2^2, \text{ where } X_1 + X_2 \leq 6.$$

If we try to maximize Z_1 while setting the lower bound of Z_2 to 5, this model usually picks the lowest possible value of X_2 , and thus does not give the efficient tradeoff. It will start giving interesting results as soon as the lower bound is greater than 19.5 (which corresponds to $Z_2(3)$).

As another example, consider the maximization of x_1 , x_2 , and x_3 subject to:

$$x_1^2 + x_2^2 \leq 16$$

$$x_1 + x_3 \leq 40$$

$$x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0$$

The Pareto-efficient solutions to this problem are non-negative values of x_1 and x_2 that lie on a quarter circle of radius 4 and a value of x_3 equal to 10 which is independent of the feasible values of x_1 and x_2 . It is unlikely that Pareto-efficient values of x_3 would result from the solution of the model if either of the first two objectives were selected for maximization subject to lower bounds on the other objectives (unless the lower bound on x_3 were 10). Similarly, if the third objective were selected for maximization both the other two objectives could be less than their maximum values (unless the lower bounds on x_1 and x_2 were such that $x_1^2 + x_2^2 = 16$).

9.5 A reservoir is planned for irrigation and low flow augmentation for water quality control. A storage volume of $6 \times 10^6 \text{ m}^3$ will be available for those two conflicting uses each year. The maximum irrigation demand (capacity) is $4 \times 10^6 \text{ m}^3$. Let X_1 be the allocation of water to irrigation and X_2 the allocation for downstream flow augmentation. Assume that there are two objectives, expressed as

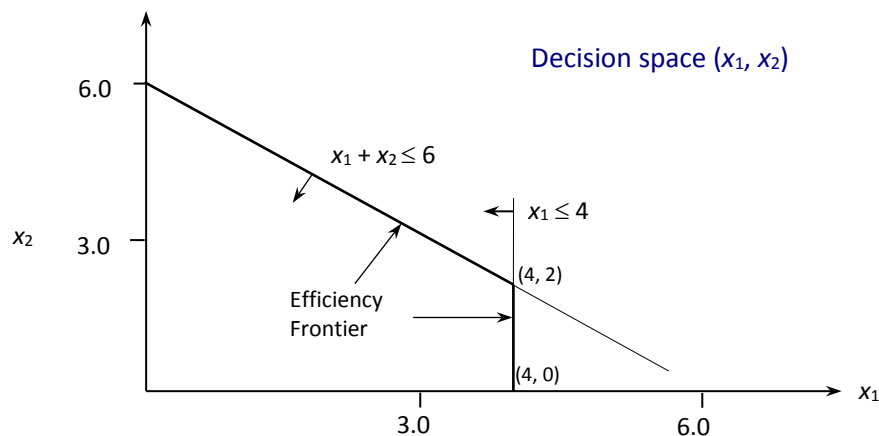
$$\begin{aligned} Z_1 &= 4X_1 - X_2 \\ Z_2 &= -2X_1 + 6X_2 \end{aligned}$$

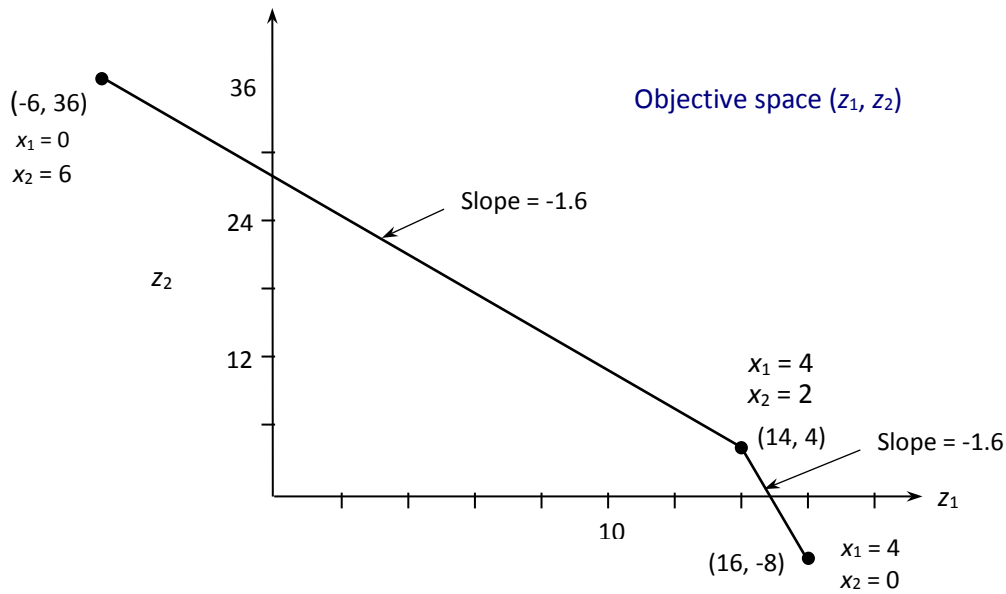
- Write the multi-objective planning model using a weighing approach and a constraint approach.
- Define the efficient frontier. This requires a plot of the feasible combinations of X_1 and X_2 .
- Assume that various values are assigned to a weight W_1 for Z_1 whereas weight W_2 for Z_2 is constant and equal to 1, verify the following solutions to the weighing model.

W_1	X_1	X_2	Z_1	Z_2
>6	4	0	16	-8
6	4	0 to 12	16 to 14	-8 to 4
<6 to >1.6	4	2	14	4
1.6	4 to 0	2 to 6	14 to -6	4 to 36
<1.6	0	6	-6	36

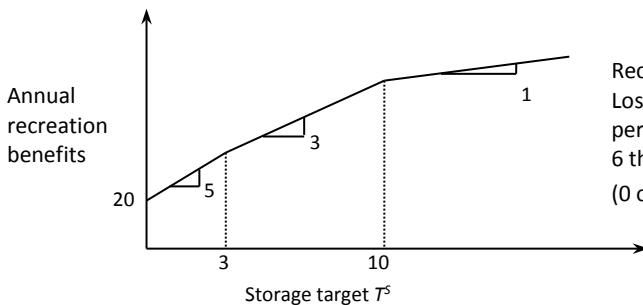
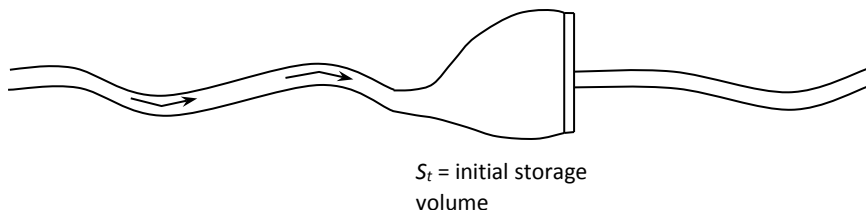
- Weighting approach objective is maximize $w_1 Z_1 + w_2 Z_2$ for various values of the weights w_1 and w_2 , where z_1 and z_2 are defined as above.
Constraint approach objective is maximize z_2 subject to $z_1 \geq L_1$ for lower bounds L_1 or L_2 , respectively.

(b)

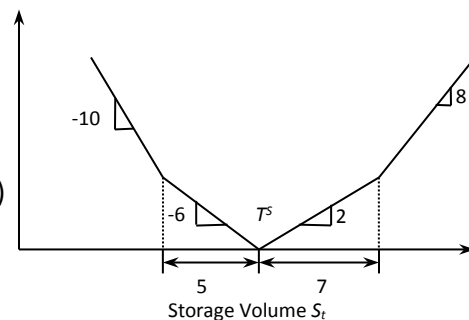


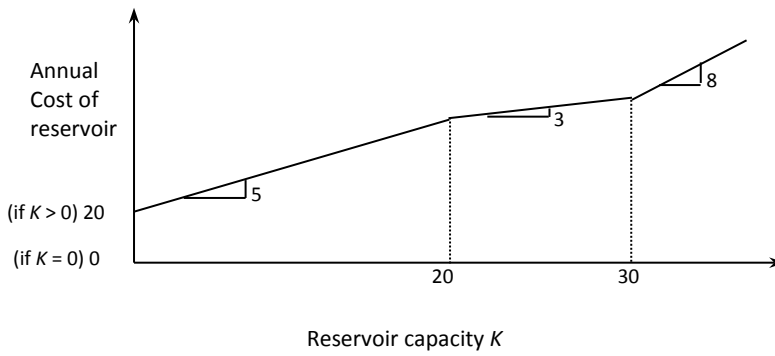
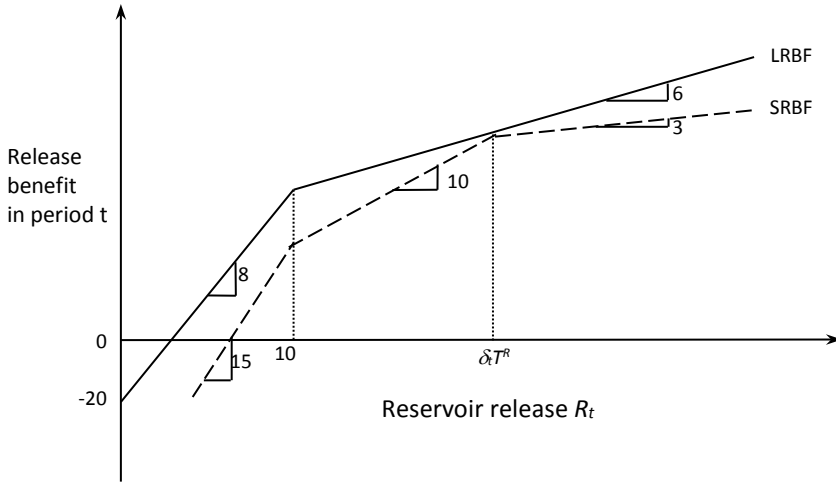


9.6 Show that the following benefit, loss, and cost functions can be included in a linear optimization problem for finding the active storage volume target T^s , annual release target T^R and the actual storage releases R_t in each within-year period t , and the reservoir capacity K . The objective is to maximize annual net benefits from the construction and operation of the reservoir. Assume that the inflows are known in each of 12 within-year periods t . Note that the loss function associated with reservoir recreation is independent of the value of T^s , unlike the loss function associated with reservoir releases. Structure the complete linear programming model. Define all variables used that are not defined below. Let $\delta_t T^R$ be the known release target in period t .



Recreation Loss in periods $t=6$ through 9 (0 otherwise)





Let Q_t be the inflow at the reservoir site in month t

$$\text{Maximize } 20 + 5T_1^S + 3T_2^S + T_3^S \quad (\text{Recreation Benefits})$$

$$- \frac{1}{2} \sum_{t=6}^9 [6(D_{1t}^S + D_{1,t+1}^S) + 10(D_{2t}^S + D_{2,t+1}^S) + 2(E_{1t}^S + E_{1,t+1}^S) + 8(E_{2t}^S + E_{2,t+1}^S)] \quad (\text{Recreation Losses})$$

$$-20 + 8T_1^R + 6T_2^R - \sum_{t=1}^{12} (15D_{1t}^R + 10D_{2t}^R - 3E_t^R) \quad (\text{Release Benefits})$$

$$-20z_1 - 5K_1 - 120z_2 - 3K_2 - 8K_3 \quad (\text{Reservoir capacity cost})$$

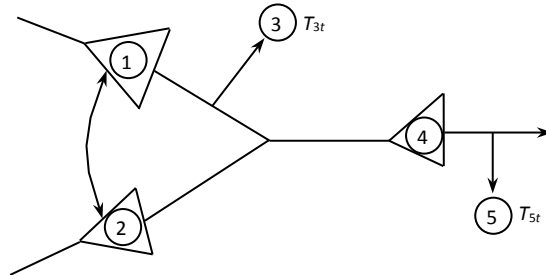
Subject to:

$$\begin{aligned}
T_1^S + T_2^S + T_3^S &= T^S & D_{1t}^S &\leq 5 \\
T_1^S &\leq 3 & E_{1t}^S &\leq 7 \\
T_2^S &\leq 7 \\
S_t &= T^S - D_{1t}^S - D_{2t}^S + E_{1t}^S + E_{2t}^S & t &= 6, 7, 8, 9, 10
\end{aligned}$$

$$\begin{aligned}
R_t &= R_{1t} + R_{2t} + R_{3t} & \forall t \\
T_{1t}^R + T_{2t}^R &= \delta_t T^R & \forall t \\
T_{1t}^R - R_{1t} &= D_{1t}^R & \forall t \\
T_{2t}^R - R_{2t} &= D_{2t}^R & \forall t \\
T_{1t}^R &\leq 10 & \forall t
\end{aligned}$$

$$\begin{aligned}
S_t + Q_t - R_t &= S_{t+1} & \forall t \text{ if } t+1 = 13, \text{ set to } 1 \\
S_t &\leq K & \forall t \\
K &= K_1 + K_2 + K_3 + 20z_2 \\
z_1 + z_2 &\leq 1 \\
z_1, z_2 &\text{ integer} \\
K_1 \leq 20z_1; K_2 \leq 10z_2; K_3 \leq Mz_2 & & (M \text{ a large number})
\end{aligned}$$

9.7 For the river basin shown, potential reservoirs exist at sites $i = 1, 2$, and 4 and a diversion can be constructed between sites 1 and 2. The cost $C_i(K_i)$ of each reservoir i is a function of its active storage capacity K_i . The cost of the diversion canal is $C_d(Q_d)$ where Q is the flow capacity of the canal. The cost of diverting a flow Q_{ijt} from site i to site j is $C_{ij}(Q_{ijt})$. The two users at sites 3 and 5 have known target allocations (demands) T_{it} in each period t . The return flow from use 3 is 40% of that allocated to use 3. Construct a model for finding the least cost of meeting various percentages of the target demands. Assume that the natural streamflows Q_t^i at each site i in each period t , are known.



$$\begin{aligned} \text{Minimize} \quad & C_1(K_1) + C_2(K_2) + C_4(K_4) + C(Q) \\ & + \sum_{t=1}^T [C_{12}(Q_{12t}) + C_{21}(Q_{21t})] \end{aligned}$$

Subject to:

$$\begin{aligned} \text{Site 1} \quad & S_t^1 + Q_t^1 - R_t^1 - Q_{12t} + Q_{21t} = S_{t+1}^1 & \forall t & \quad T+1=1 \\ & S_t^1 = K_1 & \forall t \\ & Q \geq Q_{12t} + Q_{21t} & \forall t \end{aligned}$$

$$\begin{aligned} \text{Site 2} \quad & S_t^2 + Q_t^2 - R_t^2 - Q_{21t} + Q_{12t} = S_{t+1}^2 & \forall t \\ & S_t^2 = K_2 & \forall t \end{aligned}$$

$$\begin{aligned} \text{Site 3} \quad & q_t^3 \leq R_t^1 + (Q_t^3 - Q_t^1) & \forall t \\ & q_t^3 \leq F_{3t} T_{3t} & \forall t \end{aligned}$$

$$\begin{aligned} \text{Site 4} \quad & S_t^4 + Q_t^4 - Q_t^1 - Q_t^2 + R_t^1 + R_t^2 - 0.6q_t^3 - R_t^4 = S_{t+1}^4 & \forall t \\ & S_t^4 \leq K_4 & \forall t \end{aligned}$$

$$\begin{aligned} \text{Site 5} \quad & q_t^5 \leq R_t^5 + (Q_t^5 - Q_t^4) & \forall t \\ & q_t^5 \leq F_{5t} T_{5t} & \forall t \end{aligned}$$

Note: One or both of the flows Q_{12t} and Q_{21t} will be zero, and pumping costs $C_{12}(0) = C_{21}(0) = 0$. The known percentage of the specified targets T_{it} to be met at sites $i = 3$ and 5 are expressed by the fractions F_{it} .

9.8 Suppose that there exist two polluters, A and B, who can provide additional treatment, X_A and X_B , at a cost of $C_A(X_A)$ and $C_B(X_B)$, respectively. Let W_A and W_B be the waste produced at sites A and B, and $W_A(1 - X_A)$ and $W_B(1 - X_B)$ be the resulting waste discharges at site A and B. These discharges must be no greater than the effluent standards $E_A^{\max}$ and $E_B^{\max}$. The resulting pollution concentration $a_{Aj}(W_A(1 - X_A)) + a_{Bj}(W_B(1 - X_B)) + q_j$ at various sites j must not exceed the stream standards $S_j^{\max}$. Assume that total cost and cost inequity [i.e., $C_A(X_A) + C_B(X_B)$ and $|C_A(X_A) - C_B(X_B)|$] are management objectives to be determined

- Discuss how you would model this multi-objective problem using the weighting and constraint (or target) approaches.
- Discuss how you would use the model to identify efficient, non-inferior (Pareto-optimal) solutions.
- Effluent standards at sites A and B and ambient stream standards at sites j could be replaced by other planning objectives (e.g., the minimization of waste discharged into the stream). What would these objectives be, and how could they be included in the multi-objective model?

(a) Minimize $w_1[C_A(X_A) + C_B(X_B)] + w_2(\Delta)$

or

Minimize $C_A(X_A) + C_B(X_B)$

Subject to: $\square \leq$ maximum allowable inequity
and

$$\Delta = U_1 + U_2$$

$$C_A(X_A) - C_B(X_B) = U_1 - U_2$$

$$W_A(1 - X_A) \leq E_A^{\max}$$

$$W_B(1 - X_B) \leq E_B^{\max}$$

$$a_{Aj}W_A(1 - X_A) + a_{Bj}W_B(1 - X_B) + q_j \leq s_j^{\max}$$

$$0 \leq X_A \leq 1$$

$$0 \leq X_B \leq 1$$

(a) To identify the Pareto-efficient solutions, vary weights (or maximum allowable inequity bound) systematically to obtain non-inferior solutions.

(b) Other objectives could be: minimize $s_j^{\max}$, or minimize $E_A^{\max}$, or minimize $E_B^{\max}$, and assign weights to each objective or assign upper limits on all but one of the objectives.

Note: It may be convenient to scale these objectives so that weights can vary from 0 to 1 for all objectives.

9.9 (a) What conditions must apply if the goal attainment method is to produce only non-inferior alternatives for each assumed target T_k and weight w_k ?

(b) Convert the goal programming objective deviation components $w_i(z_i^* - z_i(\bar{x}))$ to a form suitable for solution by linear programming.

(a) The target goals must be equal to or greater than the maximum obtainable values of the maximization objectives.

(b) Let each term $z_i^* - z_i(\bar{x})$ in the objective equal $z_{1i} - z_{2i}$ then the overall objective is to minimize $\sum_i w_i(z_{1i} - z_{2i})$.

9.10 Water quality objectives are sometimes difficult to quantify. Various attempts have been made to include the many aspects of water quality in single water quality indices. One such index was proposed by Dinius (Social Accounting Systems for Evaluating Water Resources, Water Resources Research, Vol. 8, 1972. pp. 1159-1177). Water quality, Q , measured in percent is given by

$$Q = \frac{w_1 Q_1 + w_2 Q_2 + \dots + w_n Q_n}{w_1 + w_2 + \dots + w_n}$$

where Q_i is the i^{th} quality constituent (dissolved oxygen, chlorides, etc.) and w_i is the weight or relative importance of the i^{th} quality constituent. Write a critique on the use of such an index in multi-objective water resources planning.

The development of any index requires some means of combining numerous data describing a system or the environment into one, or at most only a few, numbers. The method of combining these data is subjective and it may not reflect the preferences of all individuals responsible for making decisions based on the values of each of these variables. Knowing a single index value rarely substitutes for a knowledge of the value of each of its components in spite of the desire by many to work with the minimum quantity of the data necessary.

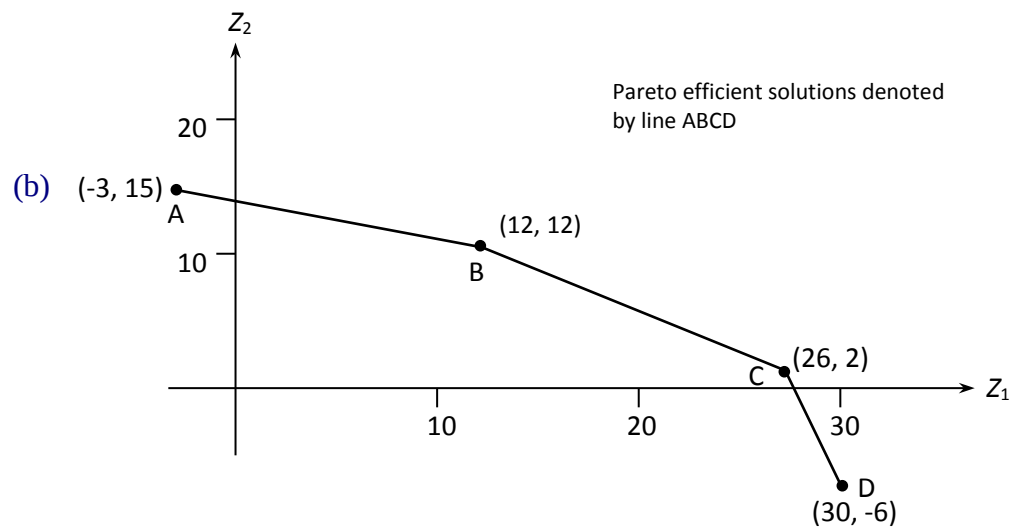
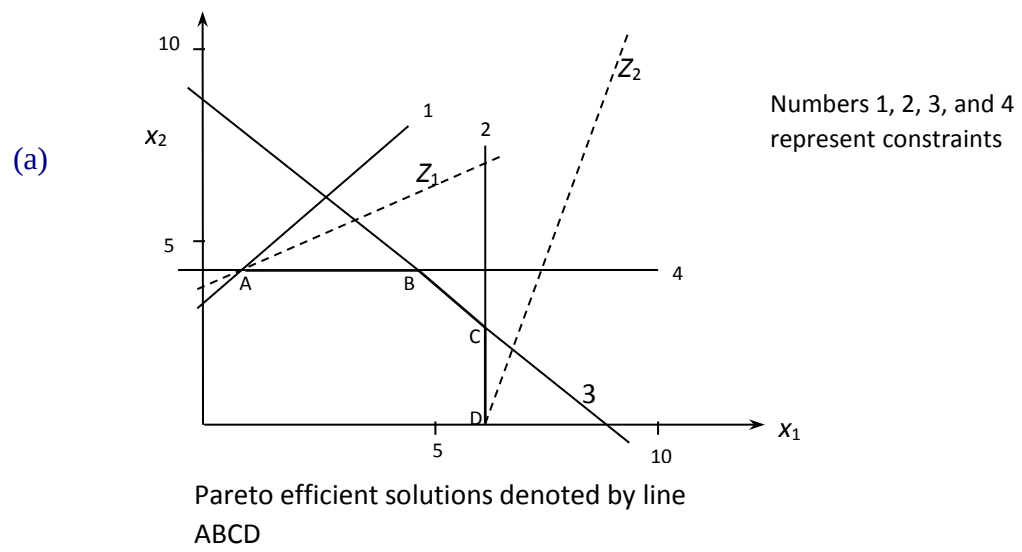
9.11 Let objective $Z_1(\mathbf{X}) = 5X_1 - 2X_2$ and objective $Z_2(\mathbf{X}) = -X_1 + 4X_2$. Both are to be maximized. Assume that the constraints on variables X_1 and X_2 are:

1. $-X_1 + X_2 \leq 3$
2. $X_1 \leq 6$
3. $X_1 + X_2 \leq 8$
4. $X_2 \leq 4$
5. $X_1, X_2 \geq 0$

- (a) Graph the Pareto-optimal or non-inferior solutions in decision space.
- (b) Graph the efficient combination of Z_1 and Z_2 in objective space.
- (c) Reformulate the problem to illustrate the weighting method for defining all efficient solutions of part (a) and illustrate this method in decision and objective space.
- (d) Reformulate the problem to illustrate the constraint method of defining all efficient solutions of part (a) and illustrate this method in decision and objective space.
- (e) Solve for the compromise set of solutions using compromise programming as defined by

$$\text{Minimize } [w_1(Z_1^* - Z_1)^{\square} + w_2(Z_2^* - Z_2)^{\square}]^{1/\square}$$

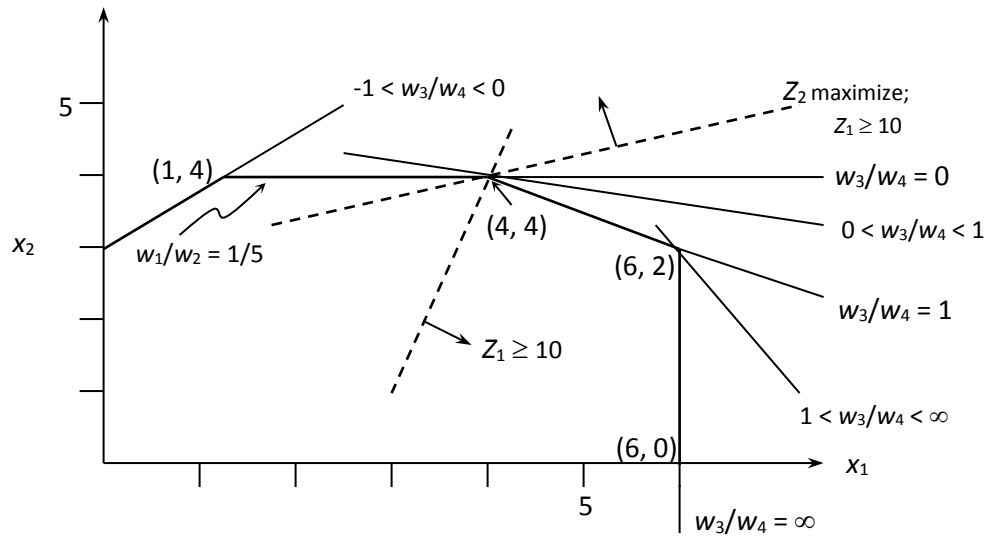
where Z_i^* represents the best value of objective i with all weights w equal to 1 and $\square$ equal to 1, 2, and ∞ .



(c)

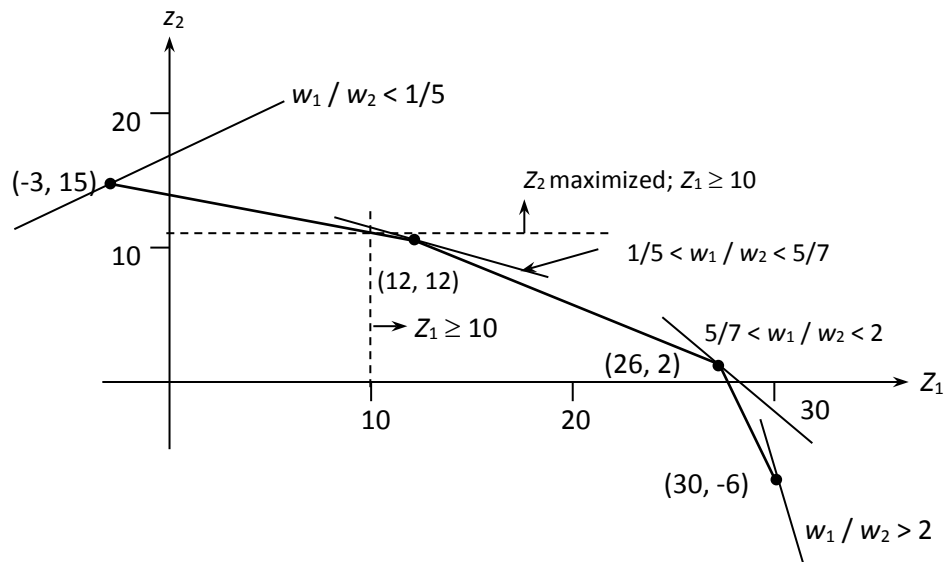
Maximize $w_1(5x_1 - 2x_2) + w_2(-x_1 + 4x_2)$

or $\max w_3x_1 + w_4x_2$ in decision space and w_3, w_4 may be negative.

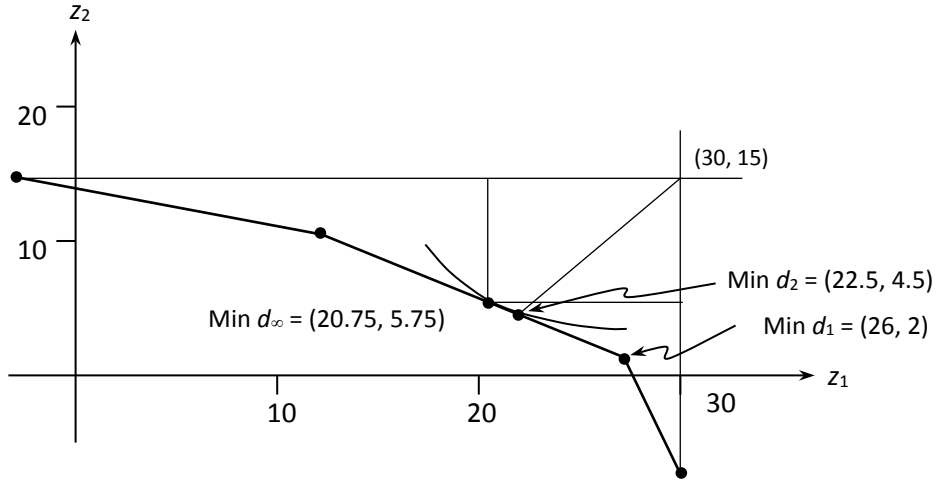


(d)

or $\max w_1z_1 + w_2z_2$ in objective space



(e)



For $\alpha = 1$, the objective is maximize $Z_1 + Z_2$, and its solution from the above graphs is $x_1 = 26, x_2 = 2$.

For $\alpha = 2$, the objective is to maximize the sum of squared deviations from the ideal $(Z_1^*, Z_2^*) = (30, 15)$, or

$$\text{Minimize } [(Z_1^* - Z_1)^2 + (Z_2^* - Z_2)^2]^{1/2}$$

which is the radius of a circle centered at (Z_1^*, Z_2^*) . This circle will be tangent to the line

having a slope $-\frac{w_1}{w_2} = -\frac{5}{7}$. This line can be defined by the equation

$$(Z_2 - 12) = -\frac{5}{7}(Z_1 - 12). \text{ T the point of tangency the radius will be 90o to the line,}$$

hence the slope of the radius will be $\frac{7}{5}$ and its equation will be

$$(Z_2 - 15) = -\frac{7}{5}(Z_1 - 30). \text{ Solving these equations simultaneously yields } (Z_1, Z_2) = (22.5, 4.5).$$

For $\alpha = \infty$ the objective becomes minimize $(\max[|Z_1^* - Z_1|, |Z_2^* - Z_2|])$ which is the set of squares centered at (Z_1^*, Z_2^*) .

Hence $(Z_1^* - Z_1) = (Z_2^* - Z_2)$ Solving simultaneously $(15 - Z_2) = (30 - Z_1)$ and

$$(Z_2 - 12) = -\frac{5}{7}(Z_1 - 12) \text{ yields the solution}$$

$$(Z_1, Z_2) = (20.75, 5.75).$$

9.12 Illustrate the procedure for selecting among three plans, each having three objectives, using indifference analysis. Let Z_{ji} represent the value of objective i for plan j . The values of each objective for each plan are given below. Assume that each objective is to be maximized. Assume that an identical indifference function for all trade-offs between pairs of objectives, namely one that implies you are willing to give up twice as many units of your higher (larger) objective value to gain one unit of your lower (smaller) objective value. [For example, you would be indifferent to two plans having as their three objective values (30, 5, 10) and (20, 5, 15).] Rank these three plans in order of preference.

Plan 1: (5, 25, 15); Plan 2: (10, 20, 10); Plan 3: (15, 10, 15)

These indifferent solutions are not unique but merely illustrate the pair-wise tradeoff process.)

Plan 1: (5, 25, 15) $\sim$ (10, 15, 15) $\sim$ (11 + 2/3, 11 + 2/3, 15)
 Plan 2: (10, 20, 10) $\sim$ (10, 15, 12 + 1/2) $\sim$ (11 + 2/3, 11 + 2/3, 12 + 1/2) (least preferred)
 Plan 3: (15, 10, 15) $\sim$ (11 + 2/3, 11 + 2/3, 15)

Another Solution

Plan 1:	(5, 25, 15) $\sim$ (7 + 1/2, 20, 15)	} <u>all tie</u>
Plan 2:	(10, 20, 10) $\sim$ (7 + 1/2, 20, 15)	
Plan 3:	(15, 10, 15) $\sim$ (11 + 2/3, 11 + 2/3, 15) $\sim$ (7 + 1/2, 20, 15)	

Alternatively:

Plan 1: (5, 25, 15) $\sim$ (5, 35, 10) $\sim$ (15, 15, 10)
 Plan 2: (10, 20, 10) $\sim$ (12 + 1/2, 15, 10) (least preferred)
 Plan 3: (15, 10, 15) $\sim$ (25, 10, 10) $\sim$ (15, 15, 10)

However:

Plan 1: (5, 25, 15) $\sim$ (7 + 1/2, 25, 10) $\sim$ (12 + 1/2, 15, 10)
 Plan 2: (10, 20, 10) $\sim$ (12 + 1/2, 15, 10) (least preferred)
 Plan 3: (15, 10, 15) $\sim$ (25, 10, 10) $\sim$ (15, 15, 10) (most preferred)

This exercise shows that this type of linear trade (giving up two high values for one additional low value) among values of pairs of objectives when three or more objectives exists may not result in a unique ranking of plans. This does not negate indifference analyses, only this particular tradeoff procedure that results in inconsistent solutions. An alternative approach to plan selection is to select the plan that minimizes the variances among the three objective values (Plan 3), assuming these values are comparable or appropriately scaled.

Chapter 10

10.1 The common version of the Streeter-Phelps equations for predicting biochemical oxygen demand BOD and dissolved oxygen deficit D concentrations are based on the following two differential equations:

$$(a) \frac{d(BOD)}{d\tau} = -K_d (BOD)$$
$$(b) \frac{dD}{d\tau} = K_d (BOD) - K_a D$$

where K_d is the deoxygenating rate constant (T^{-1}), K_a is the reaeration-rate constant (T^{-1}), and τ is the time of flow along a uniform reach of stream in which dispersion is not significant. Show the integrated forms of (a) and (b).

Integrating (a)

$$BOD_\tau = BOD_o e^{-K_d \tau} \quad \text{where } BOD_o = BOD (ML^{-3}) \text{ at } \tau = 0$$

or $BOD_o = W_o/Q = \text{mass waste discharge per unit time} / \text{streamflow rate}$, and time of flow $\tau = X/U$, if the flow in the reach is constant.

Integrating (b) after substituting the right-hand side of the above equation into (b),

$$D_\tau = BOD_o \frac{K_d}{K_a - K_d} (e^{-K_d \tau} - e^{-K_a \tau}) + D_o e^{-K_a \tau}$$

10.2 Based on the integrated differential equations in Exercise 10.1:

- (a) Derive the equation for the distance X_c downstream from a single point source of BOD that for a given streamflow will have the lowest dissolved oxygen concentration.
- (b) Determine the relative sensitivity of the deoxygenation-rate constant K_d and the reaeration-rate constant K_a on the critical distance X_c and on the corresponding critical deficit D_c . For initial conditions, assume that the reach has a velocity of 2 m/s (172.8 km/day), a K_d of 0.30 per day, and a K_a of 0.4 per day. Assume that the DO saturation concentration is 8 mg/l, the initial deficit is 1.0 mg/l, and the BOD concentration at the beginning of the reach (including that discharged into the reach at that point) is 15 mg/l.

- (a) To find X_c , the critical distance from an input of BOD at $X = 0$, which equals $\tau_c U$, differentiate D_t with respect to τ and equate to zero to calculate τ_c and hence X_c .

$$\begin{aligned}\frac{dD_\tau}{d\tau} &= BOD_o \left(\frac{K_d}{K_a - K_d} \right) (-K_d e^{-K_d \tau} + K_a e^{-K_a \tau}) - K_a D_o e^{-K_a \tau} \\ &= \frac{K_a}{K_d} \left[1 - \frac{(K_a - K_d) D_o}{BOD_o K_d} \right] e^{-K_a \tau} - e^{-K_d \tau}\end{aligned}$$

and since $-e^{-K_d \tau} / e^{-K_a \tau} = -e^{(K_a - K_d) \tau}$

$$\ln e^{(K_a - K_d) \tau} = \ln \left\{ \frac{K_a}{K_d} \left[1 - \frac{(K_a - K_d) D_o}{BOD_o K_d} \right] \right\}$$

therefore

$$\tau_c = \frac{1}{K_a - K_d} \ln \left\{ \frac{K_a}{K_d} \left[1 - \frac{(K_a - K_d) D_o}{BOD_o K_d} \right] \right\}$$

and

$$X_c = U \tau_c$$

Note that X_c and τ_c are independent of BOD_o if $D_o = 0$.

$$\begin{aligned}\text{(b)} \quad D_{\tau_c} &= 15 \left[\frac{K_d}{K_a - K_d} (e^{-K_d \tau_c} - e^{-K_a \tau_c}) \right] + 1.0 e^{-K_a \tau_c} \\ X_c &= (172.8 \text{ km/day})(\tau_c \text{ days})\end{aligned}$$

K_d (days ⁻¹)	K_a (days ⁻¹)	τ_c (days)	$D \tau_c$ (mg/l)	X_c (km)	Remarks	ΔD_c (mg/l)	$\Delta \tau_c$ (days)
0.3	0.4	2.65	5.1	458	Base condition		
0.15	0.4	3.45	3.3	596	50% increase in K_d	-1.8	0.8
0.6	0.4	1.92	7.1	331	100% increase in K_d	2.0	-0.7
0.3	0.2	3.84	7.1	663	50% increase in K_a	2.0	1.2
0.3	0.8	1.73	3.3	298	100% increase in K_a	-1.8	-0.9

The larger the value of D_o , the more sensitive D_c will be to changes in K_a .

10.3 To account for settling of BOD , in proportion to the BOD concentration, and for a constant rate of BOD addition R due to runoff and scour, and oxygen production ($A > 0$) or reduction ($A < 0$) due to plants and benthic deposits, the following differential equations have been proposed:

$$\frac{d(BOD)}{d\tau} = -(K_d + K_s)BOD + R \quad (1)$$

$$\frac{dD}{d\tau} = K_d BOD - K_a D - A \quad (2)$$

where K_s is the settling rate constant (T^{-1}) and τ is the time of flow. Integrating these two equations results in the following deficit equation:

$$D_\tau = \frac{K_d}{K_a - (K_d + K_s)} \left[(BOD_o - \frac{R}{K_d + K_s}) \{ \exp[-(K_d + K_s)\tau] - \exp(-K_a\tau) \} \right] + \frac{K_d}{K_a} \left\{ \left(\frac{R}{K_d + K_s} - \frac{A}{K_d} \right) [1 - \exp(-K_a\tau)] \right\} + D_o \exp(-K_a\tau) \quad (3)$$

where BOD_o and D_o are the BOD and DO deficit concentrations at $\tau = 0$.

- Compare this equation with that found in Exercise 10.1 if K_s , R , and A are 0.
- Integrate equation (1) to predict the BOD_τ at any flow time τ .

(a) The integrated “oxygen sag” equations are identical if $K_s = R = A = 0$.

(b) Note that $\int \frac{dx}{a + bx} = \frac{1}{b} \ln(a + bx) + c_0$ for constants a , b , and c_0 .

$$\text{Thus } \int \frac{dBOD}{R - (K_d + K_s)BOD} = \frac{1}{-(K_d + K_s)} \ln[R - (K_d + K_s)BOD] + c_0$$

$$\ln[R - (K_d + K_s)BOD] = -(K_d + K_s)(\tau - c_0)$$

$$R - (K_d + K_s)BOD = c_1 \exp[-(K_d + K_s)\tau]$$

$$BOD_\tau = c_2 \exp[-(K_d + K_s)\tau] + \frac{R}{K_d + K_s}$$

$$\text{If } \tau = 0, BOD_\tau = BOD_0 \quad \text{hence } c_2 = BOD_0 - \frac{R}{K_d + K_s}$$

$$BOD_\tau = [BOD_0 - (\frac{R}{K_d + K_s})] \exp[-(K_d + K_s)\tau] + \frac{R}{K_d + K_s}$$

10.4 Develop finite difference equations for predicting the steady-state nitrogen component and DO deficit concentrations D in a multi-section one-dimensional estuary. Define every parameter or variable used.

Define the following variables

N_{ik}	= concentration of nitrogen component i in section k (ML^{-3})
K_{ik}	= first order decay constant for N_{ik} (T^{-1})
$K_{i, i+1, k}$	= forward reaction coefficient for N_{ik} (T^{-1})
$\alpha_{k-1, k}$	= $L_k / (L_{k-1} + L_k)$
L_k	= length of reach or section k (L)
V_k	= volume of reach or section (L^3)
$E_{k-1, k}$	= dispersion coefficient between reaches $k-1$ and k ($L^2 T^{-1}$)
$E'_{k-1, k}$	= $E_{k-1, k} A_{k-1, k} / 0.5(L_{k-1} + L_k)$ ($L^3 T^{-1}$) = eddy exchange or bulk dispersion rate coefficient
$A_{k-1, k}$	= cross sectional area of interface between sections $k-1, k$ (L^2)
$Q_{k-1, k}$	= flow from reach or section $k-1$ to k ($L^3 T^{-1}$)
W_{ik}	= direct mass discharge of nitrogen component i into reach k (MT^{-1})

Steady-state model: For each section or reach k and for nitrogen forms $i = 1, 2, 3, 4$

$$\begin{aligned}
 0 = & Q_{k-1, k} [\alpha_{k-1, k} N_{i, k-1} + (1 - \alpha_{k-1, k}) N_{ik}] \\
 & - Q_{k, k+1} [\alpha_{k, k+1} N_{i, k} + (1 - \alpha_{k, k+1}) N_{i, k+1}] \\
 & + E'_{k-1, k} (N_{i, k-1} - N_{i, k}) - E'_{k, k+1} (N_{ik} - N_{i, k+1}) \\
 & - V_k [K_{ik} N_{i, k} - K_{i-1, i, k} N_{i-1, k}] \\
 & - W_{ik}
 \end{aligned}
 \left. \begin{array}{l} \\ \\ \\ \\ \end{array} \right\} \begin{array}{l} \text{mass transport to and} \\ \text{from section } k \\ \\ \text{net mass dispersion} \\ \\ \text{Decay, settling and conversion} \\ \text{From } i-1 \text{ to } i \text{ in } k \\ \\ \text{direct addition of form } i \end{array}$$

for predicting the dissolved oxygen deficit concentration D_k resulting from ammonia oxidation (NH_3 to NO_2) having a reaction rate $3.43K_{23k}$, and from nitrate oxidation (NO_2 to NO_3) having a reaction rate $1.14K_{34k}$:

$$\begin{aligned}
 0 = & Q_{k-1, k} [\alpha_{k-1, k} D_{k-1} + (1 - \alpha_{k-1, k}) D_k] \\
 & - Q_{k, k+1} [\alpha_{k, k+1} D_k + (1 - \alpha_{k, k+1}) D_{k+1}] + E'_{k-1, k} (D_{k-1} - D_k) \\
 & + E'_{k, k+1} (D_k - D_{k+1}) - V_k [K_k^a D_k + 3.43K_{23k} N_{2k} + 1.14K_{34k} N_{3k}] \quad \forall k
 \end{aligned}$$

where K_k^a is the reaeration rate constant (T^{-1}) for section or reach k .

10.5 Using Michaelis-Menten kinetics develop equations for

- (a) Predicting the time rate of change of a nutrient concentration N (dN/dt) as a function of the concentration of bacterial biomass B ;

- (b) Predicting the time rate of change in the bacterial biomass $B(dB/dt)$ as a function of its maximum growth rate $\mu_B^{\max}$, temperature T , B , N , and the specific-loss rate of bacteria ρ_B ; and
- (c) Predicting the time rate of change in dissolved oxygen deficit (dD/dt) also as a function of N , B , ρ_B , and the reaeration-rate constant K_a (T^{-1}).

How would these three equations be altered by the inclusion of protozoa P that feed on bacteria, and in turn require oxygen? Also write the differential equations for the time rate of change in the concentration of protozoa $P(dP/dt)$.

(a) Define for any nutrient N :

a_{NB}	= mass of nutrient N consumed per unit increase in bacterial biomass (MM^{-1})
$\mu_{BN}^{\max}$	= maximum growth (increase) of bacterial biomass resulting from a unit mass of nutrient N per unit time ($MM^{-1}T^{-1}$) (for a given temperature)
K_N	= nutrient half-saturation concentration for bacterial growth (ML^{-3})

then

$$\frac{dN}{dt} = -a_{NB}\mu_{BN}^{\max}\left(\frac{N}{K_N + N}\right)B$$

- (b) B = endogenous respiration rate of bacteria
= mass decrease of a unit mass of bacteria per unit time ($MM^{-1}T^{-1}$)

then

$$\frac{dB}{dt} = \mu_{BN}^{\max}\left(\frac{N}{K_N + N}\right)B - \rho_B B$$

(c) For dissolved oxygen, let:

K_a	= reaeration rate coefficient (T^{-1})
DO^S	= saturation concentration of dissolved oxygen (ML^{-3})
a_{OB}^N	= mass of oxygen required per unit increase in bacteria biomass resulting from nutrient N (MM^{-1})
a_{OB}	= mass of oxygen required per unit increase in bacteria biomass due to endogenous respiration (MM^{-1})

then

$$\frac{dDO}{dt} = K_a(DO^S - DO) - a_{OB}^N\mu_{BN}^{\max}\left(\frac{N}{K_N + N}\right)B - a_{OB}\rho_B B$$

In the above three-equation model, note that the term $\mu_{BN}^{\max}\left(\frac{N}{K_N + N}\right)B$ equals

$a_{BN}\frac{dN}{dt}$ where a_{BN} is the mass of bacteria increase resulting from a unit mass of nutrient N (i.e., $a_{BN} = 1/a_{NB}$). Hence dB/dN is assumed constant and equal to -

a_{BN} . In addition, oxygen consumption in nutrient degradation is assumed proportional to dN/dt . Oxygen consumption in endogenous respiration is assumed constant per unit bacterial biomass.

If protozoa are to be considered, then

$$\begin{aligned}\frac{dB}{dt} &= \mu_{NB}^{\max} \left(\frac{N}{K_N + N} \right) B - a_{BP} \mu_{NB}^{\max} \left(\frac{B}{K_B + B} \right) P - \rho_B B \\ \frac{dP}{dt} &= \mu_P^{\max} \left(\frac{B}{K_B + B} \right) P - \rho_P P \\ \frac{dDO}{dt} &= K_a (DOS - DO) - a_{OB}^N \mu_{NB}^{\max} \left(\frac{N}{K_N + N} \right) B - a_{OB}^P \mu_P^{\max} \left(\frac{B}{K_B + B} \right) P \\ &\quad - a_{OB}(\rho_B) B - a_{OP}(\rho_P) P\end{aligned}$$

where

a_{BP}	= mass of bacteria consumed per unit increase in protozoa mass (MM^{-1})
$\mu_P^{\max}$	= maximum growth (increase) of protozoa mass resulting from a unit mass feeding on bacteria per unit time ($MM^{-1}T^{-1}$) (for a given temperature)
K_B	= bacterial half-saturation concentration for protozoa growth (ML^{-3})
a_{OB}^P	= mass of oxygen required per unit increase in protozoa mass resulting from feeding on bacteria N (MM^{-1})
a_{OP}	= mass of oxygen required per unit increase in protozoa mass due to endogenous respiration (MM^{-1})

10.6 Many equations for predicting stream temperature use Eulerian coordinates. The actual behavior of the stream temperature is more easily demonstrated if Lagrangian coordinates (i.e., time of flow t rather than distance X) are used. Assuming insignificant dispersion, the “time-of-flow” rate of temperature change of a water parcel as it moves down-stream is

$$\frac{dT}{d\tau} = \frac{\lambda}{\rho c D} (T_E - T)$$

- Assuming that λ , D , and T_E are constant over interval of time of flow $t_2 - t_1$, integrate the equation above to derive the temperature T_1 at locations X_1 .
- Develop a model for predicting the temperature at a point in a non-dispersive stream downstream from multiple point sources (discharges) of heat.

(a) Let $\alpha = \lambda/\rho c D$

then
$$\int \frac{dT}{T_E - T} = \int \alpha d\tau = \alpha \tau$$

since $\int \frac{dx}{a+bx} = \frac{1}{b} \ln(a+bx) + c_0,$

$$\int \frac{dT}{T_E - T} = \ln(T_E - T) + c_0 = \alpha \tau$$

hence $T_E - T = c_1 e^{-\alpha \tau}$ or $T = T_E - c_1 e^{-\alpha \tau}$

if $\tau = \tau_1 = 0, T = T_1$ and if $\tau = \infty, T = T_E$, thus $c_1 = T_E - T_1$

and $T_\tau = T_E - (T_E - T_1) \exp[-\alpha(\tau - \tau_1)]$

or $T_2 = T_E - (T_E - T_1) \exp[-\lambda(\tau_2 - \tau_1) / \rho c D]$

(b) Let q_i be the wastewater flow discharge into the stream at site i , H_i the temperature of the flow and Q_i the stream flow just above the site i having temperature T_i^- . After mixing, the temperature at site i will be T_i^+ . For any two adjacent sites i and j , i upstream of j , the initial temperature in the reach from i to j will be:

$$T_i^+ = (T_i^- Q_i + H_i q_i) / (Q_i + q_i)$$

Let τ_{ij} be the time of flow of Q_j ($Q_j = Q_i + q_i$) from site i to j . Then given parameters T_E, λ, ρ, c , and D for the reach from i to j .

$$T_j^- = T_E - (T_E - T_i^+) \exp(-\lambda \tau_{ij} / \rho c D)$$

$$T_j^+ = (T_j^- Q_j + H_j q_j) / (Q_j + q_j) \quad \text{and so on.}$$

10.7 Consider three well-mixed bodies of water that have the following constant volumes and freshwater inflows:

Water Body	Volume (m ³)	Flow (m ³ /s)	Displacement Time
1	3×10^{12}	3×10^3	3.17 years
2	3×10^8	3×10^2	11.6 days
3	3×10^4	3	2.8 hours

The first body is representative of the Great Lakes in North America, the second is characteristic in size to the upper New York harbor with the summer flow of the Hudson River, and the third is typical of a small bay or cove. Compute the time required to achieve 99% of the equilibrium concentration, and that concentration, of a substance having an initial concentration, and that concentration, of a substance having an initial concentration of 0 (at time = 0) and an input of N (MT^{-1}) for each of the three water bodies. Assume that the decay-rate constant K is 0, 0.01, 0.05, 0.25, 1.0, and 5.0 days⁻¹ and compare the results.

From the appropriate equations in the text chapter (10)

$$t_{0.99} = \frac{-V}{Q + KV} \ln(1 - 0.99) = \frac{4.61V}{Q + KV}$$

$$c(t_{0.99}) = \frac{N}{Q + KV} \{1 - \exp[+\frac{V}{Q + KV} \ln(1 - 0.99)(\frac{Q + KV}{V})]\}$$

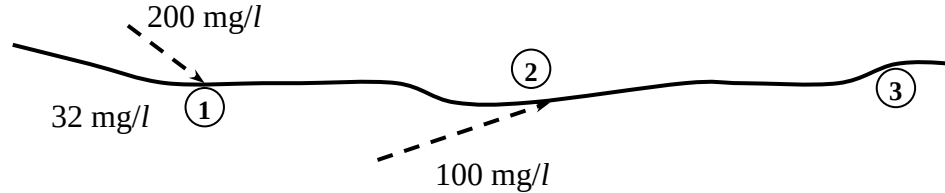
$$= \frac{N}{Q + KV} \{1 - (1 - 0.99)\} = \frac{N}{Q + KV} (0.99)$$

Water Body	K	$t_{0.99}$	$c(t_{0.99})$
1	0	5.3×10^4	$3.8 \times 10^{-3}N$
	0.01	4.6×10^2	$3.3 \times 10^{-5} N$
	0.05	91.9	$6.6 \times 10^{-6} N$
	0.25	18.4	$1.3 \times 10^{-6} N$
	1.00	4.6	$3.3 \times 10^{-7} N$
	5.00	0.9	$6.6 \times 10^{-8} N$
2	0	53.3	$3.8 \times 10^{-2}N$
	0.01	47.8	$3.4 \times 10^{-2} N$
	0.05	33.8	$2.4 \times 10^{-2} N$
	0.25	13.7	$9.8 \times 10^{-3} N$
	1.0	4.2	$3.0 \times 10^{-3} N$
	5.0	0.9	$6.5 \times 10^{-4} N$
3	0	0.53	3.8 N
	0.01	0.53	3.8 N
	0.05	0.53	3.8 N
	0.25	0.52	3.7 N
	1.0	0.48	3.4 N
	5.0	0.34	2.4 N

If N is expressed in g/day , Q and KV are expressed in m^3/day , then $c(t)$ will be in $g/m^3 = mg/l$.

Thus for each water body i , $Q_i = \text{Flow in } m^3/sec \text{ times } 8.64 \times 10^4 \text{ sec/day}$. The solutions for $t_{0.99}$ (days) and $c(t_{0.99})$ (mg/l) are given above; assuming N is expressed in $10^6 g/day$.

10.8 Consider the water pollution problem as shown in the Figure below. There are two sources of nitrogen, 200 mg/l at site 1 and 100 mg/l at site 2, going into the river, whereas the nitrogen concentration in the river just upstream of site 1 is 32 mg/l.



The unknown variables are the fraction of nitrogen removal at each of those sites that would achieve concentrations no greater than 20 mg/l and 25 mg/l just upstream of site 2 and at site 3 respectively, at a total minimum cost. Let those nitrogen removal fractions be X_1 and X_2 . Assuming unit costs of removal as \$30 and \$20 at site 1 and site 2 respectively, the model can be written as:

$$\begin{array}{ll} \text{Minimize} & 30 X_1 + 20 X_2 \\ \text{Subject to:} & 200(1 - X_1)0.25 + 8 \leq 20 \\ & 200(1 - X_1)0.15 + 100(1 - X_2)0.60 + 5 \leq 25 \\ & X_1 \leq 0.9, X_2 \leq 0.9 \end{array}$$

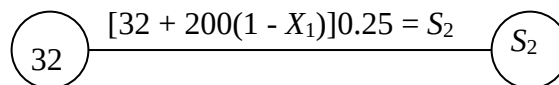
Another way to write the two quality constraints of this model is to define variables S_i ($i=1,2,3$) as the concentration of nitrogen just upstream of site i . Beginning with a concentration of 32 mg/l just upstream of site 1, the concentration of nitrogen just upstream of site 2 will be

$$[32 + 200(1 - X_1)]0.25 = S_2 \text{ and } S_2 \leq 20.$$

The concentration of nitrogen at site 3 will be

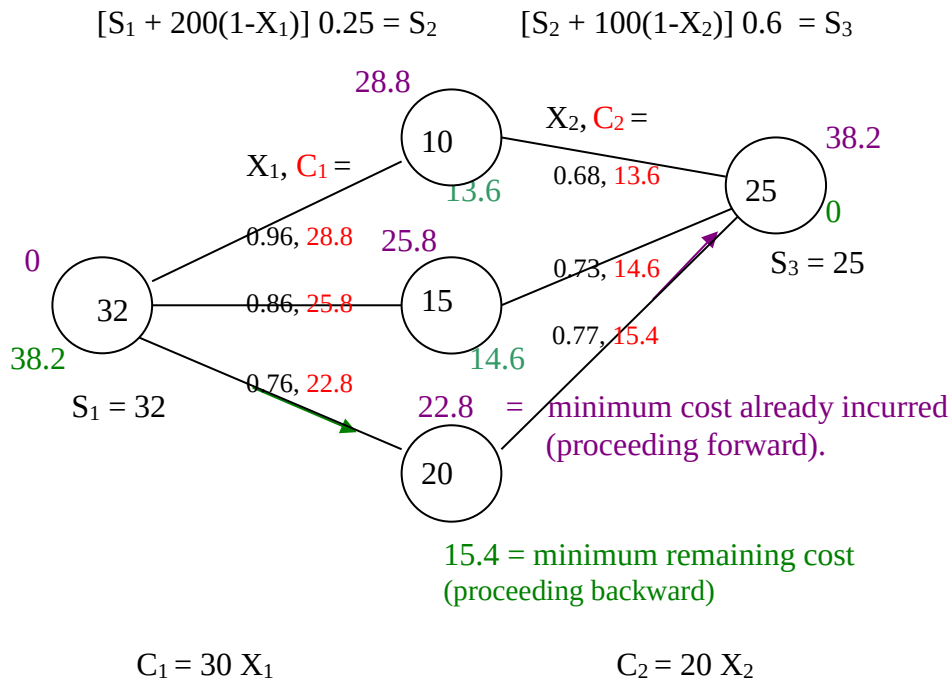
$$[S_2 + 100(1 - X_2)]0.60 = S_3 \text{ and } S_3 \leq 25.$$

This makes the problem easier to solve using discrete dynamic programming. The nodes or states of the network can be discrete values of S_i , the concentration of nitrogen in the river at sites i (just upstream of sites 1 and 2 and at site 3). The links represent the decision variable values, X_i that will result in the next discrete concentration, S_{i+1} given S_i . The stages i are the different source sites or river reaches. A section of the network in stage 1 (reach from site 1 to site 2) will look like:



So if S_2 is 20, X_1 will be 0.76; if S_2 is 15, X_1 will be 0.86. For S_2 values of 10 or less X_1 must exceed 0.90 and these values are infeasible. The cost associated with the link or decision will be $30 X_1$.

Setup the dynamic programming network. It begins with a single node representing the state (concentration) of 32 mg/l just upstream of site 1. It will end with a single node representing the state (concentration) 25 mg/l. The maximum possible state (concentration value just upstream of site 2 must be no greater than 20 mg/l. You can use discrete concentration values in increments of 5 mg/l. This will be a very simple network. Find the least-cost solution using both forward and backward moving dynamic programming procedures. Please show your work.



10.9 Identify a three alternative sets (feasible solutions) of storage lagoon volume capacities V and corresponding land application areas A and irrigation volumes Q_{2t} in each month t within a year that satisfy a 10 mg/l maximum $\text{NO}_3\text{-N}$ content in the drainage water of a land disposal system. In addition to the data listed below, assume that the influent nitrogen n_{1t} is 50 mg/l each month, with 10% ($\alpha = 0.1$) of the nitrogen in organic form. Also assume that the soil is a well-drained silt loam containing 4500 kg/ha of organic nitrogen in the soil above the drains. The soil has a monthly drainage capacity d of 60 cm and has a field capacity moisture content M of 10 cm. Maximum plant nitrogen uptake values $N_t^{\max}$ are 35 kg/ha during April through October, and 70 kg/ha during May through September. Finally, assume that

because of cold temperatures, no wastewater irrigation is permitted during November through March. December, January, and February's precipitation is in the form of snow and will melt and be added to the soil moisture inventory in March.

The following three solutions were obtained by trail and error. When the equations are solved exactly, slightly different concentrations may be obtained.

#1 $V = 4000 \times 10^3 \text{ m}^3$
 $A = 385.3 \text{ ha}$

t :	1	2	3	4	5	6	7	8-12
$IR_t \text{ (cm)}$:	30	30	30	30	30	30	10	0
$Q_{2t} (10^3 \text{ m}^3)$:	1156	1156	1156	1156	1156	1156	385	0

Max. $n_{2t} = 9.3 \text{ mg/l}$ in month 5 (Aug)

#2 $V = 5000 \times 10^3 \text{ m}^3$
 $A = 252.4 \text{ ha}$

t :	1	2	3	4	5	6	7	8-12
$IR_t \text{ (cm)}$:	50	50	50	50	50	20	20	0
$Q_{2t} (10^3 \text{ m}^3)$:	1262	1262	1262	1262	1262	505	505	0

Max. $n_{2t} = 8.7 \text{ mg/l}$ in month 5 (Aug)

#3 $V = 4500 \times 10^3 \text{ m}^3$
 $A = 305 \text{ ha}$

t :	1	2	3	4	5	6	7	8-12
$IR_t \text{ (cm)}$:	50	50	50	50	50	20	20	0
$Q_{2t} (10^3 \text{ m}^3)$:	1262	1262	1262	1262	1262	505	505	0

Max. $n_{2t} = 9.6 \text{ mg/l}$ in month 5 (Aug)

10.10 Consider the problem of estimating the minimum total cost of waste treatment in order to satisfy quality standards within a stream. Let the stream contain seven homogenous reaches r , reach $r = 1$ being at the upstream end and reach $r = 7$ at the downstream end. Reaches $r = 2$ and 4 are tributaries entering the mainstream at the beginning of 1, 3, 5, 6, and 7. Point sources of BOD enter the stream at the beginning of reaches 1, 2, 3, 4, 6, and 7. Assuming that at least 60% BOD removal is required at each discharge site, solve for the least-cost solution given the data in the accompanying table.

Can you identify more than one type of model to solve this problem? How would this model be expanded to specifically include both carbonaceous BOD and nitrogenous BOD and nonpoint waste discharges?

Reach No.	Design BOD Load (mg/l)	Present % Removal Load	ANNUAL COSTS OF VARIOUS DESIGN BOD REMOVALS			
			60%	75%	85%	90%
1	248	67	0	22,100	77,500	120,600
2	408	30	630,000	780,000	987,000	1,170,000
3	240	30	210,000	277,500	323,000	378,000
4	1440	30	413,000	523,000	626,000	698,000
6	2180	30	500,000	638,000	790,000	900,000
7	279	30	840,000	1,072,000	1,232,500	1,350,000

Reach No.	Time of Flow (days)	Waste Water Discharge (10 ³ m ³ /day)	Entering Reach Flow (10 ³ m ³ /day)	Total Reach Flow (10 ³ m ³ /day)	DO saturation conc. (mg/l)	Maximum allowable DO Deficit (mg/l)	DO Deficit of waste-water (mg/l)	DO Conc at begi -nning of Reach (mg/l)	BOD Conc at begi -nning of Reach (mg/l)	Av. Deoxgn rate constant for Reach (days ⁻¹)	Reaeration-Rate constant (days-1)
1	0.235	19	5,129	5,148	10.20	3.20	1.0	9.50	1.66	0.31	1.02
2	1.330	140	4,883	5,023	9.95	2.45	1.0	8.00	0.68	0.41	0.60
3	1.087	30	10,171	10,201	9.00	2.00	1.0	?	?	0.36	0.63
4	2.067	53	1,120	1,173	9.70	3.75	1.0	9.54	1.0	0.35	0.09
5	0.306	0	11,374	11,374	9.00	2.50	--	?	?	0.34	0.72
6	1.050	98	11,374	11,472	8.35	2.35	1.0	--	--	0.35	0.14
7	6.130	155	11,472	11,627	8.17	4.17	1.0	--	--	0.30	0.02

One method for modeling this problem is to compute values of coefficients d_{ij} and b_{ij}^a for converting BOD and DO deficit mass (MT⁻¹) at site i to DO deficit concentrations at site j can be calculated. Alternatively one can omit the flow terms in those coefficients so that they define the mass of DO deficit at site j rather than the concentration. If t_{ij} is the time of flow between i and j ,

$$d_{ij} = \frac{K_d}{K_a - K_d} (e^{-K_d \tau_{ij}} - e^{-K_a \tau_{ij}})$$

$$b_{ij}^a = e^{-K_a \tau_{ij}}$$

for the appropriate deoxygenation and reaeration constants K_d and K_a between i and j . These are calculated and are listed below for this problem. Let each site i be the upstream end of reach i , and let site 8 be the end of reach 7

		Site j				
Site i	b_{ij}	3	5	6	7	8
	1	0.79	0.40	0.32	0.27	0.24
	2	0.45	0.23	0.18	0.16	0.14
	3		0.50	0.40	0.35	0.31
	4		0.83	0.67	0.57	0.51
	5			0.80	0.69	0.61
	6				0.86	0.76
	7					0.88

		Site j				
Site i	d_{ij}	3	5	6	7	8
	1	0.06	0.24	0.25	0.38	0.64
	2	0.28	0.27	0.25	0.32	0.47
	3		0.23	0.24	0.38	0.67
	4		0.46	0.42	0.48	0.66
	5			0.09	0.33	0.78
	6				0.28	0.79
	7					0.78

Let W_{ik} be an unknown weight associated with a particular BOD removal percent k at site i (beginning of reach i) so that $\sum_k W_{ik} = 1$ for all i . Let C_{ik} be the cost and

R_k be the fraction (percent/100) BOD removal as shown in the above table for each site i . Define W_i as the BOD concentration (mg/l) in the wastewater flow Q_i^W (10^6 m³/day) before treatment at site i , and BOD_i^0 and D_i the BOD and DO deficit concentrations (mg/l) in the entering flows Q_i^0 (10^6 m³/day) at sites 1, 2, and 4. The problem is to find the cost-effective BOD removal fractions P_i at each waste source site that satisfy the stream quality standards $D_j^{\max}$, at each site j having a streamflow Q_j (10^6 m³/day).

The problem can be modeled as:

$$\begin{aligned}
 &\text{Minimize} && \sum_i \sum_k C_{ik} W_{ik} \\
 &\text{Subject to} && \sum_k W_{ik} = 1 && i = 1, 2, 3, 4, 6, 7 \\
 &&& \sum_k R_k W_{ik} = P_i && i = 1, 2, 3, 4, 6, 7 \\
 &&& D_j = \frac{1}{Q_j} \left[\sum_k W_i Q_i^W (1 - P_i) d_{ij} + \sum_{i=1,2,4} (BOD_i^0 Q_i^0 d_{ij} + D_i^0 Q_i^0 b_{ij}^a) \right] && \forall j \\
 &&& D_j \leq D_j^{\max} && \forall j
 \end{aligned}$$

An alternate model involves a mass balance of BOD and DO deficit concentrations at each junction, and the use of coefficients b_i , d_i , and b_i^a for converting initial reach BOD and DO deficit concentrations (or masses) for each reach i .

Letting t_i be the time of flow in reach i , for appropriate deoxygenation and reaeration rate constants K_d and K_a for each reach i :

$$b_i = e^{-K_d \tau_i}$$

$$d_i = \frac{K_d}{K_a - K_d} (e^{-K_d \tau_i} - e^{-K_a \tau_i})$$

$$b_i^a = e^{-K_a \tau_i}$$

Reach	b_i	d_i	b_i^a
1	0.93	0.06	0.79
2	0.58	0.28	0.45
3	0.68	0.23	0.50
4	0.49	0.46	0.83
5	0.90	0.09	0.80
6	0.69	0.28	0.86
7	0.16	0.78	0.88

Let Q_i^W be the wastewater flow, Q_i^T be the tributary flow, and Q_i^- be the initial streamflow entering reach i . After mixing

$$\sum_W Q_i^W + \sum_T Q_i^- + Q_i^T = Q_i$$

These superscripts will also apply to the BOD and DO deficits in these flows. Plus (+) superscripts will denote end-of-reach concentrations.

$$\begin{aligned} \text{Minimize} \quad & \sum_k \sum_i C_{ik} W_{ik} \\ \text{Subject to} \quad & \sum_k W_{ik} = 1 \quad i = 1, 2, 3, 4, 6, 7 \\ & \sum_k R_k W_{ik} = P_i \quad i = 1, 2, 3, 4, 6, 7 \\ & B_i^W = W_i (1 - P_i) \quad i = 1, 2, 3, 4, 6, 7 \end{aligned}$$

$$B_i = (B_i^- Q_i^- + B_i^W Q_i^W) / Q_i \quad i = 1, 2, 4$$

$$D_i = (D_i^- Q_i^- + D_i^W Q_i^W) / Q_i \leq D_i^{\max} \quad i = 1, 2, 4$$

$$B_3 = (B_1^+ Q_1 + B_2^+ Q_2 + B_3^W Q_3^W) / Q_3$$

$$B_5 = (B_3^+ Q_3 + B_4^+ Q_4) / Q_5$$

$$D_5 = (D_3^+ Q_3 + D_4^+ Q_4) / Q_5 \leq 2.5$$

$$B_6 = (B_5^+ Q_5 + B_6^W Q_6^W) / Q_6$$

$$D_6 = (D_5^+ Q_5 + D_6^W Q_6^W) / Q_6 \leq 2.35$$

$$B_7 = (B_6^+ Q_6 + B_7^W Q_7^W) / Q_7$$

$$D_7 = (D_6^+ Q_6 + D_7^W Q_7^W) / Q_7 \leq 4.17$$

$$B_i^+ = B_i b_i \quad i = 1, 2, 3, 4, 6, 7$$

$$D_i^+ = B_i d_i + D_i b_i^a \leq D_i^{\max} \quad i = 1, 2, 3, 4, 6, 7$$

The solutions to these two models are essentially the same since each reach flow is assumed constant.

Minimum total cost = 372×10^3

BOD removal fractions and DO concentrations

Site i	P_i	DO _{i} (end of reach)
1	0.67	9.52
2	0.75	8.10
3	0.60	7.86
4	0.90	6.12
5	---	7.75
6	0.90	6.32
7	0.78	4.00

Carbonaceous and nitrogenous BOD could be included in these models by replacing each BOD term with two terms, one for BOD^C and another for BOD^N . Two removal fractions, R_k^C and R_k^N would have to be known for each k , so that

$$\sum_k R_k^C W_{ik} = P_i^C \quad \text{and} \quad \sum_k R_k^N W_{ik} = P_i^N$$

Each term $W_i Q_i^W (1 - P_i) d_{ij}$ in the first model would be replaced by

$W_i^C Q_i^W (1 - P_i^C) d_{ij}^C + W_i^N Q_i^W (1 - P_i^N) d_{ij}^N$, and for the second model each $W_i(1 -$

$W_i)$ constraint would become two involving $W_i^C (1 - P_i^C)$ and $W_i^N (1 - P_i^N)$ to

define both B_i^{WC} and B_i^{WN} and B_i^{C+} and B_i^{N+} would have to be calculated, and the deficit equations would have to include both of these concentrations, e.g.

$$D_i^+ = B_i^C d_i^C + B_i^N d_i^N + D_i b_i^a$$

To include nonpoint uniform BOD or DO deficit mass inputs from runoff into any reach, one must estimate the mass input per unit length along the reach per unit

time. Let BOD_i and DOD_i ($ML^{-1}T^{-1}$) be these mass inputs along a reach i . Then the total mass from this runoff at some distance X within the reach, for BOD, is

$$B_{X_i}^R = \int_0^X (BOD_i e^{-K_d x/U}) dx = \frac{UBOD_i}{K_d} (1 - e^{-K_d X/U})$$

and for DO deficit:

$$\begin{aligned} D_{X_i}^R &= \int_0^X [BOD_i (\frac{K_d}{K_a - K_d}) (e^{-K_d x/U} - e^{-K_a x/U}) + DOD_i e^{-K_a x/U}] dx \\ &= U \{ BOD_i [\frac{1 - e^{-K_a X/U}}{K_a} - \frac{e^{-K_d X/U} - e^{-K_a X/U}}{K_a - K_d}] + DOD_i (\frac{1 - e^{-K_a X/U}}{K_a}) \} \end{aligned}$$

If X is the reach length, then these masses should be considered just as if they were point sources at the reach end in order to calculate downstream total BOD or DO deficit concentrations.

10.11 Discuss what would be required to analyze flow augmentation alternatives in Exercise 10.8. How would the costs of flow augmentation be defined and how would you modify water quality models to include flow augmentation alternatives?

Flow augmentation would require reservoir storage or some type of water diversion upstream of the area in which water quality is to be managed. The cost of the structures required for flow augmentation, plus the net loss of benefits from other water uses, if, should be compared to the costs of alternative measures for water quality control. A more discussion of flow augmentation operating policies and reliabilities, and the analysis of flow augmentation benefits and costs are contained in Dorfman et al. (1972).

The model developed in exercise 10.8 could be used to examine flow augmentation alternatives for discrete values of design stream flows, since most of the rate coefficients, and certainly the flows themselves, are dependent on the extent of flow augmentation. To include flow as an explicit decision variable would result in a highly nonlinear set of simultaneous equations.

10.12 Develop a dynamic programming model to estimate the least-cost number, capacity, and location of artificial aerators to ensure meeting minimum allowable dissolved oxygen standards where they would otherwise be violated during an extreme low-flow design condition in a non-branching section of a stream. Show how wastewater treatment alternatives, and their costs, could also be included in the dynamic programming model.

First one could identify particular sites along a river where artificial aeration is feasible and likely to be of value. Let j be an index denoting these site locations, which will be the stages of the system, and let D_j , the oxygen deficit concentration just upstream of site j , be the state of the system. Finally, define $F_j(D_j)$ as the minimum total cost of aerators at the remaining sites downstream of site j required to meet the DO standards, given D_j at site j .

Assuming the last site is site J , clearly

$$F_J(D_J) = 0 \quad \forall D_J$$

For the next-to-last site, called $J - 1$,

$$F_{j-1}(D_{j-1}) = \min_{P_{j-1}} [C_{j-1}(P_{j-1}) + F_j(D_j)] \quad \forall D_{j-1}$$

$$O_{j-1} = O_{j-1}(D_{j-1}, P_{j-1})$$

$$D_j = B_{j-1}d_{j-1,j} + (D_{j-1}Q_{j-1} - O_{j-1})b_{j-1,j}^a$$

$$D_j \leq DO_j^S - DO_j^{\min}$$

In general:

$$F_j(D_j) = \min_{P_j} [C_j(P_j) + F_{j+1}(D_{j+1})]$$

$$O_j = O_j(D_j, P_j)$$

$$D_{j+1} = B_j d_{j,j+1} + (D_j Q_j - O_j) b_{j,j+1}^a$$

$$D_{j+1} \leq DO_{j+1}^S - DO_{j+1}^{\min}$$

The above recursive equations assume site j is just upstream of site $j + 1$, and that all variables and parameters are defined. After solving for the optimal P_j 's and examining the DO profile, it may be appropriate to see if the model should again be solved for a different set of locations j . If the treatment of BOD is also a known variable, say R_j at site j (or any non-aerator site), the recursive equation become a function of two state variables, the BOD and DO deficit concentrations B_j and D_j .

$$F_j(B_j, D_j) = \min_{R_j, P_j} [C_j^P(P_j) + C_j^R(R_j) + F_{j+1}(B_{j+1}, D_{j+1})]$$

$$R_j^{\min} \leq R_j \leq R_j^{\max}$$

$$D_{j+1} = [B_j + W_j(1 - R_j)]d_{j,j+1} + (D_j Q_j + O_j)b_{j,j+1}^a$$

$$D_{j+1} \leq DO_{j+1}^{SAT} - DO_{j+1}^{\min}$$

$$O_j = O_j(D_j, P_j)$$

$$B_{j+1} = [B_j + W_j(1 - R_j)]b_{j,j+1}$$

where R_j is the fraction of BOD removed at site j and $b_{j,j+1}$ is the BOD concentration (ML^{-3}) at site $j + 1$ per unit of BOD input (MT^{-1}) at site j .

10.13 Using the data provided, find the steady-state concentrations C_t of a constituent in a well-mixed lake of constant volume $30 \times 10^6 \text{ m}^3$. The production N_{ti} of the constituent occurs at three sites i , and is constant in each of four seasons in the year. The required fractions of constituent removal P_i at each site i are to be set so that they are equal at all sites i and the maximum concentration in the lake in each period t must not exceed 20 mg/l .

Period, t	Days in Period	Flow, Q_t ($10^3 \text{ m}^3/\text{day}$)	Constituent Decay Rate, constant, K_t (days^{-1})
1	100	90	0.02
2	80	150	0.03
3	90	200	0.05
4	95	120	0.04

Constituent Discharge Site, i	Constituent Production (kg/day)
1	38000
2	25000
3	47000

This problem has many solutions, ranging from all P_i equal to the minimum of all $P_i^{\max}$ to a minimum allowable treatment $P^{\min}$ found by minimizing $P^{\min}$ where $C^{\max}$ is 20 mg/l or 0.02 kg/m^3 , and all $P_i = P^{\min}$. This is an LP problem:

$$\begin{aligned}
 &\text{Minimize} && P^{\min} \\
 &\text{Subject to:} && \frac{N_t}{Q_t + KV}(1 - e_t) + C_{t-1}e_t \leq C_t^{\max} = 0.02 && \forall t \\
 & && N_t = (38000 + 25000 + 47000)(1 - P^{\min}) && \forall t \\
 & && && (\text{kg/m}^3) \\
 & && 0 \leq P^{\min} \leq \min_i(P_i^{\max}) && \forall t \\
 & && && (\text{kg/day})
 \end{aligned}$$

The unknown variables are $P^{\min}$ and all C_t . The known parameters e_t are

$$e_t = \exp[-\Delta t(\frac{Q_t}{V} + Q_t)] \quad \forall t$$

Assume the optimal $P^{\min}$ is less than all $P_i^{\max}$ which are less than 1. The values of the terms in the above model are listed below. These values assume as optimal value of $P^{\min} = 0.87$, and hence a total discharge N_t of $110,000 (0.13) = 14,300$ kg/day. These values are rounded off appropriately and hence are not the exact values of the LP solution.

<i>Period</i>	<i>Δt days</i>	<i>Q_t + K_tV m³/day</i>	<i>e_t</i>	<i>1 - e_t</i>
1	100	690,000	0.100	0.900
2	80	1050,000	0.061	0.939
3	90	1700,000	0.006	0.994
4	95	1320,000	0.015	0.985

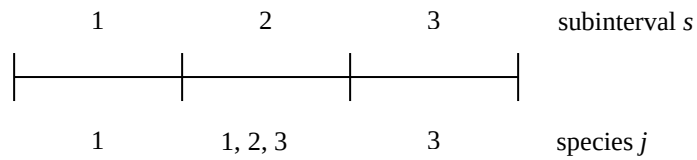
<i>Period</i>	<i>C_t kg/m³</i>	=	<i>N_t(1 - e_t)/(Q_t + K_tV)</i>	+	<i>C_{t-1}(e_t)</i>
1	0.020	=	14300(0.900)/ 690,000	+	(0.011)(0.100)
2	0.014	=	14300(0.939)/1050,000	+	(0.020)(0.061)
3	0.008	=	14300(0.994)/1700,000	+	(0.014)(0.006)
4	0.011	=	14300(0.985)/1320,000	+	(0.008)(0.015)

- 10.14 Suppose that the solution of a model such as that used in Exercise 10.13, or measured data, indicated that for a well-mixed portion of a saltwater lake, the concentrations of nitrogen ($i = 1$), phosphorus ($i = 2$), and silicon ($i = 3$) in a particular period t were 1.1, 0.1, and 0.8 mg/l, respectively. Assume that all other nutrients required for algal growth are in abundance. The algal species of concern are three in number and are denoted by $j = 1, 2, 3$. The data required to estimate the probable maximum algal bloom biomass concentration are given in the accompanying table. Compute this bloom potential for all k_i and k equal to 0, 0.8, and 1.0

Parameter (Algae Species Index j)	PARAMETER VALUE		
	1	2	3
a_{1j} = mg N/mg dry wt of algae j	0.04	0.01	0.20
a_{2j} = mg P/mg dry wt of algae j	0.06	0.02	0.10
a_{3j} = mg Si/mg dry wt of algae j	0.08	0.01	0.03
D_j = mortality and grazing rate constant (days ⁻¹)	0.6	0.4	0.20
d_j = mortality rate constant, (days ⁻¹)	0.3	0.1	0.10
v = extinction reduction rate constant for dead algae, (days ⁻¹)	0.07	0.07	0.07
$\eta_j^{\max}$ = max. extinction coef. (m ⁻¹)	0.07	0.07	0.10
$\eta_j^{\min}$ = min. extinction coef. (m ⁻¹)	0.01	0.03	0.03
η_j = increase in extinction coef. per unit increase in mg/l (g/m ³) of dry wt of species j (m ² /g)	0.05	0.164	0.04
Nutrient Index i :	1	2	3
Nutrient:	N	P	Si
μi = mineralization rate constant, (days ⁻¹)	0.02	0.69	0.62
Extinction Subinterval s :	1	2	3
Extinction coefficient range:	0.01-0.03	0.03-0.07	0.07-0.10
Algae species $j \in S_s$:	1	1, 2, 3	3
Extinction coefficient without algae = $\eta_0 = 0.01 \text{ m}^{-1}$			

Note: Since there are three extinction subintervals, there are three models to solve for each value of $k_i = k$

Solution:



$$B^{\max} = \max_s B^s$$

$$B^s = \max \sum_{j \in S_s} x_j$$

Subject to:

$$N_i = w_i + \sum_{j \in S_s} a_{ij} [1 + k_i (\frac{D_j}{u_i})] x_j \quad \forall i$$

$$\eta_s^{\min} \leq \eta_0 + \sum_{j \in S_s} \eta_j [1 + k(\frac{D_j}{v})] x_j \leq \eta_s^{\max}$$

$$x_i \geq 0 \quad \forall j$$

$$w_i \geq 0 \quad \forall I$$

Note that grazing can be omitted by substituting d_j in place of D_j in the above constraints, if one assumes that mortality due to grazing will not result in a release of nutrients, and does not significantly decrease the bloom potential.

Using d_j (in place of D_j) equal to 0.3, 0.1 and 0.1 ($j = 1, 2, 3$ respectively) and for the three sets of $k_i = k$ values, the following algal biomass concentrations x_j , in mg/l, are obtained.

$k_i = k$	s	x_1	x_2	x_3	B^s	B^{ma}
0	1	0.40	-	-	0.40	1.31
	2	0.77	0	0.54	1.31	
	3			infeasible		
0.8	1	0.09	-	-	0.09	0.70
	2	0	0	0.70	0.70	
	3	-	-	0.35	0.35	
1.0	1	0.08	-	-	0.08	0.87
	2	0	0	0.62	0.62	
	3	-	-	0.87	0.87	

Using D_j equal to 0.6, 0.4, and 0.2 for $j = 1, 2, 3$ respectively, the solutions are:

$k_i = k$	s	x_1	x_2	x_3	B^s	B^{ma}
0	1	0.40	-	-	0.40	1.31
	2	0.77	0	0.54	1.31	
	3			infeasible		
0.8	1	0.05	-	-	0.05	0.46
	2	0	0	0.46	0.46	
	3	-	-	0.45	0.45	
1.0	1	0.04	-	-	0.04	0.39
	2	0	0	0.39	0.39	
	3	-	-	0.38	0.38	

Chapter 11

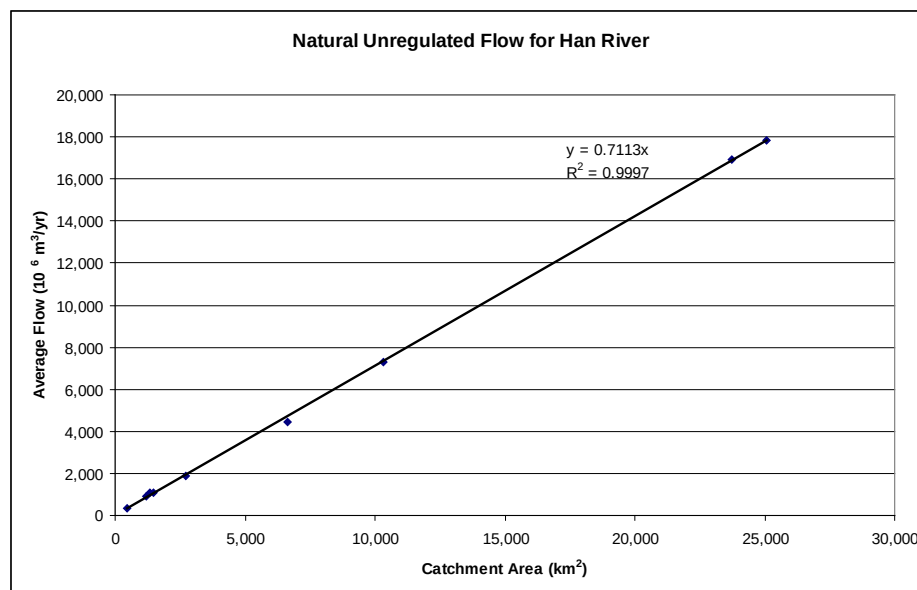
11.1 Using the following information pertaining to the drainage area and discharge in the Han River in South Korea, develop an equation for predicting the natural unregulated flow at any site in the river, by plotting average flow as a function of catchment area. What does the slope of the function equal?

Gage Point	Catchment Area (km ²)	Average Flow (10 ⁶ m ³ /yr)
First bridge of the Han River	25,047	17,860
Pal Dang dam	23,713	16,916
So Yang dam	2,703	1,856
Chung Ju dam	6,648	4,428
Yo Ju dam	10,319	7,300
Hong Chun dam	1,473	1,094
Dal Chun dam	1,348	1,058
Kan Yun dam	1,180	926
Im Jae dam	461	316

A plot of the average flow Q^S as a function of the catchment area A^S upstream of each site s shows an essentially constant slope. This indicates the flow runoff per unit catchment area is approximately constant throughout the watershed.

A plot of average flow vs. catchment area is located on the following page. From the graph, it is clear that a linear relationship can be fitted to the data with a very high R^2 value (0.997). The line passes through the origin and has a slope of 0.7113.

$$\text{Flow} = 0.7113 * \text{Area}$$



11.2 In watersheds characterized by significant elevation changes, one can often develop reasonable predictive equations for average annual runoff per hectare as a function of elevation. Describe how one would use such a function to estimate the natural average annual flow at any gage in a watershed which is marked by large elevation changes and little loss of water from stream channels due to evaporation or seepage.

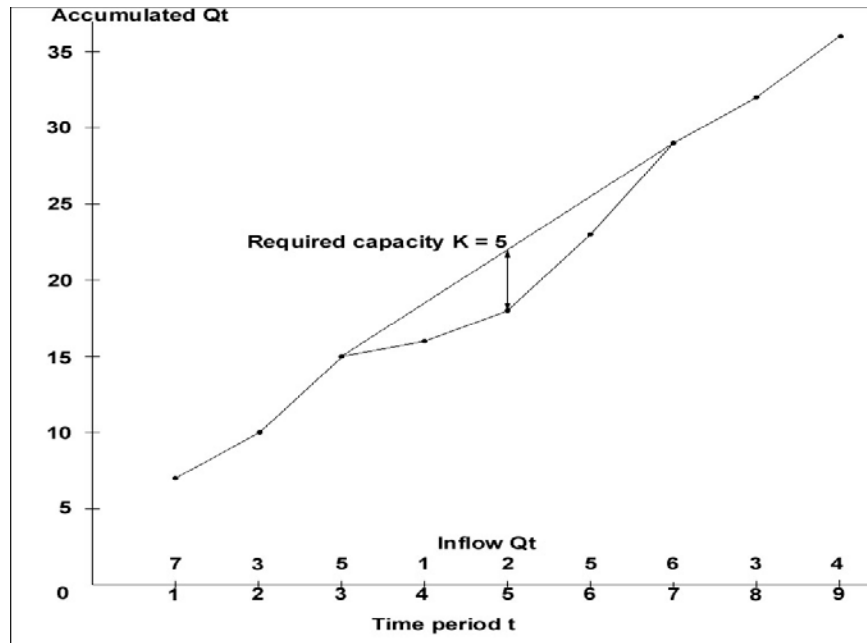
Based on various average annual flows Q^S at sites s throughout the watershed, a plot of Q^S/A^S and the corresponding elevation E^S of the sites s will define Q^S/A^S as a function of elevation, $f(E^S)$. (If such a function is not well defined it may be necessary to consider Q^S/A^S as some function of E^S and E^S , as in some watersheds in Romania where it appears Q^S/A^S is a function of $E^S/\sqrt{A^S}$. To define seasonal or monthly flows $Q_t^{S'}$ at some ungaged site s' , assume for some known seasonal flow Q_t^S at a representative gage site

$$Q_t^{S'} = Q_t^S \frac{Q^{S'}}{Q^S} = Q_t^S \frac{A^{S'} f(E^{S'})}{A^S f(E^S)}$$

This equation may or may not be a very sufficient predictor. The best approach is to discuss this problem with a local hydrologist and develop the most appropriate relationship for the particular site.

11.3 Compute the storage yield function for a single reservoir system by the mass diagram and modified sequent peak methods given the following sequences of annual flows: (7, 3, 5, 1, 2, 5, 6, 3, 4). Next assume that each year has two distinct hydrologic seasons, one wet and the other dry, and that 80% of the annual inflow occurs in season $t = 1$ and 80% of the yield is desired in season $t = 2$. Using the modified sequent peak method, show the increase in storage capacity required for the same annual yield resulting from within-year redistribution requirements.

Mass diagram analysis showing the estimated required storage capacity (K) for given inflow $Q_t = \{7, 3, 5, 1, 2, 5, 6, 3, 4\}$ and release $R_t = 4$ in each period



Sequent peak analysis showing the estimated required storage capacity (K) for given inflow $Q_t = \{7, 3, 5, 1, 2, 5, 6, 3, 4\}$ and release $R_t = 4$ in each period t .

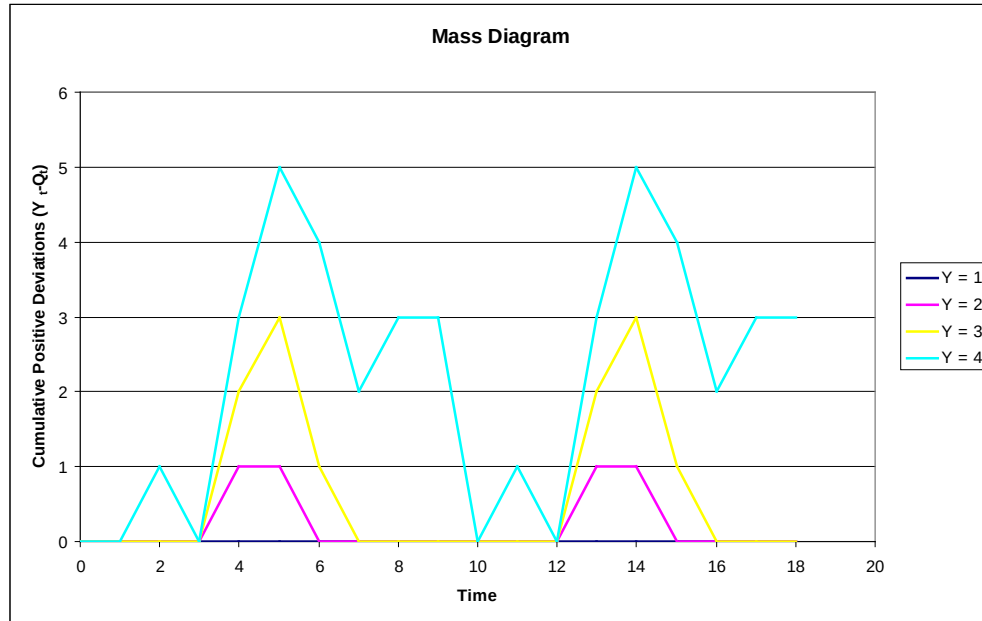
					$K_0 = 0$
Year (t)	[$R_t - Q_t + K_{t-1}$]	=			K_t
1	4 - 7 + 0	=			0
2	4 - 3 + 0	=			1
3	4 - 5 + 1	=			0
4	4 - 1 + 0	=			3
5	4 - 2 + 3	=			5 = Required K
6	4 - 5 + 5	=			4
7	4 - 6 + 4	=			2
8	4 - 3 + 2	=			3
9	4 - 4 + 3	=			3

Assume that each year has wet season = 1 and dry season = 2 and 80 % of inflow occurs in season 1, while 80 % of the yield is desired in season 2. Using the modified sequent peak method to show the increase in storage capacity required for the same annual yield resulting from within-year redistribution requirements.

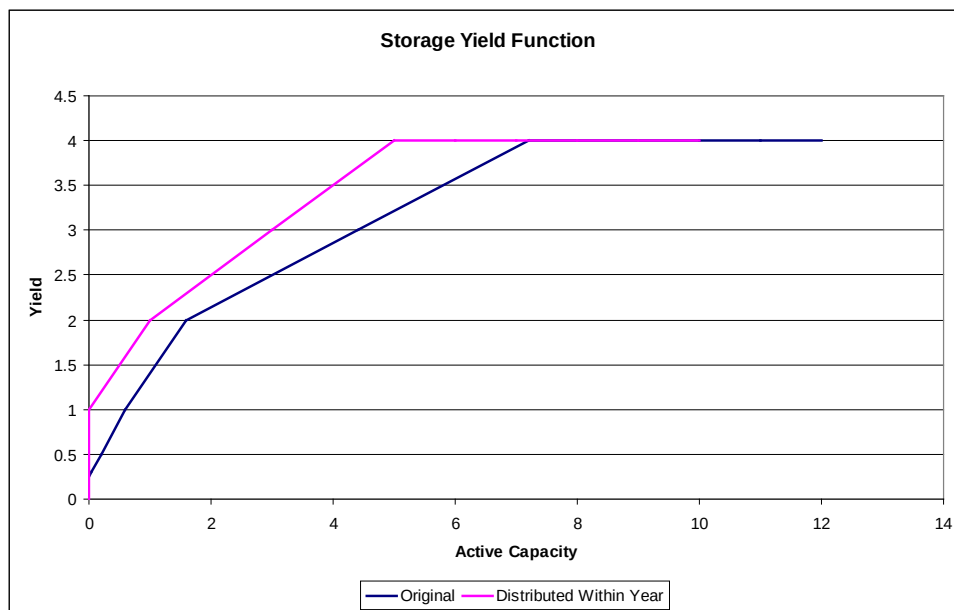
Year (t)	Season	[Rt -	Qt +	Kt-1]	=	Kt
1	1	0.8	5.6	0.0		0.0
	2	3.2	1.4	0.0		1.8
2	1	0.8	2.4	1.8		0.2
	2	3.2	0.6	0.2		2.8
3	1	0.8	4.0	2.8		0.0
	2	3.2	1.0	0.0		2.2
4	1	0.8	0.8	2.2		2.2
	2	3.2	0.2	2.2		5.2
5	1	0.8	1.6	5.2		4.4
	2	3.2	0.4	4.4		7.2= required K
6	1	0.8	4.0	7.2		4.0
	2	3.2	1.0	4.0		6.2
7	1	0.8	4.8	6.2		2.2
	2	3.2	1.2	2.2		4.2
8	1	0.8	2.4	4.2		2.6
	2	3.2	0.6	2.6		5.2
9	1	0.8	3.2	5.2		2.8
	2	3.2	0.8	2.8		5.2

(1) Reservoir Yield	(2) Capacity Required for Annual Yield	(3) Capacity Required for Two-Season Yield	(4) Required Capacity Increase
1	0	0.6	0.6
2	1	1.6	0.6
3	3	4.4	1.4
3.5	4	5.8	1.8
4	5	7.2	2.2

Use of the mass diagram approach resulted in the same storage yield function. A plot of the cumulative positive deviation vs. time, used in the mass diagram approach is shown below. The required storage for a given yield is the maximum cumulative positive deviation between yield and inflow.



A graph of the new storage yield function, with the yield distributed within the year follows. From the plot, it is clear that more storage is required to achieve the yield when it is distributed within the year.



11.4 Write two different linear programming models for estimating the maximum constant reservoir release or yield Y given a fixed reservoir capacity K , and for estimating the minimum reservoir capacity K required for a fixed yield Y . Assume that there are T time periods of historical flows available. How could these models be used to define a

storage capacity-yield function indicating the yield Y available from a given capacity K ?

Maximize $Y = \text{Yield (unknown)}$

Subject to: $S_t + Q_t - Y \leq S_{t+1} \quad \forall t, T + 1 = 1$
 $S_t \leq K = \text{Capacity (fixed)} \quad \forall t$

A storage-yield function can be defined by solving this LP model for different yields $Y \leq \text{mean flow } (1/T)(\sum_t^T Q_t)$. All variables are non-negative.

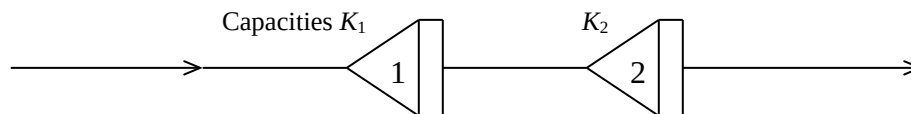
Minimize $K = \text{capacity (unknown)}$

Subject to: $S_t + Q_t - Y (\text{fixed}) \leq S_{t+1} \quad \forall t, T + 1 = 1$
 $S_t \leq K = \text{Capacity (unknown)} \quad \forall t$

A storage-yield function can be defined by solving this LP model for different values of K given different yields Y . All variables are non-negative.

11.5 (a) Construct an optimization model for estimating the least-cost combination of active storage capacities, K_1 and K_2 , of two reservoirs located on a single stream, used to produce a reliable constant annual flow or yield (or greater) downstream of the two reservoirs. Assume that the cost functions $C_s(K_s)$ at each reservoir site s are known and there is no dead storage and no evaporation. (Do not linearize the cost functions; leave them in their functional form.) Assume that 10 years of monthly unregulated flows are available at each site s .

(b) Describe the two-reservoir operating policy that you would incorporate into a simulation model to check the solution obtained from the optimization model.



(a)

natural unregulated flows	Q_{ty}^1	Q_{ty}^2
initial storage volumes	S_{ty}^1	S_{ty}^2
reservoir releases	R_{ty}^1	Y (+ excess)
Constant yield	Y	

$$\begin{array}{ll}
\text{Minimize} & C_1(K_1) + C_2(K_2) \\
\text{Subject to:} & S_{ty}^1 + Q_{ty}^1 - R_{ty}^1 = S_{t+1,y}^1 \quad \forall t, y \\
& S_{ty}^1 \leq K_1 \quad \forall t, y \\
& S_{ty}^2 + Q_{ty}^2 - Q_{ty}^1 + R_{ty}^1 - Y \leq S_{t+1,y} \quad \forall t, y
\end{array}$$

The constant yield Y is fixed (known), as are all natural flows Q_{ty}^S . When $t = 12$, $t + 1 = 1$ and y is $y + 1$. If $y = 10$, $y + 1 = 1$. All variables K_s , S_{ty}^S , S_{ty}^S are non-negative. Do you need $S_{ty}^2 \leq K_2$?

(b) Release water from the downstream reservoir, $s = 2$, first and release water from the upstream reservoir, $s = 1$, only if the lower reservoir is empty. This minimizes the chance of excess releases (spills) from the lower reservoir that are wasted, and cannot be stored for later use.

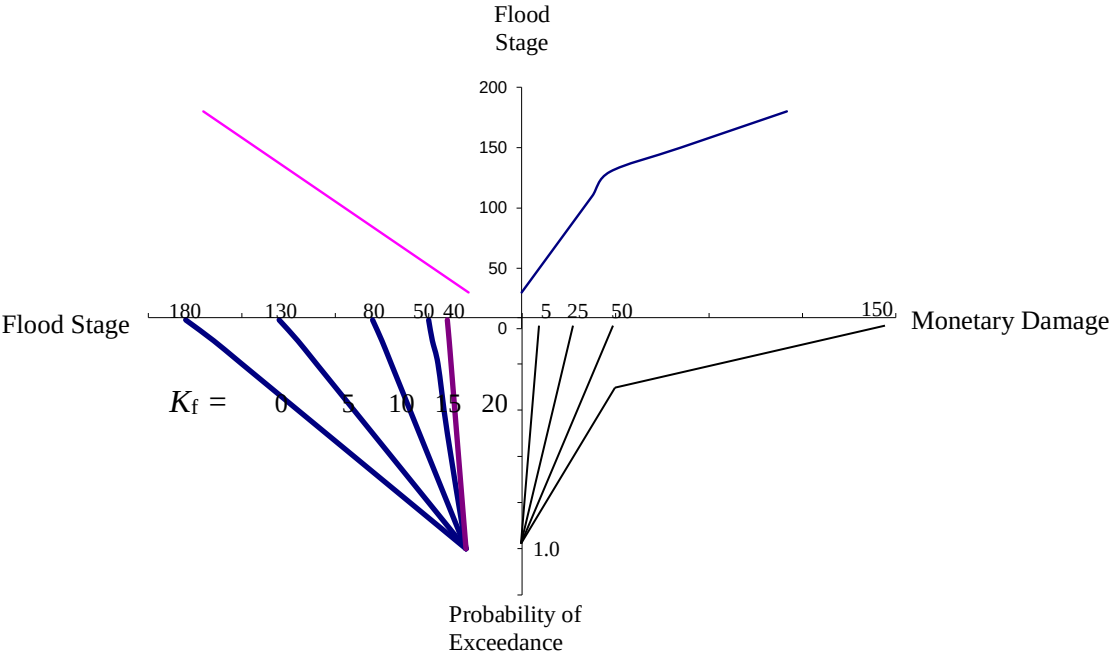
11.6 Given the information in the accompanying tables, compute the reservoir capacity that maximizes the net expected flood damage reduction benefits less the annual cost of reservoir capacity.

Reservoir Capacity	FLOOD STAGE FOR FLOOD OF RETURN PERIOD T					Annual Capacity Cost
	$T = 1$	$T = 2$	$T = 5$	$T = 10$	$T = 100$	
0	30	105	150	165	180	10 ^a
5	30	80	110	120	130	25
10	30	55	70	75	80	30
15	30	40	45	48	50	40
20	30	35	38	39	40	70

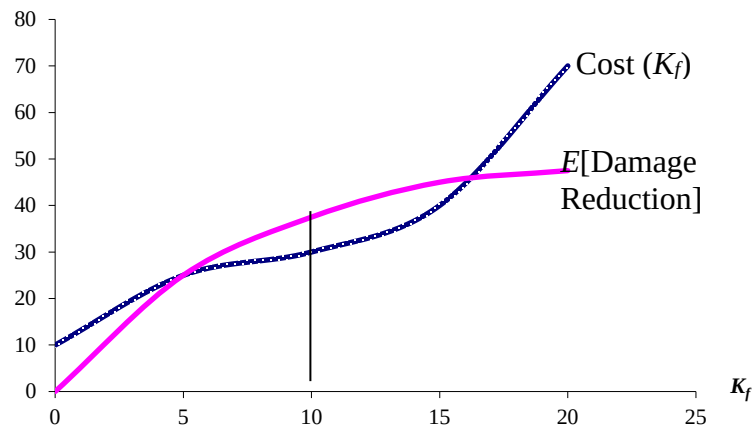
^a10 is fixed cost if capacity > 0 ; otherwise, it is 0.

Flood Stage	Cost of Flood Damage
30	0
50	10
70	20
90	30
110	40
130	50
150	90
180	150

This is a simplified example all the flow exceedance probability distributions are linear, making the calculation of their areas easy. The axis in the 4- way graph for flood stage can begin at 0 or at 30; the results will be the same.

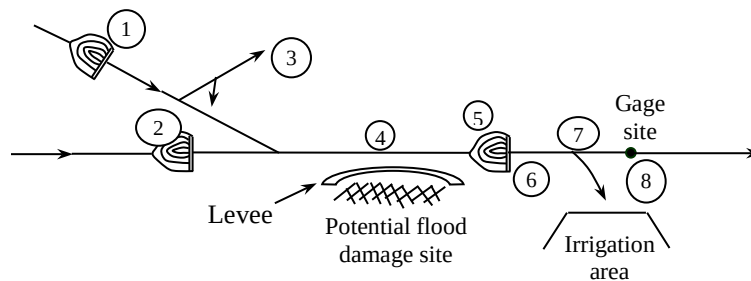


Capacity K_f	$E[\text{Flood Damage}]$	$E[\text{Damage Reduction}]$	Cost (K_f)	$E[\text{Net Benefits}]$
0	50	0	0	0
5	25	25	25	0
10	12.5	37.5	30	7.5 = maximum
15	50.	45.0	40	5
20	2.5	47.5	70	-22.5



Maximum expected annual net benefits result when $K_f = 10$, or approximately so if the net benefit function is smoothed rather than piecewise linear.

11.7 Develop a deterministic, static, within-year model for evaluating the development alternatives in the river basin shown in the accompanying figures. Assume that there are $t = 1, 2, 3, \dots, n$ within-year periods and that the objective is to maximize the total annual net benefits in the basin. The solution of the model should define the reservoir capacities (active + flood storage capacity), the annual allocation targets, the levee capacity required to protect site 4 from a T -year flood, and the within-year period allocations of water to the uses at sites 3 and 7. Clearly define all variables and functions used, and indicate how the model would be solved to obtain the maximum-net-benefit solution.



Site	Fraction of Gage Flow	
1	0.	Potential reservoir for water supply,
2	0.3	Potential reservoir for water supply,
3	0.15	Diversion to a use, 60% of allocation
4	-	Existing development, possible flood
5	0.6	Potential reservoir for water supply,
6	-	Hydropower; plant factor = 0.30
7	0.9	Potential diversion to an irrigation
8	1.0	Gage site

For simplicity assume no evaporation losses or dead storage requirements. Omitting the appropriate subscripts t for time periods and s for site, let T , K , D , E , and P be the target, reservoir capacity, deficit, excess, and power plant capacity variables, respectively. Let Q_t and R_t be the natural streamflows and reservoir releases, and S_t be the initial reservoir storage volumes in period t . K_f will denote the flood storage capacity at site 2 that will contain a peak flow of QS and QR is the downstream channel flood flow capacity. The relationship between QS and K_f is defined by the function $\kappa(QS)$. The unregulated design flood peak flow for which protection is required is QN . KWH will be the kilowatt hours of energy, H will be net storage head, h_t the hours in a period t . The variable q will be the water supply allocation. Benefit functions will be $B(\cdot)$, $L(\cdot)$ will denote loss functions and $C(\cdot)$ will denote the cost functions. If $t = T$, then $T + 1 = 1$.

Objective: Maximize

$$\begin{aligned}
& B_3(T_3) - \sum_{t=1}^T L_{3t}(D_{3t}) - C_1(K_1) - C_2(K_2) - C_4(QR_4) \\
& - C_5(K_5) + B_5(T_5) - \sum_{t=1}^T [L_{5t}^D(D_{5t}) + L_{5t}^E(E_{5t})] \\
& + B_6(T_6) - \sum_{t=1}^T [L_{6t}^D(D_{6t}) + L_{6t}^E(E_{6t})] - C_6(P_6) + B_7(T_7)
\end{aligned}$$

Subject to:

$$\begin{aligned}
\text{Site 1} \quad & S_t^1 + 0.1Q_t^8 - R_t^1 = S_{t+1}^1 & \forall t & \quad T + 1 = 1 \\
& S_t^1 \leq K_1 & \forall t &
\end{aligned}$$

$$\begin{aligned}
\text{Site 2} \quad & S_t^2 + 0.3Q_t^8 - R_t^2 = S_{t+1}^2 & \forall t & \\
& S_t^2 + K_f^2 \leq K_2 & \forall t \text{ in flood season} & +1 \\
& S_t^2 \leq K_2 & \forall t \text{ not in flood season} &
\end{aligned}$$

$$\begin{aligned}
\text{Site 3} \quad & q_t^3 \leq 0.15Q_t^8 - 0.1Q_t^8 + R_t^1) & \forall t & \\
& q_t^3 = T_3 - D_{3t} + E_{3t} & \forall t &
\end{aligned}$$

$$\begin{aligned}
\text{Site 4} \quad & QN^4 = QS_4 + QR_4 \\
& K_f \geq \kappa(QS)
\end{aligned}$$

$$\begin{aligned}
\text{Site 5} \quad & S_t^5 + (0.6 - 0.1 - 0.3)Q_t^8 + R_t^1 + R_t^2 - 0.4q_t^3 - R_t^5 - q_t^5 = S_{t+1}^5 & \forall t & \\
& S_t^5 \leq K_5 & \forall t &
\end{aligned}$$

$$\begin{aligned}
& S_t^5 = T_5 - D_{5t} + E_{5t} & \forall t & \\
\text{Site 6} \quad & KWH_t^6 = kq_t^6 H_t^5 & \forall t &
\end{aligned}$$

$$\begin{aligned}
KWH_t^6 &= T_6 - D_{6t} + E_{6t} & \forall t \\
KWH_t^6 &\leq 0.3h_t P_6 & \forall t \\
\text{Site 7} \quad q_t^7 &\leq (0.9 - 0.6)Q_t^8 + R_t^5 + q_t^6 & \forall t \\
q_t^7 &\geq f_t^7 T_7 = \text{specified fraction of season target } \forall t \text{ in growing} \\
&\quad \text{season}
\end{aligned}$$

11.8 List the potential difficulties involved when attempting to structure models for defining:

- (a) Water allocation policies for irrigation during the growing season.
 - (b) Energy production and capacity of hydroelectric plants.
 - (c) Dead storage volume requirements in reservoirs.
 - (d) Active storage volume requirements in reservoir.
 - (e) Flood storage capacities in reservoirs.
 - (f) Channel improvements for flood damage reduction.
 - (g) Evaporation and seepage losses from reservoirs.
 - (h) Water flow or storage targets using long-run benefits and short- run loss functions.
- (a) Appropriate allocation in each period t depends on previous allocations, and on future allocations in growing season. Data are not always available to determine effects of marginal changes in allocations as a function of previous allocations. Also benefits derived are a function of each allocation with in growing period as together they influence crop yield at the end of period. It is difficult to capture this relationship, even if known, in linear models.
 - (b) Energy production is a nonlinear no separable function of unknown flow and storage heads. The cost of power capacity depends upon the storage heads as well as on the kilowatts of capacity.
 - (c) Dead storage volumes are generally increasing from sediment deposition throughout the life of reservoir. It is not always obvious which value to assume in a static model. The inclusion of dead storage in models may also require some trial and error solutions so that reservoir capacities are not in any solution if only equal to dead storage (or mixed integer programming can be used).
 - (d) Active storage requirements may include that needed for over-year as well as within-year water supply storage needs. This may result in numerous continuity and capacity constraints, especially if hydrologic uncertainty is considered.
 - (e) The effectiveness of flood storage capacity depends on the actual storage volume in addition to K_f . Non-linear flood routing requires trial and error solutions coupled with simulation studies and assumptions regarding the location and intensity of the design flood rainfall. Potential flood reduction benefits are hard to estimate. Non-structural alternatives should also be considered.

- (f) Flood peaks downstream of any channel improvements will be affected by this improvement, and this effect is not considered in the proposed approach to incorporating flood control into multipurpose models. These downstream impacts may be detrimental.
- (g) Evaporation is a nonlinear function of storage volume, and hence can only be approximated in LP models. Seepage data are often poor or unavailable.
- (h) Benefit and loss functions are difficult to define for some uses. The loss functions are also dependent on the target, and may change over time if individuals adapt to changing conditions and to actual allocations that differ from target allocations. If because of linearity restrictions computed losses exceed total benefits, this may be unrealistic and in error. Hence calculated optimal targets may be non-optimal.

11.9 Assume that demand for water supply capacity is expected to grow as $t(60 - t)$, for t in years. Determine the minimum present value of construction cost of some subset of water supply options described below so as to always have sufficient capacity to meet demand over the next 30 years. Assume that the water supply network currently has no excess capacity so that some project must be built immediately. In this problem, assume project capacities are independent and thus can be summed. Use a discount factor equal to $\exp(-0.07 t)$. Before you start, what is your best guess at the optimal solution?

Project Number	Construction Cost	Capacity
1	100	200
2	115	250
3	190	450
4	270	700

To solve ignore clearly inferior combinations of projects such as {1, 2, 4} because either constructing 1 then 4 or 2 then 4 or 1, 2 and then 3 would always be cheaper. Note, given existing capacity, the next facility must be built at time

$$\tau(Z) = 30 - \sqrt{900 - Z}.$$

Set X	Optimal Sequence	Discounted Cost C(X)	Total Capacity Z
1	1	100	200
2	2	115	250
3	3	190	450
4	4	270	700
{1, 2}	2, 1	188.0	450
{1, 3}	3, 1	244.1	650
{2, 3}	3, 2	252.2	700
{1, 4}	4, 1	303.0	900
{2, 4}	4, 2	307.9	950
{1, 2, 3}	3, 2, 1	285.1	900

Clearly the least-cost solution that gives a capacity of 900 is the sequence 3, 2, 1. Note that 3, 1, 2 (which is not listed above) has a discounted cost of 286.7. This is less than the cost of the second best solution above.

Sample calculation. For the set {2, 3}

$$\text{Cost of 2, 3 is } 115 + (e^{-0.07[30 - \sqrt{900 - 250}]})190 = 253.61$$

$$\text{Cost of 3, 2 is } 190 + (e^{-0.07[30 - \sqrt{900 - 450}]})115 = 252.17$$

So that 3, 2 has a lower discounted cost than 2, 3.

Minimum cost of {1, 2, 3} is minimum of

$$\{1,2\} + 3 = 187.95 + (190)e^{-0.07\tau(450)} = 290.66$$

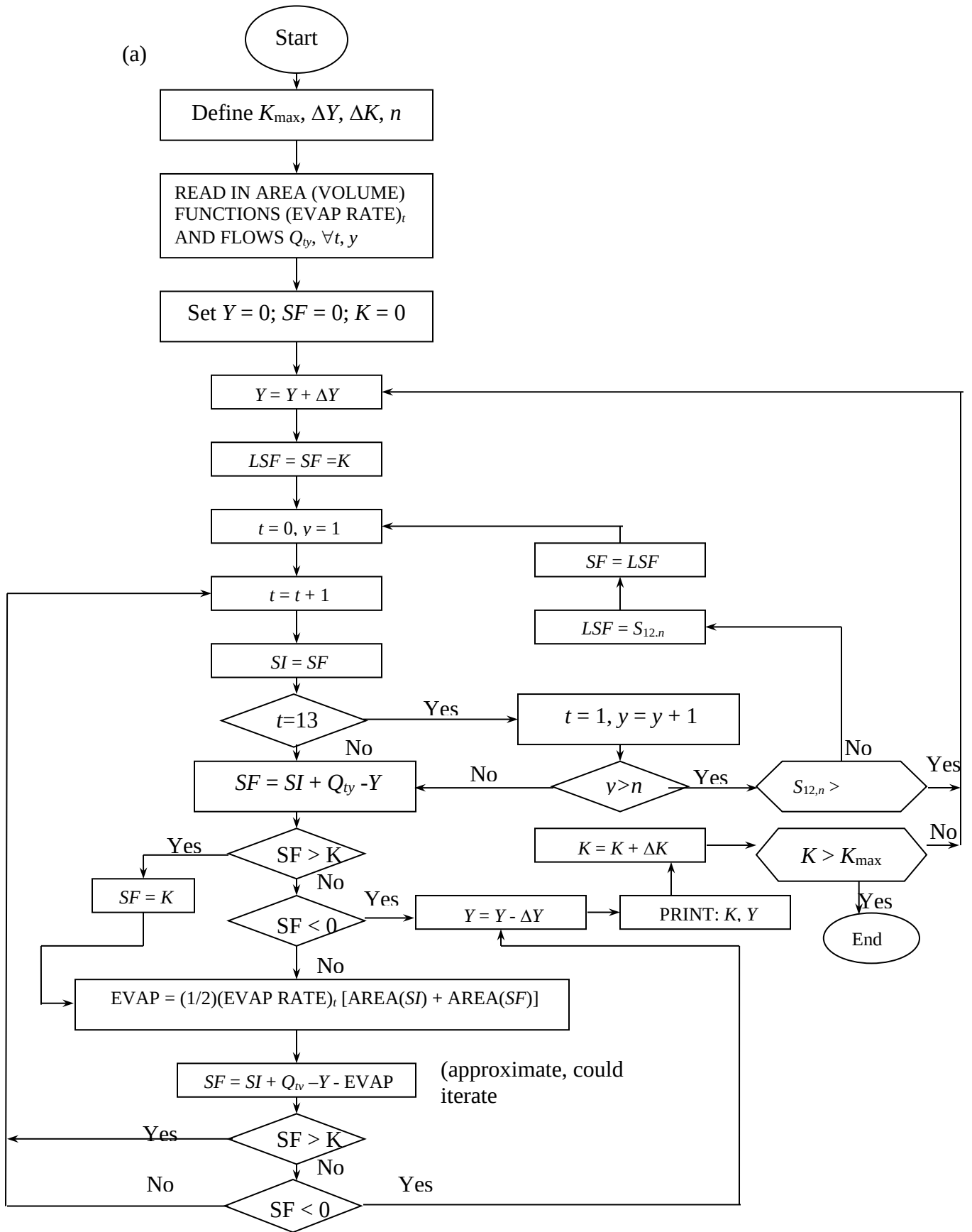
$$\{1,3\} + 2 = 244.06 + (115)e^{-0.07\tau(650)} = 286.65$$

$$\{2,3\} + 1 = 252.17 + (100)e^{-0.07\tau(700)} = 285.12$$

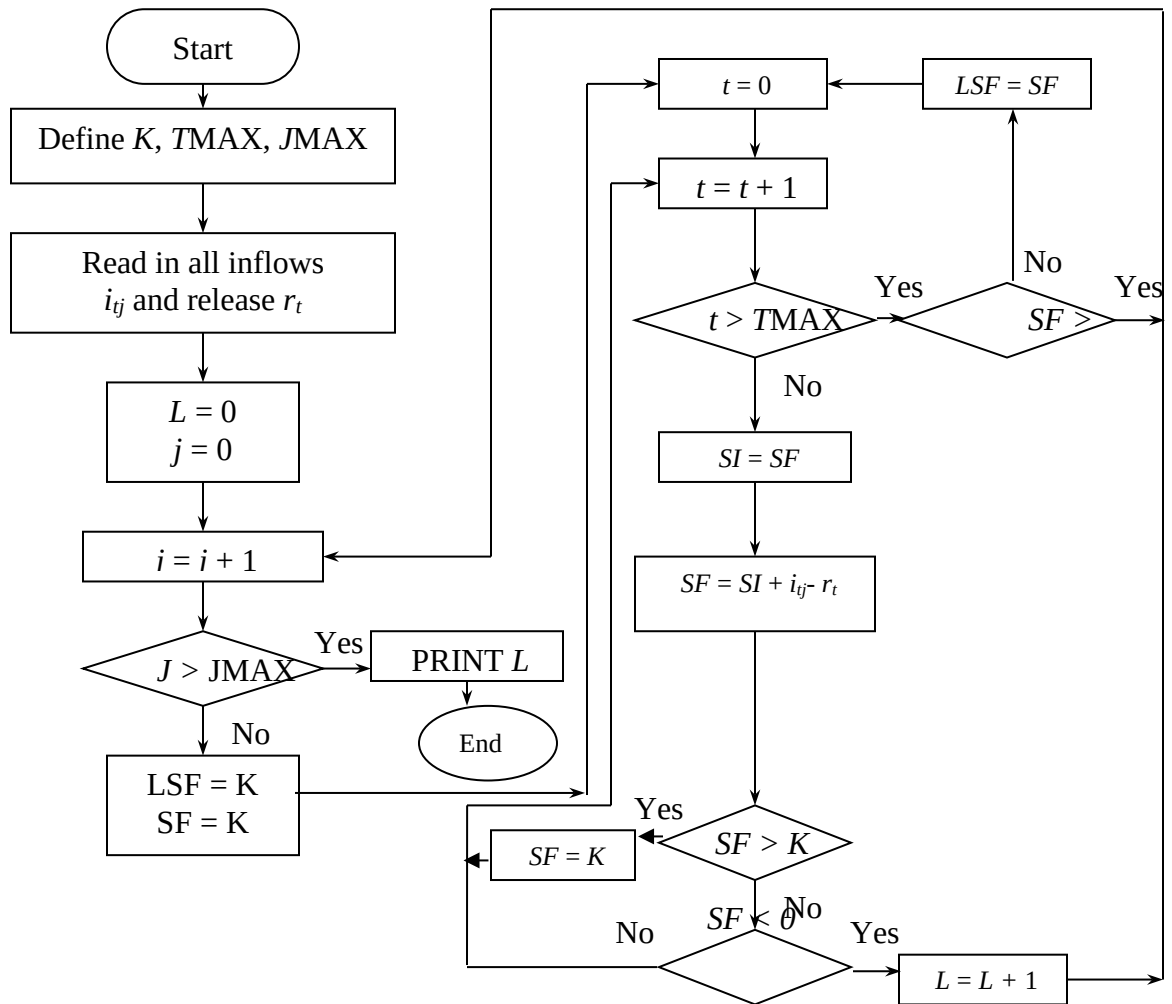
11.10 (a) Construct a flow diagram for a simulation model designed to define a storage-yield function for a single reservoir given known inflows in each month t for n years. Indicate how you would obtain a steady-state solution not influenced by an arbitrary initial storage volume in the reservoir at the beginning of the first period. Assume that evaporation rates (mm per month) and the storage volume/surface area functions are known.

(b) Write a flow diagram for a simulation model to be used to estimate the probability that any specific reservoir capacity, K , will satisfy a series of known release demands, r_t , downstream given unknown future inflows, i_t . You need not discuss how to generate possible future sequences of streamflows, only how to use them to solve this problem.

The following diagram is one of many possible approaches. Note that a steady-state is required, i.e., a solution or simulation sequence that ends with the same or greater storage than initially assumed. The following model assumes a full reservoir (initial storage volume = capacity) at the beginning of each new capacity simulation and finds the maximum yield that insures a final storage LSF at the end of the simulation sequence no less than the initial storage at the beginning of the simulation. $K_{\max}$ is the maximum storage capacity to be considered.



(b) Let j be the index denoting a particular sequence of streamflows i_{tj} and L be a counter of the number of sequences in which one or more of the releases r_t could not be made during a sequence. As in (a), the final storage at the end of the sequence of inflows i_{tj} cannot be than the initial assumed storage. The approximate reliability of a capacity K will equal L divided by the total number of sequences $JMAX$. The magnitude and frequency of failure, while possibility of interest, could but will not be considered here. The following program merely computes the number of failures L .



- 11.11 (a) Develop an optimization model for finding the cost-effective combination of flood storage capacity at an upstream reservoir and channel improvements at a downstream potential damage site that will protect the downstream site from a pre-specified design flood of return period T . Define all variables and functions used in the model.

(b) How could this model be modified to consider a number of design floods T and the benefits from protecting the potential damage site from those design floods? Let BF_T be the annual expected flood protection benefits at the damage site for a flood having return periods of T .

(c) How could this model be further modified to include water supply requirements of A_t to be withdrawn from the reservoir in each month t ? Assume known natural flows Q_t^s at each site s in the basin in each month t .

(d) How could the model be enlarged to include recreation benefits or losses at the reservoir site? Let T^s be the unknown storage volume target and D_t^s be the difference between the storage volume S_t^s and the target T^s if $S_t^s - T^s > 0$, and E_t^s be the difference if $T^s - S_t^s > 0$. Assume that the annual recreation benefits $B(T^s)$ are a function of the target storage volume T^s and the losses $L^D(D_t^s)$ and $L^E(E_t^s)$ are associated with the deficit D_t^s and excess E_t^s storage volumes.

(a) Minimize:

$$C_f(K_f) + C_R(QR)$$

$$\left[\begin{array}{c} \text{annual cost of flood} \\ \text{storage capacity} \end{array} \right] + \left[\begin{array}{c} \text{annual cost of} \\ \text{channel improvement} \end{array} \right]$$

Subject to:

$$QN_T \leq QS_T + QR \quad (1)$$

$$\left[\begin{array}{c} \text{peak flood} \\ \text{flow of} \\ \text{contained} \\ \text{return period} \\ T \end{array} \right] \leq \left[\begin{array}{c} \text{portion} \\ \leq \text{produced by} \\ \text{upstream} \\ \text{storage} \end{array} \right] + \left[\begin{array}{c} \text{portion} \\ + \\ \text{in} \\ \text{channel} \end{array} \right]$$

$$K_f \geq K_{fT}(QS_T)$$

$$\left[\begin{array}{c} \text{flood storage} \\ \text{capacity} \\ \text{in reservoir} \end{array} \right] \geq \left[\begin{array}{c} \text{function of} \\ \text{flood peak} \\ \text{reduction at} \\ \text{damage site} \end{array} \right]$$

(b) One would subtract $\sum_T BF_t X_T$ from the objective function and replace (1)

$$X_T Q N_T \leq Q S_T + Q R \quad \forall T$$

and add

$$\sum_T X_T \leq 1 \quad X_T = \text{integer} \geq 0$$

(c) Add:

$$\begin{array}{ccccccc} S_t & + & Q_t & - & R_t & = & S_{t+1} \quad \forall t \\ \left[\begin{array}{c} \text{Initial} \\ \text{storage} \end{array} \right] & + & \left[\begin{array}{c} \text{inflow} \end{array} \right] & - & \left[\begin{array}{c} \text{release} \end{array} \right] & \left[\begin{array}{c} = \\ \end{array} \right] & \left[\begin{array}{c} \text{storage} \end{array} \right] \end{array}$$

Define K as total active plus flood control storage capacity at site s assuming no dead storage capacity,

$$S_t + K_f - K \leq 0 \quad \text{for all flood season } t$$

$$S_t^S - K^S \leq 0 \quad \text{for all non-flood season } t$$

Finally

$$R_t \geq A_t \quad \text{specified requirements} \quad \forall t$$

and replace objective function terms $C_f(K_f)$ with $C_K(K)$

(d) Add to the objective function

$$-B(T^S) + \sum_t \frac{1}{2} [L_t^D(D_t) + L_{t+1}^D(D_{t+1}) + L_t^E(E_t) + L_{t+1}^E(E_{t+1})]$$

for all t in recreational season

and add to the constraints

$S_t = T^s - D_t + E_t \quad \forall t \text{ of the recreation season plus the next } t.$

It should be obvious that computing the average loss, as done above, in each period t , is preferable to computing the loss based on the average storage volume, $(S_t + S_{t+1})/2$, in each period t .

11.12 Given the hydrologic and economic data listed below, develop and solve a linear programming model for estimating the reservoir capacity K , the flood storage capacity K_f , and the recreation storage volume target T that maximize the annual expected flood control benefits, $B_f(K_f)$, plus the annual recreation benefits, $B(T)$, less all losses $L^D(D_t)$ and $L^E(E_t)$ associated with deficits D_t or excesses E_t in the periods of the recreation season, minus the annual cost $C(K)$ of storage reservoir capacity K . Assume that the reservoir must also provide a constant release or yield of $Y = 30$ in each period t . The flood season begins at the beginning of period 3 and lasts through period 6. The recreation season begins at the beginning of period 4 and lasts through period 7.

Period t	1	2	3	4	5	6	7	8	9
Inflows to reservoir	50	30	20	80	60	20	40	10	70

$$B_f(K_f) = \begin{cases} 12 K_f & \text{if } K_f \leq 5 \\ 60 + 8(K_f - 5) & \text{if } 5 \leq K_f \leq 15 \\ 140 + 4(K_f - 15) & \text{if } K_f \geq 15 \end{cases}$$

$$C(K) = \begin{cases} 45 + 10 K & \text{if } 0 \leq K_f \leq 5 \\ 95 + 6(K - 5) & \text{if } 5 \leq K_f \leq 20 \\ 185 + 10(K - 20) & \text{if } 20 \leq K_f \leq 40 \\ 385 + 15(K - 40) & \text{if } K_f \geq 40 \end{cases}$$

$$B(T) = 9T$$

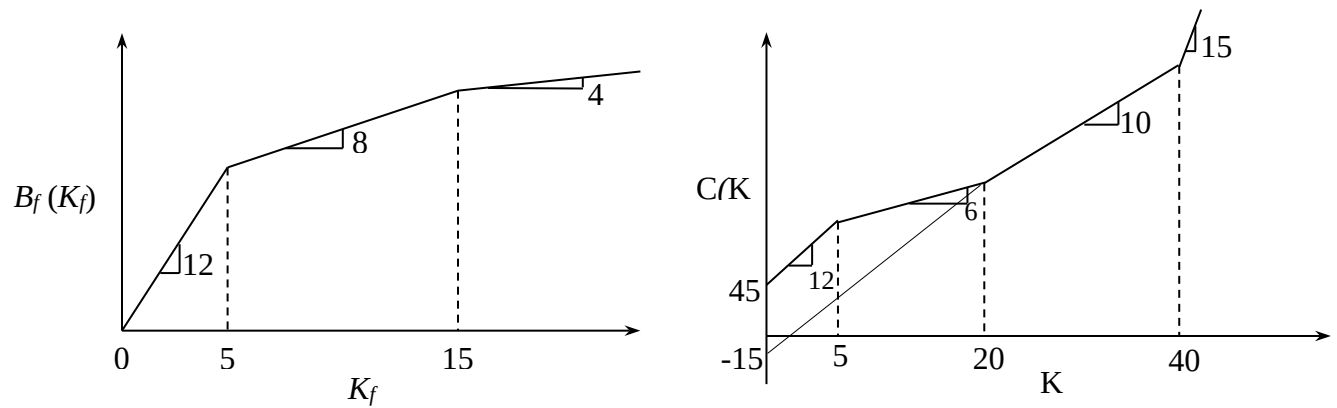
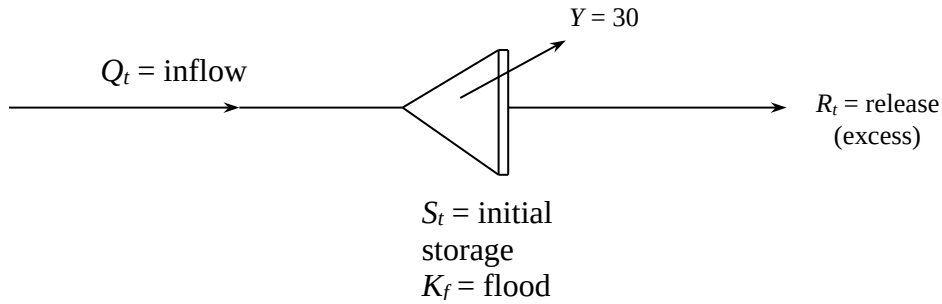
where T is a particular unknown value of -
reservoir storage

$$L^D(D_t) = 4D_t$$

where D_t is $T - S_t$ if $T \geq S_t$

$$L^E(E_t) = 2 E_t$$

where E_t is $S_t - T$ if $S_t \geq T$



$$\text{Maximize } [12K_f^1 + 8K_f^2 + 4K_f^3] + [9T - \frac{1}{2} \sum_{t=4}^7 (4(D_t + D_{t+1}) + 2(E_t + E_{t+1}))]$$

[Flood Storage Benefits] + [Recreation Benefits – Losses]

$$- (-15 + 10K_1 + 15K_2)$$

Modified Reservoir
Cost Function since
K will be ≥ 20

Subject to:

$$\begin{aligned} \text{Continuity} \quad & S_{t+1} - S_t + Y \leq Q_t \quad \forall t (t+1=1) \\ \text{Capacity} \quad & S_t + K_f - K \leq 0 \quad t = 3, 4, 5, 6, 7 \\ & S_t - K \leq 0 \quad t = 1, 2, 8, 9 \\ \text{Yield Requirement} \quad & Y = 30 \end{aligned}$$

Linearization

$$S_t - T + D_E - E_t = 0 \quad t = 4, 5, 6, 7, 8$$

$$K_f^1 + K_f^2 + K_f^3 - K_f = 0$$

$$K_1 + K_2 - K = 0$$

$$K_1 \leq 40$$

$$K_f^1 \leq 5$$

$$K_f^2 \leq 15 - 5 = 10$$

Partial Solution; $K = 25$; $K_f = 5$; $T = 20$.

11.13 The optimal operation of multiple reservoir systems for hydropower production presents a very nonlinear and often difficult problem.

Use dynamic programming to determine the operating policy that maximizes the total annual hydropower production of a two-reservoir system, one downstream of the other. The releases R_{1t} from the upstream reservoir plus the unregulated incremental flow ($Q_{2t} - Q_{1t}$) constitute the inflow to the downstream dam. The flows Q_{1t} into the upstream dam in each of the four seasons along with the incremental flows ($Q_{2t} - Q_{1t}$) and constraints on reservoir releases are given in the accompanying two tables:

Upstream Dam (flow in 10^6)			
Season t	Inflow Q_{1t}	Minimum Release	Maximum Release Through Turbines
1	60	20	90
2	40	30	90
3	80	20	90
4	120	20	90

Downstream Dam (flow in 10^6)			
Season t	Incremental Flow, ($Q_{2t} - Q_{1t}$)	Minimum Release	Maximum Release Through Turbines
1	50	30	140
2	30	40	140
3	60	30	140
4	90	30	140

Note that there is a limit on the quantity of water that can be released through the turbines

for energy generation in any season due to the limited capacity of the power plant and the desire to produce hydropower during periods of peak demand.

Additional information that affects the operation of the two reservoirs are the limitations on the fluctuations in the pool levels (head) and the storage-head relationships:

Data	Upstream Dam	Downstream Dam
Maximum head, $H^{\max}$	70 m	90 m
Minimum head, $H^{\min}$	30 m	60 m
Maximum storage Volume, $S^{\max}$	$150 \times 10^6 \text{ m}^3$	$400 \times 10^6 \text{ m}^3$
Storage-net head relationship	$H = H^{\max}(S/S^{\max})^{0.64}$	$H = H^{\max}(S/S^{\max})^{0.62}$

In solving the problem, discretize the storage levels in units of $10 \times 10^6 \text{ m}^3$. Do a preliminary analysis to determine how large a variation in storage might occur at each reservoir. Assume that the conversion of potential energy equal to the product $R_i H_i$ to electric energy is 70% efficient independent of R_i and H_i . In calculating the energy produced in any season t at reservoir i , use the average head during the season

$$\bar{H}_i = \frac{1}{2}[H_i(t) + H_i(t+1)]$$

Report your operating policy and the amount of energy generated per year. Find another feasible policy and show that it generates less energy than the optimal policy.

Show how you could use linear programming to solve for the optimal operating policy by approximating the product term $R_i \bar{H}_i$ by a linear expression.

(a) Using the recursion equation

$$f_t^n(S_{1t}, S_{2t}) = \max_{S_{1,t+1}, S_{2,t+1}} \{R_{1t} \bar{H}_{1t} + R_{2t} \bar{H}_{2t} + f_{t+1}^{n-1}(S_{1,t+1}, S_{2,t+1})\}$$

$$\text{where } R_{1t} = S_{1t} + Q_{1t} - S_{1,t+1} \quad 10^6 \text{ m}^3$$

$$R_{2t} = S_{2t} + (Q_{2t} - Q_{1t}) + R_{1t} - S_{2,t+1} \quad 10^6 \text{ m}^3$$

$$\bar{H}_{it} = \frac{1}{2}[H_i(t) + H_i(t+1)] \quad \text{meters}$$

$$H_i(t) = H^{\max} (S_{it} / S_i^{\max})^{\beta_i} \quad \text{meters}$$

$$KWH_{it} = (2730)(0.65)(R_{it} \bar{H}_{it}) \quad \text{kwh}$$

subject to restrictions on releases through turbines and storage and capacity limitations.

Solution:

Season t	Q_{1t}	S_{1t}	$\bar{H}_{1t}$	R_{1t}	10^6 KWH_{1t}	$(Q_{2t} - Q_{1t})$	S_{2t}	$\bar{H}_{2t}$	R_{2t}	10^6 KWH_{2t}
1	60	150	70.0	60	7.45	50	400	90.0	110	17.57
2	40	150	63.7	80	9.04	30	400	87.9	140	21.84
3	80	110	59.0	70	7.33	60	375	85.0	140	21.12
4	120	120	65.4	90	10.45	90	360	87.2	140	21.66

Annual $KWH = 116.45 \times 10^6 \text{ kwh}$

(b) For LP: Assuming average heads and then adjusting them to the actual ones resulting from solutions, after 2 or 3 iterations, our solution converged to a slightly different but essentially equivalent one found by DP.

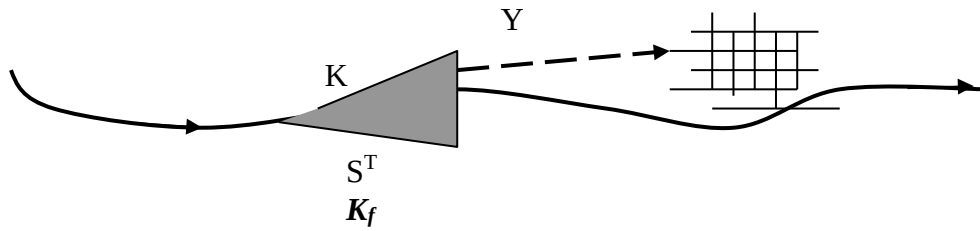
Season t	Q_{1t}	S_{1t}	$\bar{H}_{1t}$	Q_{2t}	10^6 S_{2t}	$(Q_{2t} - Q_{1t})$	R_{1t}	$\bar{H}_{2t}$	R_{2t}	10^6 S_{2t}
1	60	150	70.0	60	7.45	50	400	90.0	110	17.57
2	40	150	62.0	90	9.90	30	400	88.6	140	22.01
3	80	110	57.4	60	6.11	60	375	85.7	140	21.29
4	120	120	65.4	90	10.45	90	360	87.2	140	21.66

Annual $KWH = 116.44 \times 10^6 \text{ kwh}$

11.14 You are responsible for planning a project that may involve the building of a reservoir to provide water supply benefits to a municipality, recreation benefits associated with the water level in the lake behind the dam, and flood damage reduction benefits. First you need to determine some design variable values, and after doing that you need to determine the reservoir operating policy.

The design variables you need to determine include:

- the total reservoir storage capacity (K),
- the flood storage capacity (K_f) in the first season that is the flood season,
- the particular storage level where recreation facilities will be built, called the storage target (S^T) that will apply in seasons 3, 4 and 5 – the recreation seasons, and finally
- the dependable water supply or yield (Y) for the municipality.



Assume you can determine these design variable values based on average flows at the reservoir site in six seasons of a year. These average flows are 35, 42, 15, 3, 15, and 22 in the seasons 1 to 6 respectively.

The objective is to design the system to maximize the total annual net benefits derived from

- flood control in season 1,
- recreation in seasons 3 through 5, and
- water supply in all seasons,

less the annual cost of the

- reservoir and
- any losses resulting from not meeting the recreation storage targets in the recreation seasons.

The flood benefits are estimated to be $2 K_f^{0.7}$.

The recreation benefits for the entire recreation season are estimated to be $8 S^T$.

The water supply benefits for the entire year are estimated to be $20 Y$.

The annual reservoir cost is estimated to be $3 K^{1.2}$.

The recreation loss in each recreation season depends on whether the actual storage volume is lower or higher than the storage target. If it is lower the losses are 12 per unit average deficit in the season, and if they are higher the losses are 4 per unit average excess in the season. It is possible that a season could begin with a deficit and end with an excess, or vice versa.

Develop and solve a non-linear optimization model for finding the values of each of the design variables: K , K_f , S^T , and Y and the maximum annual net benefits. (There will be other variables as well. Just define what you need and put it all together in a model.)

Does the solution give you sufficient information that would allow you to simulate the system using a sequence of inflows to the reservoir that are different than the ones used to get the design variable values? If not how would you define a reservoir operating policy? After determining the system's design variable values using optimization, and then determining the reservoir operating policy, you would then simulate this system over many years to get a better idea of how it might perform.

The optimization model developed for solution by LINGO is:

$$\begin{aligned} \max = & 2*(Kf^{0.7}) + 8*Starget - \\ & 12*(D3/2+D4+D5+D6/2) - 4*(E3+E4+E5+E6/2) - \\ & 3*(K^{1.2}) + 20*Y; \end{aligned}$$

$$\begin{aligned} S1 + Q1 - R1 - Y &= S2; \\ S2 + Q2 - R2 - Y &= S3; \\ S3 + Q3 - R3 - Y &= S4; \\ S4 + Q4 - R4 - Y &= S5; \\ S5 + Q5 - R5 - Y &= S6; \\ S6 + Q6 - R6 - Y &= S1; \end{aligned}$$

$$\begin{aligned} S3 &= Starget - D3 + E3; \\ S4 &= Starget - D4 + E4; \\ S5 &= Starget - D5 + E5; \\ S6 &= Starget - D6 + E6; \end{aligned}$$

$$\begin{aligned} S1 + Kf &< K; \\ S2 + Kf &< K; \\ S3 &< K; \\ S4 &< K; \\ S5 &< K; \\ S6 &< K; \end{aligned}$$

data:
Q1=35;
Q2=42;
Q3=15;
Q4=3;
Q5=15;
Q6=22;
enddata

Variable definitions:

St = storage at the beginning of time period t

Qt = inflow in time period t

Rt = excess release in time period t

Y = yield, constant over all time periods, defined as the minimum of all possible yields

Starget = target storage for recreation,

Dt = deficit with respect to Starget, for t = 3, 4, 5, 6 (the recreation season)

Et = excess with respect to Starget, for t = 3, 4, 5, 6 (the recreation season)

Kf = flood capacity of the reservoir, constant and required in t = 1 and 2

Results of the model:

Objective value: 224

Variable	Value
KF	27.00000
TARGET	42.19228
K	54.19228
Y	15.00000

Kf is 27, which indicates that the benefits from flood protection during the flood season are important compared to other benefits considered in the objective function. Yield = 15, which is guaranteed under these flow conditions (Q1 to Q6). Recreation and Yield appear to be driving the results. Target storage (Starget) is maximized to be equal to total storage, and the maximum deviation from the target is in $t = 5$. Excess are all zero. Releases only occur in periods 5 and 6 in order to decrease storage in period 1 for the flood season.

11.15 Suppose you have 19 years of monthly flow data at a site where a reservoir could be located. How could you construct a model to estimate what the required over-year and within year storage needed to produce a specified annual yield Y that is allocated to each month t by some known fraction δ_t . What would be the maximum reliability of those yields? If you wanted to add to that an additional secondary yield having only 80% reliability, how would the model change? Make up 19 annual flows and assume the average monthly flows are specified fractions of those annual flows. Just using these annual flows and the average monthly fractions, solve your model.

The maximum reliability associated with 19 years is 19/20 or 0.95 or 95 %.

The general model for this problem can be written as follows:

Minimize K_a

Subject to

$$S_y + Q_y - Y - R_y = S_{y+1} \quad \forall y \quad \text{if } y = n, y + 1 = 1$$

$$S_y \leq K_a^o \quad \forall y$$

$$s_t + \beta_t Y - \delta_t Y = s_{t+1} \quad \forall t \quad \text{if } t = T, t + 1 = 1$$

$$s_t \leq K_a^w \quad \forall t$$

$$K_a \geq K_a^o + K_a^w$$

or combining the last two constraints: $K_a \geq s_t + K_a^0$ for all t .

In the model above, δ_t refers to the fraction of the annual yield dedicated to a given month, and β_t refers to the fraction of the annual flow that arrives in a given month. Both variables sum to one over an entire year.

The LINGO code for this model is shown below. Values were generated for the annual flows and the average monthly fractions for yield and flow. The model was run using various values of specified yield and used to generate a plot of the required active storage vs. yield.

SETS:

```
YEARS/1..19/:Q,R;  
YEAR_NUMBERS/1..20/:S;  
MONTHS/1..12/:D,B;  
MONTH_NUMBERS/1..13/:MS;  
ENDSETS
```

```
Min=K;
```

```
!Initial Storage;
```

```
S(1)= S(TN+1);
```

```
@FOR(YEARS(t):
```

```
!Continuity Equations;
```

```
S(t+1)= S(t)+Q(t)-R(t)-Y;
```

```
!Storage Constraint;
```

```
S(t)<= K0;);
```

```
!Initial monthly storage;
```

```
MS(1)= MS(TM+1);
```

```
@FOR(MONTHS(m):
```

```
!Continuity;
```

```
MS(m+1)=MS(m)+B(m)*Y-D(m)*Y;
```

```
!Storage Constraint;
```

```
MS(m)<=KW;);
```

```
K0+KW<=K;
```

Data:

```
TN=19;
```

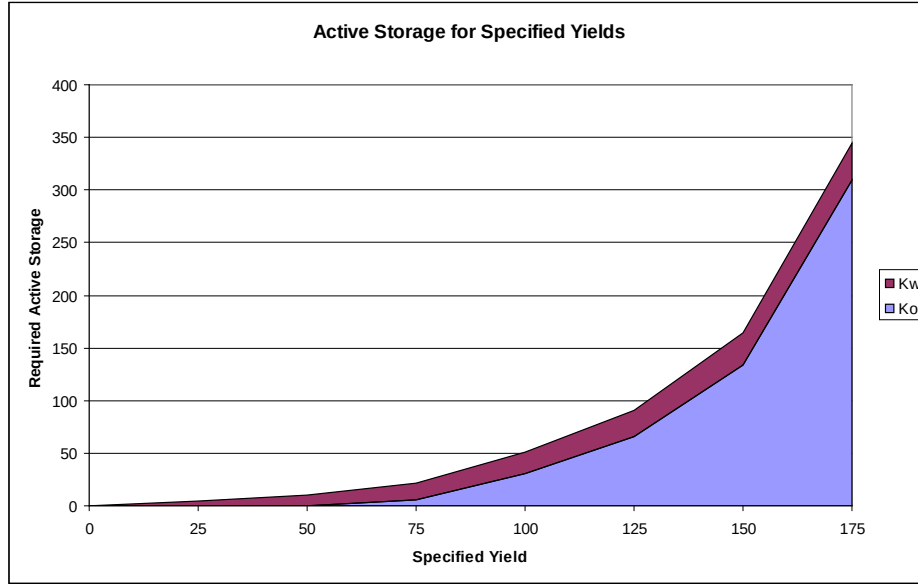
```
TM=12;
```

```
Q=121,235,84,100,224,139,69,191,109,440,146,165,  
108,145,86,149,367,206,340;
```

```
D=0.025,0.025,0.025,0.1,0.1,0.15,0.20,0.20,0.1,
```

0.025, 0.025, 0.025;
 B=0.025, 0.025, 0.1, 0.1, 0.15, 0.15, 0.1, 0.1, 0.1, 0.1,
 0.025, 0.025;
 Y=?;

EndData



If we want to add an additional secondary yield having 80% reliability, the model can be written as follows:

Minimize K_a

Subject to

$$S_y + Q_y - Y - \alpha_y^{0.8} Y_{0.8} - R_y = S_{y+1} \quad \forall y \quad \text{if } y = n, y+1 = 1$$

$$S_y \leq K_a^o \quad \forall y$$

$$s_t + \beta_t(Y + Y_{0.8}) - \delta_t(Y + Y_{0.8}) = s_{t+1} \quad \forall t \quad \text{if } t = T, t+1 = 1$$

$$s_t \leq K_a^w \quad \forall t$$

$$K_a \geq K_a^o + K_a^w$$

Since we have 19 years of data, a yield of 80% reliability will not be met in 3 out of the 19 years. In order to accomplish this, we set the 80% yield variable to zero in the three most critical periods (e.g., with the lowest three-year average flow), which for our given set of flows, is years 13 - 15. We will do this by setting the multiplier $\alpha_y^{0.8}$ to a value of less than 1 (in this problem zero) for those years. The same annual flows and monthly fractions from the previous model were reused in the LINGO model below. Note that we can run this model for many different combinations of primary and secondary yield, but the sum of the two values can never be greater than the average annual inflow, which is about 180 for our data

set. For this problem, let us consider a fixed primary yield of 100 and various secondary yields from 0 to 80.

SETS:

YEARS/1..19/:Q,R,A;
YEAR_NUMBERS/1..20/:S;
MONTHS/1..12/:D,B;
MONTH_NUMBERS/1..13/:MS;
ENDSETS

Min=K;

!Initial Storage;
S(1)=S(TN+1);

@FOR(YEARS(t):
!Continuity Equations;
S(t+1)=S(t)+Q(t)-R(t)-Y-(A(t)*SY);

!Storage Constraint;
S(t)<=K0;);

!Initial monthly storage;
MS(1)=MS(TM+1);

@FOR(MONTHS(m):
!Continuity;
MS(m+1)=MS(m)+(B(m)*(Y+SY))-(D(m)*(Y+SY));

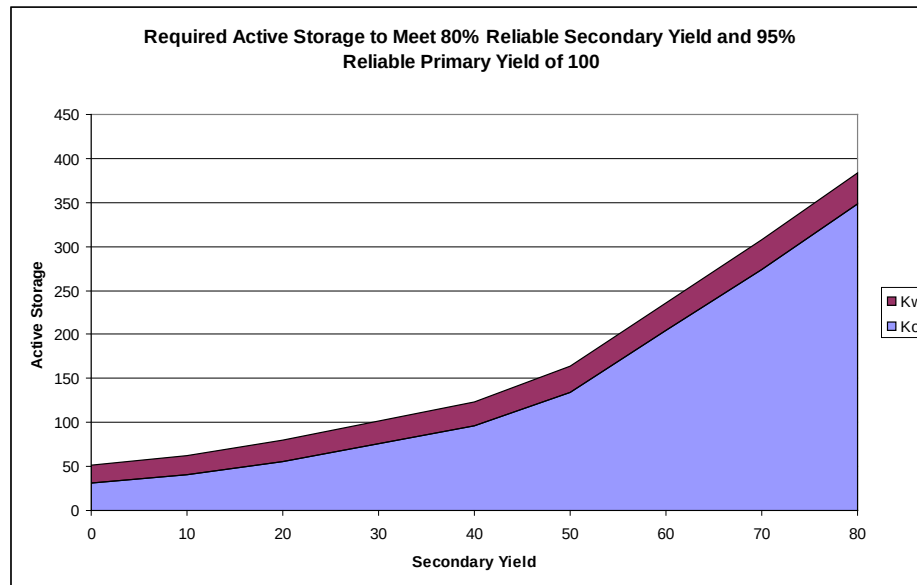
!Storage Constraint;
MS(m)<=KW;);

K0+KW<=K;

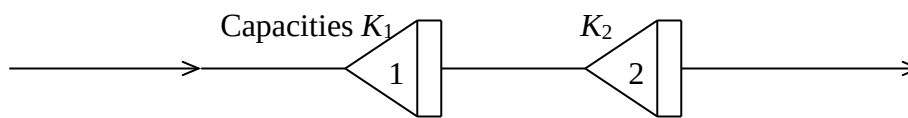
Data:

TN=19;
TM=12;
Q=121,235,84,100,224,139,69,191,109,440,146,165,
108,145,86,149,367,206,340;
D=0.025,0.025,0.025,0.1,0.1,0.15,0.20,0.20,0.1,0.
025,0.025,0.025;
B=0.025,0.025,0.1,0.1,0.15,0.15,0.1,0.1,0.1,0.1,
0.025,0.025;
A=1,1,1,1,1,1,1,1,1,1,1,1,0,0,0,1,1,1,1;
Y=100;
SY=?;
EndData

See the plot below for a graphical display of the storage required to meet various secondary yields in combination with a primary yield of 100. From the results of the two models, we can see that, for example, it takes less storage to meet a primary yield of 100 and a 70% reliable secondary yield of 80 ($K=384$), than it does to meet only a primary yield of 180 ($K=405$). Therefore, using a combination of yields with varying reliabilities can result in cost savings when compared to using only one annual yield.



11.16



a) Develop an optimization model for estimating the least-cost combination of active storage K^1 and K^2 capacities at two reservoir sites on a single stream that are used to produce a reliable flow or yield downstream of the downstream reservoir. Assume 10 years of monthly flow data at each reservoir site. Identify what other data are needed.

b) Describe the two-reservoir operating policy that could be incorporated into a simulation model to check the solution obtained from the optimization model

Define $C^s(K^s)$ = cost of active storage capacity at site s ; where $s = 1, 2$
 K_d^s = dead storage capacity of reservoir at site s ; $K_d^s = 0$

S_t^s	= storage volume at beginning of period t at site s.
L^s	= loss of water due to evaporation at site s; $L^s = 0$
R_t^{12}	= release from reservoir at site 1 to site 2 in period t
Y_t	= yield to downstream in period t
Q_t^s	= 10 years of monthly natural flows available at each site s
a_o^s	= area associated with dead storage volume at site s
a^s	= area per unit storage volume at site s
e_t^s	= evaporation depth in period t at site s

Model:

$$\text{Min: } C^1(K^1) + C^2(K^2)$$

S.T:

Site 1

$$S_t^1 + Q_t^1 - R_t^{12} - L_t^1 = S_{t+1}^1 \quad \forall t$$

$$S_t^1 + K_d^1 \leq K^1 \quad \forall t$$

$$L_t^1 = a_o^1 e_t^1 + a^1 e_t^1 (S_{t+1}^1 + S_t^1) 0.5 \quad \forall t$$

Site 2

$$S_t^2 + R_t^{12} + Q_t^2 - Q_t^2 - L_t^2 - Y_t \geq S_{t+1}^2 \quad \forall t$$

$$S_t^2 + K_d^2 \leq K^2 \quad \forall t$$

$$L_t^2 = a_o^2 e_t^2 + a^2 e_t^2 (S_{t+1}^2 + S_t^2) 0.5 \quad \forall t$$

All variables are nonnegative

(b) Release yield from downstream reservoir and if needed release flow from upstream reservoir to make up any deficit. Otherwise let the upstream reservoir fill and spill, and release only more than the yield from the downstream reservoir if necessary due to no more storage capacity.

11.17 Given inflows to an effluent storage lagoon that can be described by a simple first-order Markov chain in each of T periods t , and an operating policy that defines the lagoon discharge as a function of the initial volume and inflow, indicate how you would estimate the probability distribution of lagoon storage volumes.

Let the subscripts i and j represent discrete streamflow intervals, in period t and $t+1$ respectively.

Let the subscripts k and l represent respective initial storage volumes in periods t and $t+1$.

Let the subscript t be the time period.

Assume we have the Markov chain of streamflows:

$CP(i,j,t)$ = the known conditional probability of j in $t+1$ given i in period t .

$JP(k,i,t)$ = the unknown joint probability of k and i in period t .
 $l(k,i,t)$ = the known operating policy function for defining l in period $t+1$ based on k and i in period t .

Define for all values of l and j :

$JP(l,j,t+1) = \sum_{l(k,i,t)} JP(k,i,t) CP(i,j,t)$ which is the sum of all combinations of k and i that result in a final storage volume l in period t , or equivalently the initial storage volume in $t+1$. This equation results in joint probabilities of l and j in period $t+1$.

Define the probabilities of storage for each k for each period t .

$$\sum_i JP(k,i,t) = SP(k,t)$$

The steady state probabilities of each streamflow i in each period t can also be defined if of interest.

$$\sum_k JP(k,i,t) = QP(i,t)$$

All $SP(k,t)$ must sum to 1 in each period t , or all $QP(i,t)$ must sum to 1 in each period t .

$$\sum_i QP(i,t) = 1 \quad \text{or} \quad \sum_k SP(k,t) = 1 \quad \text{for each } t.$$

Thus the probability of storage volumes represented by the subscripts k in each period t are the values of $SP(k,t)$. Notice you don't even have to know what those actual volumes are to determine their probabilities.

11.18 (a) Using the inflow data in the table below, develop and solve a yield model for estimating the storage capacity of a single reservoir required to produce a yield of 1.5 that is 90% reliable in both of the two within-year periods t , and an additional yield of 1.0 that is 70% reliable in period $t = 2$.

(b) Construct a reservoir-operating rule that defines reservoir release zones for these yields.

(c) Using the operating rule, simulate the 18 periods of inflow data to evaluate the adequacy of the reservoir capacity and storage zones for delivering the required yields and their reliabilities. (Note that in this simulation of the historical record the 90% reliable yield should be satisfied in all the 18 periods, and the incremental 70% reliable yield should fail only two times in the 9 years.)

(c) Compare the estimated reservoir capacity with that which is needed using the sequent peak procedure.

(d)

Year	Period	Inflow
1	1	1.0
	2	3.0
2	1	0.5
	2	2.5
3	1	1.0
	2	2.0
4	1	0.5
	2	1.5
5	1	0.5
	2	0.5
6	1	0.5
	2	2.5
7	1	1.0
	2	5.0
8	1	2.5
	2	5.5
9	1	1.5
	2	4.5

The deterministic model for this problem can be written as follows:

Minimize K

Subject to

$$S_{2,y} = S_{1,y} + Q_{1,y} - R_{1,y} - Y_{0.9} \quad \forall y \quad \text{for period 1.}$$

$$S_{1,y+1} = S_{2,y} + Q_{2,y} - R_{2,y} - Y_{0.9} - \alpha_y^{0.7} Y_{0.7} \quad \forall y \quad \text{if } y = n, y + 1 = 1 \\ \text{for period 2.}$$

$$S_{1,y} \leq K \quad \forall y$$

$$S_{2,y} \leq K \quad \forall y$$

The LINGO code for this deterministic model is located below. The model solution shows 6.5 as the estimated capacity needed to meet the specified yields at the required levels of reliability.

SETS:

YEARS/1..9/:Q1,Q2,R1,R2,A,S2;

NUMBERS/1..10/:S1;

ENDSETS

Min=K;

!Initial Storage;

S1(1)=S2(TN);

@SUM(YEARS: A)=7;

@FOR(YEARS(t):


```

!Continuity Equations;
S2(t)=S1(t)+Q1(t)-R1(t)-Y;
S1(t+1)=S2(t)+Q2(t)-R2(t)-Y-A(t)*SY;

!Constraint on A;
A(t)<=1;

!Storage Constraints;
S1(t)<=K;
S2(t)<=K);
Data:
TN=9;
Y=1.5;
SY=1;
Q1=1, 0.5, 1, 0.5, 0.5, 0.5, 1, 2.5, 1.5;
Q2=3, 2.5, 2, 1.5, 0.5, 2.5, 5, 5.5, 4.5;
EndData

```

Note: For this solution to be valid, the non-zero values of A must be 1.

For comparison, we can also consider the approximate over-year/within-year model, which can be written as follows:

$$\begin{aligned}
& \text{Minimize } K_a \\
& \text{Subject to} \\
& S_y + Q_y - Y_{0.9} - \alpha_y^{0.7} Y_{0.7} - R_y = S_{y+1} \quad \forall y \quad \text{if } y = n, y+1 = 1 \\
& S_y \leq K_a^o \quad \forall y \\
& s_t + \beta_t (Y + Y_{0.7}) - \delta_t (Y + Y_{0.7}) = s_{t+1} \quad \forall t \quad \text{if } t = T, t+1 = 1 \\
& s_t \leq K_a^w \quad \forall t \\
& K_a \geq K_a^o + K_a^w
\end{aligned}$$

In this model, the value for β_t is calculated as the ratio of the average inflow in period t divided by the average annual inflow. Likewise, the value for δ_t is calculated as the ratio of the yield requested in period t divided by the total annual yield. The value for $Y_{0.9}$ is 3, since a yield of 1.5 will be required in both of the two periods and the value for $Y_{0.7}$ will be 1. See the LINGO code for this model below:

```

SETS:
YEARS/1..9/:Q,R,A;
YEAR_NUMBERS/1..10/:S;
PERIODS/1..2/:D,B;
PERIOD_NUMBERS/1..3/:PS;
ENDSETS

```

```

Min=K;

!Initial Storage;
S(1)=S(TN+1);

@FOR(YEARS(y):
!Continuity Equations, PY is primary yield, SY is
secondary yield;
S(y+1)=S(y)+Q(y)-R(y)-PY-(A(y)*SY);

A(y)<=1;

!Storage Constraint;
S(y)<=K0;);

!Reliability Constraint;
@SUM(YEARS: A)=7;

!Initial monthly storage;
PS(1)= PS(TP+1);

@FOR(PERIODS(t):
!Continuity;
PS(t+1)=PS(t)+(B(t)*(PY+SY))-(D(t)*(PY+SY));

!Storage Constraint;
PS(t)<=KW;);

K0+KW<=K;

Data:
TN=9;
TP=2;
Q=4, 3, 3, 2, 1, 3, 6, 8, 6;
D=0.625, 0.375;
B=0.266, 0.734;
PY=3;
SY=1;
EndData

```

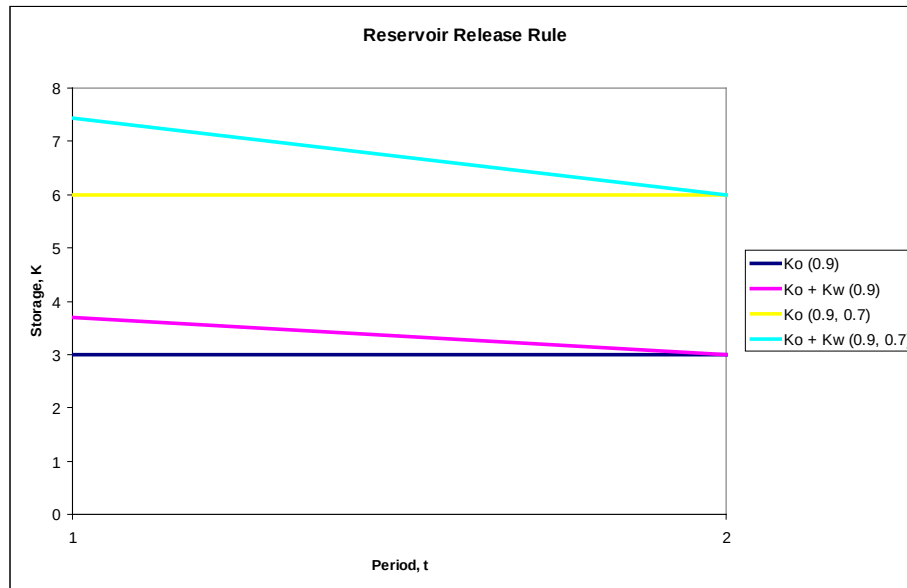
Solution of the model estimates that the required storage to meet the requested yields at the specified reliabilities is 7.4, which exceeds the estimate from the deterministic model by 0.9, or 14%.

(b) In order to construct the operating rule, we need to run the over-year/within-year model twice: once with only the primary yield, and once with both yields.

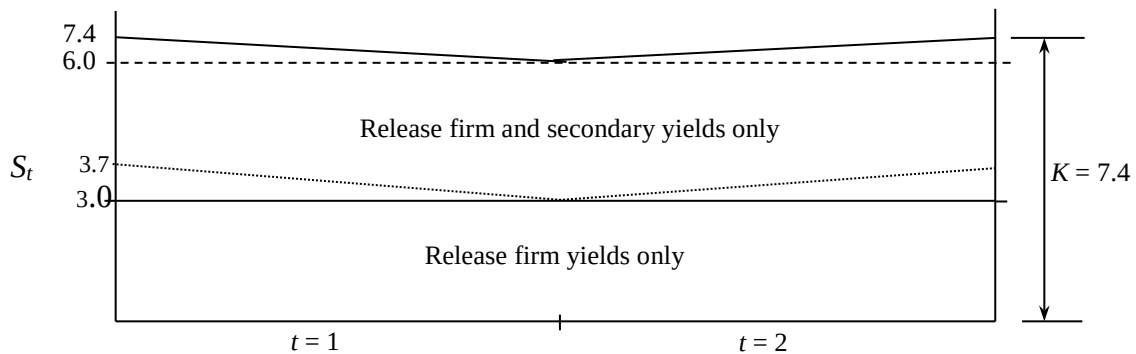
One should note that the values for δ_t need to be recalculated when we add or remove the secondary yield. See the table on the following page, which shows the required over-year and within year storages corresponding to the two different yield scenarios.

Period	$Y_{0.9}$			$Y_{0.9} + Y_{0.7}$		
	K_o	K_w	$K_o + K_w$	K_o	K_w	$K_o + K_w$
1	3	0.7	3.7	6	1.44	7.44
2	3	0	3	6	0	6

See the plot below for a graphical representation of the table above.



This plot could be a rule curve by including the entire period 2 – ending where period 1 begins. In this case period 2 is just the mirror image of the above plot for period 1.



(c) Using the operating rule, 18 periods of inflow data were simulated to evaluate the adequacy of the reservoir capacity and storage zones for delivering the

required yields and their reliabilities. (Note that in this simulation of the historical record the 90% reliable yield should be satisfied in all the 18 periods, and the incremental 70% reliable yield should fail only two times in the 9 years. A more realistic simulation would result from simulating multiple sequences of hydrologic data whose averages are as used in this model.)

The table below contains the results of a simulation using a capacity of 3.7, when considering the primary yield of 1.5 only.

Year	S1	Q1	Y1	R1	S2	Q2	Y2	R2
1	3.7	1	1.5	0	3.2	3	1.5	1
2	3.7	0.5	1.5	0	2.7	2.5	1.5	0
3	3.7	1	1.5	0	3.2	2	1.5	0
4	3.7	0.5	1.5	0	2.7	1.5	1.5	0
5	2.7	0.5	1.5	0	1.7	0.5	1.5	0
6	0.7	0.5	1.2	0	0	2.5	1.5	0
7	1	1	1.5	0	0.5	5	1.5	0.3
8	3.7	2.5	1.5	1	3.7	5.5	1.5	4
9	3.7	1.5	1.5	0	3.7	4.5	1.5	3
10	3.7							

We can see from the table that in period 1 of year 6, that the reservoir could only release a yield of 1.2. The yield of 1.5 should have been reliable in all periods and all years. However, the over-year within year model is an approximation of the true model. If we use a storage value of 4, calculated from the deterministic model, the yield is reliable in all periods of all years. See the results in the table below.

Year	S1	Q1	Y1	R1	S2	Q2	Y2	R2
1	3.7	1	1.5	0	3.2	3	1.5	0.7
2	4	0.5	1.5	0	3	2.5	1.5	0
3	4	1	1.5	0	3.5	2	1.5	0
4	4	0.5	1.5	0	3	1.5	1.5	0
5	3	0.5	1.5	0	2	0.5	1.5	0
6	1	0.5	1.5	0	0	2.5	1.5	0
7	1	1	1.5	0	0.5	5	1.5	0
8	4	2.5	1.5	1	4	5.5	1.5	4
9	4	1.5	1.5	0	4	4.5	1.5	3
10	4							

Next, the reservoir was simulated using both yields. For this case, the approximate model calculated a storage capacity of 7.4. See the results in the table below. Note that the primary yield was released in all periods and that the secondary yield for period 2 was not released in periods 2 and 5.

Year	S1	Q1	Y1	R1	S2	Q2	Y2	R2
1	7.4	1	1.5	0	6.9	3	2.5	0
2	7.4	0.5	1.5	0	6.4	2.5	1.5	0
3	7.4	1	1.5	0	6.9	2	2.5	0
4	6.4	0.5	1.5	0	5.4	1.5	2.5	0
5	4.4	0.5	1.5	0	3.4	0.5	1.5	0
6	2.4	0.5	1.5	0	1.4	2.5	2.5	0
7	1.4	1	1.5	0	0.9	5	2.5	0
8	3.4	2.5	1.5	0	4.4	5.5	2.5	0
9	7.4	1.5	1.5	0	7.4	4.5	2.5	2
10	7.4							

The deterministic model calculated a required storage capacity of 6.5. If we simulate the reservoir with this smaller capacity, the reservoir is able to meet primary yield in all periods and again fails to meet the secondary yield in years 2 and 5 only. See the results in the table below. Note that all of these simulations end with a storage volume at least equal to the initial storage volume, a requirement for a valid simulation. Again, a stochastic simulation of other than the inflow data used in this model would produce more information.

Year	S1	Q1	Y1	R1	S2	Q2	Y2	R2
1	6.5	1	1.5	0	6	3	2.5	0
2	6.5	0.5	1.5	0	5.5	2.5	1.5	0
3	6.5	1	1.5	0	6	2	2.5	0
4	5.5	0.5	1.5	0	4.5	1.5	2.5	0
5	3.5	0.5	1.5	0	2.5	0.5	1.5	0
6	1.5	0.5	1.5	0	0.5	2.5	2.5	0
7	0.5	1	1.5	0	0	5	2.5	0
8	2.5	2.5	1.5	0	3.5	5.5	2.5	0
9	6.5	1.5	1.5	0	6.5	4.5	2.5	2
10	6.5							

(d) Comparing the estimated reservoir capacity with that which is needed using the sequent peak procedure, see a table showing the sequent peak method estimation below. The estimate from the sequent peak method of 6.5 is the same as the estimate obtained using the deterministic method, as indeed it should. As mentioned previously, the estimate using the approximate model was determined to be 7.4.

Year	Season	Flow	Yield	K(t-1)	Kt
1	1	1	1.5	0.0	0.5
	2	3	2.5	0.5	0.0
2	1	0.5	1.5	0.0	1.0
	2	2.5	2.5	1.0	1.0
3	1	1	1.5	1.0	1.5
	2	2	2.5	1.5	2.0
4	1	0.5	1.5	2.0	3.0
	2	1.5	1.5	3.0	3.0
5	1	0.5	1.5	3.0	4.0
	2	0.5	1.5	4.0	5.0
6	1	0.5	1.5	5.0	6.0
	2	2.5	2.5	6.0	6.0
7	1	1	1.5	6.0	6.5
	2	5	2.5	6.5	4.0
8	1	2.5	1.5	4.0	3.0
	2	5.5	2.5	3.0	0.0
9	1	1.5	1.5	0.0	0.0
	2	4.5	2.5	0.0	0.0
1	1	1	1.5	0.0	0.5
	2	3	2.5	0.5	0.0
2	1	0.5	1.5	0.0	1.0
	2	2.5	2.5	1.0	1.0
3	1	1	1.5	1.0	1.5
	2	2	2.5	1.5	2.0
4	1	0.5	1.5	2.0	3.0
	2	1.5	1.5	3.0	3.0
5	1	0.5	1.5	3.0	4.0
	2	0.5	1.5	4.0	5.0
6	1	0.5	1.5	5.0	6.0
	2	2.5	2.5	6.0	6.0
7	1	1	1.5	6.0	6.5
	2	5	2.5	6.5	4.0
8	1	2.5	1.5	4.0	3.0
	2	5.5	2.5	3.0	0.0
9	1	1.5	1.5	0.0	0.0
	2	4.5	2.5	0.0	0.0

11.19 One possible modification of the yield model of would permit the solution algorithm to determine the appropriate failure years associated with any desired reliability instead of having to choose these years prior to model solution. This modification can provide an estimate of the extent of yield failure in each failure year and include the economic consequences of failures in the objective function. It can also serve

as a means of estimating the optimal reliability with respect to economic benefits and losses. Letting F_y be the unknown yield reduction in a possible failure year y , then in place of $\alpha_{py}Y_p$ in the over-year continuity constraint, the term $(Y_p - F_y)$ can be used. What additional constraints are needed to ensure (1) that the average shortage does not exceed $(1 - \alpha_{py})Y_p$ or (2) that at most there are f failure years and none of the shortages exceed $(1 - \alpha_{py})Y_p$.

Define: F as set of possible failure years that contains n years

F_y as the unknown yield reduction in a failure year, that could be included in an objective function with an associated economic loss

f as the maximum number of failure years permitted, if applicable ($f \leq n$).

1) The constraints

$$\frac{1}{f} \sum_{y \in F} F_y \leq (1 - \alpha_{py})Y_p \quad \text{for all } y \in F$$

together with the over-year constraints

$$S_y + Q_y - (Y_p - F_y) - R_y = S_{y+1} \quad \forall y \in F$$

would insure that the average shortage would not exceed any maximum allowable shortage in any year $y \in F$. If there are more failure years than f , reduce the smallest F_y that is greater than 0 to 0, and repeat until the number of $F_y > 0$ equals f (or use integer programming).

The constraint $F_y \leq (1 - \alpha_{py})Y_p$ together with the over-year constraints as written above, and constraining the number of failure years to be no greater than f again by reducing each smallest $F_y > 0$, in excess of f , to be 0, one at a time, until there are only f $F_y > 0$ (or by using integer programming), will insure that there are at most f failure years and each shortage does not exceed $(1 - \alpha_{py})Y_p$. A mixed integer to insure that the number of failures does not exceed f requires

a) for some large M , that $F_y \leq M z_y$,

b) that $\sum_y z_y \leq f$,

c) that each $z_y \leq 1$ and

d) that z_y is integer $\forall y \in F$.

11.20 In Indonesia there exists a wet season followed by a dry season each year. In one area of Indonesia all farmers within an irrigation district plant and grow rice during the wet season. This crop brings the farmer the largest income per hectare; thus they would all prefer to continue growing rice during the dry season. However, there is insufficient water during the dry season for irrigating all 5000 hectares of available irrigable land for rice production. Assume an available irrigation water supply of $32 \times 10^6 \text{ m}^3$ at the beginning of each dry season, and a minimum requirement of $7000 \text{ m}^3/\text{ha}$ for rice and $1800 \text{ m}^3/\text{ha}$ for the second crop.

- (a) What proportion of the 5000 hectares should the irrigation district manager allocate for rice during the dry season each year, provided that all available hectares must be given sufficient water for rice or the second crop?
- (b) Suppose that crop production functions are available for the two crops, indicating the increase in yield per hectare per m^3 of additional water, up to $10,000 \text{ m}^3/\text{ha}$ for the second crop. Develop a model in which the water allocation per hectare, as well as the hectares allocated to each crop, is to be determined, assuming a specified price or return per unit of yield of each crop. Under what conditions would the solution of this model be the same as in part (a)?

- (a) Let X_R be the hectares of rice to be grown during the dry season and $5000 - X_R$ be the hectares of the second crop. The total amount of water required is

$$7000 X_R + 1800(5000 - X_R) = 32 \times 10^6$$

$$\text{Hence } X_R = [(32 \times 10^6) - ((1.8)(5) \times 10^6)] / [(7 - 1.8) \times 10^3]$$

$$X_R = 4435 \text{ ha of rice}$$

- (b) Let the known parameters:

P_R, P_S equal the price per unit yield of rice and the second crop

Y_R, Y_S equal the yield per hectare given the original allocations of water (7000 and $1800 \text{ m}^3/\text{ha}$)

y_R, y_S equal the additional yield per hectare m^3 of additional water.

Define as the unknown variables

X_R, X_S = the hectares of rice and second crop

W_R, W_S = the additional water allocations per ha

Assuming the objective is to maximize total income, the model can be written:

$$\text{Maximize } P_R(Y_R + y_R, W_R)X_R + P_S(Y_S + y_S, W_S)X_S$$

Subject to:

Water availability

$$X_R (7000 + W_R) + X_S (7000 + W_S) \leq 32 \times 10^6$$

Land availability

$$X_R + X_S \leq 5000$$

Bounds

$$W_R \leq 10,000 - 7000 = 3000$$

$$W_S \leq 2,000 - 1800 = 200$$

The solution of this model will be the same as (a) if it does not pay to increase the water allocations to the higher valued rice, i.e., if $P_{RYR} < P_{SYS}$ and of course $P_{RYR} > P_{SYS}$ (which is assumed in (a)) and $4435 P_{RYR} + (5000 - 4435) P_{SYS} > 0$ or $565 P_{SYS} > 136 P_{RYR}$. If this last condition does not hold, rice will be grown on all the land that can be irrigated with $7000 \text{ m}^3/\text{ha}$, namely 4571 ha, and 429 ha will be idle.

11.21 Along the Nile River in Egypt, irrigation farming is practiced for the production of cotton, maize, rice, sorghum, full and short berseem for animal production, wheat, barley, horsebeans, and winter and summer tomatoes. Cattle and buffalo are also produced, and together with the crops that require labor, water, fertilizer, and land area (feddans). Farm types or management practices are fairly uniform, and hence in any analysis of irrigation policies in this region this distinction need not be made. Given the accompanying data develop a model for determining the tons of crops and numbers of animals to be grown that will maximize (a) net economic benefits based on Egyptian prices, and (b) net economic benefits based on international prices. Identify all variables used in the model.

Known parameters:

C_i	= miscellaneous cost of land preparation per feddan
P_i^E	= Egyptian price per 1000 tons of crop i
P_i^I	= international price per 1000 tons of crop i
v	= value of meat and dairy production per animal
g	= annual labor cost per worker
f^P	= cost of P fertilizer per ton
f^N	= cost of N fertilizer per ton
Y_i	= yield of crop i , tons/feddan
α	= feddans serviced per animal
β	= tons straw equivalent per ton of berseem carryover from winter to summer
r^w	= berseem requirements per animal in winter
s^{wh}	= straw yield from wheat, tons per feddan
s^{ba}	= straw yield from barley, tons per feddan
r^s	= straw requirements per animal in summer
μ_i^N	= N fertilizer required per feddan of crop i
μ_i^P	= P fertilizer required per feddan of crop i
l_{im}	= labor requirements per feddan in month m , man-days
w_{im}	= water requirements per feddan in month m , 1000 m^3
h_{im}	= land requirements per month, fraction (1 = full month)

Required Constraints. (Assume known resource limitations for labor, water, and land):

- (a) Summer and winter fodder (berseem) requirements for the animals.
- (b) Monthly labor limitations.
- (c) Monthly water limitations.
- (d) Land availability each month.
- (e) Minimum number of animals required for cultivation.
- (f) Upper bounds on summer and winter tomatoes (assume these are known).
- (g) Lower bounds on cotton areas (assume this is known).

Other possible constraints:

- (a) Crop balances.
- (b) Fertilizer balances.
- (c) Labor balance.
- (d) Land balance.

This model is a simplification of one developed by Dr. G. Kutcher and his colleagues at the World Bank and in Egypt for studying the agricultural economy of an area of the Nile valley.

Maximize total net income based on Egyptian or International prices (P_i^E or P_i^I)

$$\text{Maximize } \sum_i P_i Q_i + vA - TC$$

Crop income + animal Income - total cost

Where Q_i is tons of crop i produced and A is the number of animals produced.

Subject to:

Definition of total cost

$$TC = \sum_i C_i X_i + gL + f^P F^P + f^N F^N$$

$$\left[\begin{array}{c} \text{Total} \\ \text{cost} \end{array} \right] = \left[\begin{array}{c} \text{Land} \\ \text{Preparation} \\ \text{cost} \end{array} \right] + \left[\begin{array}{c} \text{labor} \\ \text{cost} \end{array} \right] + \left[\begin{array}{c} \text{Fertilizer} \\ \text{Cost} \end{array} \right]$$

where X_i is the hectares of land in crop i .
 L is the total labor required.
 F^P, F^N is the tons of fertilizer required.

Crop Production

$y_i X_i = Q_i$, the tons of crop i produced, $\forall i$ except berseem

$$y_F X_F + y_S X_S = r^w A + QB$$

$$\left[\begin{array}{c} \text{Quantity of} \\ \text{full and short} \\ \text{berseem produced} \end{array} \right] = \left[\begin{array}{c} \text{Animal requirements} \\ \text{for berseem} \\ \text{in winter} \end{array} \right] + \left[\begin{array}{c} \text{Straw} \\ \text{Requirement} \\ \text{For animals} \end{array} \right]$$

summer fodder (berseem) requirements

$$s^{wh} X_W + s^{ba} X_B \geq \beta QB + r^s A$$

$$\left[\begin{array}{c} \text{Straw yield} \\ \text{from wheat} \end{array} \right] + \left[\begin{array}{c} \text{Straw yield} \\ \text{from barley} \end{array} \right] \geq \left[\begin{array}{c} \text{equivalent} \\ \text{straw carryover} \\ \text{from winter} \end{array} \right] + \left[\begin{array}{c} \text{Straw} \\ \text{Requirement} \\ \text{for animals} \end{array} \right]$$

Fertilizer balances

$$\sum_f \mu_i^N X_i = F^N$$

$$\sum_f \mu_i^P X_i = F^P$$

Labor Constraints

$$\sum_i l_{im} X_i \leq L_m \quad \forall m$$

$$\left[\begin{array}{c} \text{Labor requirements} \\ \text{Per month} \end{array} \right] \leq \left[\begin{array}{c} \text{labor} \\ \text{supply} \end{array} \right]$$

$$\sum_i \sum_m l_{im} X_i = 240 L$$

$$\left[\begin{array}{c} \text{Labor requirements} \\ \text{in year} \end{array} \right] = \left[\begin{array}{c} \text{Man-days} \\ \text{Per year} \end{array} \right] \left[\begin{array}{c} \text{Number of} \\ \text{workers} \end{array} \right]$$

Water Availability

$$\sum_i w_{im} X_i \leq w_m \quad \forall m$$

$$\left[\begin{array}{c} \text{Water} \\ \text{requirements} \end{array} \right] \leq \left[\begin{array}{c} \text{water} \\ \text{supply} \end{array} \right]$$

Land Availability

$$\sum_i h_{im} X_i \leq \text{Land} \quad \forall m$$

$$\left[\begin{array}{c} \text{Land requirements} \\ \text{in month } m \end{array} \right] \leq \left[\begin{array}{c} \text{Land} \\ \text{available} \end{array} \right]$$

Animal Requirements

$$A \leq \frac{1}{\alpha} \sum_i X_i$$

$$\left[\begin{array}{c} \text{Number of} \\ \text{animals} \end{array} \right] \leq \left[\begin{array}{c} \text{Animal needs} \\ \text{Per feddans} \end{array} \right] \left[\begin{array}{c} \text{Number of} \\ \text{feddans} \end{array} \right]$$

Bounds

$$X_i^{\min} \leq X_i \quad \text{as required for cotton}$$

$$y_i X_i \leq y_i X_i^{\min} \quad \text{as required for tomatoes}$$

11.22 In Algeria there are two distinct cropping intensities, depending upon the availability of water. Consider a single crop that can be grown under intensive rotation or extensive rotation on a total of A hectares. Assume that the annual water requirements for the intensive rotation policy are 16000 m^3 per hectare, and for the extensive rotation policy they are 4000 m^3 per hectare. The annual net production returns are 4000 and 2000 dinars, respectively. If the total water available is $320,000 \text{ m}^3$, show that as the available land area A increases, the rotation policy that maximizes total net income changes from one that is totally intensive to one that is increasingly extensive.

Would the same conclusion hold if instead of fixed net incomes of 4000 and 2000 dinars per hectares of intensive and extensive rotation, the net income depended on the quantity of crop produced? Assuming that intensive rotation produces twice as much produced by extensive rotation, and that the net income per unit of crop Y is defined by the simple linear function $5 - 0.05Y$, develop and solve a linear programming model to determine the optimal rotation policies if A equals 20, 50, and 80. Need this net income or price function be linear to be included in a linear programming model?

Let X_I be the hectares of intensive cropping intensity and X_E be the hectares of extensive cropping intensity. Maximizing net annual production:

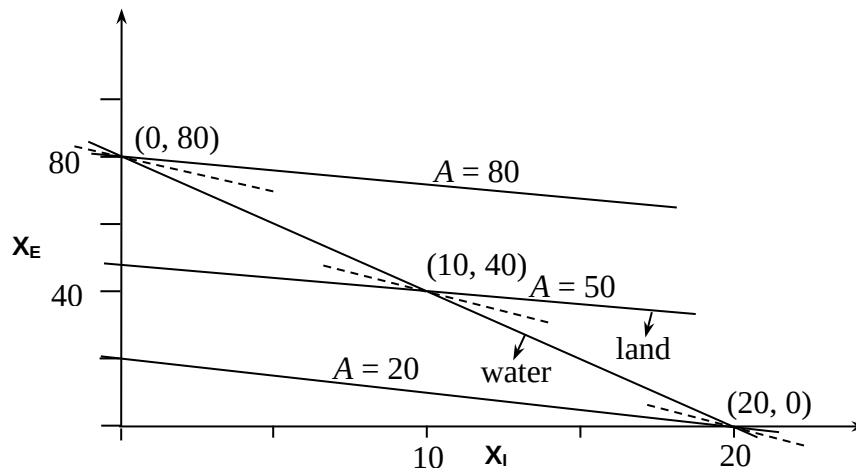
$$\text{Maximize} \quad 4000 X_I + 2000 X_E$$

$$\text{Subject to} \quad 16 X_I + 4 X_E \leq 320$$

$$X_I + X_E \leq A$$

$$X_I, X_E \geq 0$$

Solving graphically:



This graph shows that as A increases, the optimum value of X_I decreases and X_E increases.

If the price times yield per hectare ratio $P_I Y_I / P_E Y_E$ were greater than 4, then for any value of $A > 0$ all available land would be allocated to intensive irrigation up to the limit possible with the available water (in this case up to 20 hectares).

Assuming the yield per hectare of extensive rotation is Y_E then to maximize net income, one would

$$\text{Maximize} \quad (5 - 0.05Y) Y$$

$$\text{Subject to} \quad Y = Y_E(2X_I + X_E) \quad \text{which defines total yield } Y$$

$$16 X_I + 4 X_E \leq 320 \quad \text{for water}$$

$$X_I + X_E \leq A = \{20, 50, 80\} \quad \text{for land}$$

$$X_I, X_E \geq 0$$

For any value of $Y_E > 0$ and $Y \leq 50$, the value of X_I and X_E will be determined by maximizing Y subject to the water and land and non-negativity constraints. Since net income decreases for $Y > 50$, $2X_I + X_E$ will not exceed $50/Y_E$. If Y can exceed 50 it will pay to keep some land idle and given the data available, it is not clear which of all feasible solutions would be preferred.

Assuming Y_E is such that Y will not exceed 50, the optimum X_I and X_E can be found by the following model:

$$\begin{array}{ll}
 \text{Maximize} & 2X_I + X_E \\
 \text{Subject to} & 4X_I + X_E \leq 80 \\
 & X_I + X_E \leq A = \{20, 50, 80\} \\
 & X_I, X_E \geq 0
 \end{array}$$

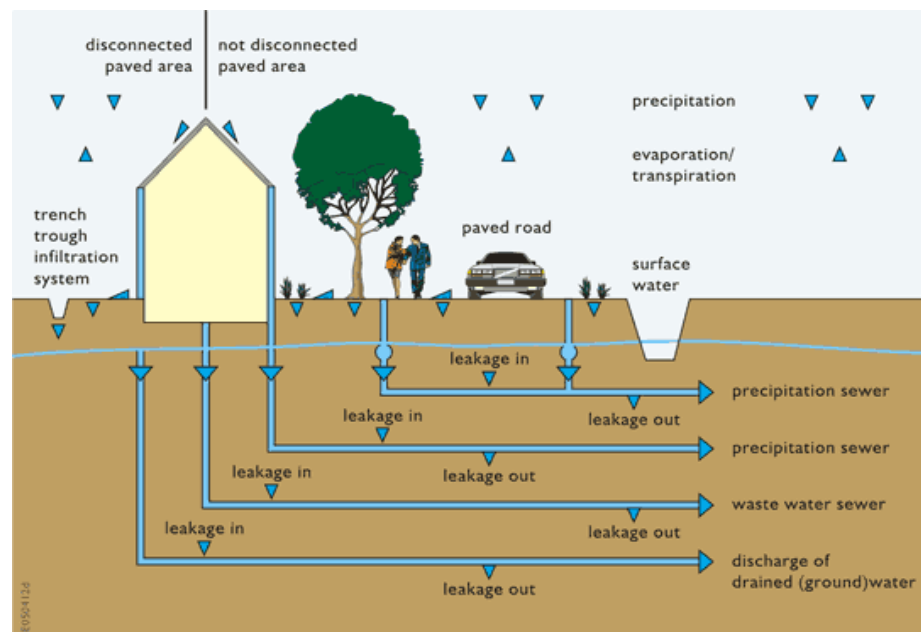
This objective has the same slope as the previous model graphed above hence the solutions (X_I, X_E) for $A = \{20, 50, 80\}$, and $(20, 0)$, $(10, 40)$, $(0, 80)$, respectively.

Chapter 12

12.1 Define the components of the infrastructure needed to bring water into your home and then collect the wastewater and treat it prior to discharging it back into a receiving water body. Draw a schematic of such a system and show how it can be modeled to determine the best design variable values. Define the data needed to model such a system and then make up values of the needed parameters and solve the model of the system.

Generally one would consider surface or ground (and perhaps desalinated) water collection, storage, transport, treatment, and then storage and distribution of treated water to the consumers as one major subsystem. The other major subsystem is the collection of this used water (wastewater – which can be divided into grey water from washing, stormwater runoff, and the like and brown or black water from toilets) in sanitary and stormwater sewers and its treatment and eventual disposal. Recycling and limited reuse is a possible feature as well.

The following figure shows some of the features that might be considered.



Once a network (conceptual model) has been drawn, a mass balance at each 'node' can be written and functions can be defined regarding treatment efficiencies and demands and amount consumed, and pollutants introduced. This will identify some of the data

required to write a simulation or optimization model of the system for examining alternative designs and operating policies.

12.2 Compare the curve number approach to the use of Manning's equation to estimate urban runoff quantities. Then define how you would predict quality and sediment runoff as well.

The SCS curve number method is a method for estimating runoff from a single rainfall event in a particular area. The curve number is based on the area's hydrologic soil group, land use treatment and hydrologic condition.

Mannings's Equation is used to estimate open channel flows based on cross sectional area, velocity, friction, and slope.

12.3 Develop a simple model for predicting the runoff of water, sediment and several chemicals from a 10-ha urban watershed in the northeastern United States during August 1976. Recorded precipitation was as follows:

Day	1	6	7	8	9	10	13	14	15	26	29
P_i (cm)	1.8	0.7	2.6	2.9	0.1	0.3	2.9	0.1	1.4	3.7	0.8

Solids (sediment) buildup on the watershed at the rate of 50 kg/ha-day, and chemical concentrations in the solids are 100 mg/kg. Assume that each runoff event washes the watershed surface clean. Assume also that there is no initial sediment buildup on August. The watershed is 30% impervious. For each storm use your model to compute:

- Runoff in cm and m^3 .
- Sediment loss (kg).
- Chemical loss (g), in dissolved and solid-phase form for chemicals with three different adsorption coefficients, $k = 5, 100, 1000$.

Daily sediment accumulation is $50(10) = 500$ kg. Table 1 below shows the computed depression storage (DS_i), the runoff (Q_i), and the sediment (X) and chemical loss (S_0X) for each precipitation event, and Table 2 shows the distribution of dissolved and solid-phase losses based on the following two equations.

$$s = \left[\frac{kS_0X}{kX + 1000Q} \right] X \quad (mg)$$

$$D = \left[\frac{s_0X}{kX + 1000Q} \right] 1000Q \quad (mg)$$

assuming Q is in m^3 , X is in kg , so is in mg/kg , and k is in l/kg .
Runoff, Q_t in cm , equals the fraction c of rainfall R_t not infiltrated minus the depression storage, DS_t in cm :

$$Q_t = \begin{cases} 0.375R_t - DS_t & \text{if } 3.75 R_t > DS_t \\ 0 & \text{otherwise} \end{cases}$$

where it is assumed that DS_t is defined by its continuity equation above.

$$DS_t = \begin{cases} \max(DS_{t-1} - 0.375R_{t-1}; 0) & R_{t-1} > 0 \\ DS^* = 0.465 \text{ cm} & \text{otherwise} \end{cases}$$

Total Q_t in m^3 is $(Q_t \text{ cm}) [m/(100 \text{ cm})] (100 \text{ m}^2 / \text{ha})(10 \text{ ha}) = (Q_t \text{ cm}) 10^3$
From the 10 ha area. The sediment loss, X in kg , is

$$X = 10^{-1} g/kg (500 \text{ kg/day}) (\text{no. of days}) \\ = (50 \text{ g/day}) (\text{no. of days})$$

Table 1

Day	Rain R_t (cm)	Depression storage DS_t (cm)	Runoff Q_t (cm)	Runoff Q_t (m^3)	Sediment Loss X (Kg)	Chemical Loss, S_0X (g)
1	1.8	0	0.68	680	500	50
6	0.7	0.465	0	0	--	--
7	2.6	0.2025	0.77	770	3000	300
8	2.9	0	1.09	109	500	50
9	0.1	0	0.04	40	500	50
10	0.3	0	0.11	110	500	50
13	2.9	0.465	0.62	620	1500	150
14	0.1	0	0.04	40	500	50
15	1.4	0	0.53	530	500	50
26	3.7	0.465	0.92	920	5500	550
29	0.8	0.465	0	0	1500	0

Table 2

Day	$k = 5$		$k = 10$		$k = 1000$	
	D	S	D	S	D	S
1	50	0	47	3	29	21
7	294	6	216	84	61	239
8	50	0	48	2	34	16
9	47	3	21	29	4	46
10	49	1	35	15	9	41
13	148	2	121	29	44	106
14	47	3	22	28	4	46
15	50	0	46	4	26	24
26	534	16	345	205	79	471
TOTAL	1269	31	901	399	290	1010

12.4 There exists a modest-sized urban subdivision of 100 ha containing 2000 people. Land uses are 60% single-family residential, 10% commercial, and 30% undeveloped. An evaluation of the effects of street cleaning practices on nutrient losses in runoff is required for this catchment.

This evaluation is to be based on the 7-month precipitation record given below. Present the results of the simulations as 7-month PO_4 and N losses as functions of street-cleaning interval and efficiency (i.e., show these losses for ranges of intervals and efficiencies). Assume a runoff threshold for washoff of $Q_0 = 0.5$ cm.

Day	PRECIPITATION (cm)						
	A	M	J	J	A	S	O
1			0.6		1.4	0.7	1.9
2	1.1	0.4				0.5	
3						0.1	0.1
4	0.1	1.5					
5	0.1	0.9			1.9	0.3	
6	0.1		1.4	1.4	1.0		0.5
7			0.1	1.1	0.7		
8	0.1	0.1		0.7			0.4
9		0.6	1.6	0.1			1.5
10					0.1		
11							
12				0.2	0.2		0.2
13	0.1			0.2		1.5	
14			0.2		0.5	3.5	0.8
15							1.0
16						4.3	2.8
17			0.7	0.8	0.5	0.8	1.9
18		0.5	0.4	0.1		0.8	0.1
19			0.4			0.4	0.9
20			0.7	0.3		2.3	
21						0.3	
22	0.1			0.1	0.4		
23	2.0						
24	3.2	0.1			0.2	4.7	
25	0.1		0.6			2.8	
26				3.0		1.6	
27							
28			0.3			0.1	
29		0.2	1.1				
30		0.1			0.6		
31					0.2		

|

This requires a simulation model using some of the equations presented in the previous exercise and referral to some of the loading equations..

Solutions for this and the remaining problems are being obtained, but are not yet available.

12.5 Managing the quantity and quality of stormwater runoff is a common urban problem. Discuss the factors to be considered when planning storm sewer networks and detention basins, and how might simulation and/or optimization methods be used to help do this.

12.6 Multiple connected pipeline networks are commonly make up urban water distribution systems. Define a simple pipeline network and develop an optimization model for finding the flows and heads in the network needed to provide required flow discharges and pressures at various nodes or junctions of the network. Discuss some of the complicating issues associated with the design of such networks that your model may not be considering.

12.7. Consider a wastewater treatment plant and associated effluent detention pond designed to release treated and stored effluent into a stream so as to adapt to varying effluent concentration standards associated with varying stream assimilative capacities as its flow and water temperature and existing pollutant loads vary over time. Develop a model for estimating the treatment plant efficiency for BOD removal, and the size of the detention basin, needed to meet the varying effluent standards of the receiving stream, at a minimum cost. Define the data you will need to do this, and all model parameters. Why might this proposed adaptive treatment scheme not be very practical?

12.8 Urban Infrastructural Asset Management is increasingly becoming a key topic in the move towards increased sustainability of water supply and wastewater systems. List some characteristics of sustainable infrastructural systems.

Sustainable infrastructure refers to the designing, building, operating and maintaining these water and wastewater systems in ways that do not diminish the social, economic and ecological processes required to maintain a functional urban environment and that meet the often changing demands for such services over time.

Is the above statement adequate, or useful, for achieving sustainability of water and wastewater systems?

Entering 'sustainable urban water systems' or similar phrases in Google will identify some current literature on this subject.

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